

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

D8 MLB

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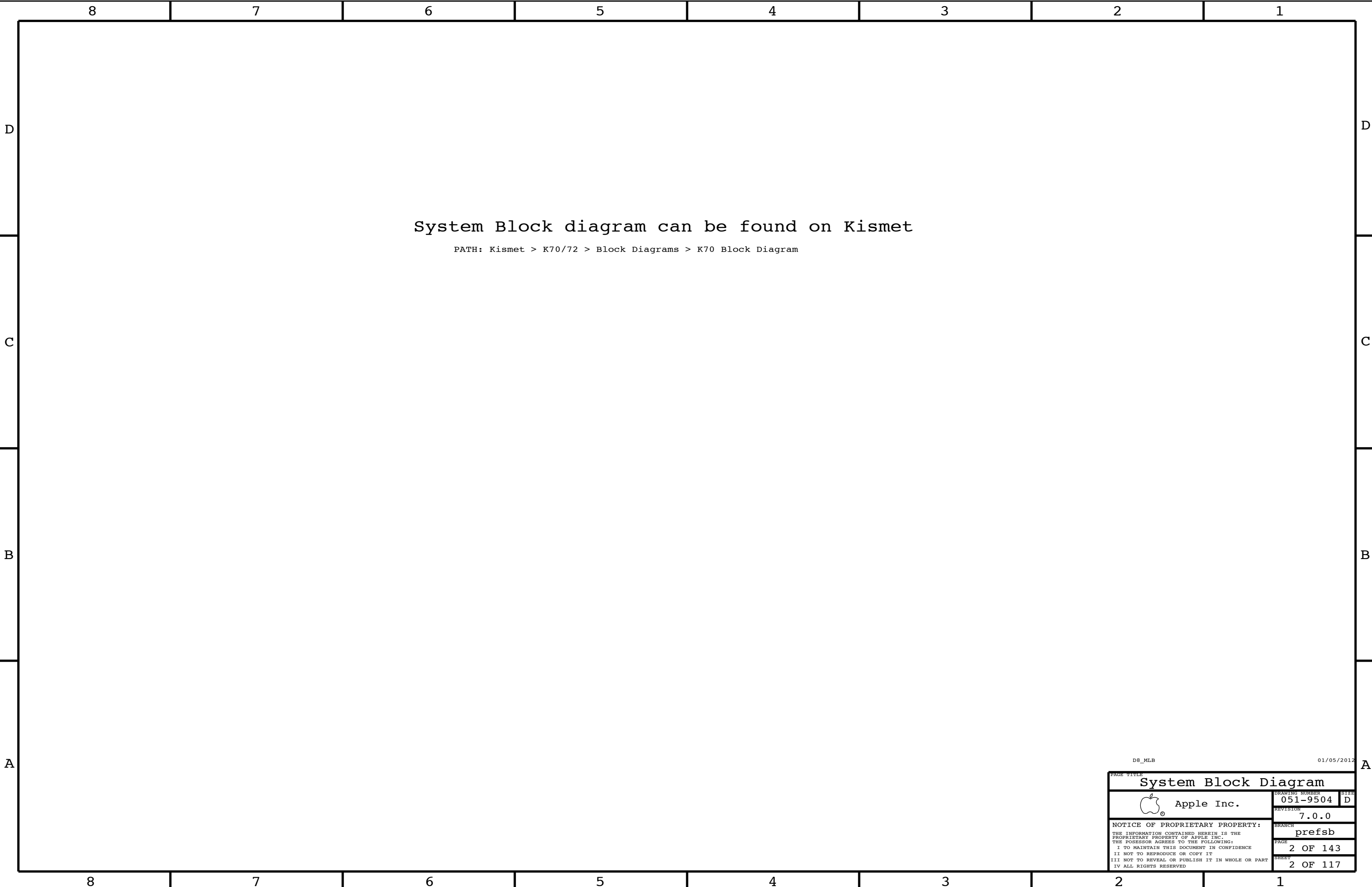
REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
7	0001607307	ENGINEERING RELEASED	2012-08-28

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21	PCH MISC	DB_MLB	N/A
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DB_MLB01/05/2012

System Block Diagram

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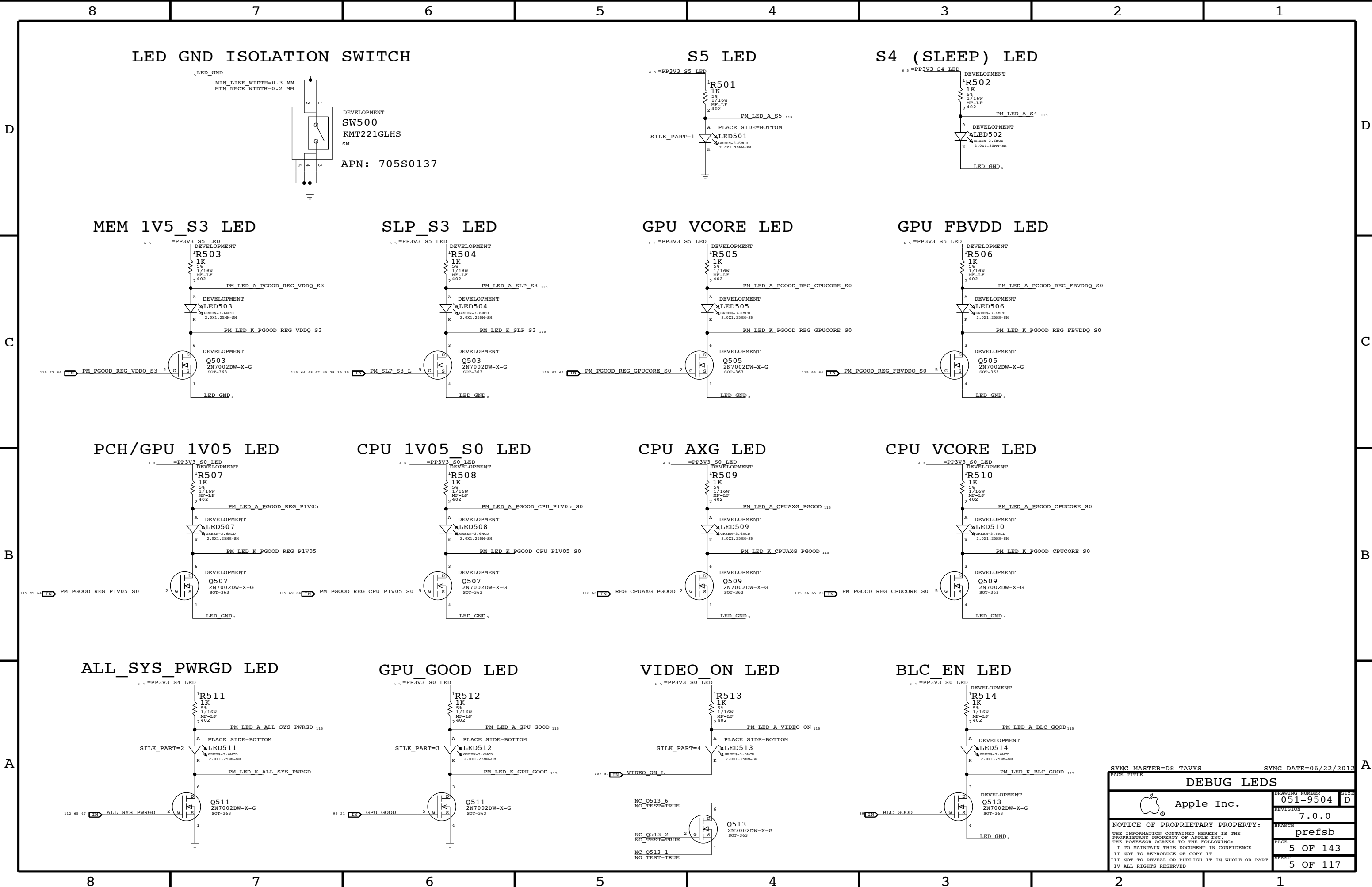
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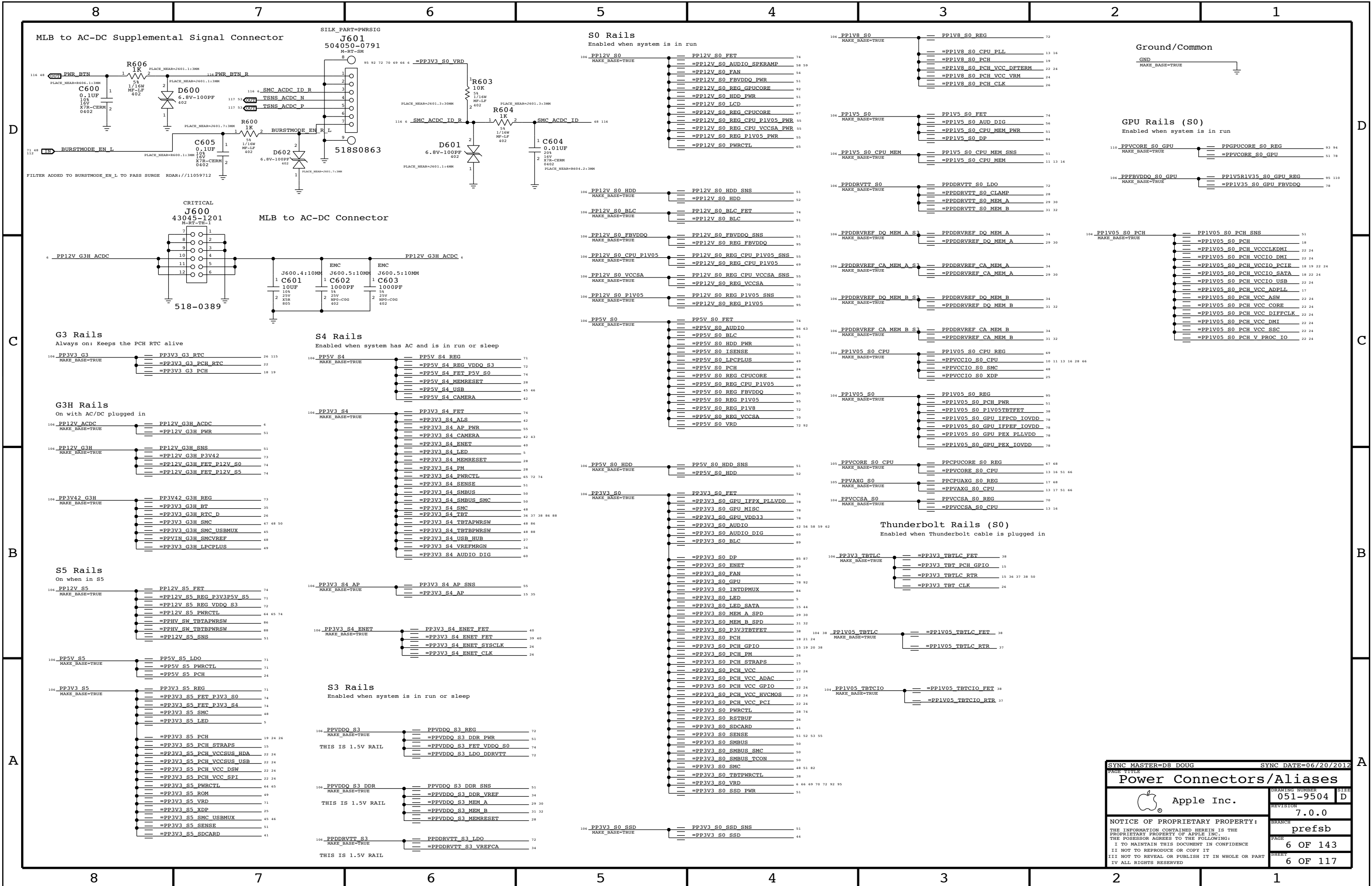
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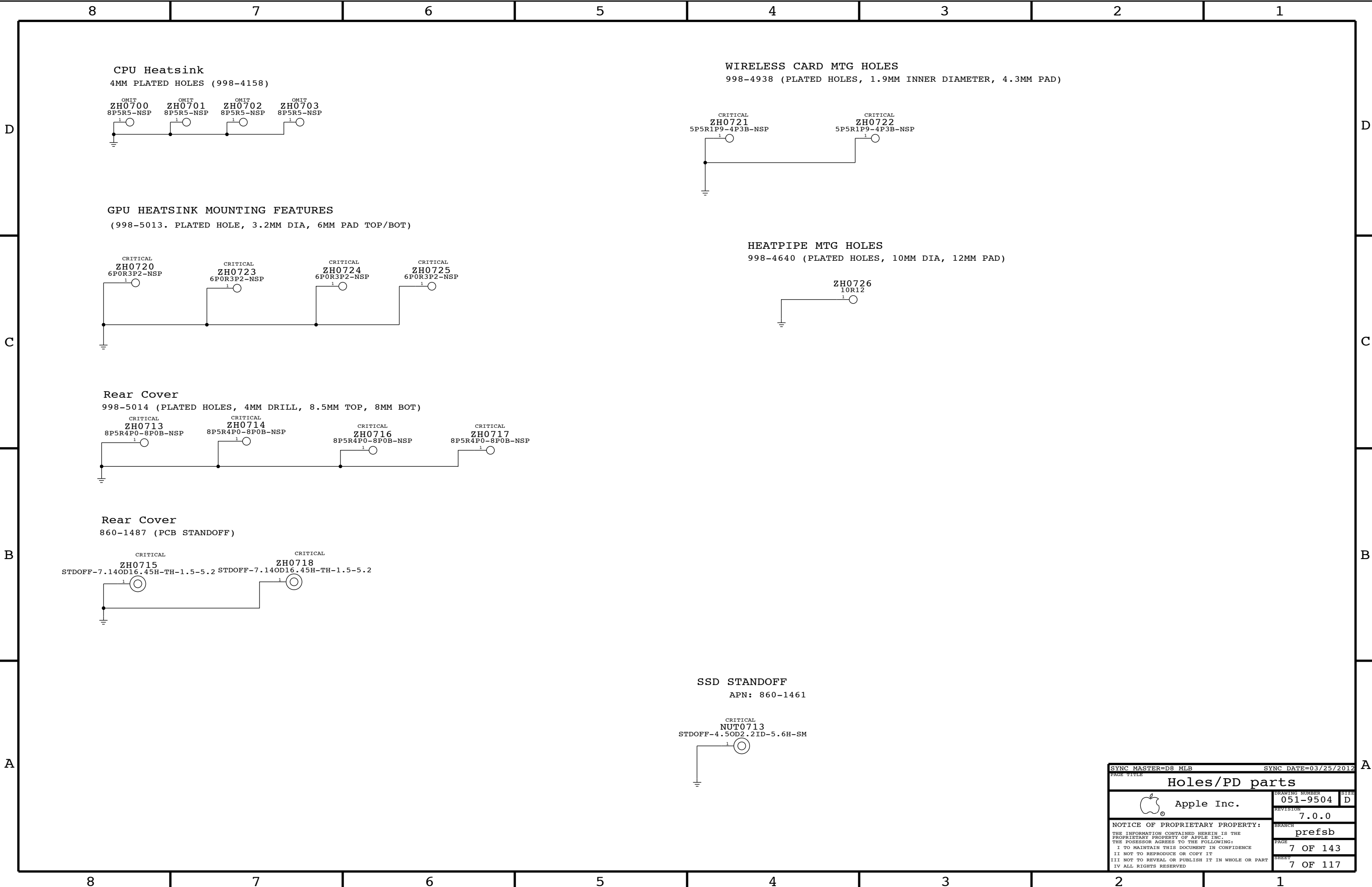
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D	BOM Variants																																																																			
	<table><tr><th>BOM NUMBER</th><th>BOM NAME</th><th>BOM OPTIONS</th></tr><tr><td>639-3816</td><td>PCBA,MLB,2.9G,4C,GK107,SAM,D8</td><td>D8_COMMON,D8,CPU:4C_2P9GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F2FR</td></tr><tr><td>639-3952</td><td>PCBA,MLB,3.1G,4C,GK107,SAM,D8</td><td>D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F49T</td></tr><tr><td>639-4092</td><td>PCBA,MLB,2.9G,4C,GK107,HYN.D8</td><td>D8_COMMON,D8,CPU:4C_2P9GHZ,GPU:107EGE,FB:BOTH_HYNIX,EEEE:F653</td></tr><tr><td>639-4093</td><td>PCBA,MLB,3.1G,4C,GK107,HYN.D8</td><td>D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_HYNIX,EEEE:F654</td></tr><tr><td>085-4433</td><td>PCBA,MLB,DEV,D8</td><td>DEVELOPMENT,D8_DEVEL</td></tr></table>								BOM NUMBER	BOM NAME	BOM OPTIONS	639-3816	PCBA,MLB,2.9G,4C,GK107,SAM,D8	D8_COMMON,D8,CPU:4C_2P9GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F2FR	639-3952	PCBA,MLB,3.1G,4C,GK107,SAM,D8	D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F49T	639-4092	PCBA,MLB,2.9G,4C,GK107,HYN.D8	D8_COMMON,D8,CPU:4C_2P9GHZ,GPU:107EGE,FB:BOTH_HYNIX,EEEE:F653	639-4093	PCBA,MLB,3.1G,4C,GK107,HYN.D8	D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_HYNIX,EEEE:F654	085-4433	PCBA,MLB,DEV,D8	DEVELOPMENT,D8_DEVEL																																										
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	639-3952	PCBA,MLB,3.1G,4C,GK107,SAM,D8	D8_COMMON,D8,CPU:4C_3P1GHZ,GPU:107EGE,FB:BOTH_SAMSUNG,EEEE:F49T																																																																	
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C	Bar Code Labels / EEEE #'s																																																																			
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B	BOM Groups																																																																			
	<table><tr><th>BOM GROUP</th><th>BOM OPTIONS</th></tr><tr><td>D8_COMMON</td><td>COMMON,ALTERNATE,D8_COMMON1,D8_PROGPARTS,D8_PRODUCTION</td></tr><tr><td>D8_COMMON1</td><td>XDP,RSMRST:GATE,SPEAKERID,VREF:CPU,TBTHV:P12V,FBA,FBB</td></tr><tr><td>D8_PROGPARTS</td><td>SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG,BLCMCU:PROG</td></tr><tr><td>D8_DEVEL</td><td>XDP_CONN,LPCPLUS,VREFMRGN:EXT,DEVEL_AUDIO,TEMPSNSDEV</td></tr><tr><td>D8_PRODUCTION</td><td>VREFMRGN:N,PRODUCTION</td></tr></table>								BOM GROUP	BOM OPTIONS	D8_COMMON	COMMON,ALTERNATE,D8_COMMON1,D8_PROGPARTS,D8_PRODUCTION	D8_COMMON1	XDP,RSMRST:GATE,SPEAKERID,VREF:CPU,TBTHV:P12V,FBA,FBB	D8_PROGPARTS	SMC:PROG,BOOTROM:PROG,TBTROM:PROG,CIVROM:PROG,CAMROM:PROG,BLCMCU:PROG	D8_DEVEL	XDP_CONN,LPCPLUS,VREFMRGN:EXT,DEVEL_AUDIO,TEMPSNSDEV	D8_PRODUCTION	VREFMRGN:N,PRODUCTION																																																
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	338S1113	1	IC,TBT,CR-4C,B1,PRO,288 PCBGA,12X12MM	U3600	CRITICAL																																																															
	337S4280	1	IC, GPU, NV GK107-GE-PS-A2	U8000	CRITICAL	GPU:107EGE																																																														
	343S0616	1	IC,BCM57766A1,ENET6SD,8X8	U3900	CRITICAL																																																															
	Programmable Parts																																																																			
	<table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>341S3672</td><td>1</td><td>IC,EEPROM,CR,V14.1 (B1),D8</td><td>U3690</td><td>CRITICAL</td><td>TBTROM:PROG</td></tr><tr><td>335S0865</td><td>1</td><td>IC,EEPROM,SERIAL,8KB,MLP8</td><td>U3690</td><td>CRITICAL</td><td>TBTROM:BLANK</td></tr><tr><td>341S3673</td><td>1</td><td>IC,PROGRMD,EFI ROM,V00FC,D7/D8</td><td>U5110</td><td>CRITICAL</td><td>BOOTROM:PROG</td></tr><tr><td>335S0807</td><td>1</td><td>IC,64 MBIT SPI SERIAL FLASH</td><td>U5110</td><td>CRITICAL</td><td>BOOTROM:BLANK</td></tr><tr><td>341S3394</td><td>1</td><td>IC,PROGRMD,SMC,A3,V2.2A32,D8</td><td>U4900</td><td>CRITICAL</td><td>SMC:PROG</td></tr><tr><td>338S1098</td><td>1</td><td>IC,SMC,LX4FS1AH5BBCIGA3</td><td>U4900</td><td>CRITICAL</td><td>SMC:BLANK</td></tr><tr><td>341S3675</td><td>1</td><td>IC,CAMERA FLASH,V7228,D7/D8</td><td>U4202</td><td>CRITICAL</td><td>CAMROM:PROG</td></tr><tr><td>335S0852</td><td>1</td><td>IC,FLASH,SPI,1MBIT,3V3</td><td>U4202</td><td>CRITICAL</td><td>CAMROM:BLANK</td></tr><tr><td>341S3645</td><td>1</td><td>IC,ENET 1MBIT, SPI,ROM, V1.13 D8</td><td>U3990</td><td>CRITICAL</td><td>CIVROM:PROG</td></tr></table>								PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	341S3672	1	IC,EEPROM,CR,V14.1 (B1),D8	U3690	CRITICAL	TBTROM:PROG	335S0865	1	IC,EEPROM,SERIAL,8KB,MLP8	U3690	CRITICAL	TBTROM:BLANK	341S3673	1	IC,PROGRMD,EFI ROM,V00FC,D7/D8	U5110	CRITICAL	BOOTROM:PROG	335S0807	1	IC,64 MBIT SPI SERIAL FLASH	U5110	CRITICAL	BOOTROM:BLANK	341S3394	1	IC,PROGRMD,SMC,A3,V2.2A32,D8	U4900	CRITICAL	SMC:PROG	338S1098	1	IC,SMC,LX4FS1AH5BBCIGA3	U4900	CRITICAL	SMC:BLANK	341S3675	1	IC,CAMERA FLASH,V7228,D7/D8	U4202	CRITICAL	CAMROM:PROG	335S0852	1	IC,FLASH,SPI,1MBIT,3V3	U4202	CRITICAL	CAMROM:BLANK	341S3645	1	IC,ENET 1MBIT, SPI,ROM, V1.13 D8	U3990	CRITICAL	CIVROM:PROG
	PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION																																																														
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	335S0807	1	IC,64 MBIT SPI SERIAL FLASH	U5110	CRITICAL	BOOTROM:BLANK																																																														
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SYNC MASTER=K70 MLB		SYNC DATE=08/23/2011	
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Signal Aliases			
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SYNC DATE=08/23/2011

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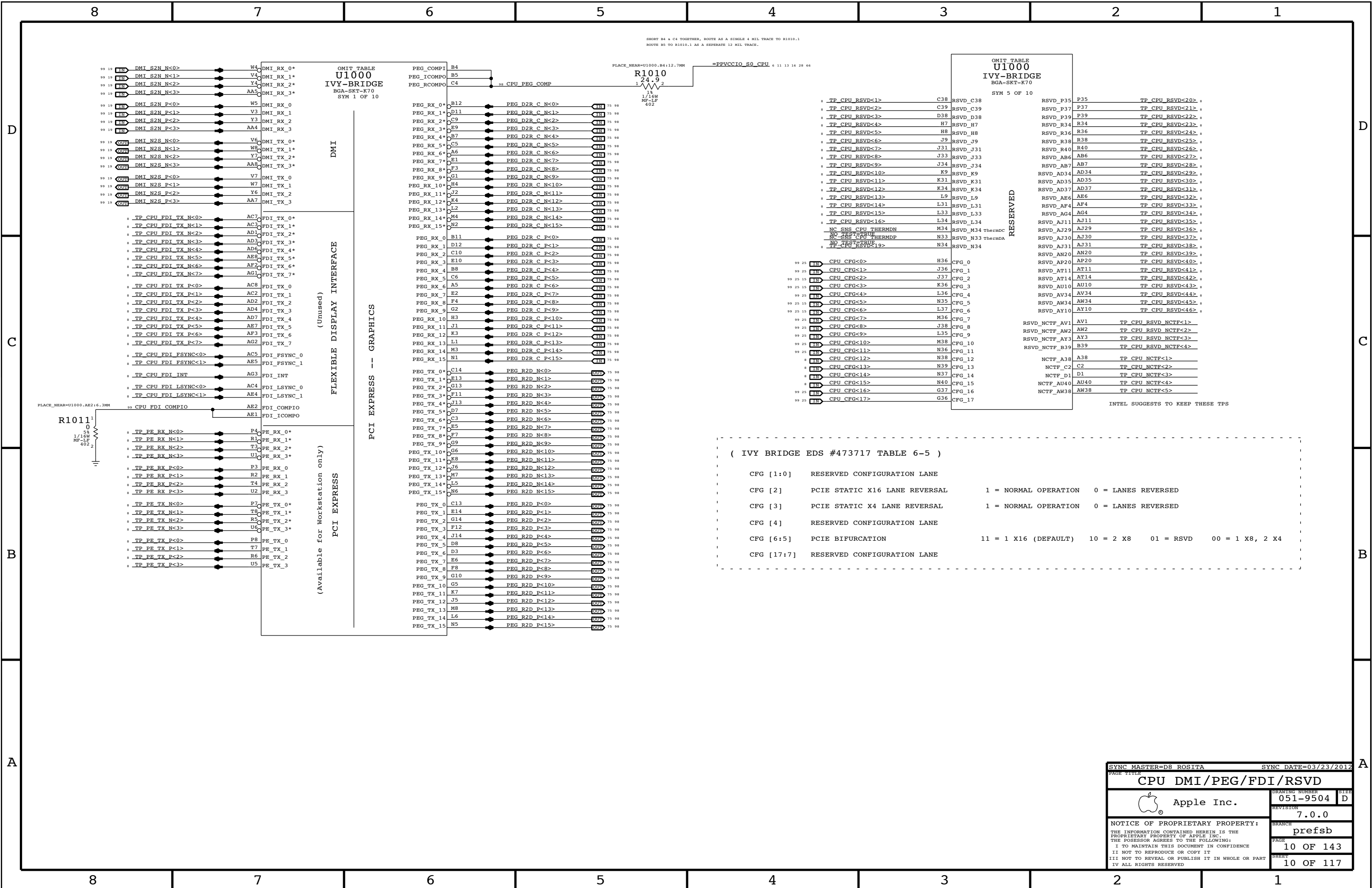
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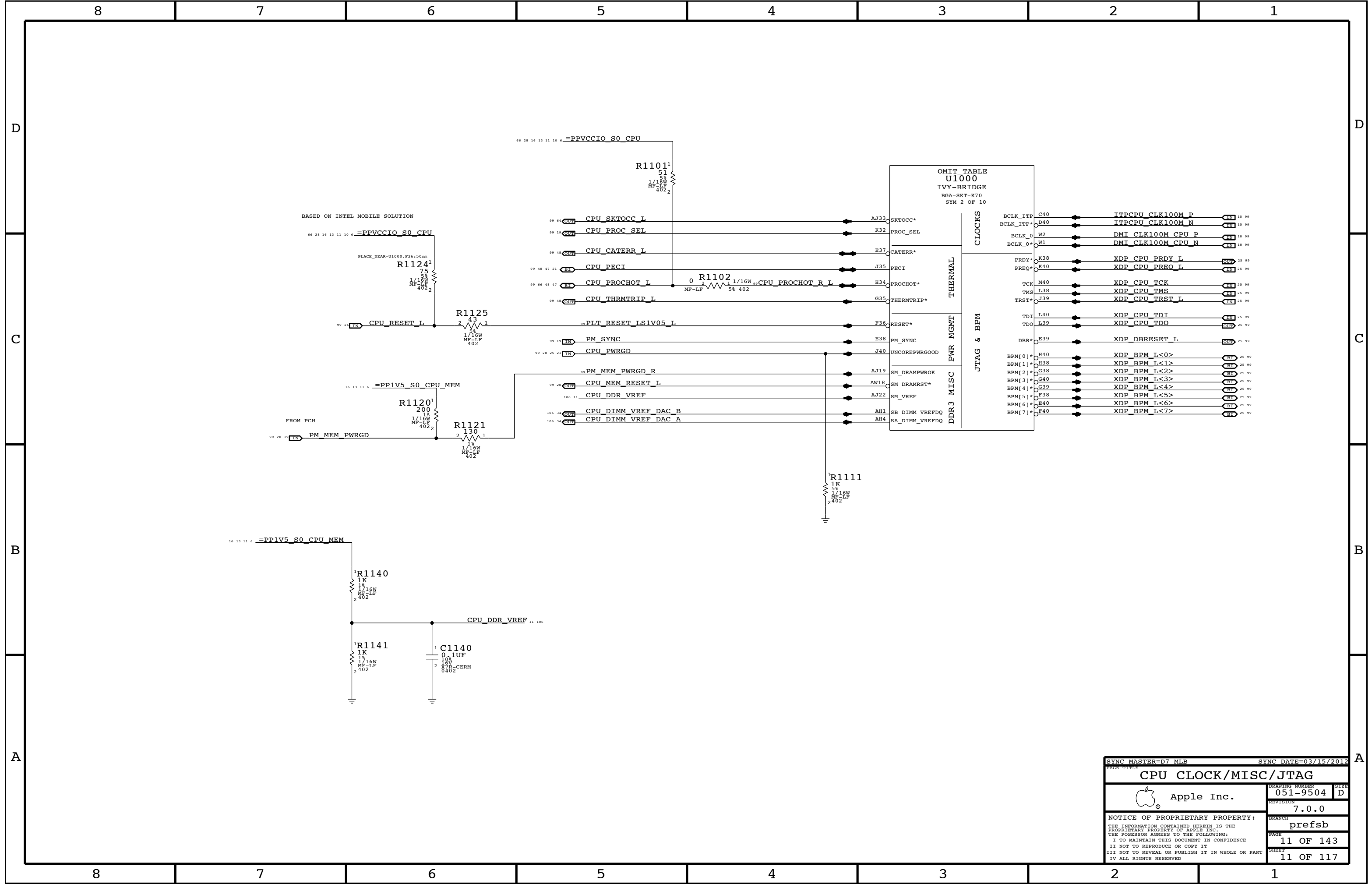
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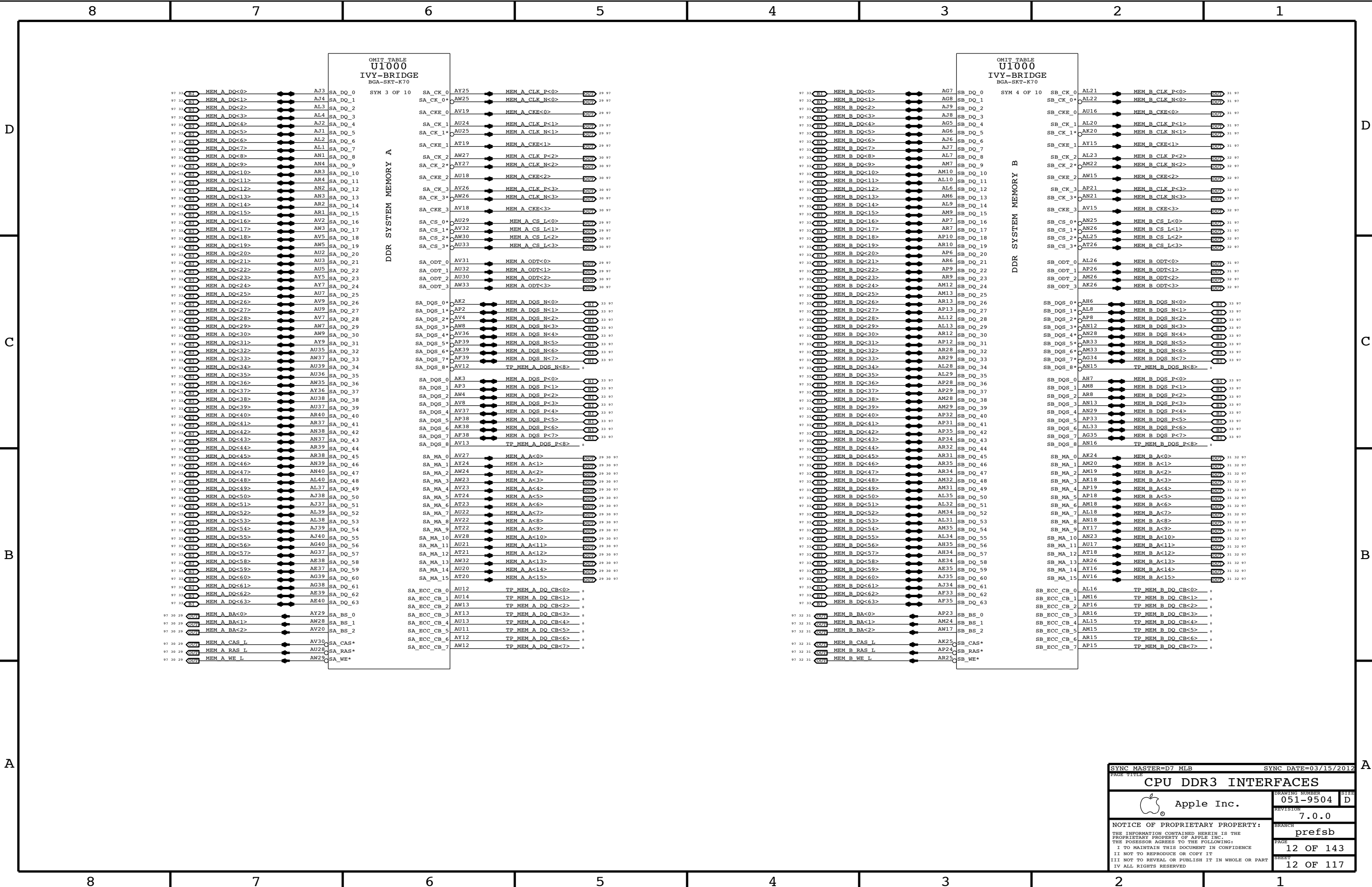
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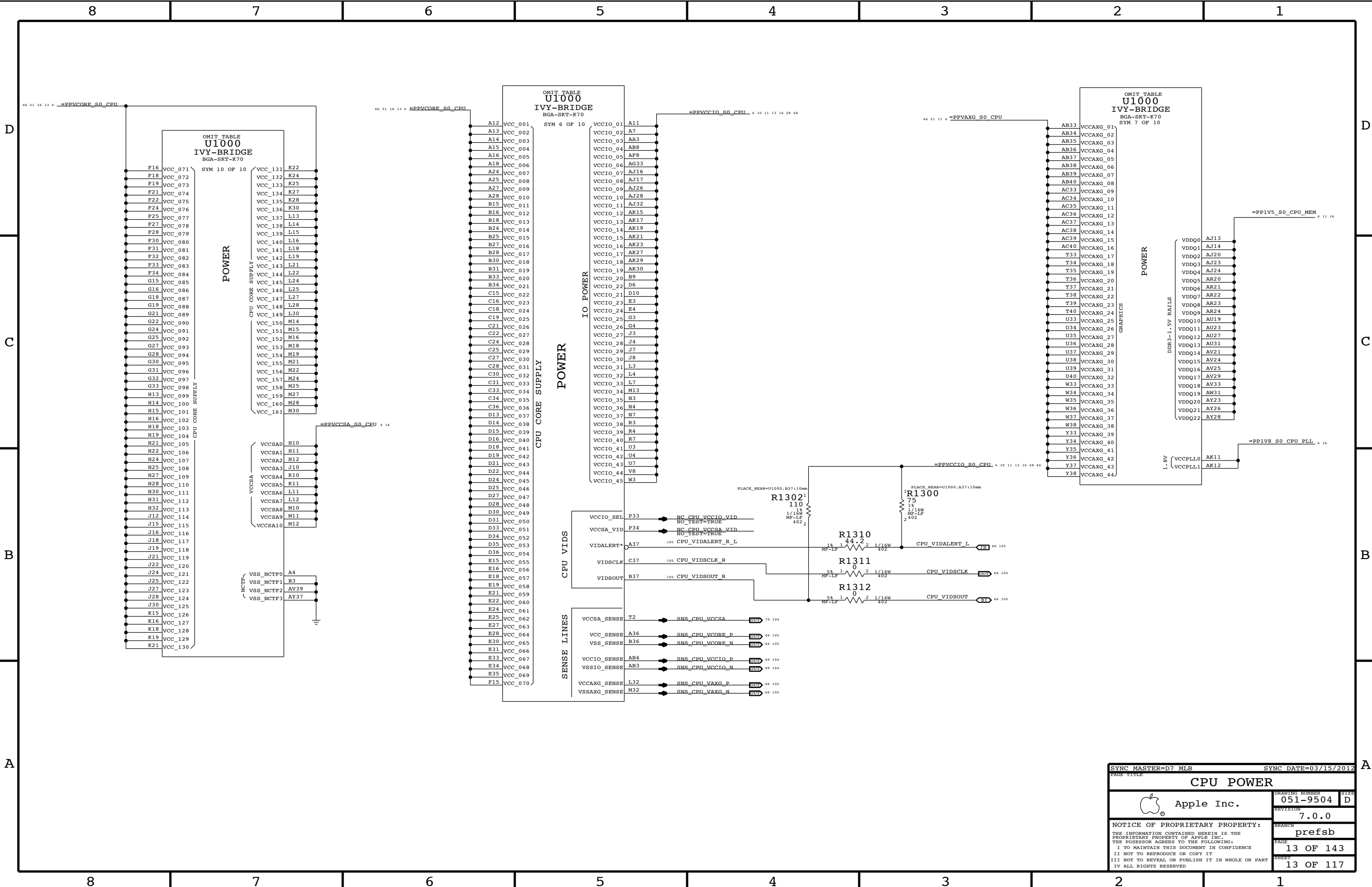
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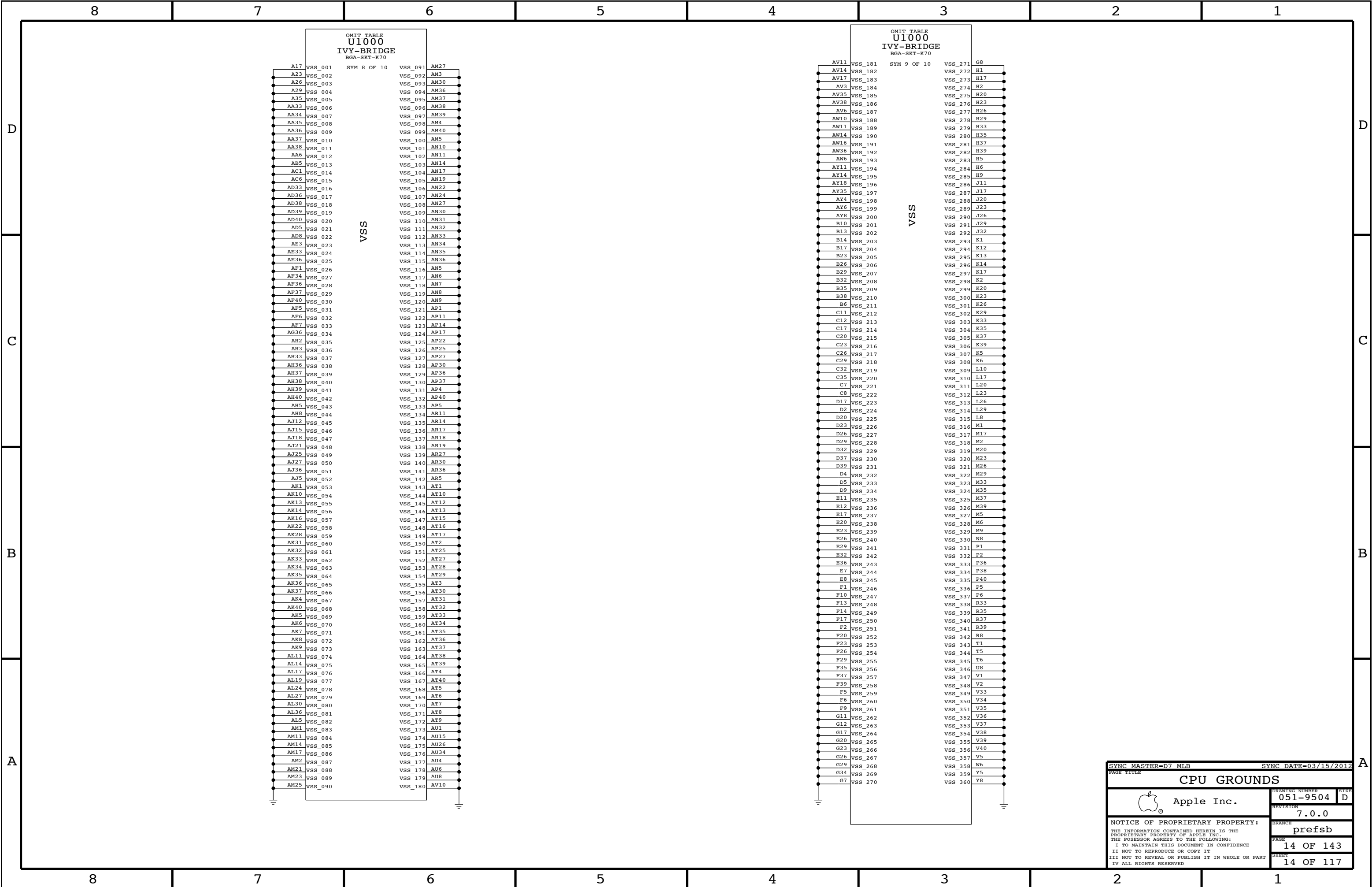
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SYNC MASTER=D7 MLB

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CPU GROUNDS

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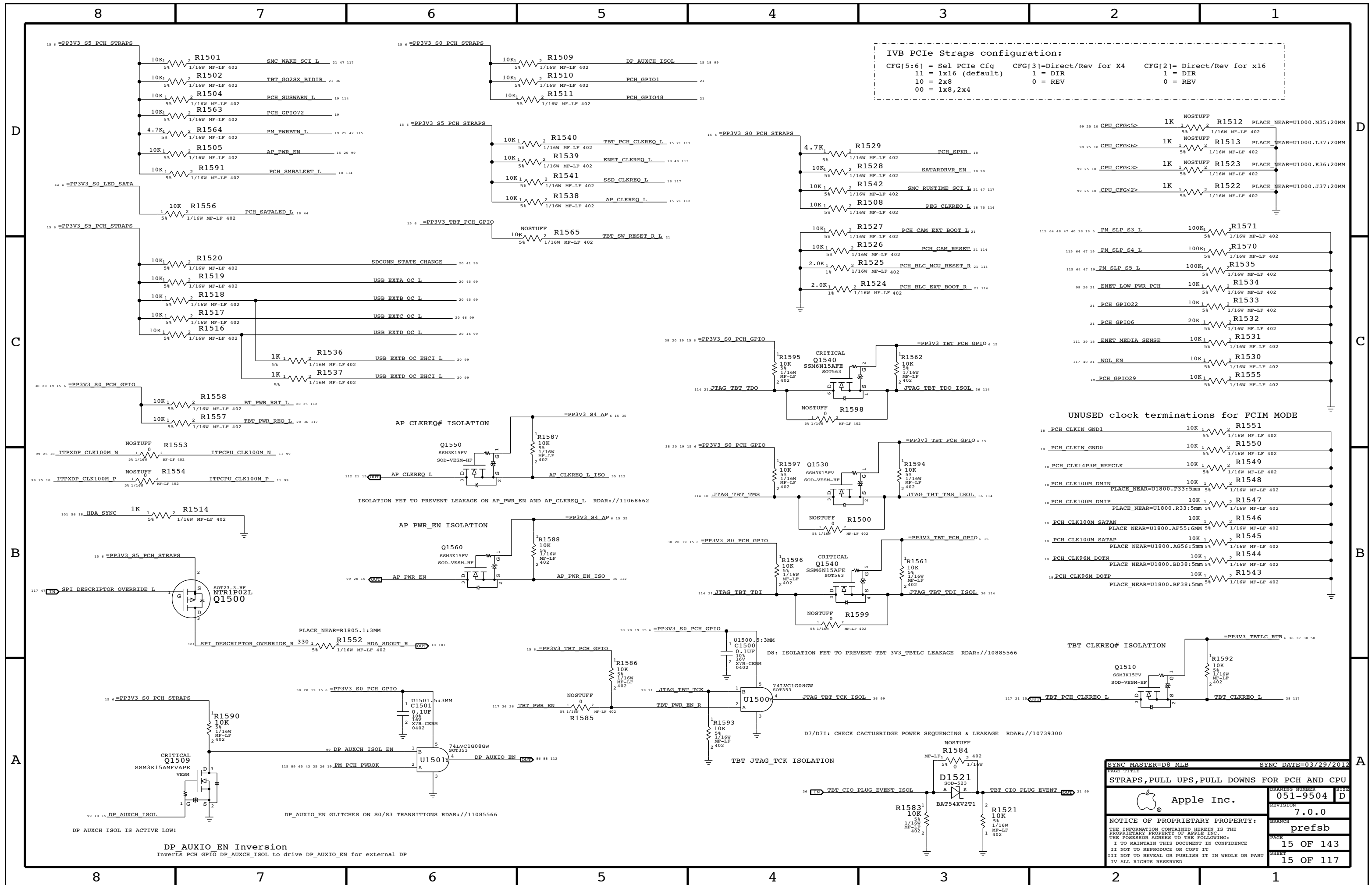
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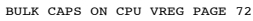
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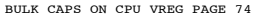


14x 22UF,0805 INTEL RECOMMENDATION 18X 22UF 0805 (14 Inside cavity and 4 North of processor)

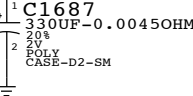
REPLACED WITH 603 PER RDAR://10700439



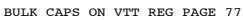
8X 22UF 0805, 6X 10UF 0805 INTEL RECOMMENDATION 9X22UF 0805,16X 0805 placeholders



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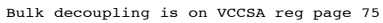
2x 47uF, 1x 22uF 0805, 1x 10uF 0603, 1x 4.7uF 0603, 1x 2.2uF 0402, 2x 1uF 0402. INTEL RECOMMENDATION 1x 10uF 0805



Place inside socket cavity



2x 10uF 0603. INTEL RECOMMENDATION 2X 10uF 0805



SYNC MASTER=D7 MLB		SYNC DATE=12/20/2011	
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CPU NON-GFX DECOUPLING			
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		REVISION	7.0.0
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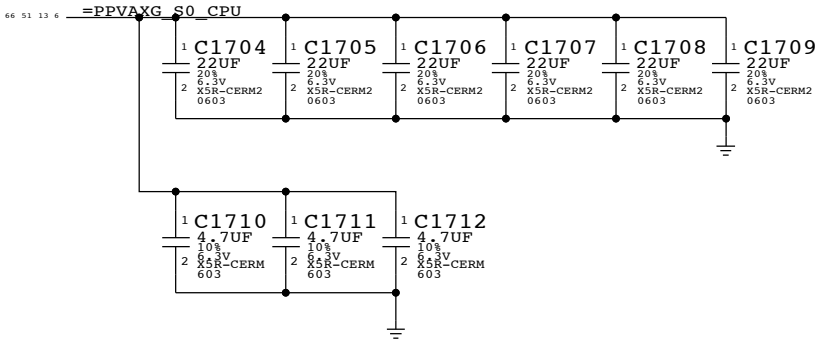
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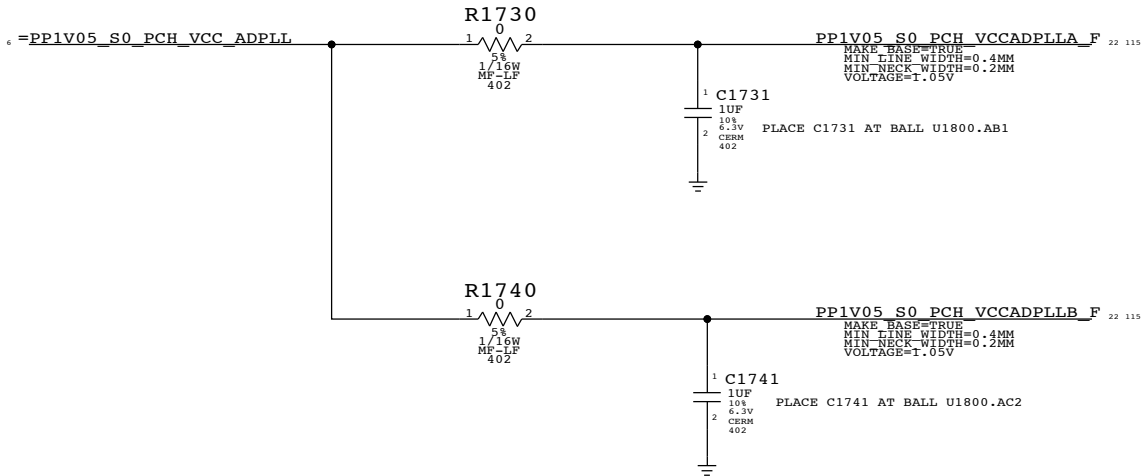
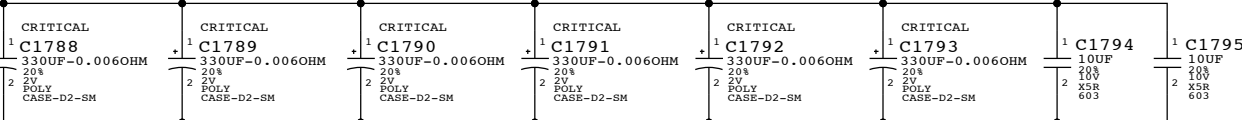
VAXG DECOUPLING

INTEL RECOMMENDATION 4X22UF 0805,3X 4.7UF

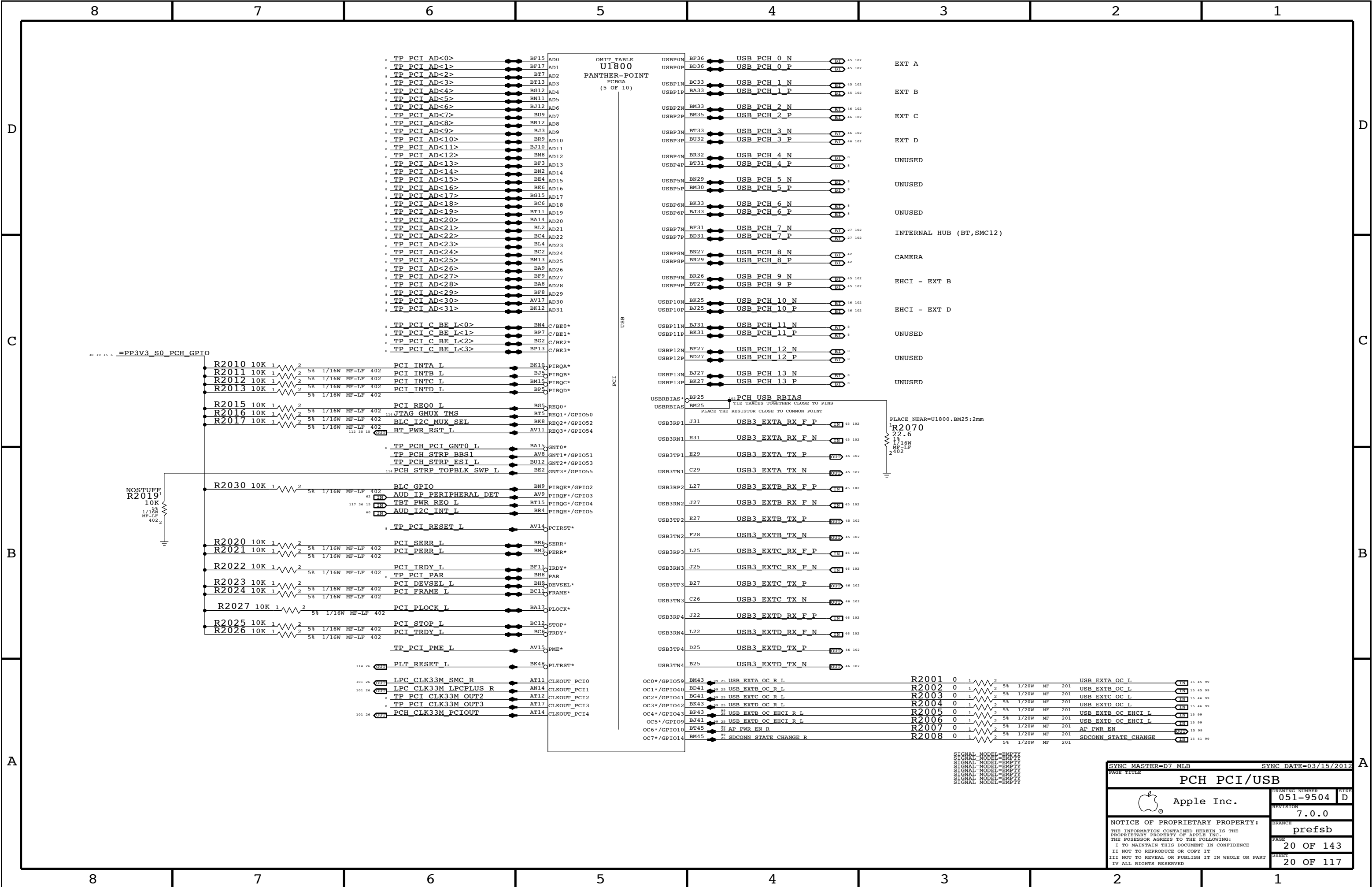
PLACEMENT_NOTE (C1704-C1709):
Place inside socket cavity



AXG BULK CAPS

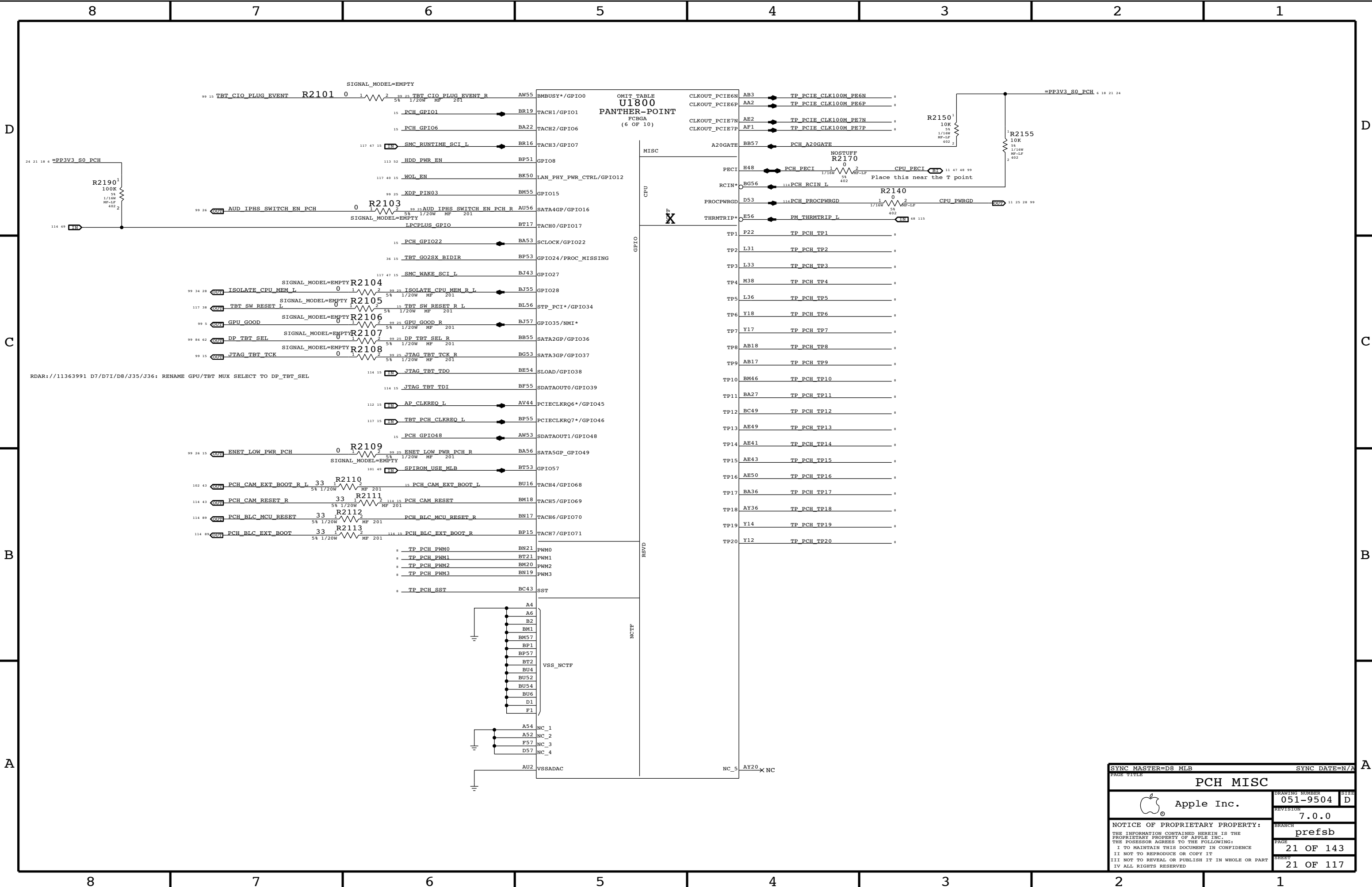


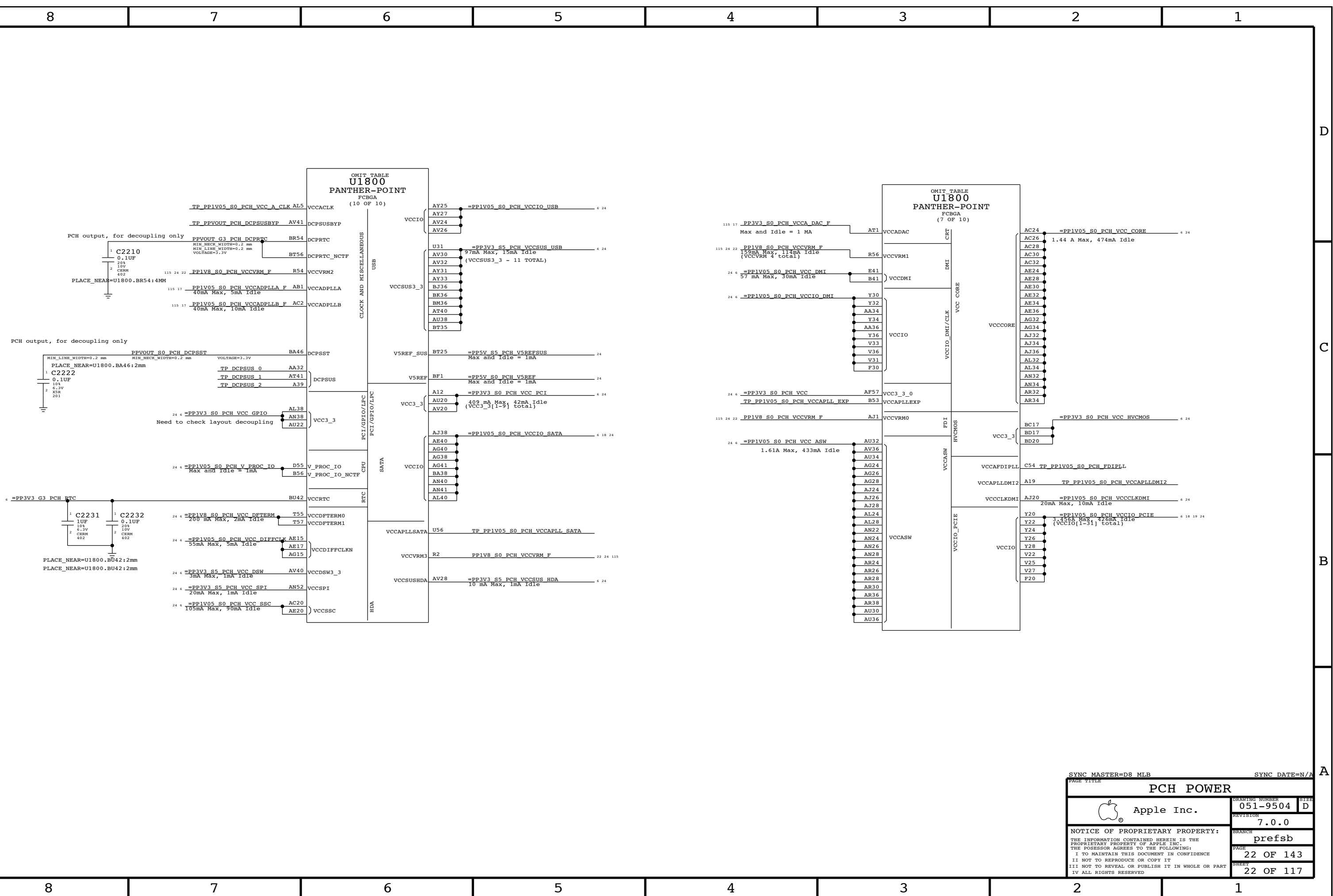
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GFX DECOUPLING & PCH PWR ALIAS			
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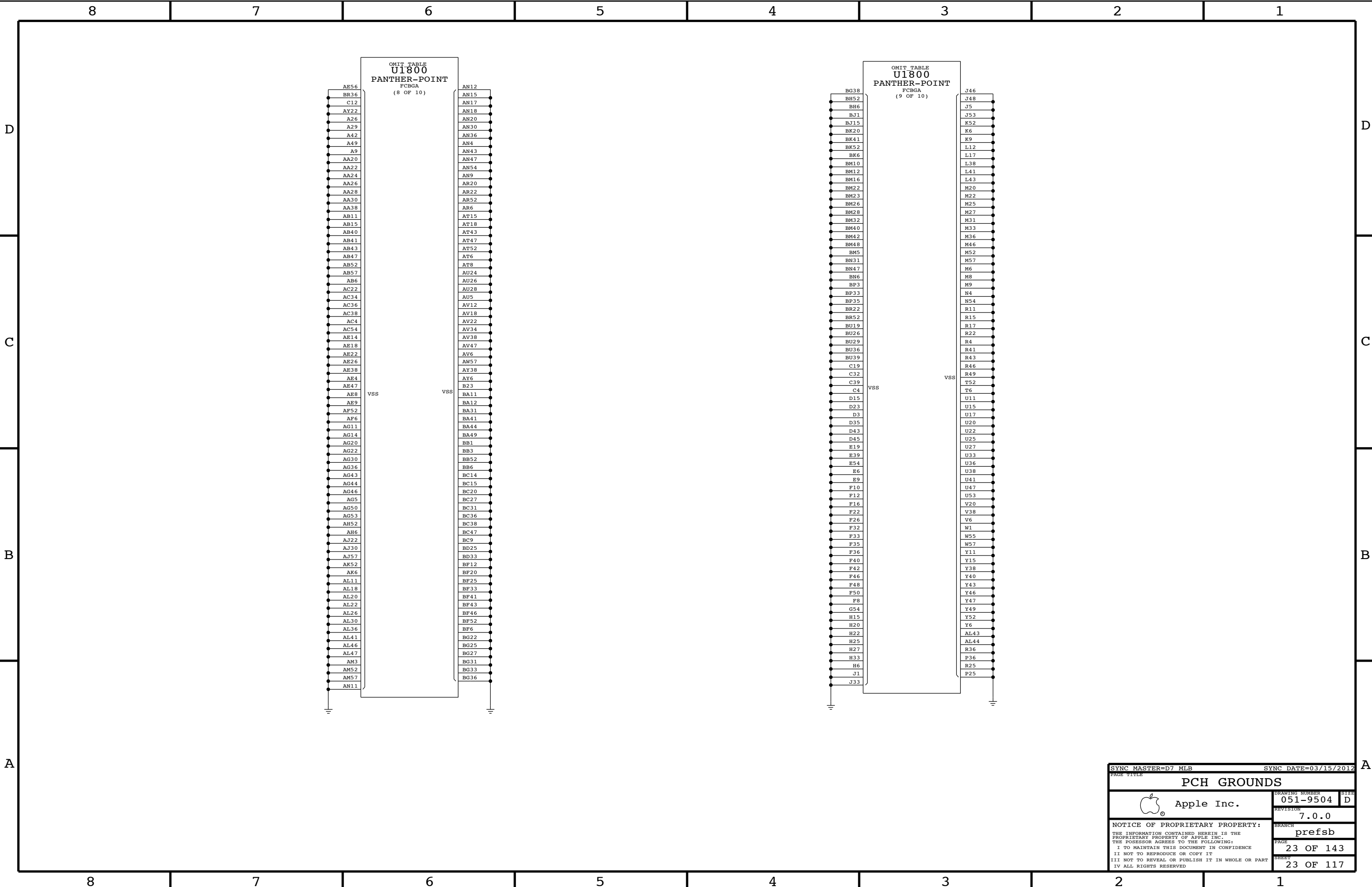


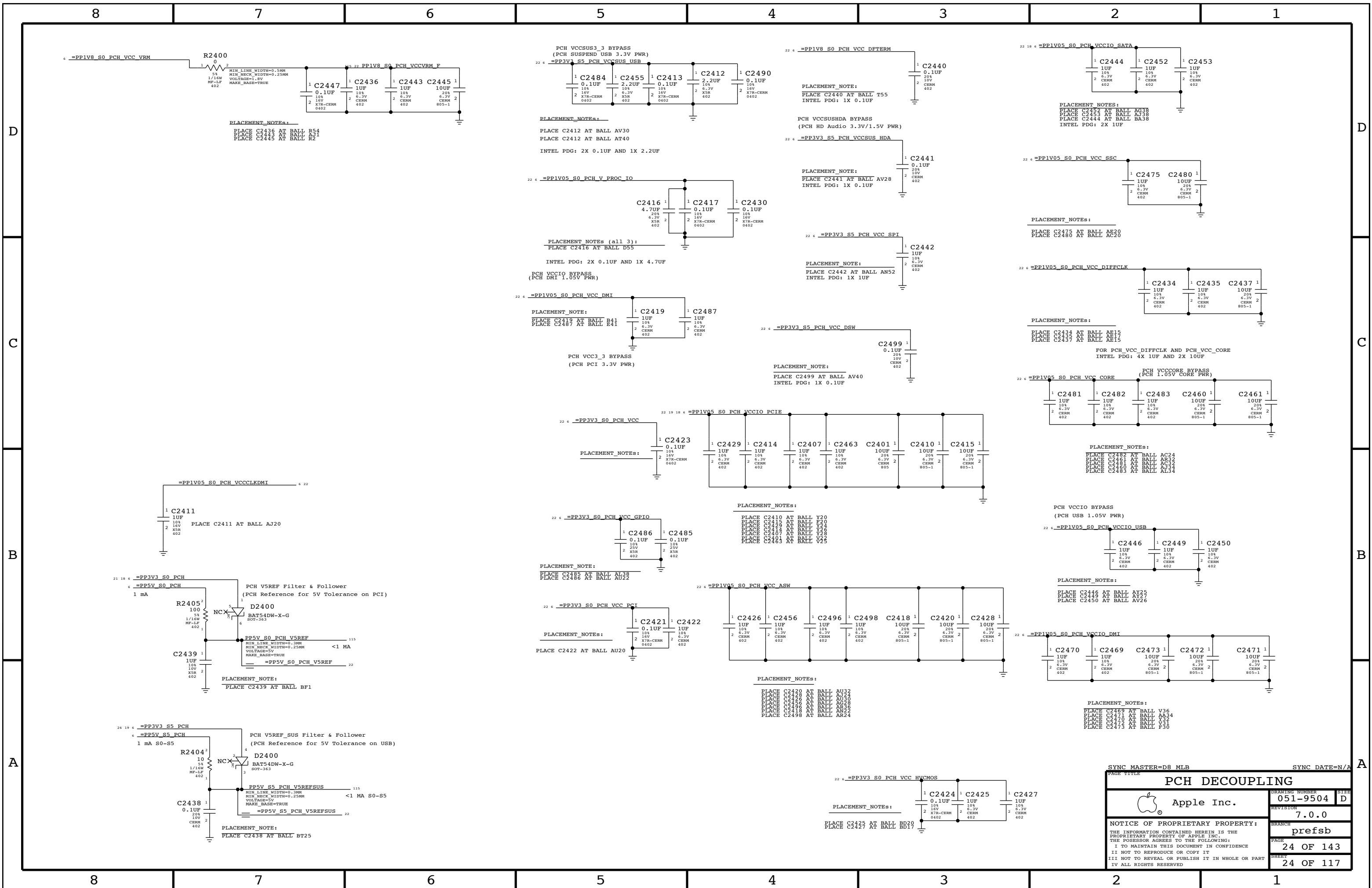
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SYNC MASTER=D7_MLB		SYNC DATE=03/15/2012	
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PCH PCI/USB		DRAWING NUMBER	051-9504
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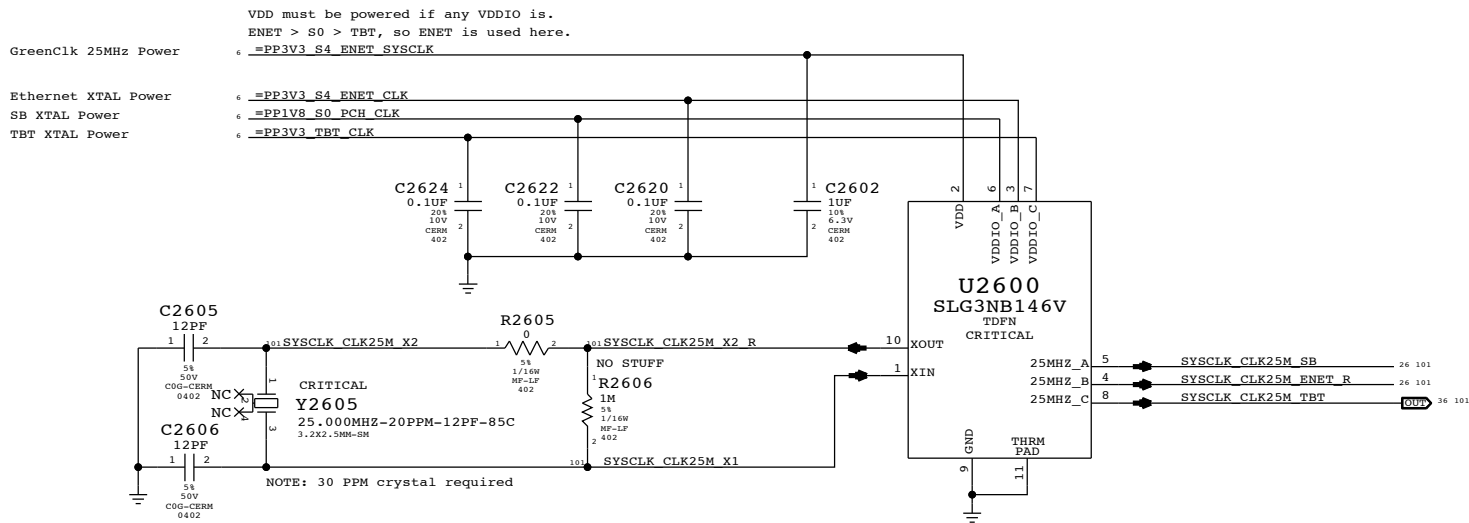




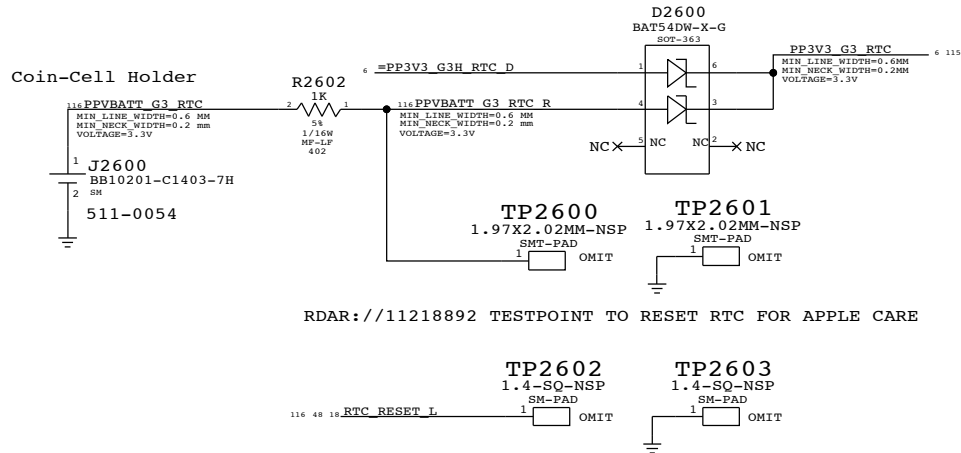


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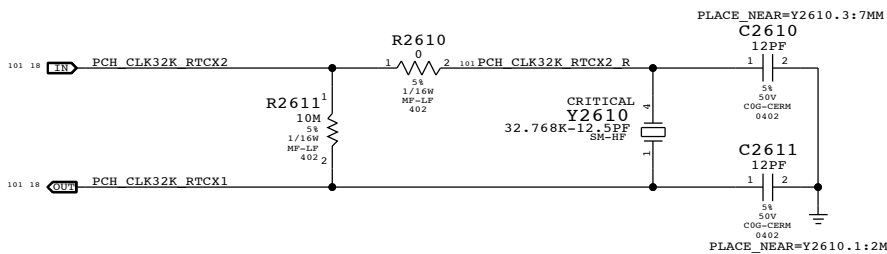
System 25MHz Clock Generator



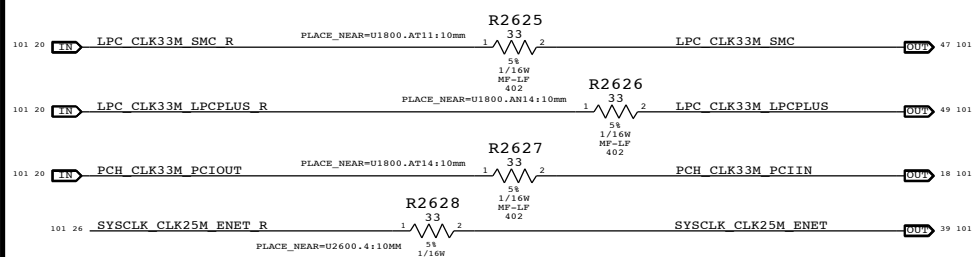
RTC Power Sources



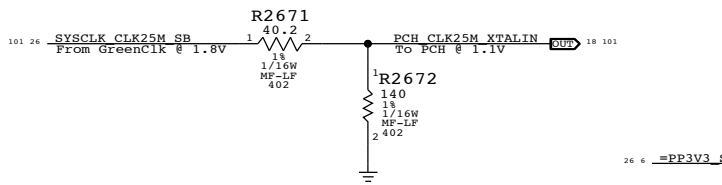
PCH RTC Crystal



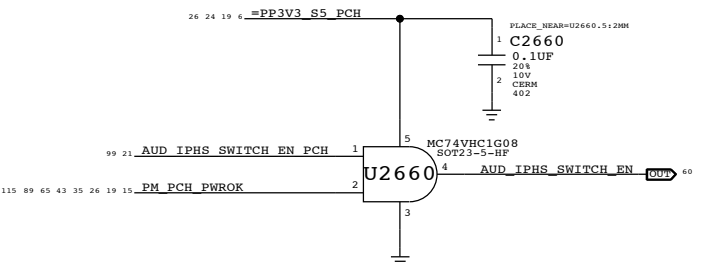
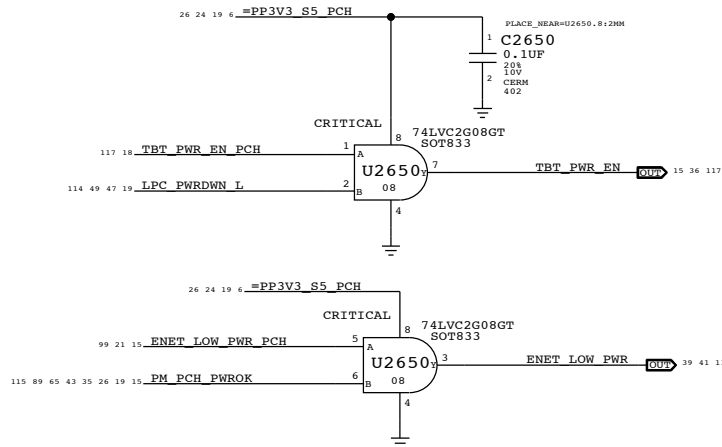
Clock series termination



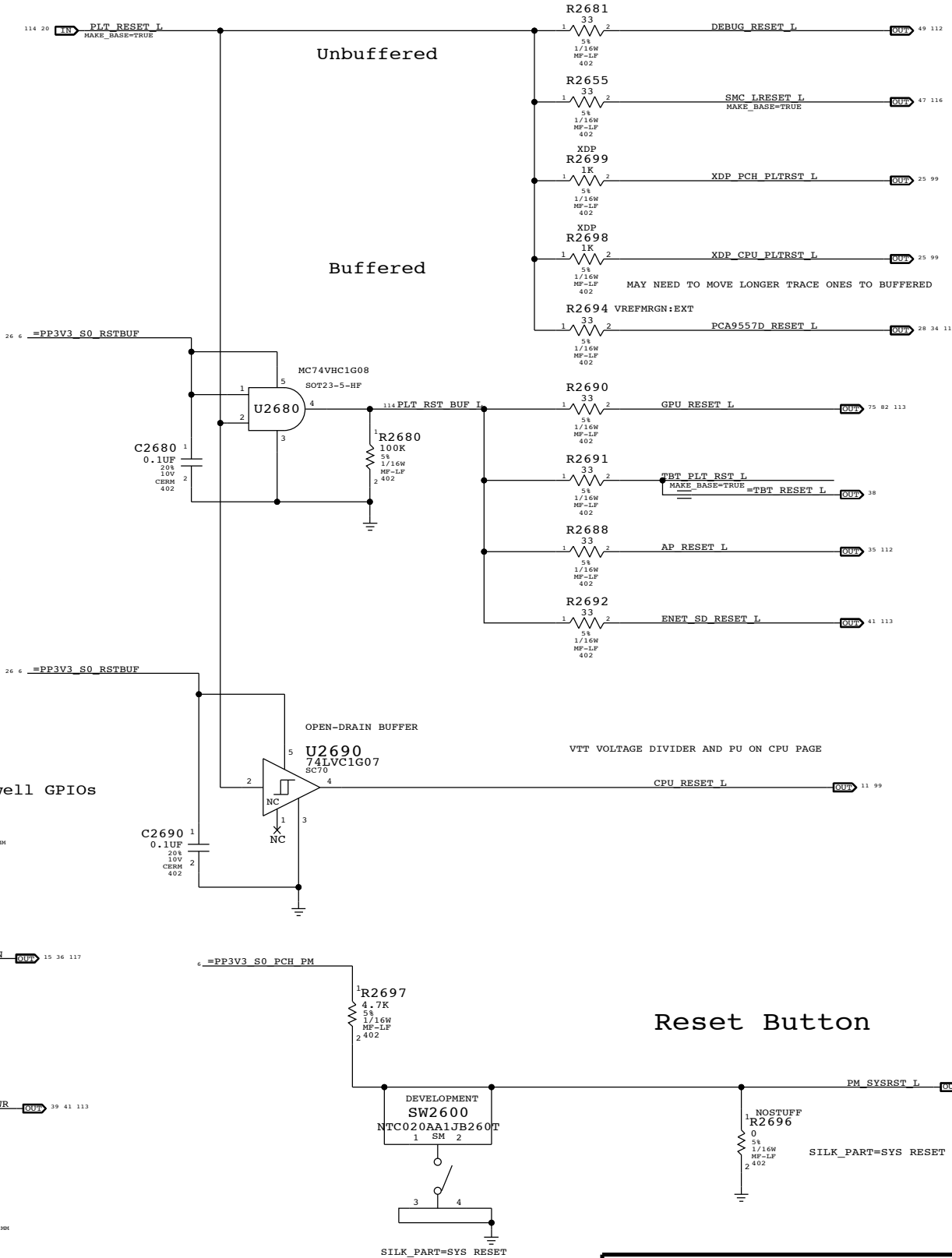
PCH 25MHZ CLOCK



GPIO Isolation to prevent glitches on critical core well GPIOs

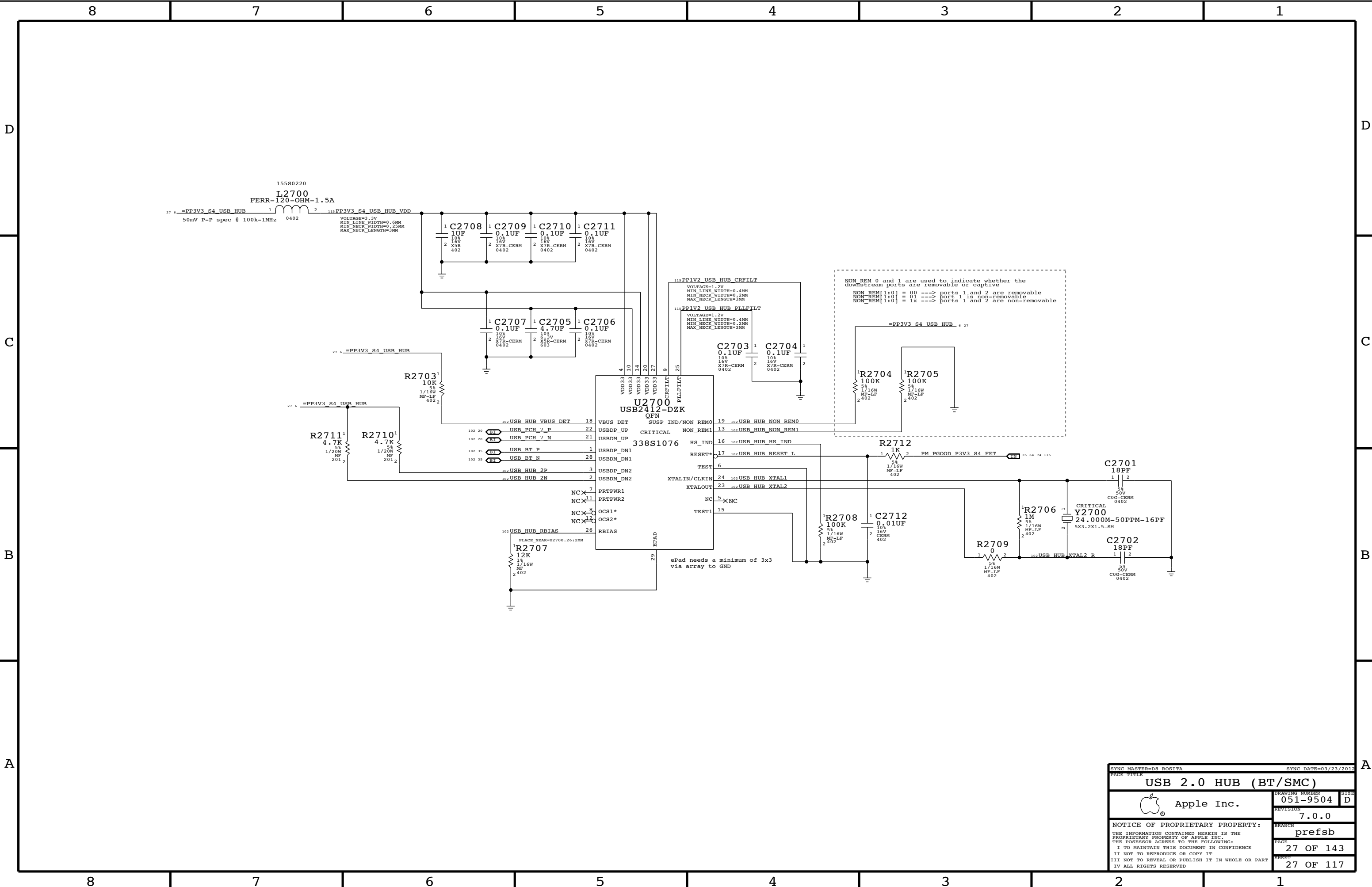


Platform Reset Connections



Reset Button

SYNC MASTER=D8 MLB		SYNC DATE=N/A	
PAGE TITLE			
CHIPSET SUPPORT			
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PAGE TITLE			
USB 2.0 HUB (BT/SMC)			
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MEM_RESET_L Generator

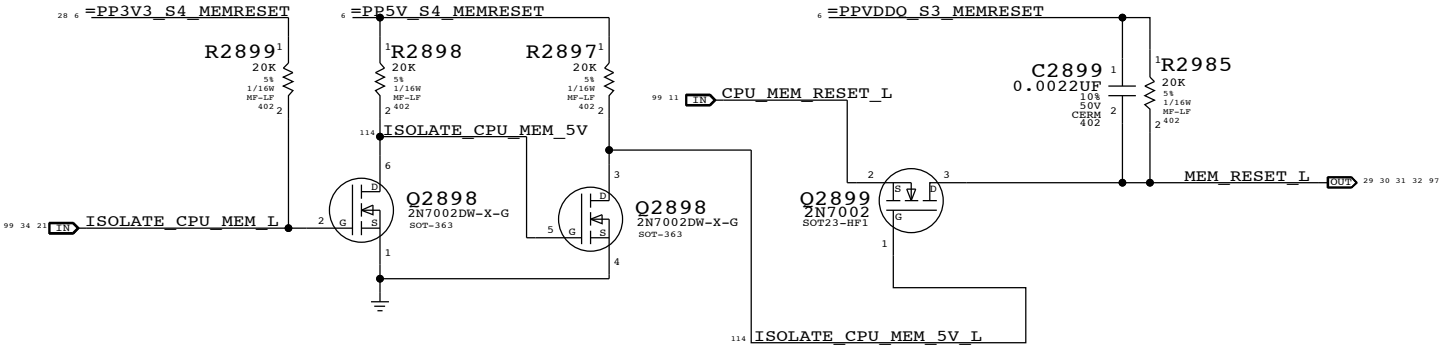
The circuits below handle MEMVTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the SO-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behaviour of signals.

WHEN HIGH: MEM_RESET_L NOT ISOLATED.

WHEN LOW: MEM_RESET_L IS ISOLATED.

MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L (Block CPU from driving MEM_RESET_L in S3)



MEMVTT_EN Generator

rdar://11117167

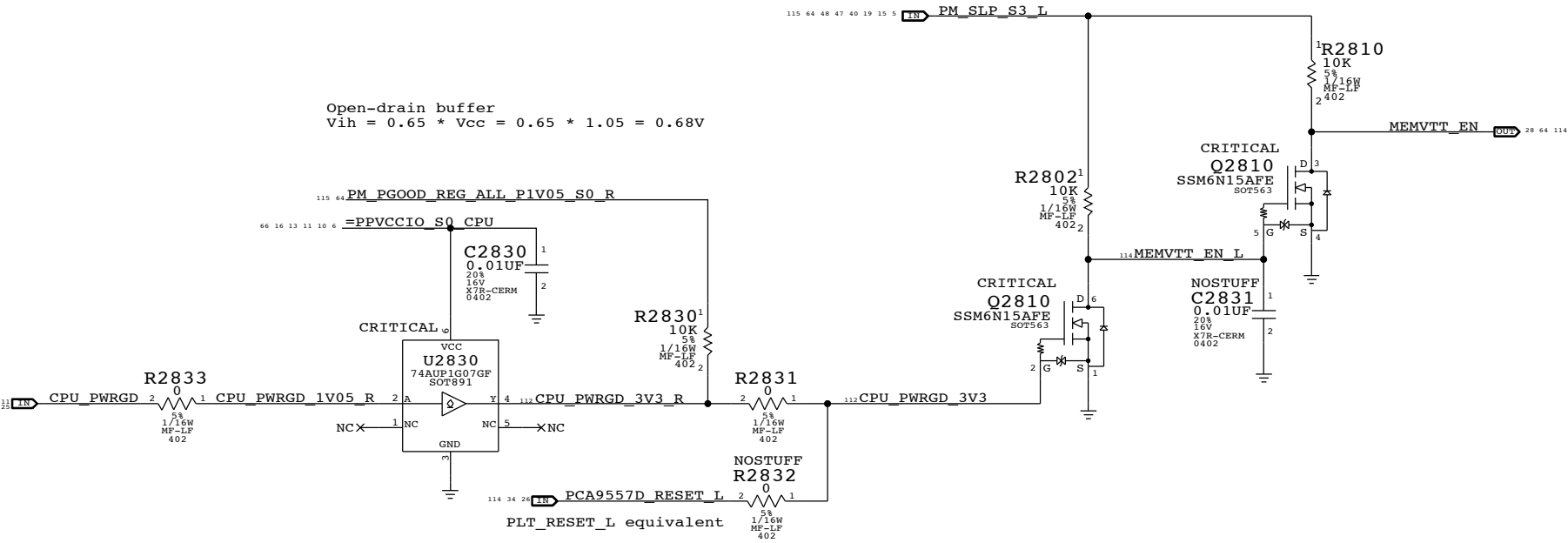
Enables MEMVTT when PCH drives CPU PWRGD.

CPU does not drive MEM_CKE until VCCORE activated but CPU 1V5 (VDDQ) leaks into it.

Clamping MEMVTT will keep the MEM_CKE low until CPU actively controls it.

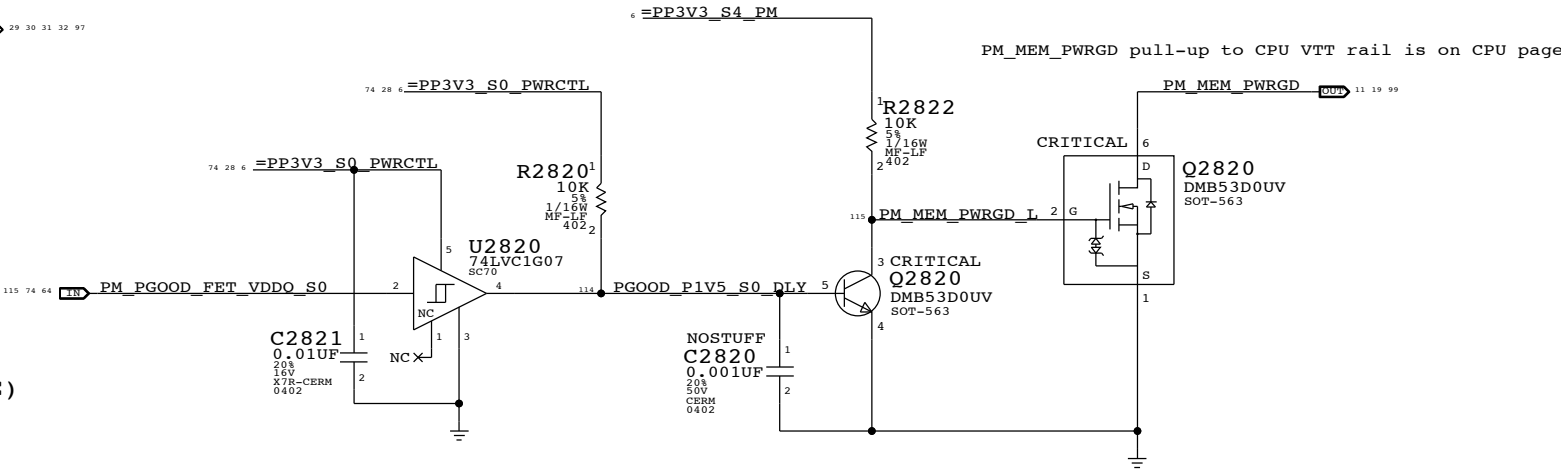
MEMVTT Clamp actively holds MEMVTT rail low until MEMVTT is enabled.

MEMVTT_EN = CPU_PWRGD * PM_SLP_S3_L (VTT is enabled when PCH tells CPU to enable VCCORE)



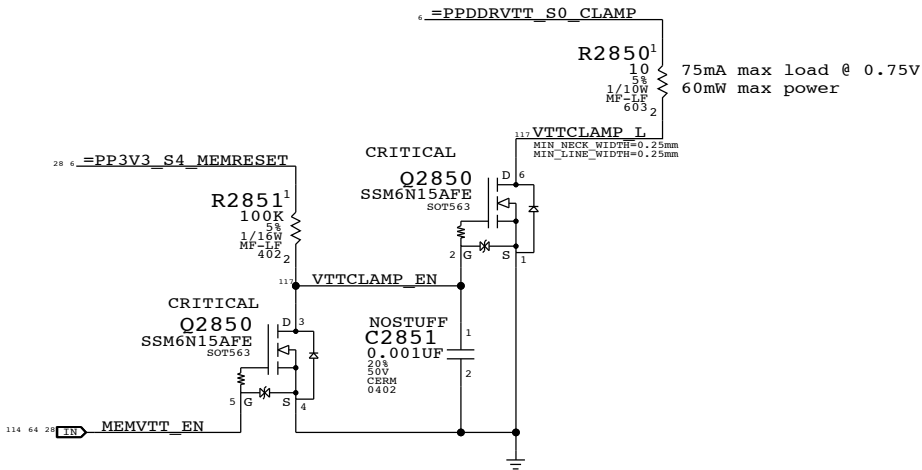
1V5 S0 "PGOOD" for CPU

With optional delay from 1V5 S0 PGOOD



MEMVTT Clamp

Ensures CKE signals are held low in S3 and in S0 before CPU PWRGD



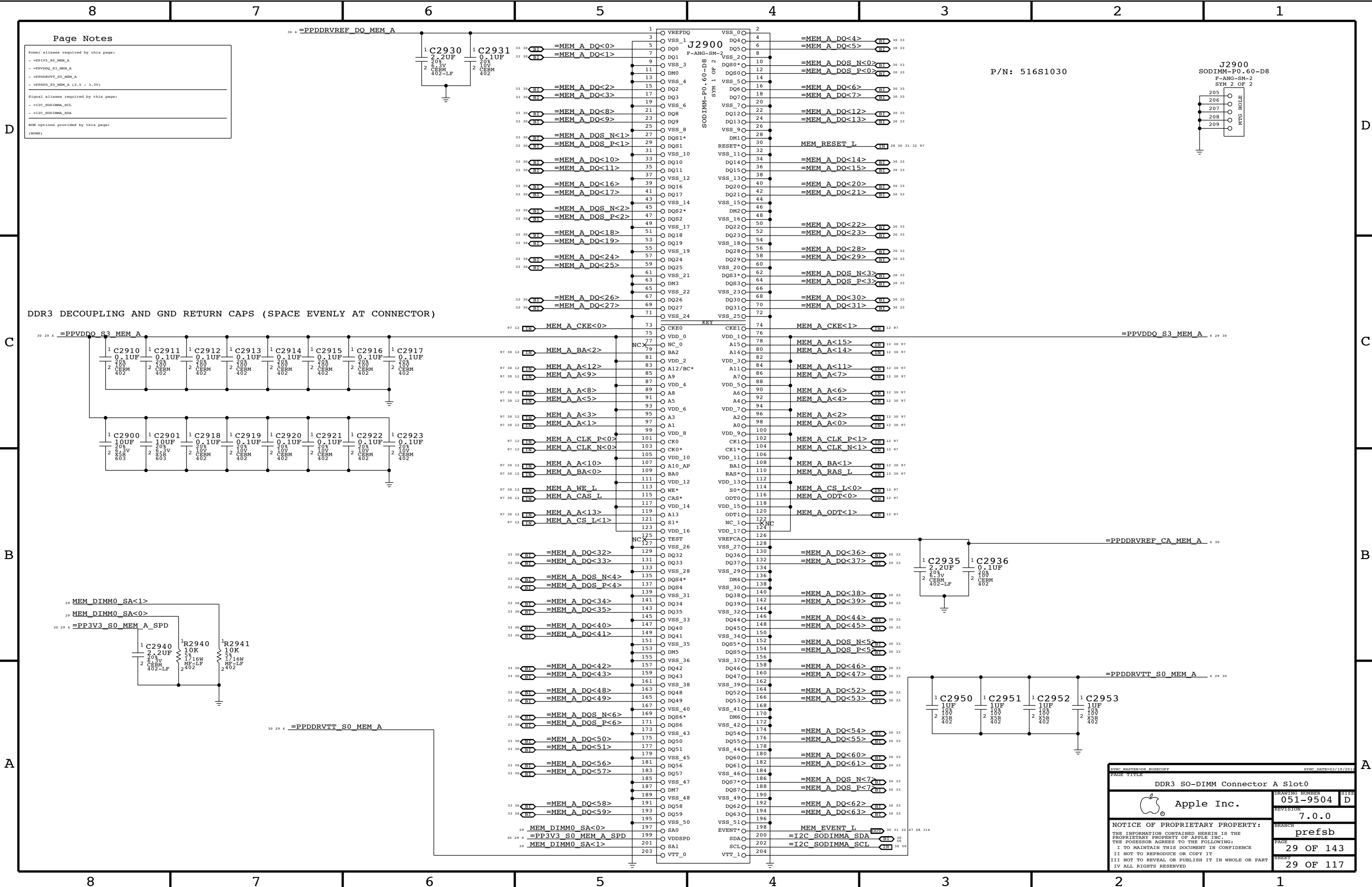
Step	ISOLATE_CPU_MEM_L	PM_SLP_S3_L	CPU_PWRGD	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN
S0	0	1	1	CPU_MEM_RESET_L	1	1
to	1	1	1	1	1	1
2	0	1	1	1	1	1
3	0	0	0	X	1	0
4	0	0	0	X	1	0
5	0	1	1	0 (*)	1	1
6	0	1	1	1	1	1
S0	7	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: On a S5->S0 transition, ISOLATE_CPU_MEM_L will default low.

Rails will power-up as if from S3, but MEM_RESET_L now needs to be asserted in S0. Software must de-assert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=D8 MARK		SYNC DATE=04/23/2012	
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CPU Memory S3 Support		DRAWING NUMBER	
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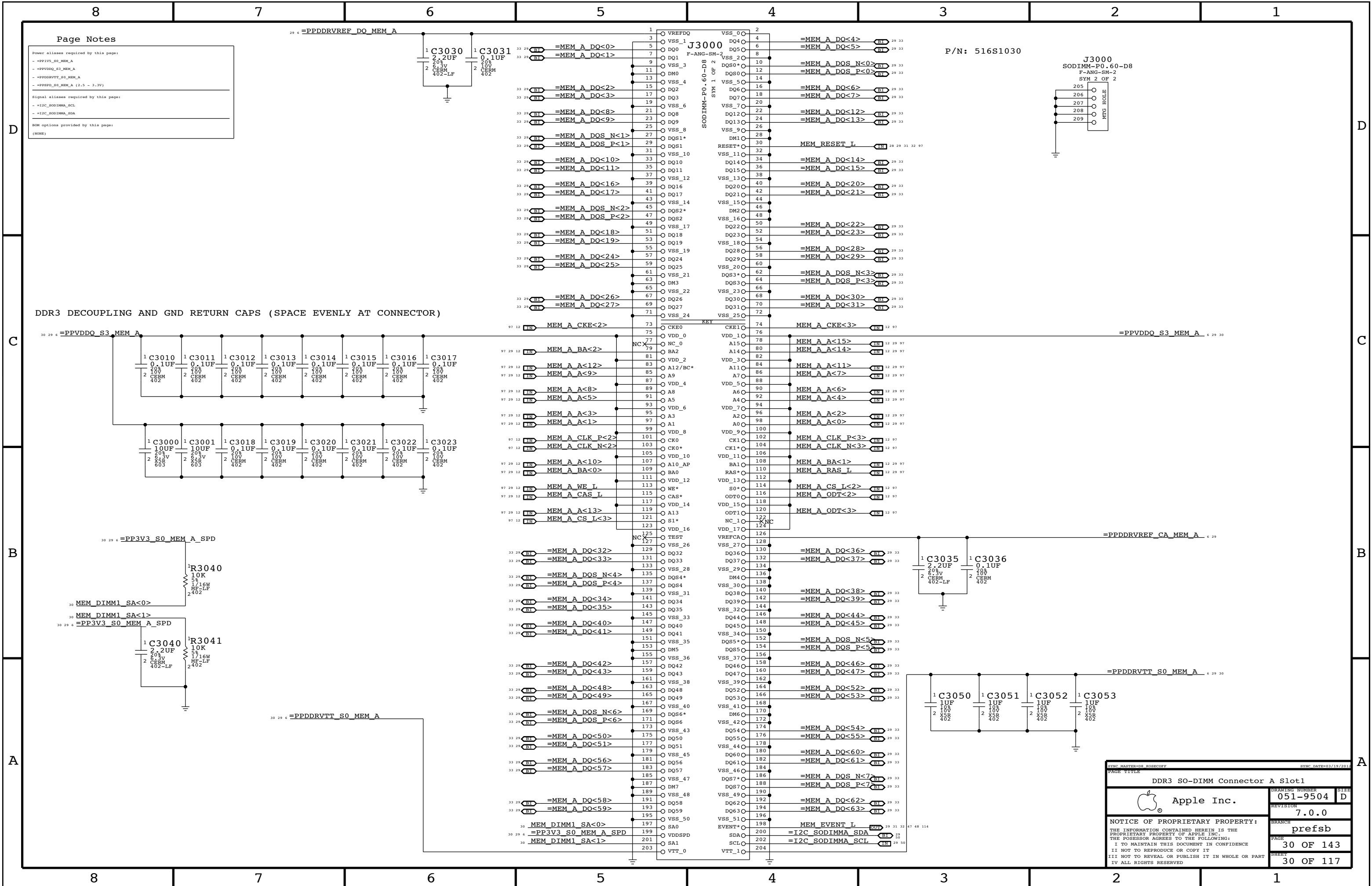
Power aliases required by this page:
- =PP3V3_S0_MEM_A
- =PPVDDO_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPDRREF_DO_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:
- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:
(NONE)

DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

PAGE TITLE		PAGE NUMBER	
DDR3 S0-DIMM Connector A Slot0		051-9504	
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Page Notes

Power aliases required by this page:

- =PP1V5_S0_MEM_A
- =PPVDDQ_S3_MEM_A
- =PPDDRVTT_S0_MEM_A
- =PPSP0_S0_MEM_A (2.5 - 3.3V)

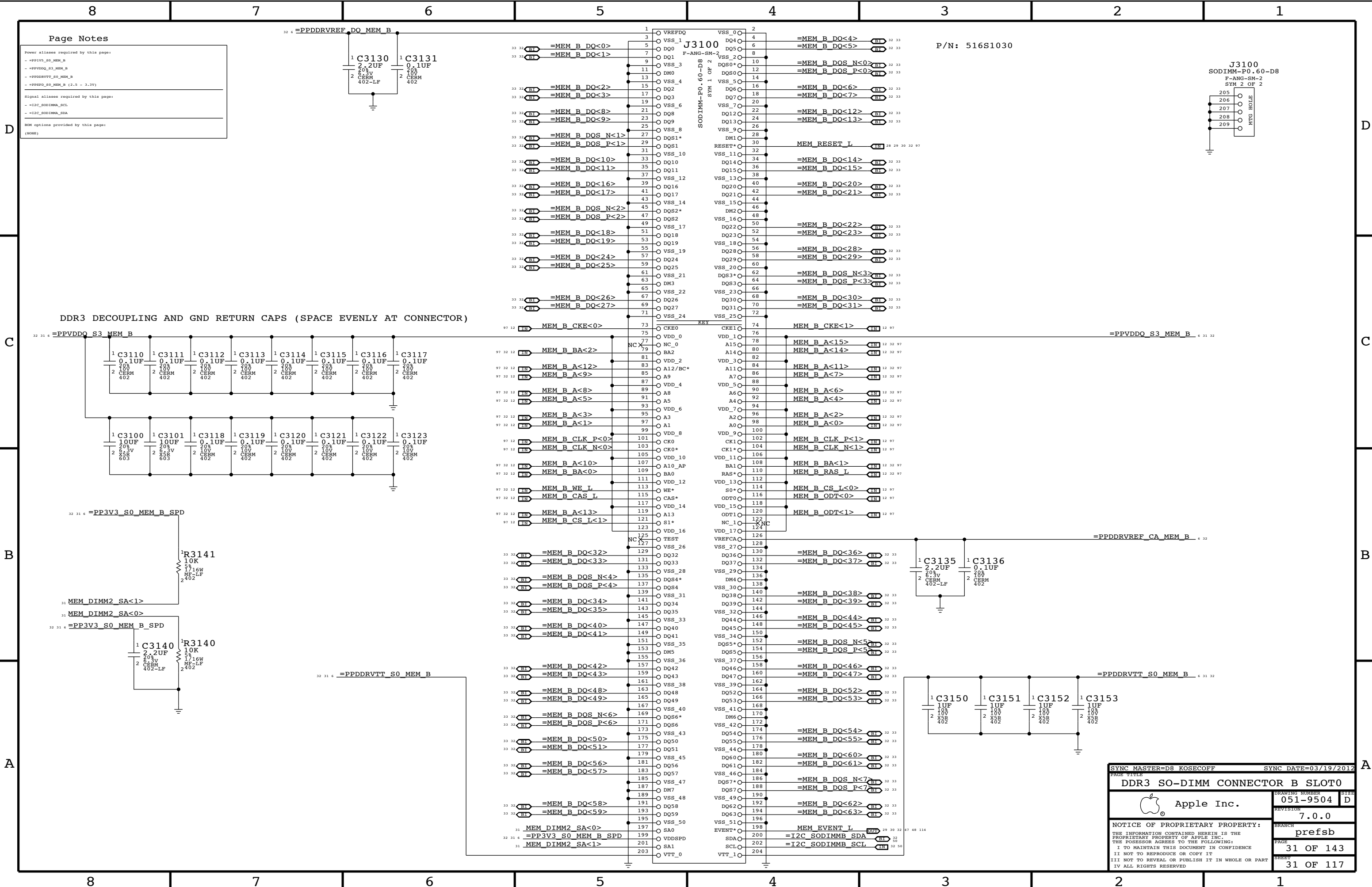
Signal aliases required by this page:

- =I2C_SODIMMA_SCL
- =I2C_SODIMMA_SDA

BOM options provided by this page:

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PAGE TITLE		PAGE TITLE	
DDR3 S0-DIMM Connector A Slot1		DRAWING NUMBER	
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Page Notes

Power aliases required by this page:

- =PP1V3_S0_MEM_B
- =PPVDDQ_S3_MEM_B
- =PPDDRVTT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

Signal aliases required by this page:

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BOM options provided by this page:

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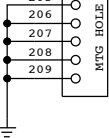
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SODIMM-P0.60-D8

F-ANG-SM-2

SYM 2 OF 2



SYNC MASTER=D8 KOSECOFF SYNC DATE=03/19/2012

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DDR3 SO-DIMM CONNECTOR B SLOT0



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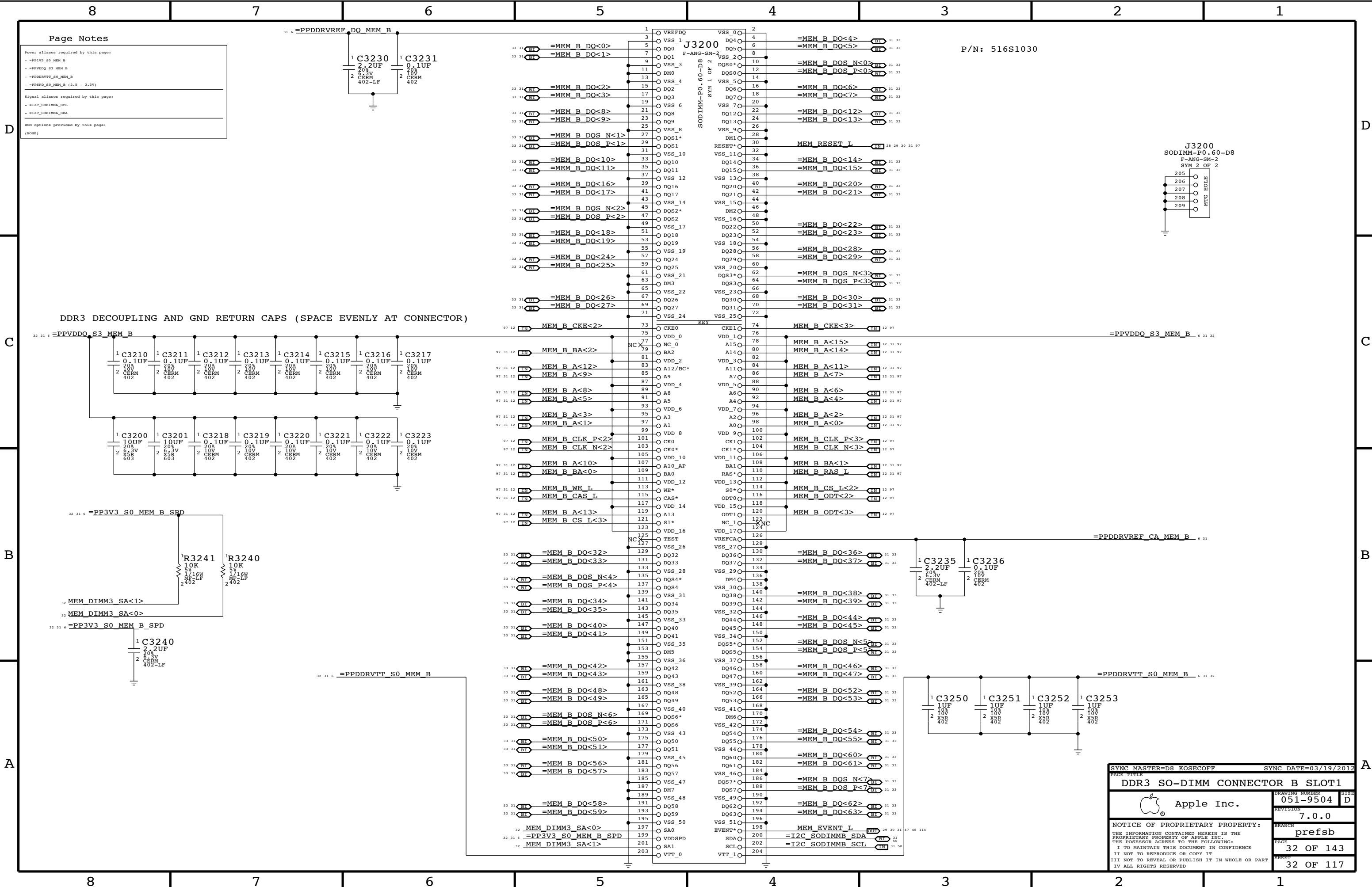
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- =PP1V5_S0_MEM_B
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- =PPDDRVTT_S0_MEM_B
- =PPSPD_S0_MEM_B (2.5 - 3.3V)

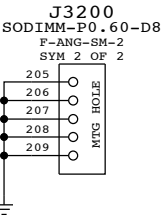
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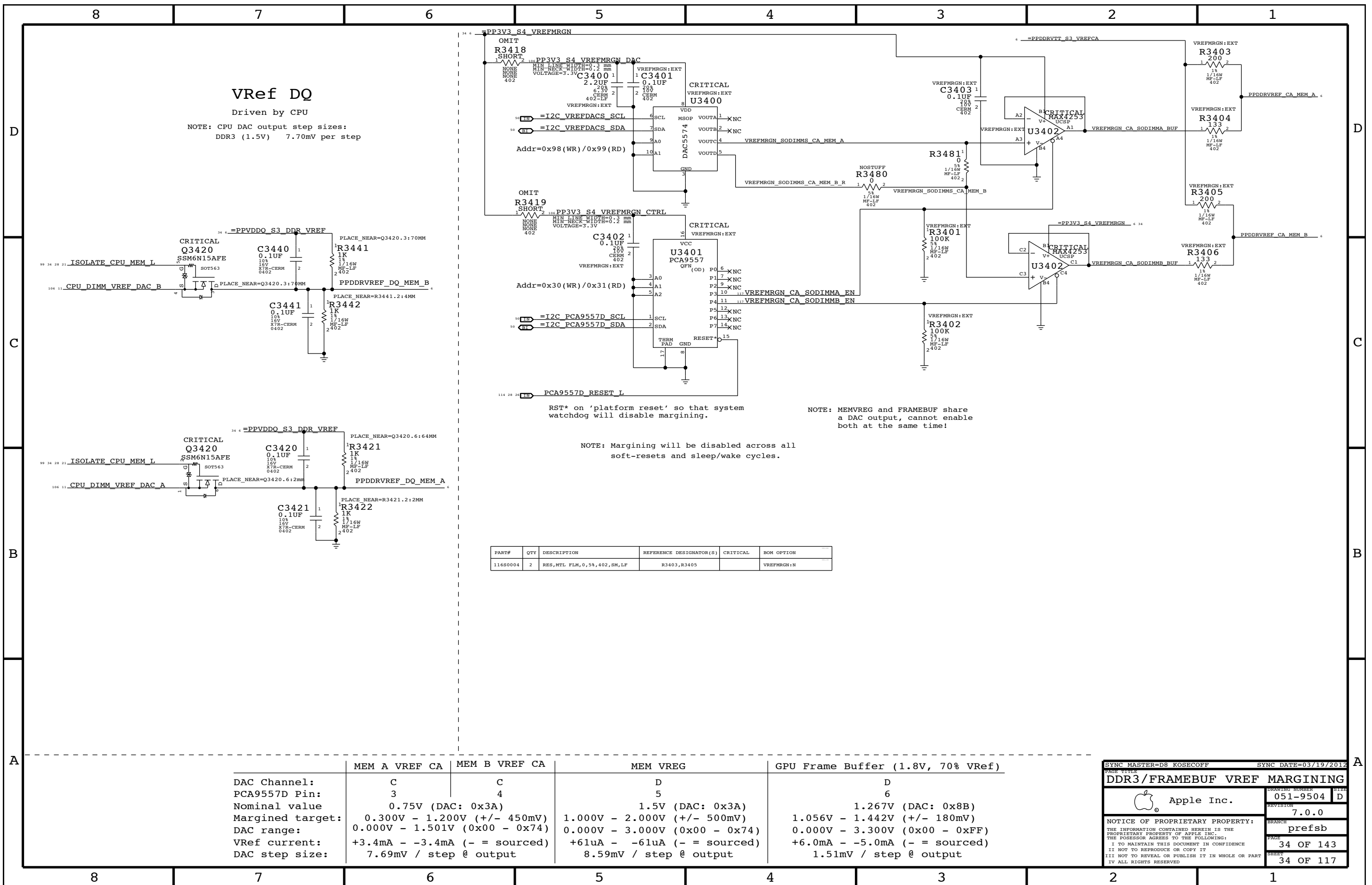
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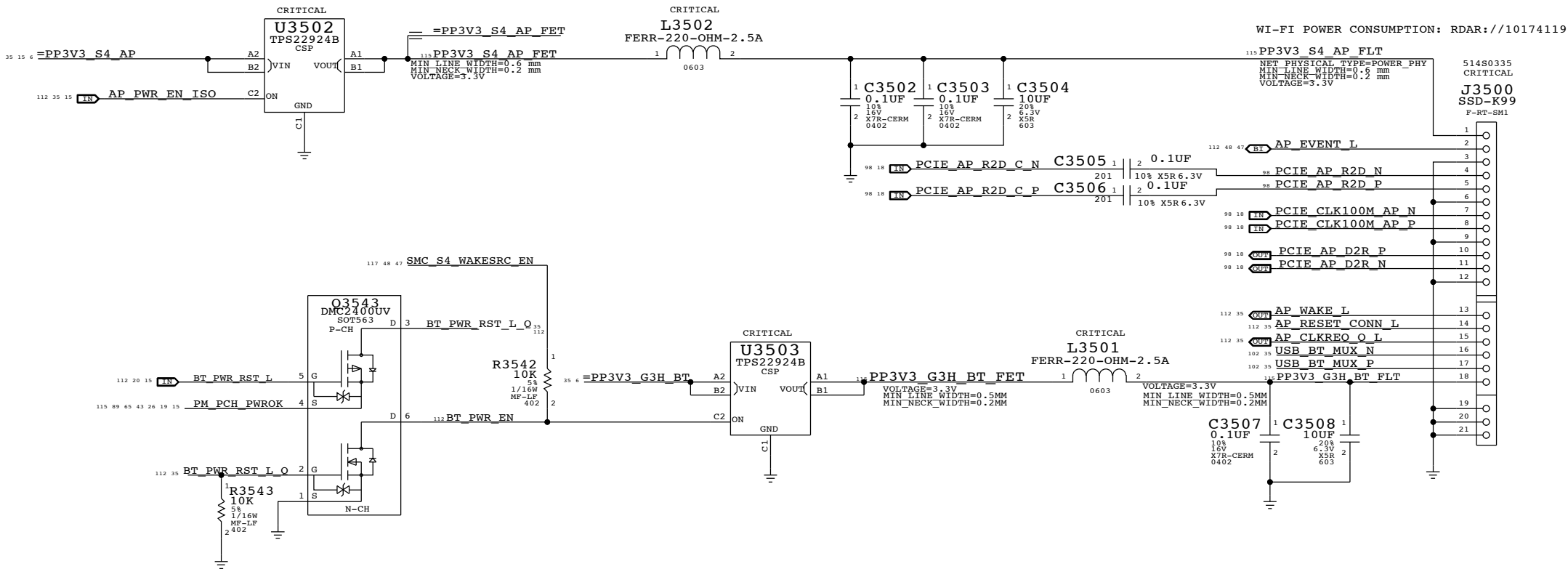


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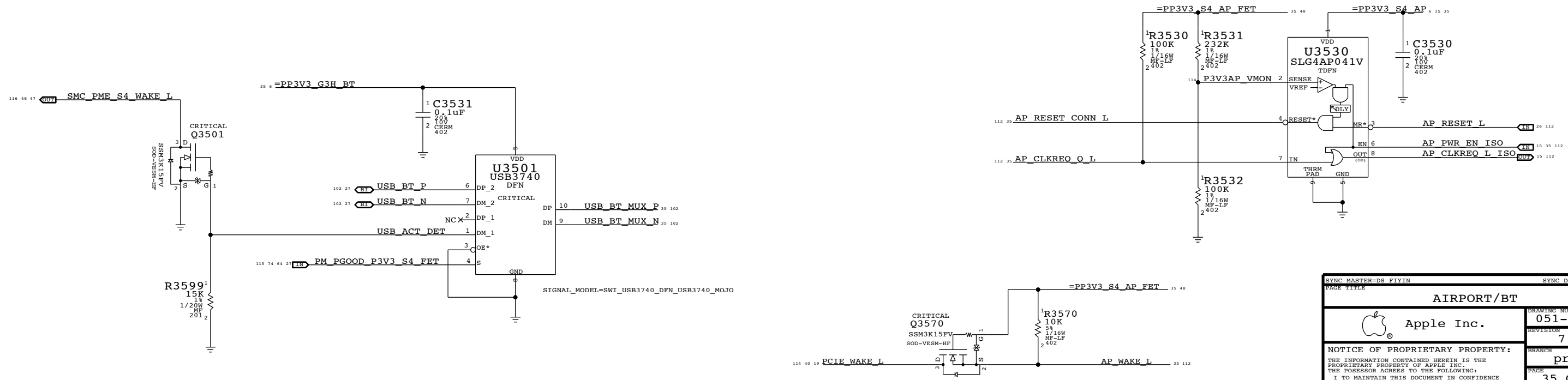
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												SYNC MASTER=D8 KOSECOFF PAGE TITLE DDR3 ALIASES AND BITSWAPS DRAWING NUMBER 051-9504 REVISION 7.0.0 BRANCH prefsb NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED SHEET 33 OF 117			
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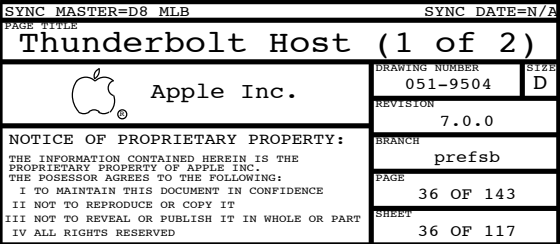
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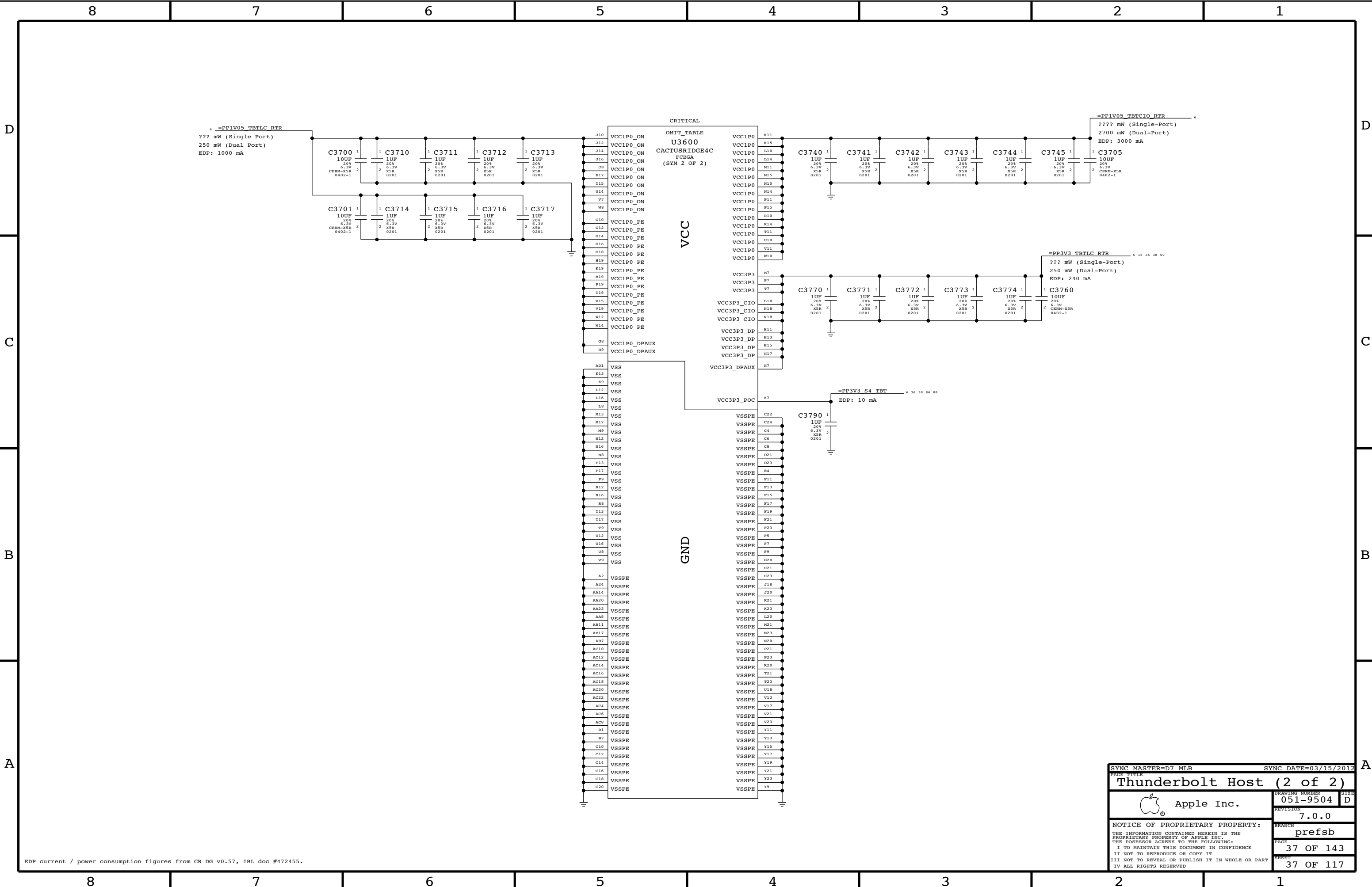


SUPERVISOR & CLKREG # ISOLATION
DELAY = 130 MS +/- 20%



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EDP current / power consumption figures from CR DG v0.57, IBL doc #472455.

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Thunderbolt Host		(2 of 2)	
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		prefsb	
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		37 OF 117	

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C

C

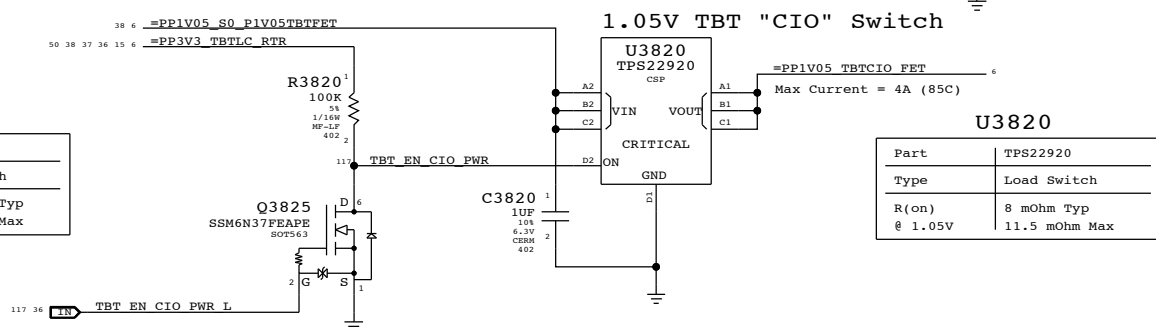
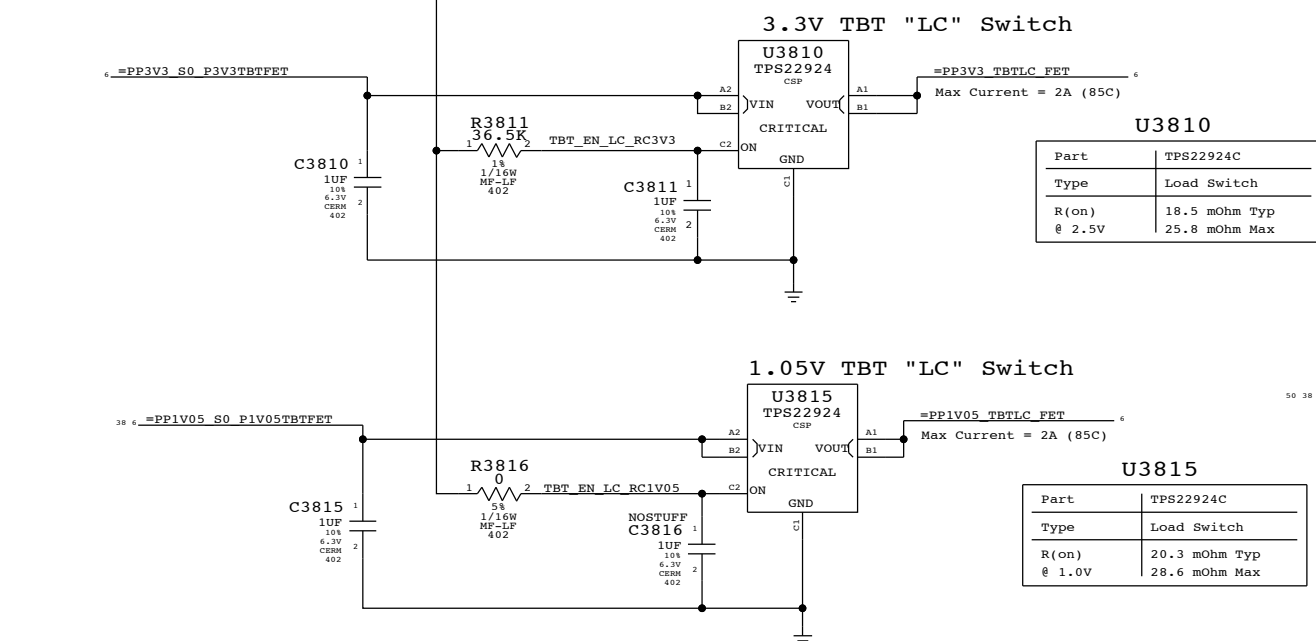
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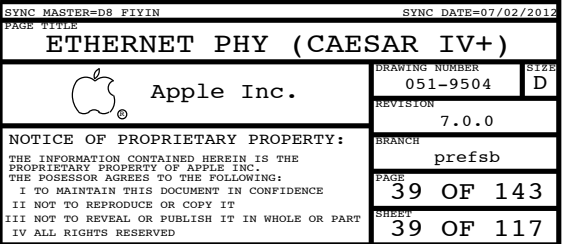
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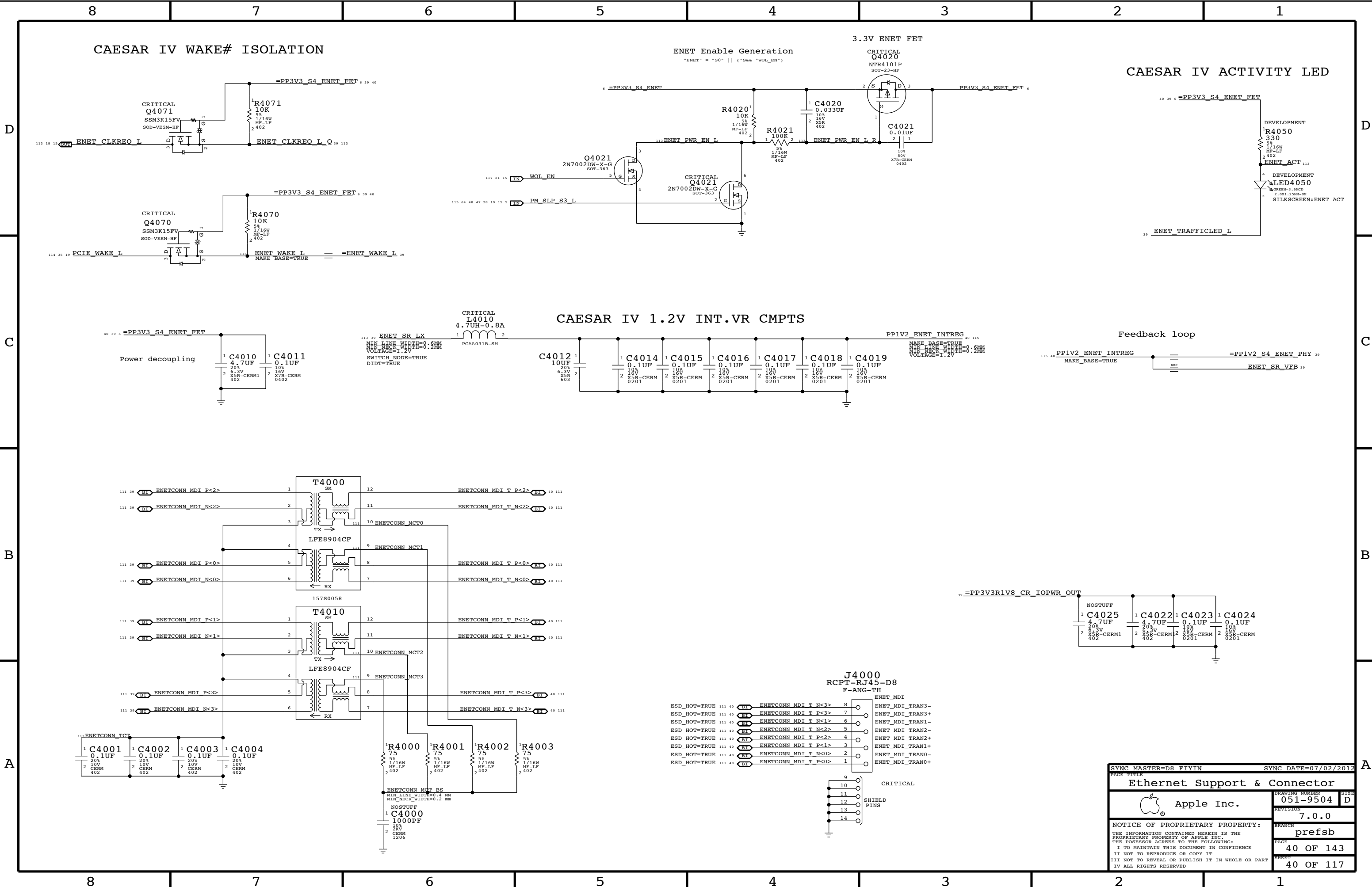
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SYNC MASTER=D8 FIYIN		SYNC DATE=07/02/2012	
PAGE TITLE		Ethernet Support & Connector	
Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	7.0.0
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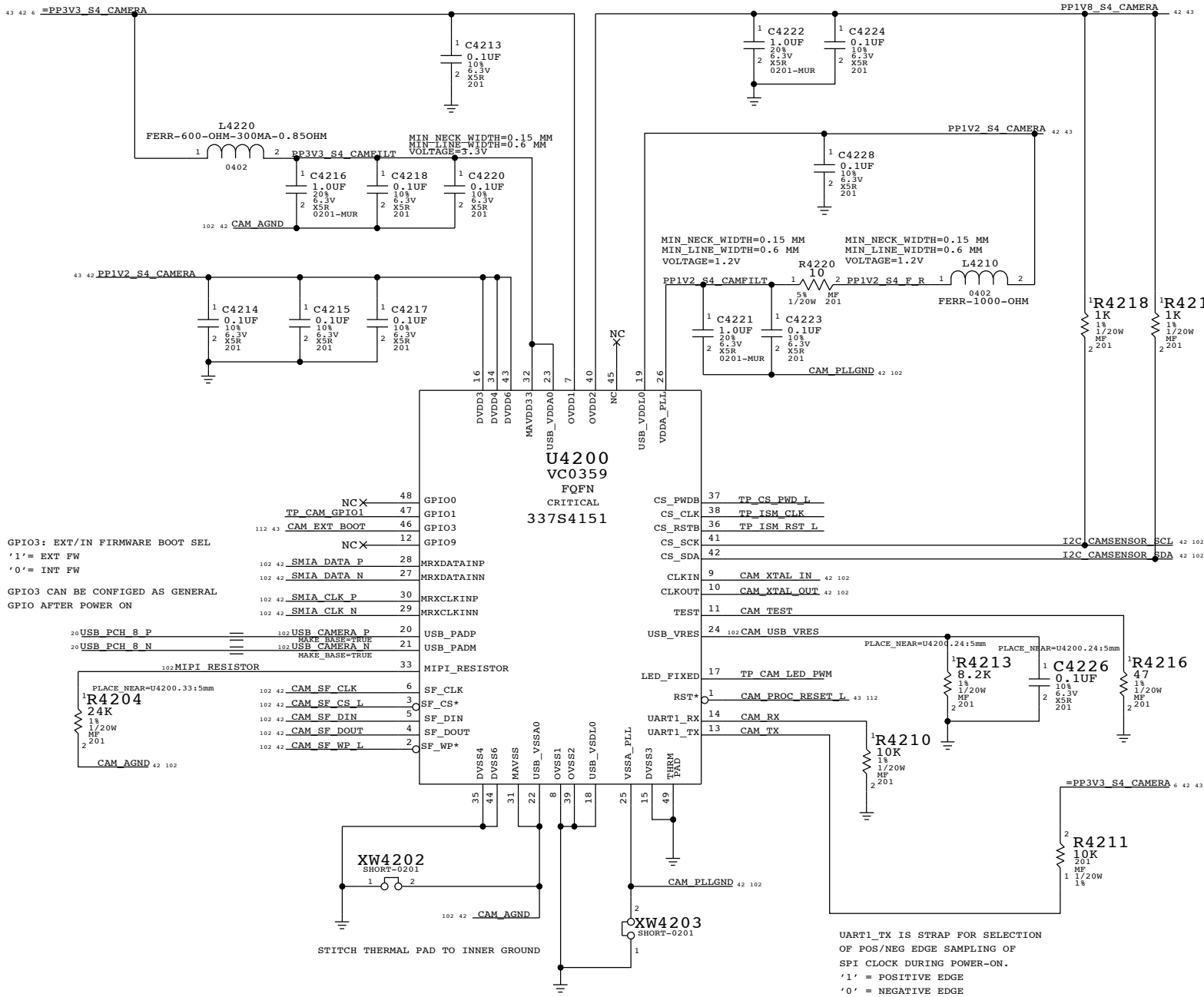
8	7	6	5	4	3	2	1
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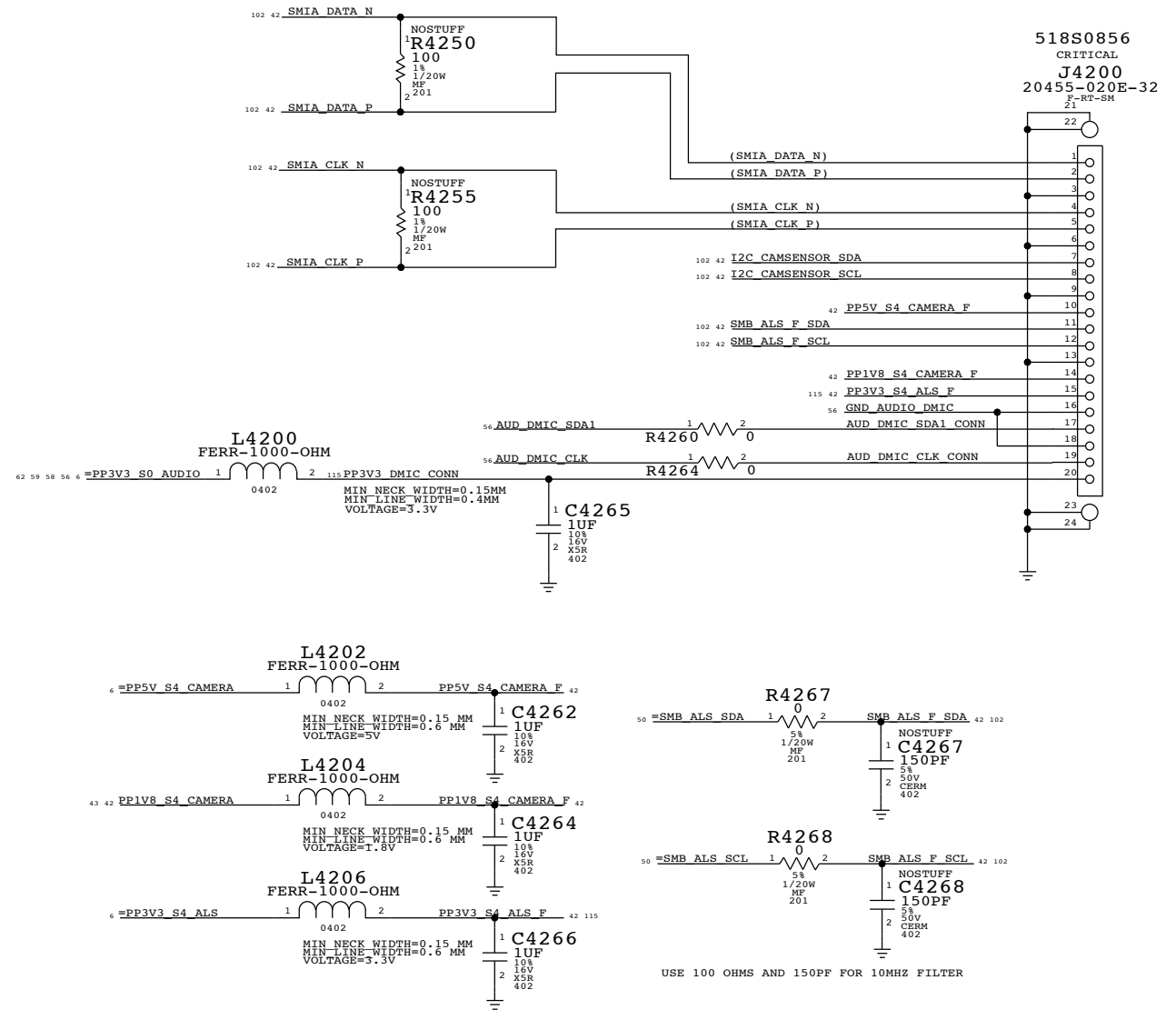
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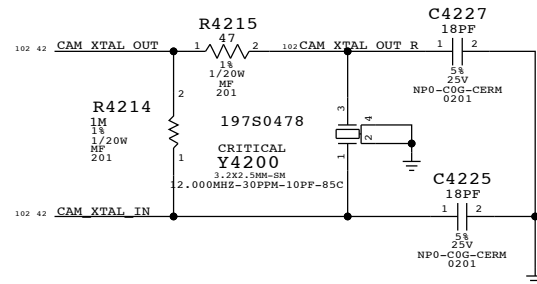
USB CAMERA CONTROLLER



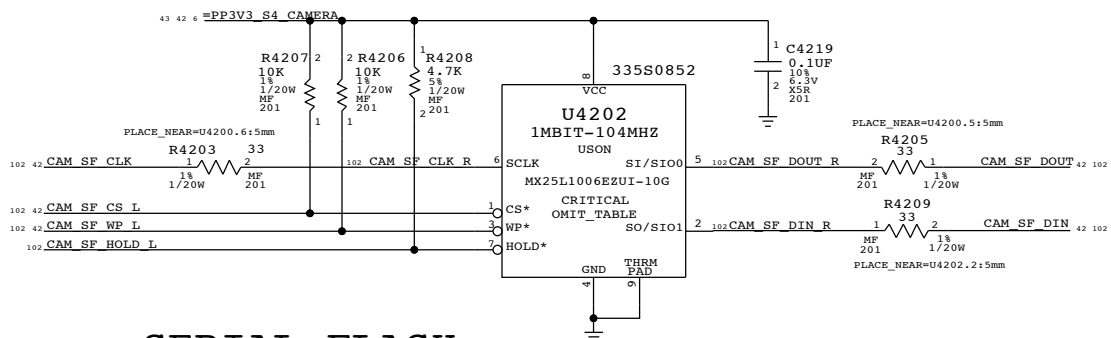
CAMERA/ALS/DMIC CONNECTOR



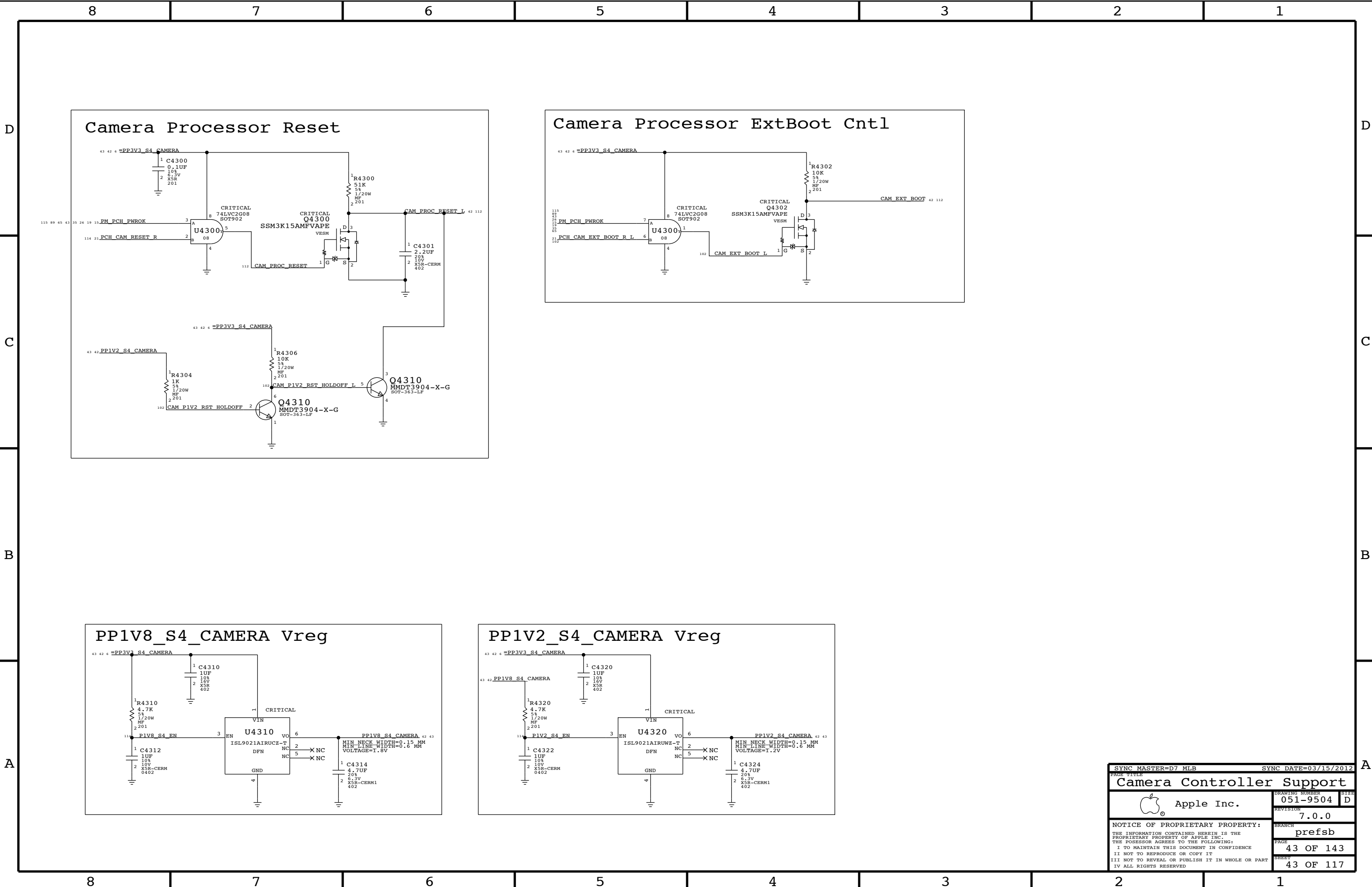
CRYSTAL



SERIAL FLASH



PAGE TITLE		PAGE NUMBER	
Camera Controller		051-9504	
Apple Inc.		7.0.0	
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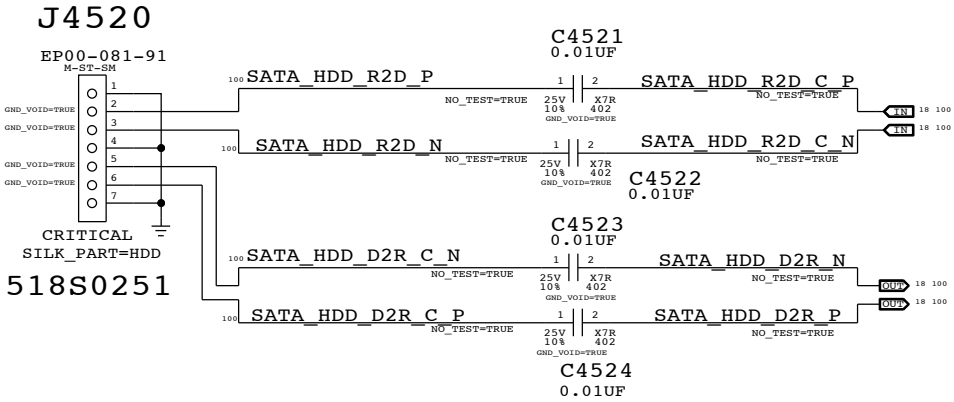
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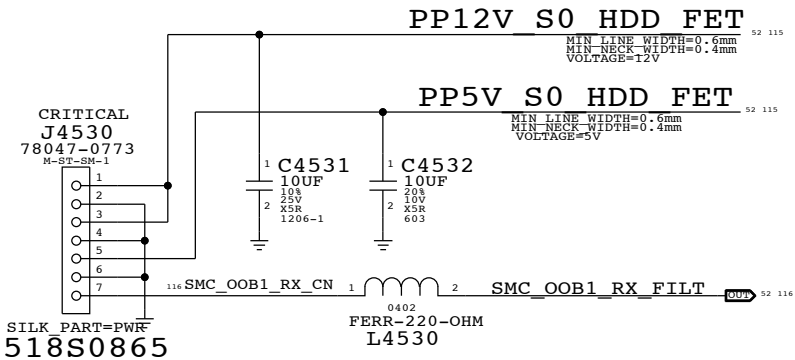
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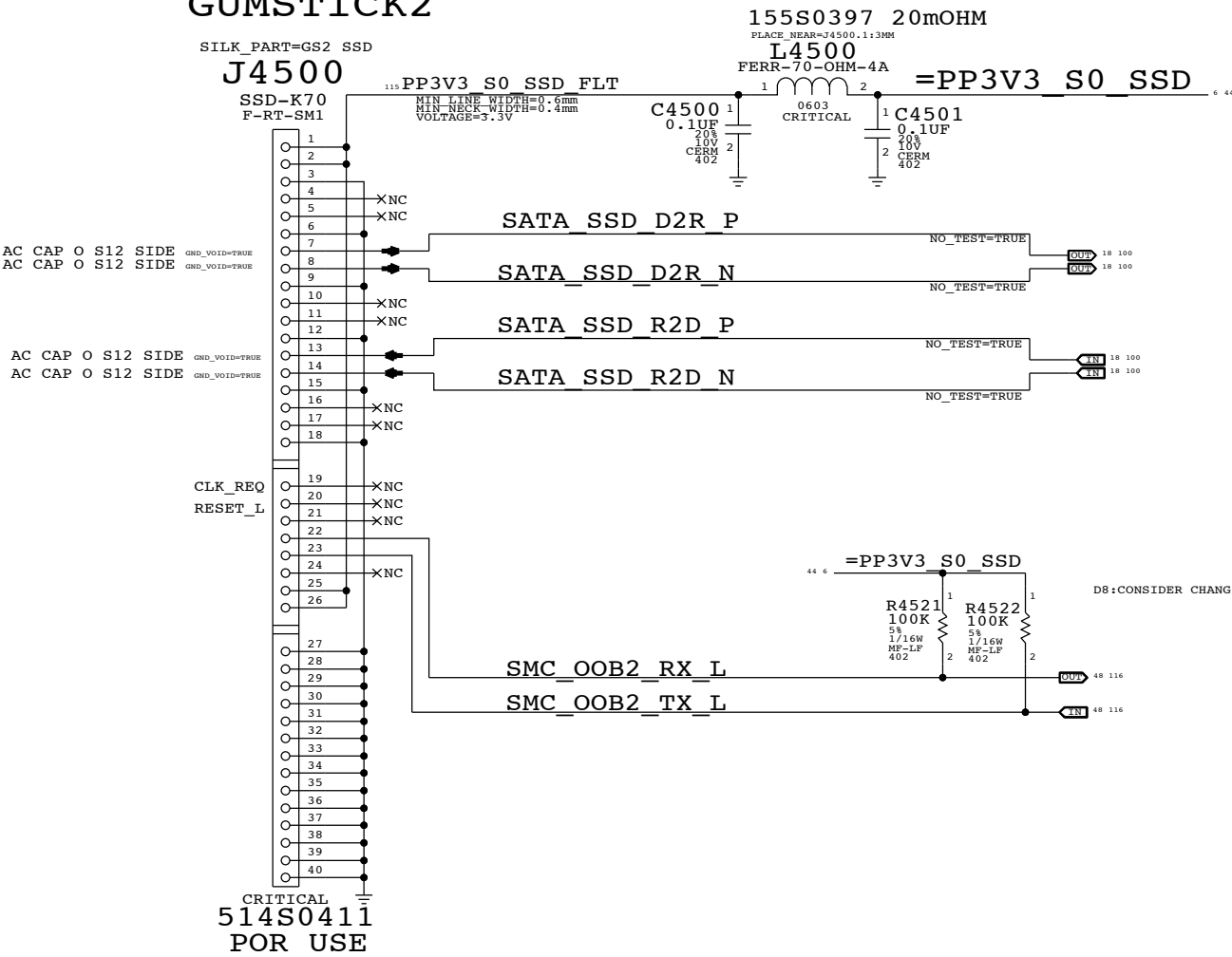
HDD DATA



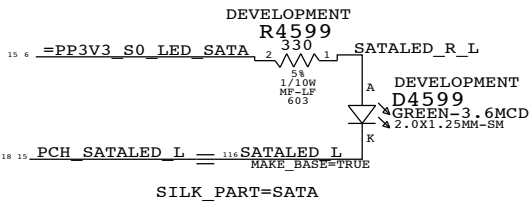
HDD POWER



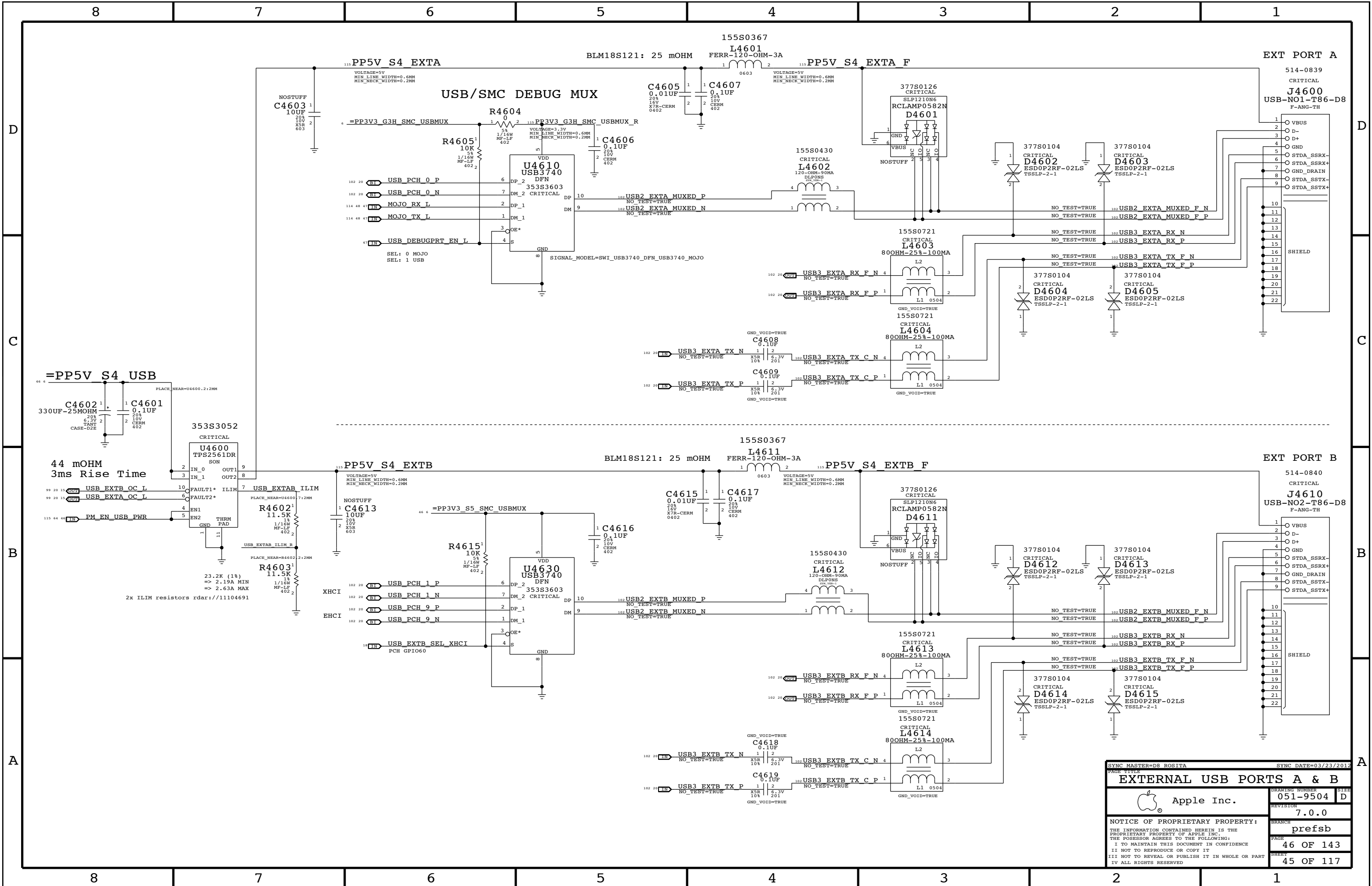
GUMSTICK2



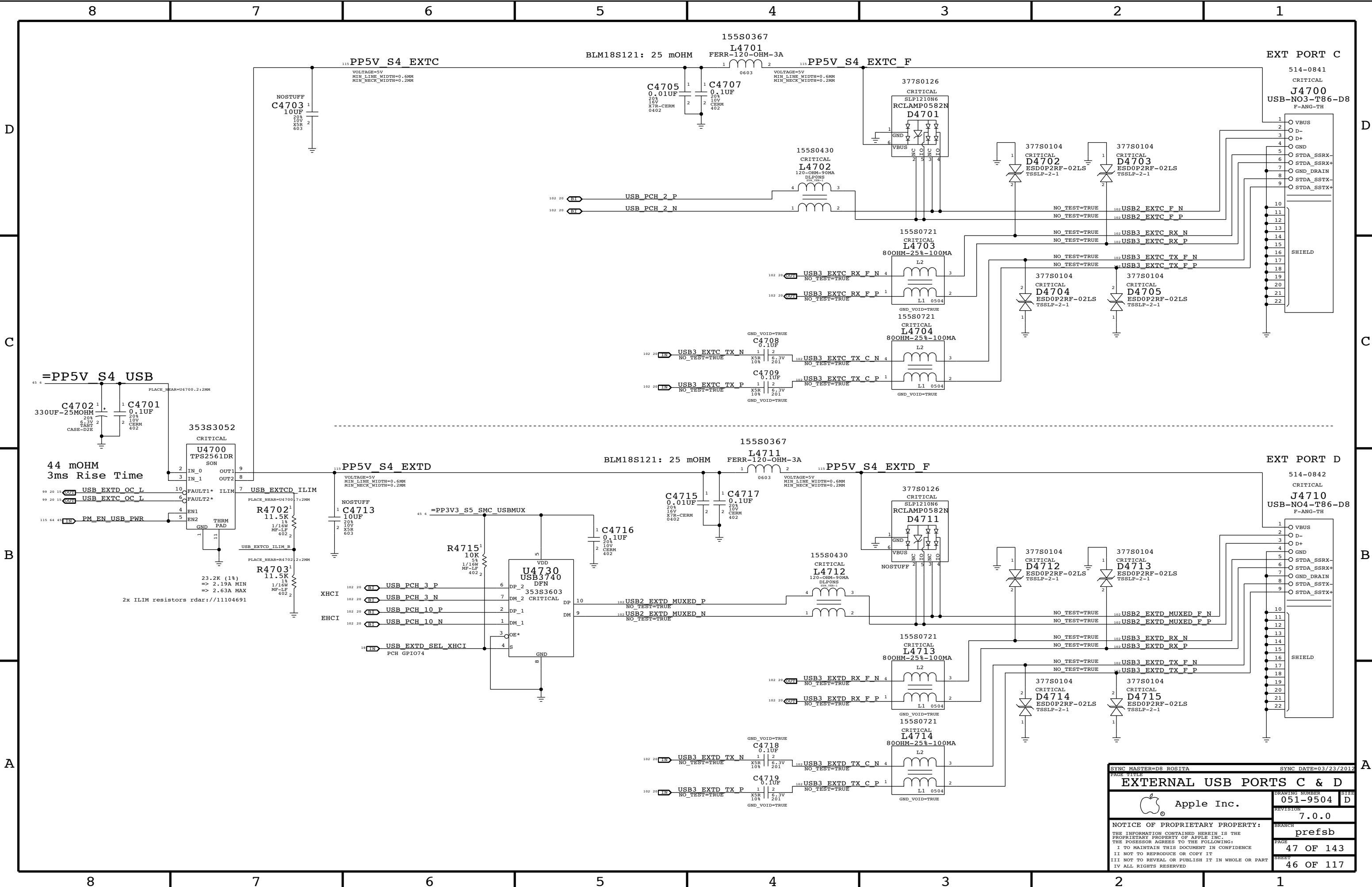
SATA Activity LED



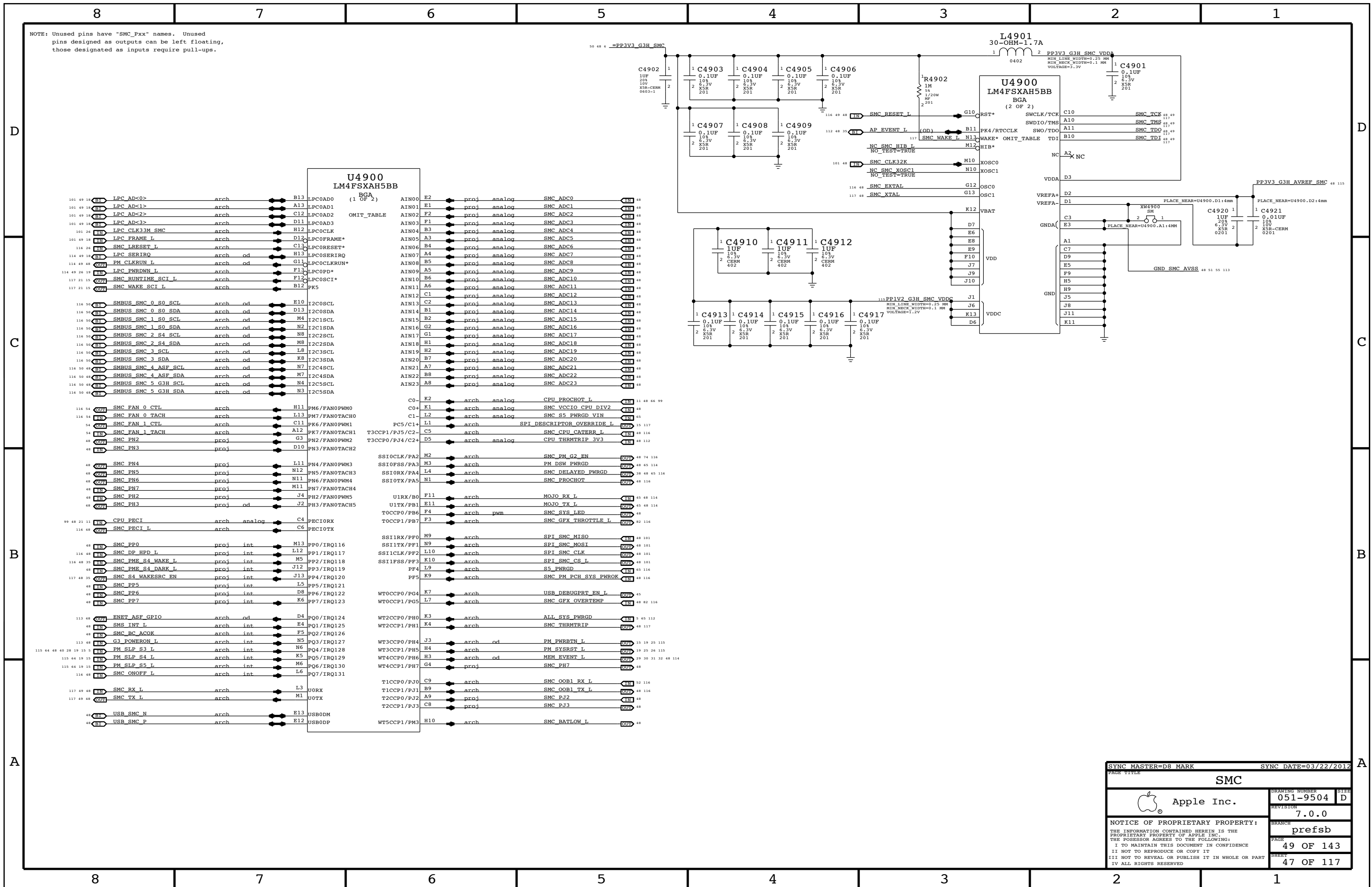
SYNC MASTER=D8_JERRY		SYNC DATE=01/31/2012	
PAGE TITLE			
SATA Connectors			
 Apple Inc.		DRAWING NUMBER	051-9504
		SIZE	D
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		BRANCH	prefsb
		PAGE	45 OF 143
		SHEET	44 OF 117

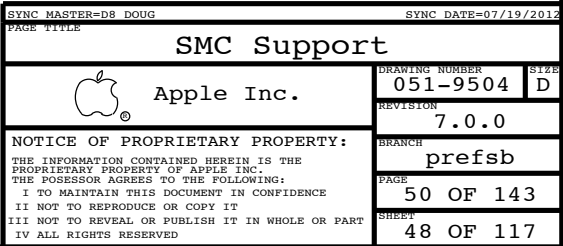


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PAGE TITLE			
EXTERNAL USB PORTS A & B		DRAWING NUMBER	
 Apple Inc.		051-9504	
		REVISION	
		7.0.0	
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SYNC MASTER=D8 ROSITA		SYNC DATE=03/23/2012	
PAGE TITLE			
EXTERNAL USB PORTS C & D			
 Apple Inc.	DRAWING NUMBER	051-9504	SIZE
	REVISION	7.0.0	
	BRANCH	prefsb	
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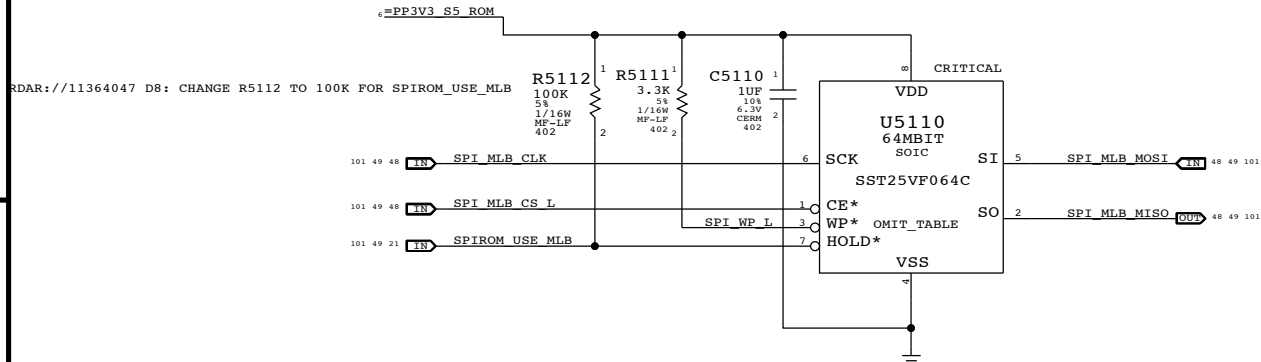
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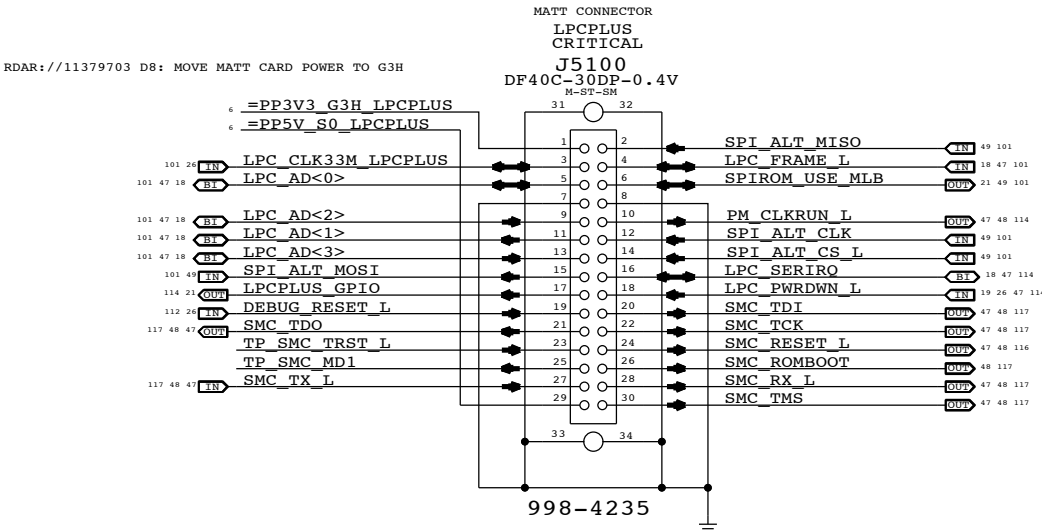
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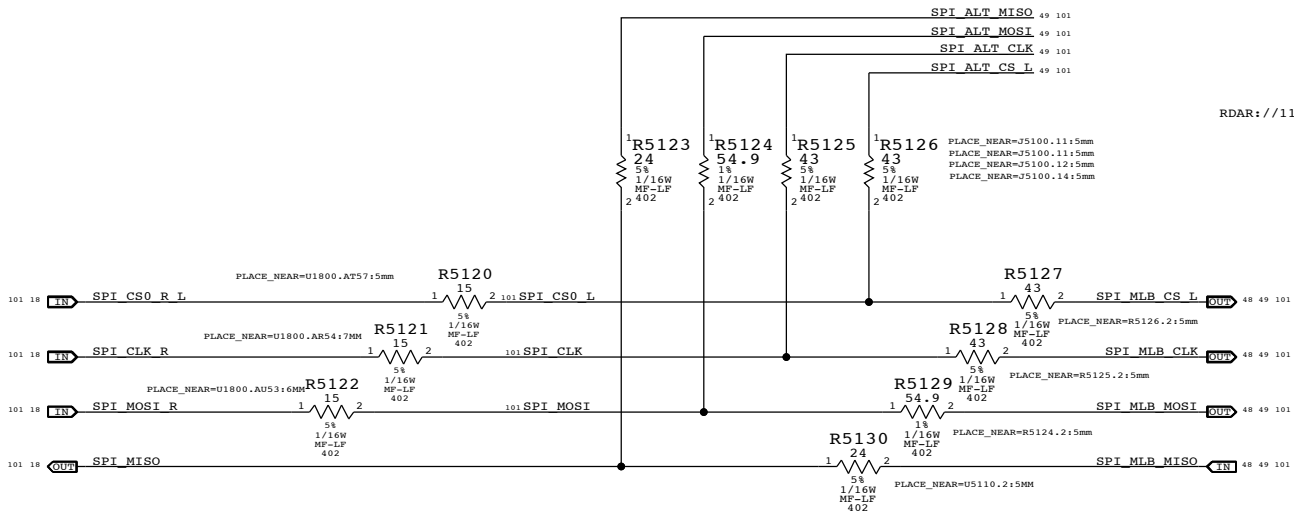
SPI BootROM



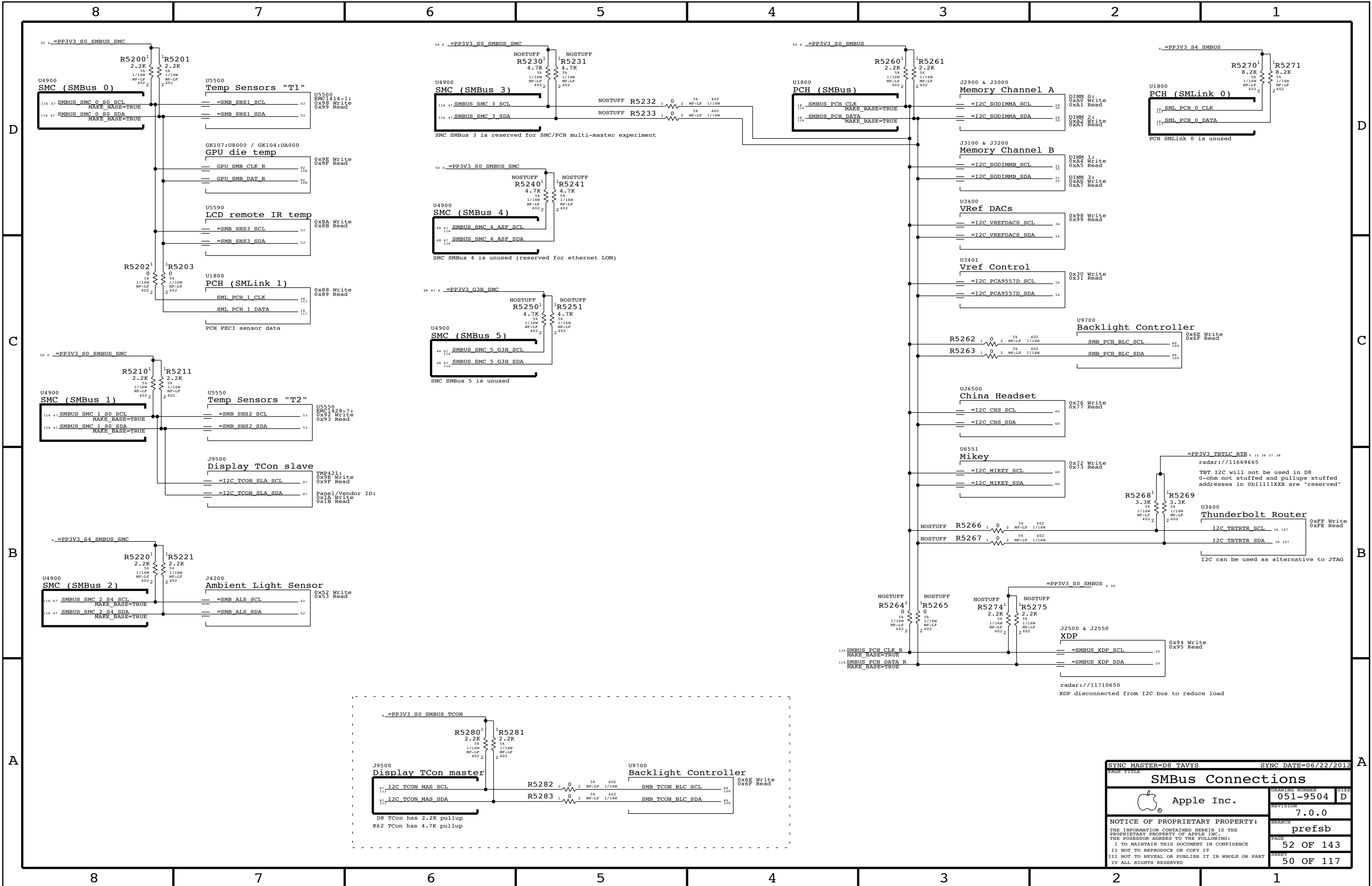
LPC+SPI Connector

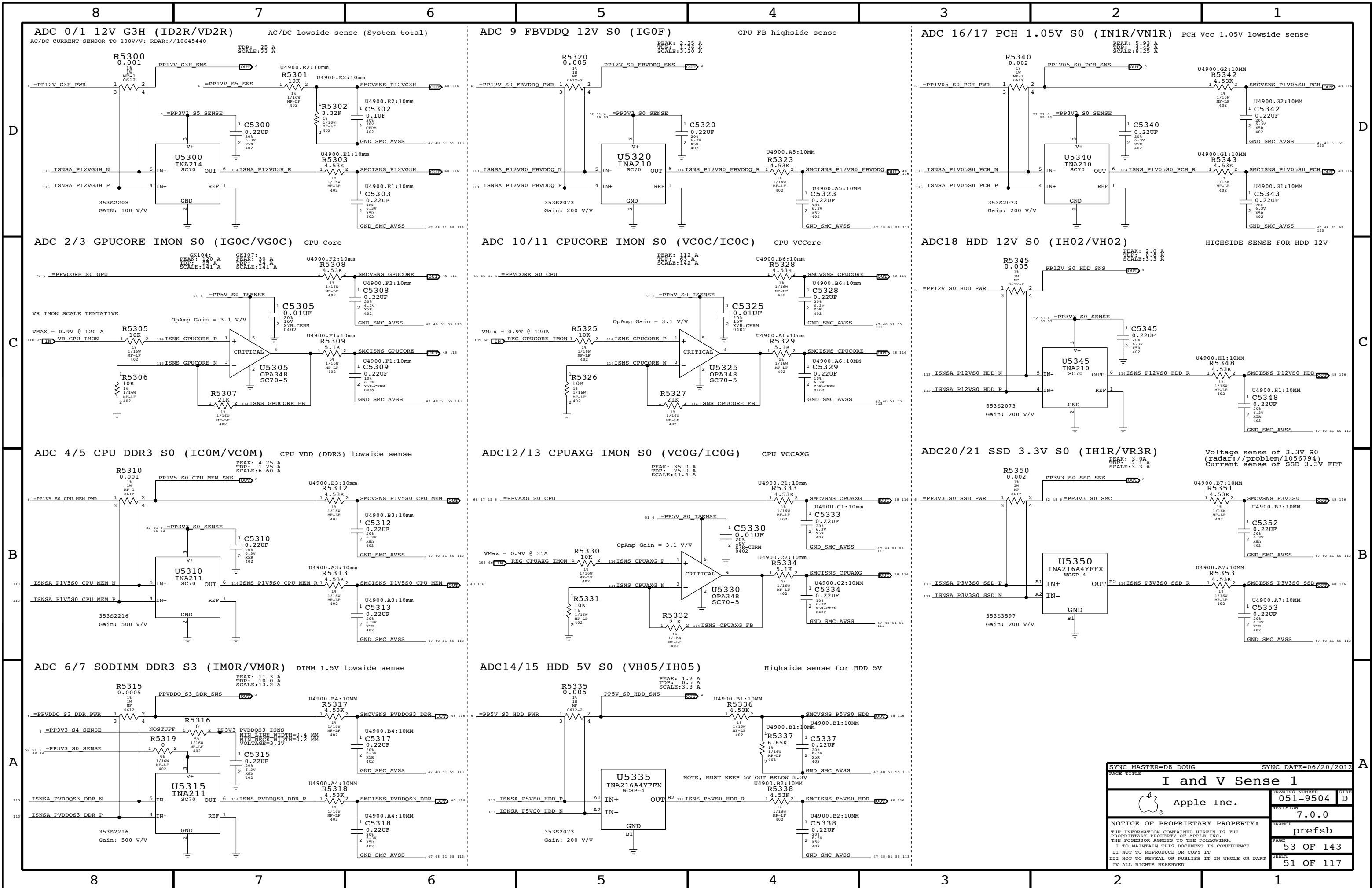


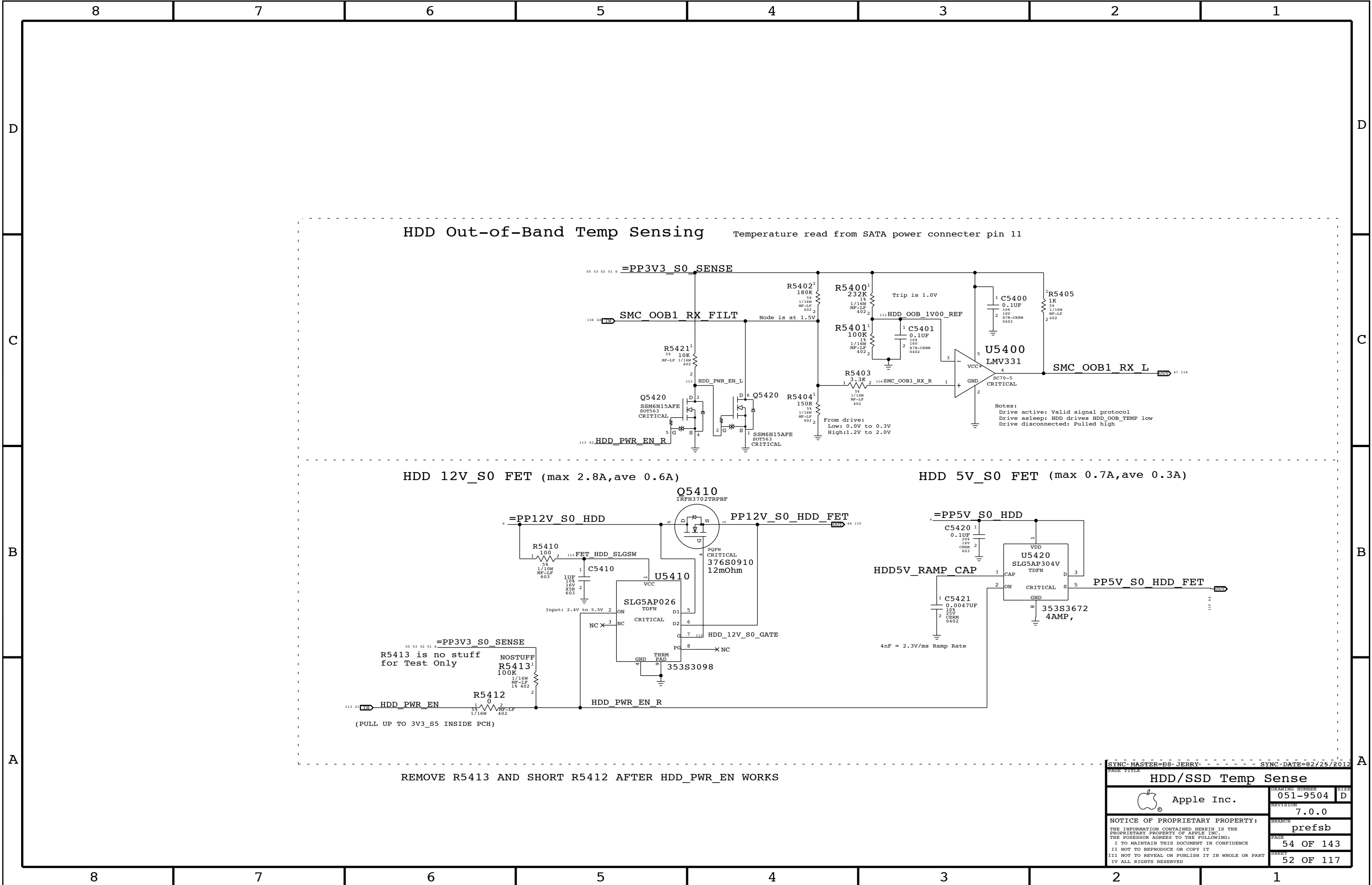
SPI Series Termination



SYNC MASTER=D8_MLB		SYNC DATE=N/A	
PAGE TITLE			
SPI and Debug Connector			
 Apple Inc.		DRAWING NUMBER	051-9504
		REVISION	7.0.0
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		PAGE	51 OF 143
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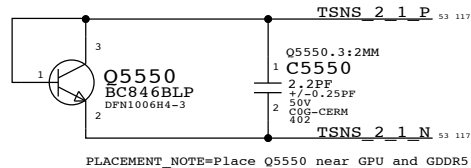


PAGE TITLE			SYNC-MASTER=D8-JERRY-	SYNC-DATE=02/25/2012
HDD/SSD Temp Sense				
DRAWING NUMBER		051-9504	SIZE	D
REVISION		7.0.0		
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		SHEET	52 OF 117	

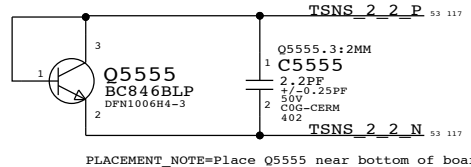
Temperature Sensor T2 EMC1428: Near GPU VR

SNS T2: TEMP SENSOR IC

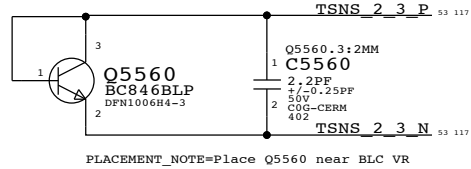
GPU Proximity



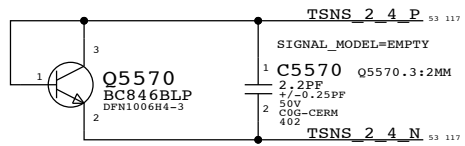
Ambient



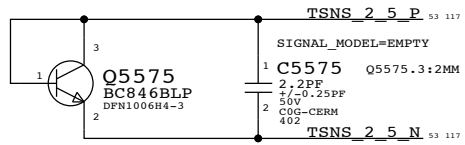
BLC Proximity



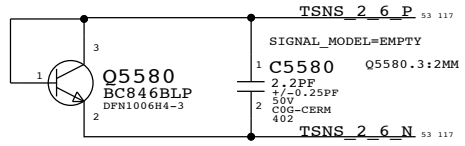
SO-DIMM Proximity 1



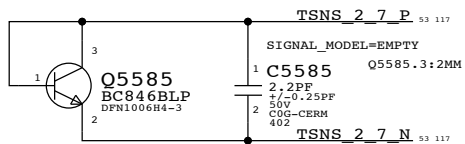
SO-DIMM Proximity 2



SO-DIMM Proximity 3



SO-DIMM Proximity 4



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
37280186	37280185		ALL	Alternate Temp Diode

GPU Prox (TG0p)

TSNS_2_1_P

TSNS_2_1_N

Ambient (TA0p)

TSNS_2_2_P

TSNS_2_2_N

SoDIMM Prox 1 (TM0p)

TSNS_2_4_P

TSNS_2_4_N

SoDIMM Prox 3 (TM2p)

TSNS_2_6_P

TSNS_2_6_N

SoDIMM Prox 4 (TM3p)

TSNS_2_7_P

TSNS_2_7_N

SoDIMM Prox 2 (TM1p)

TSNS_2_5_P

TSNS_2_5_N

BLC PROX (Tb0p)

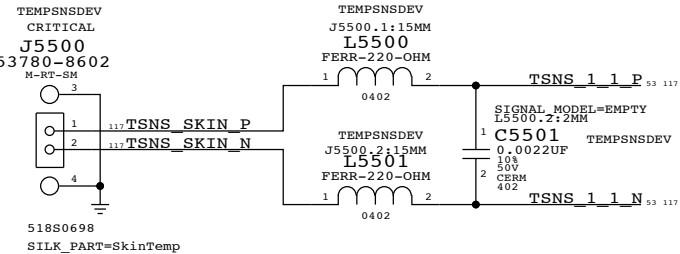
TSNS_2_3_P

TSNS_2_3_N

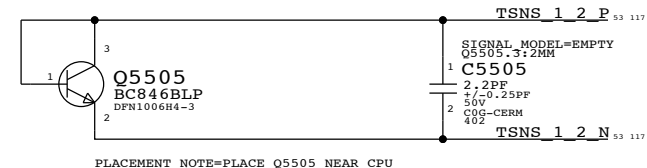
EMC1428-7: 6.8K PULL UP: I2C ADDRESS: WRITE: 0x92, READ: 0x93

Temperature Sensor T1 EMC1414: Near PSU Conn

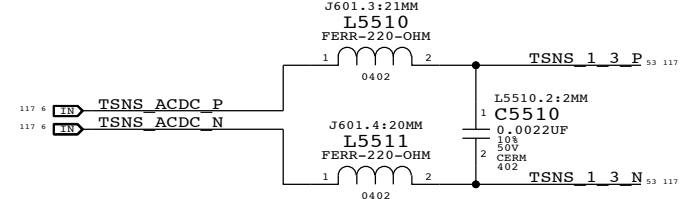
Skin



CPU Proximity



AC/DC Via connector to diode inside PSU



Skin Temp (TS0p)

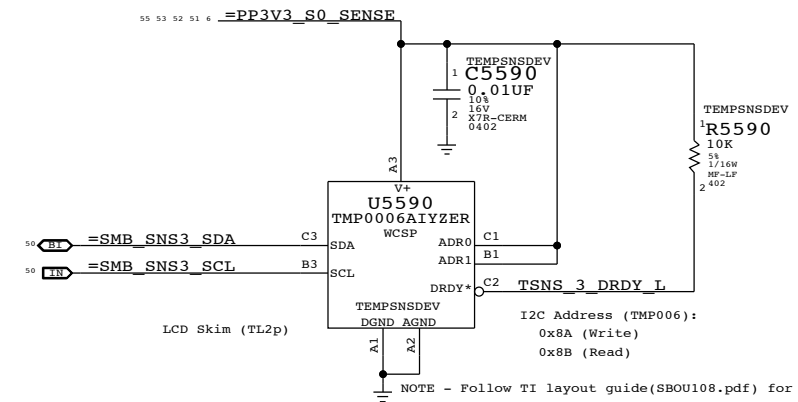
CPU Prox (TC0p)

AC/DC (Tp2p)

Note: Internal sensor of the EMC 1414 will be used as MLB sensor. MLB Prox 1 (Tm0p)

PLACEMENT_NOTE=PLACE U5500 NEAR PSU CONNECTOR

Temperature Sensor T3: LCD Remote Sensor (Dev4Now)



SYNC MASTER=D8 DOUG		SYNC DATE=06/07/2012	
PAGE TITLE			
Temperature Sensors			
 Apple Inc.		DRAWING NUMBER	051-9504
		SIZE	D
		REVISION	7.0.0
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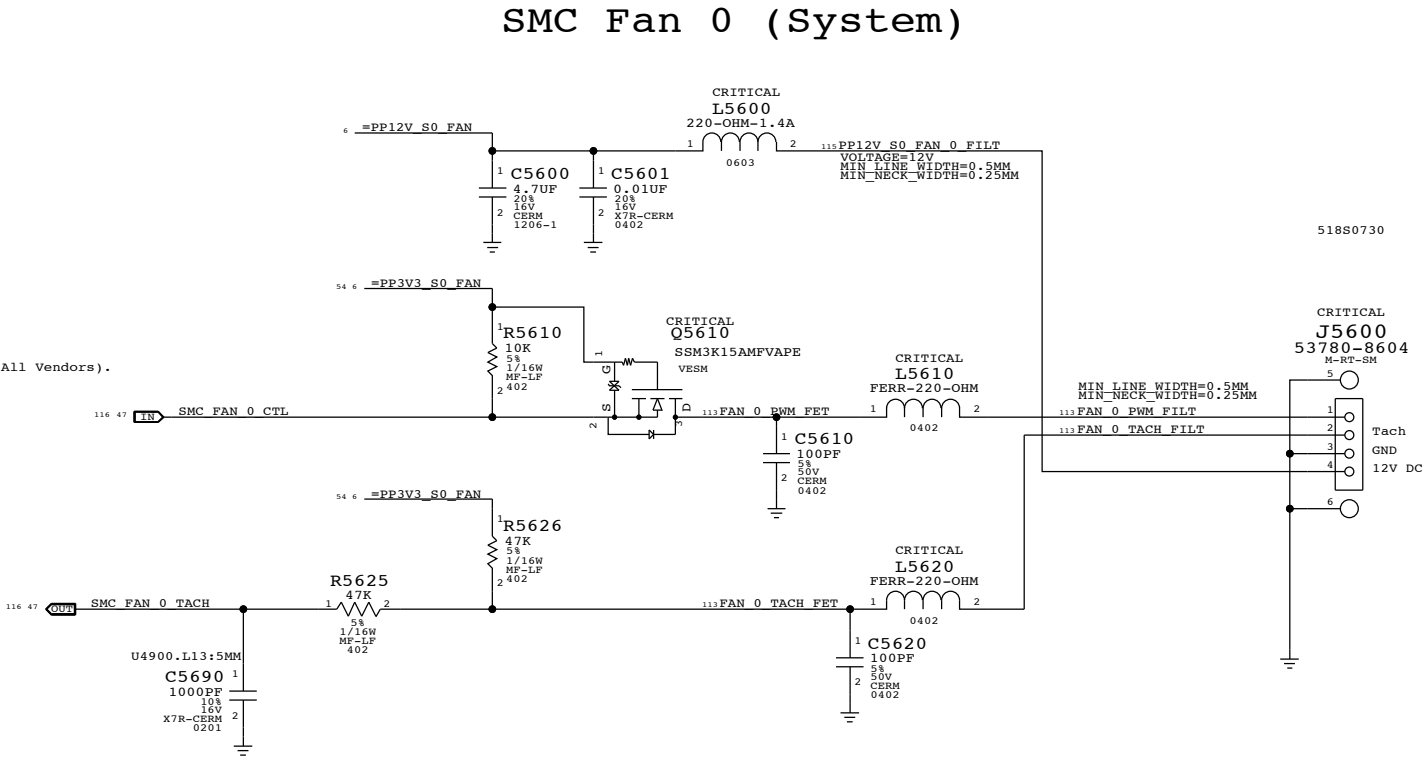
A

Note:

The circuit for the PWM input to the fan acts as a non-inverting level-shifter to protect the SMC. It is assumed there is a pull-up to 5V/12V inside the fan, otherwise when the SMC PWM goes low and Q5610 turns on, there would be 5V/12V present on the SMC pin! Then by definition, the drain of Q5610 is at common and the SMC sinks current when Q5610 is on.

This resembles an open-drain if there is a pull-up, going to a Hi-Z FET input.

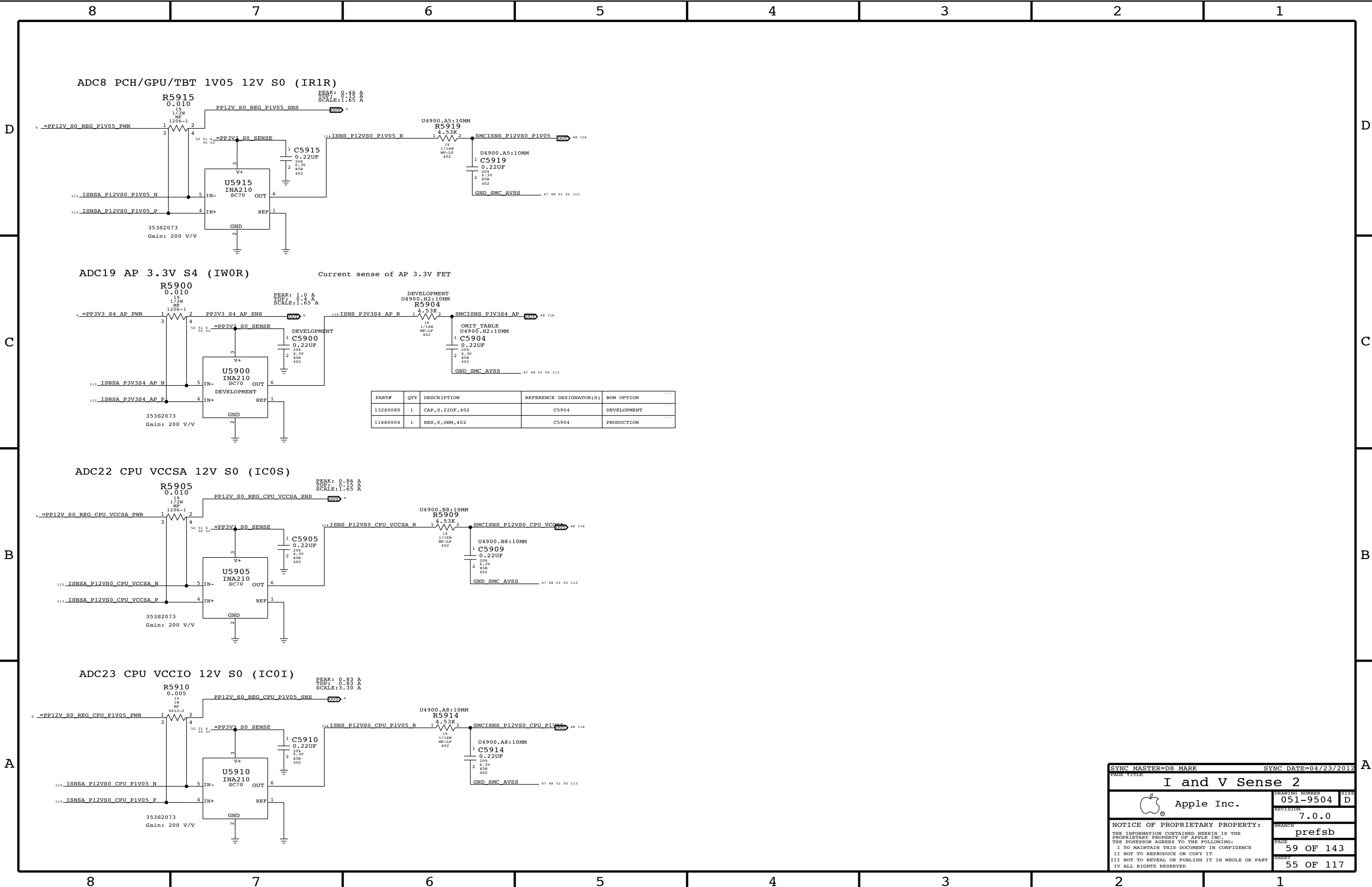
Otherwise, this is simply a pass-FET. See RADAR: 10565825- D7: Need schematic and PCB file of fan(All Vendors).



SMC Fan 1 (Unused)



SYNC MASTER=D8 DOUG		SYNC DATE=07/19/2012	
PAGE TITLE			
System Fan		DRAWING NUMBER	
	Apple Inc.	051-9504	SIZE
		REVISION	D
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		BRANCH	prefsb
		PAGE	56 OF 143
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PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
132S0080	1	CAP,0.22UF,402	C5904	DEVELOPMENT
116S0004	1	RES,0,0HM,402	C5904	PRODUCTION

SYNC MASTER=D8 MARK

SYNC DATE=04/23/2012

I and V Sense 2

 Apple Inc.

DRAWING NUMBER
051-9504

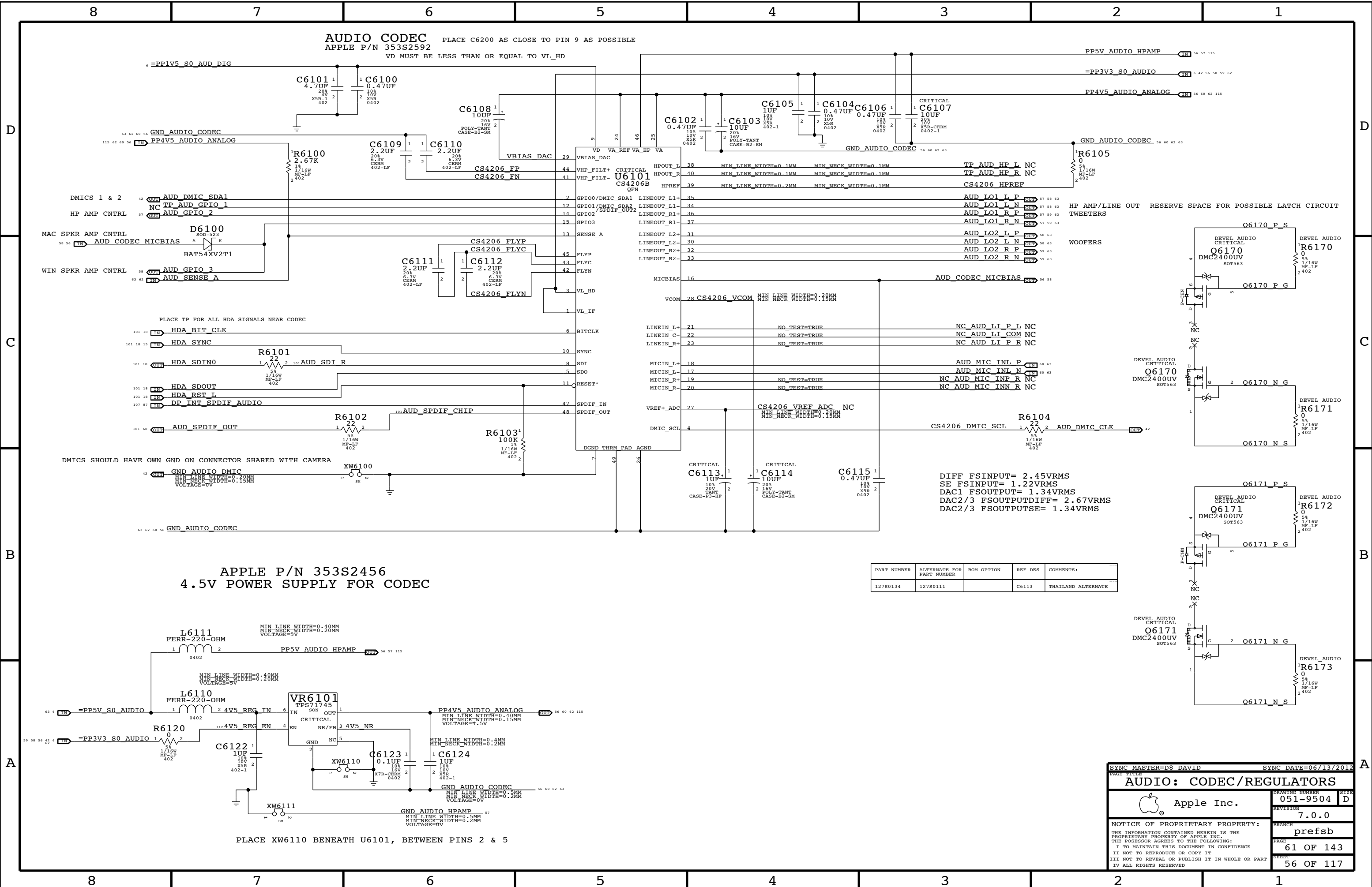
REVISION
7.0.0

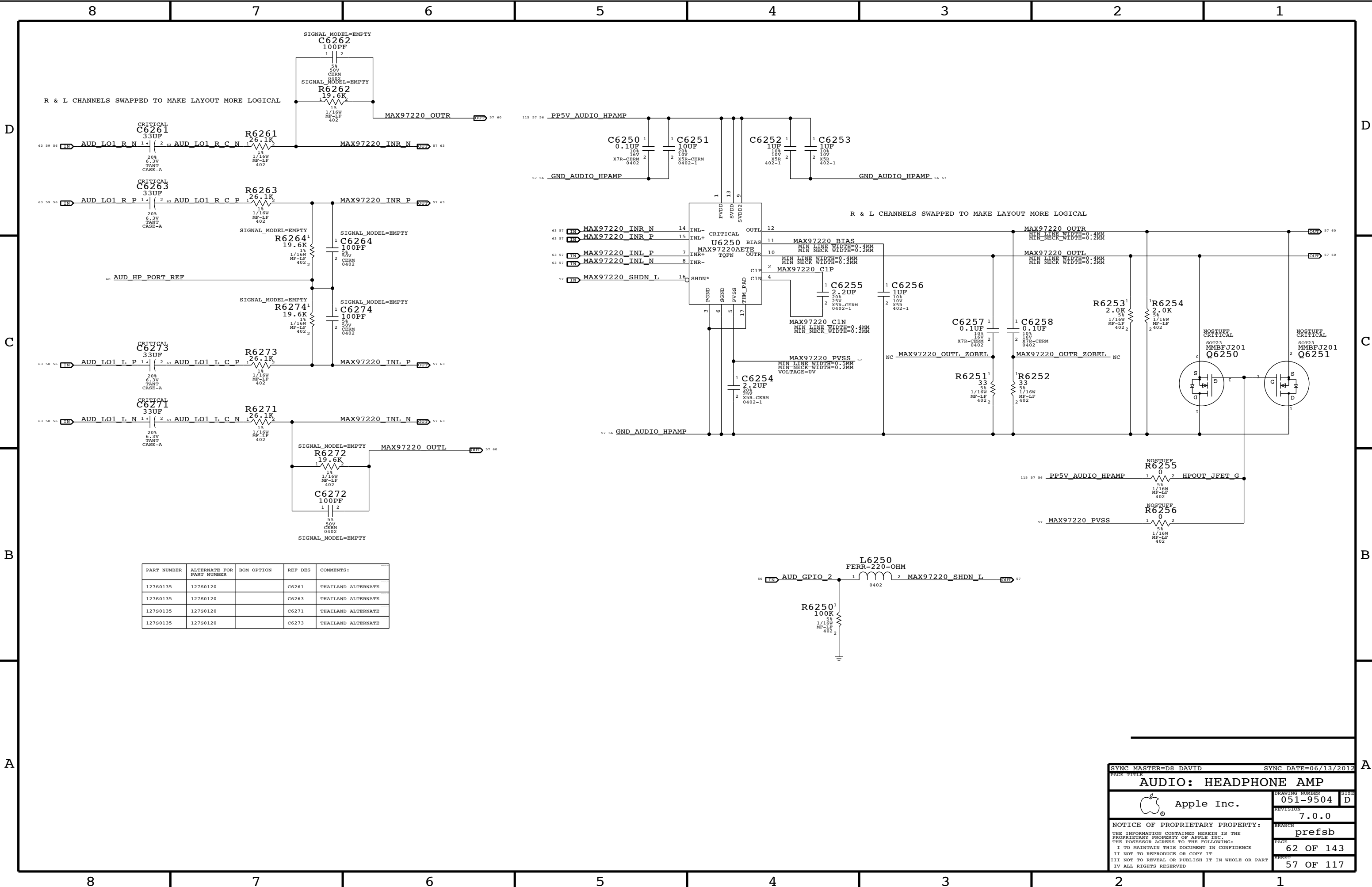
BRANCH
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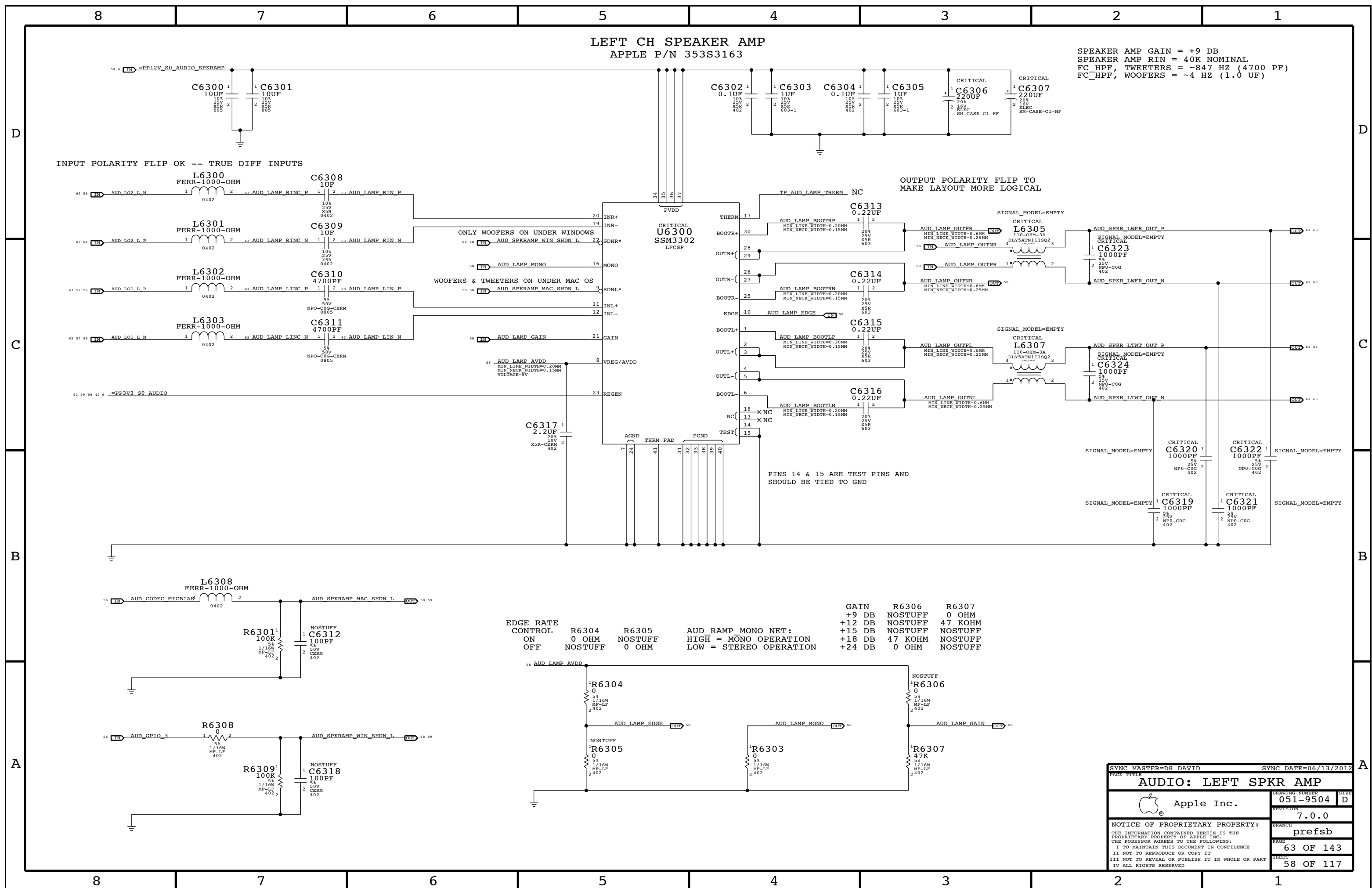
PAGE
59 OF 143

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PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
12780135	12780120		C6261	THAILAND ALTERNATE
12780135	12780120		C6263	THAILAND ALTERNATE
12780135	12780120		C6271	THAILAND ALTERNATE
12780135	12780120		C6273	THAILAND ALTERNATE



MIKEY RECEIVER CKT

WRITE: 0X72 READ: 0X73 APN 353S3231

I2C ADDRESSES				
MIKEY	U6551	READ	0111 0011	0X73
MIKEY	U6551	WRITE	0111 0010	0X72

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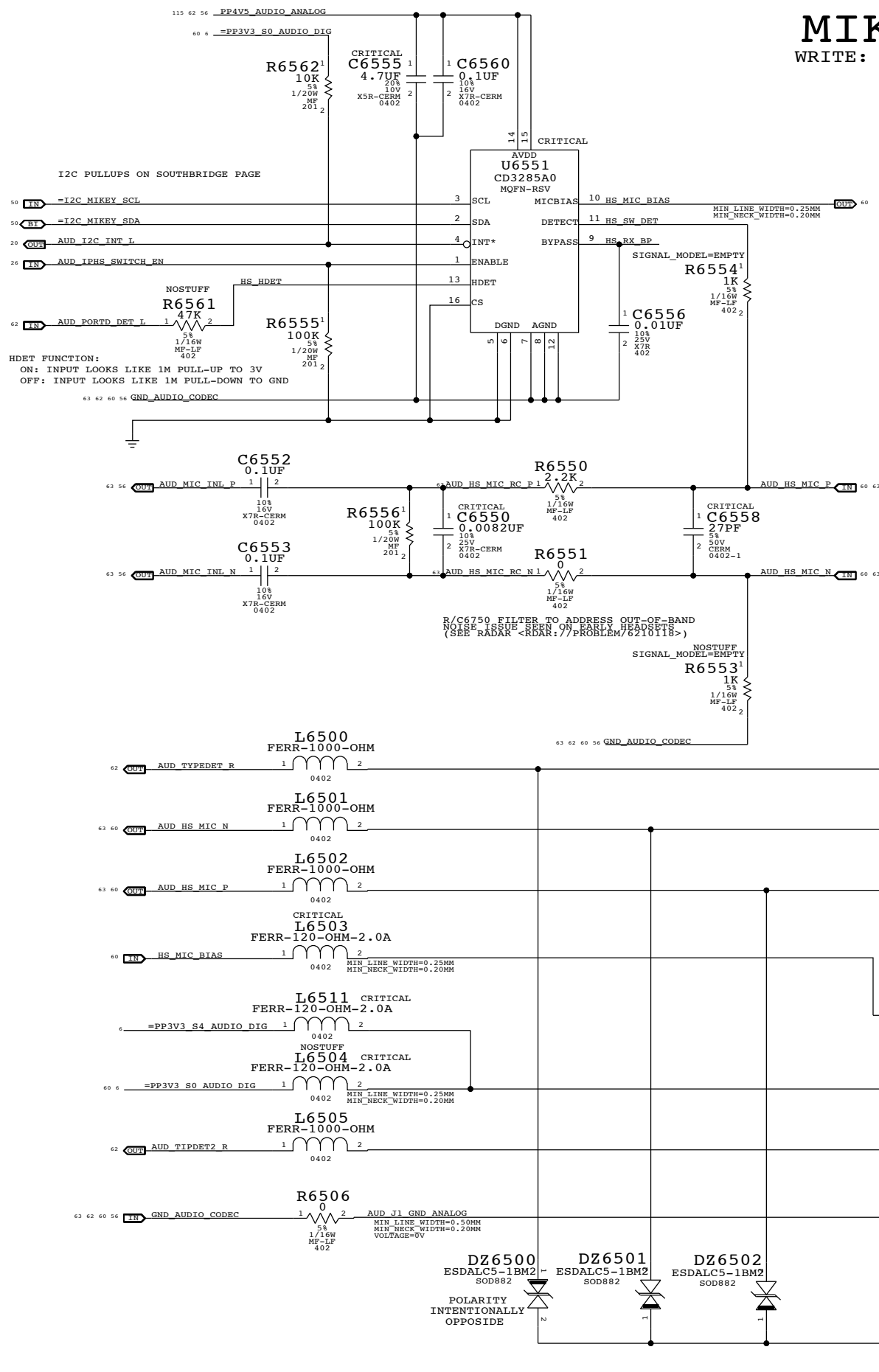
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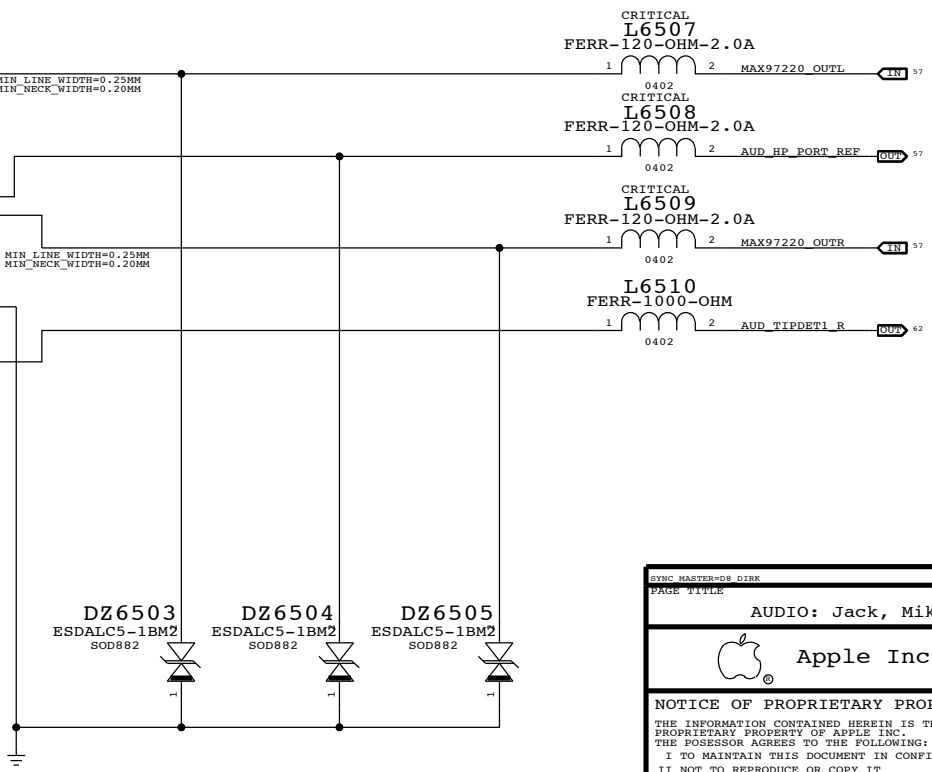
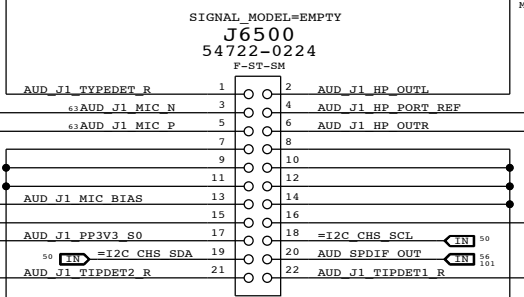
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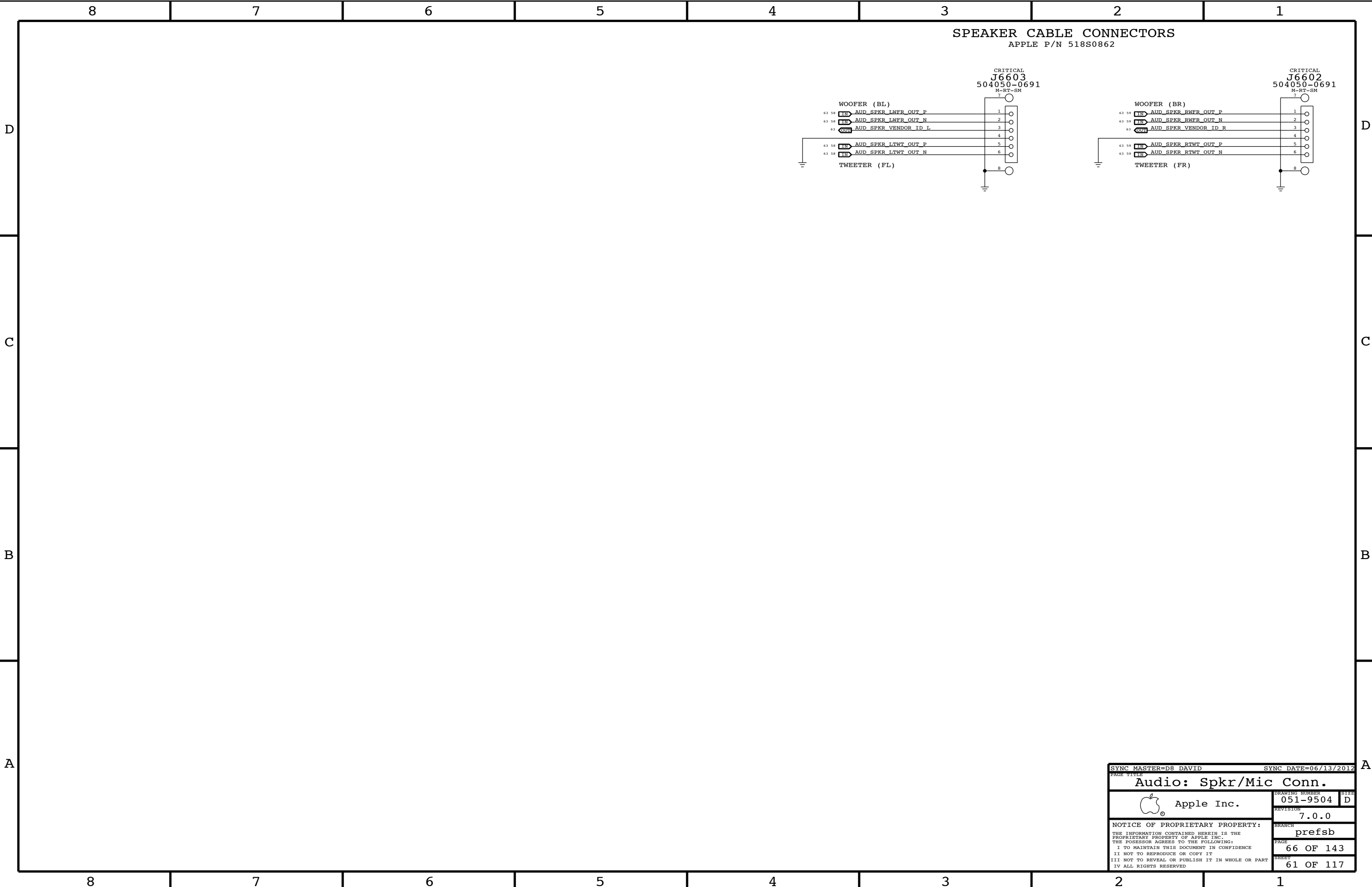
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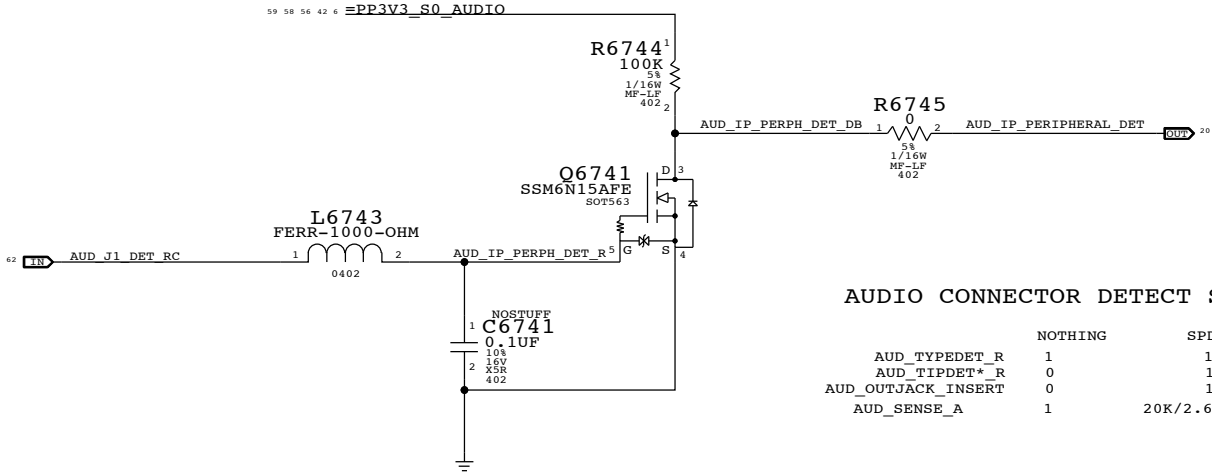
APPLE P/N 518S0687



AUDIO: Jack, Mikey, CHS Switch		DRAWING NUMBER		SIZE
Apple Inc.		051-9504		D
REVISION		7.0.0		BRANCH
prefsb		PAGE		65 OF 143
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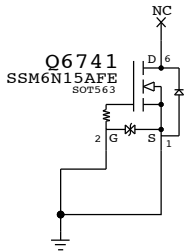


IPHS HS Detect Debounce CKT

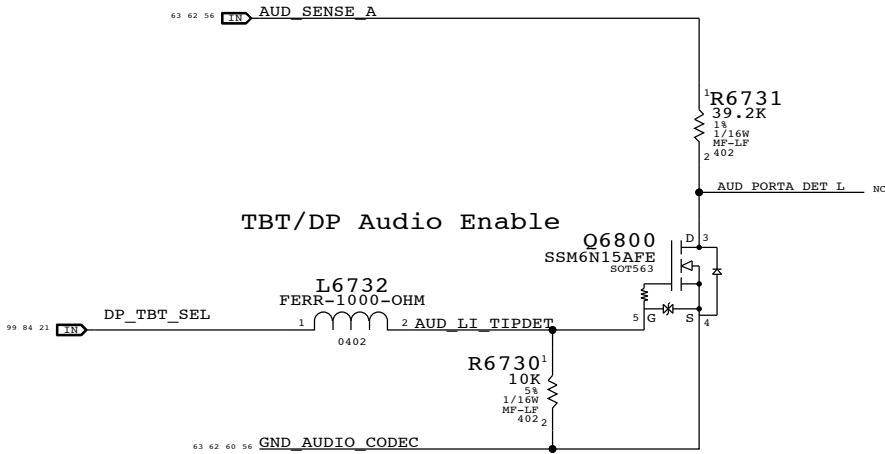


AUDIO CONNECTOR DETECT STATES

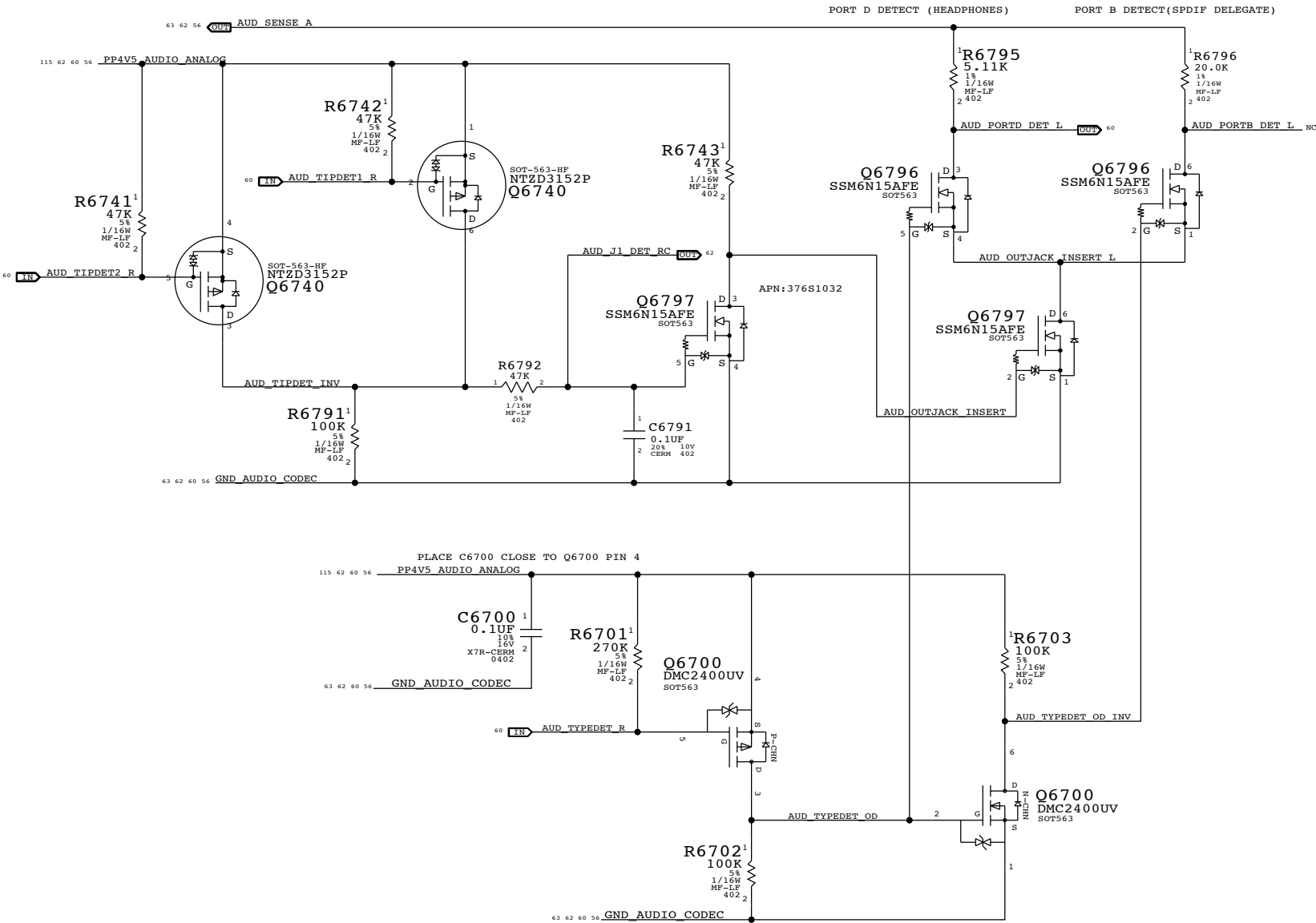
	NOTHING	SPDIF	HEADPHONE
AUD_TYPERDET_R	1	1	0
AUD_TIPDET*_R	0	1	1
AUD_OUTJACK_INSERT	0	1	1
AUD_SENSE_A	1	20K/2.67K RDIV	5.11K/2.67K RDIV



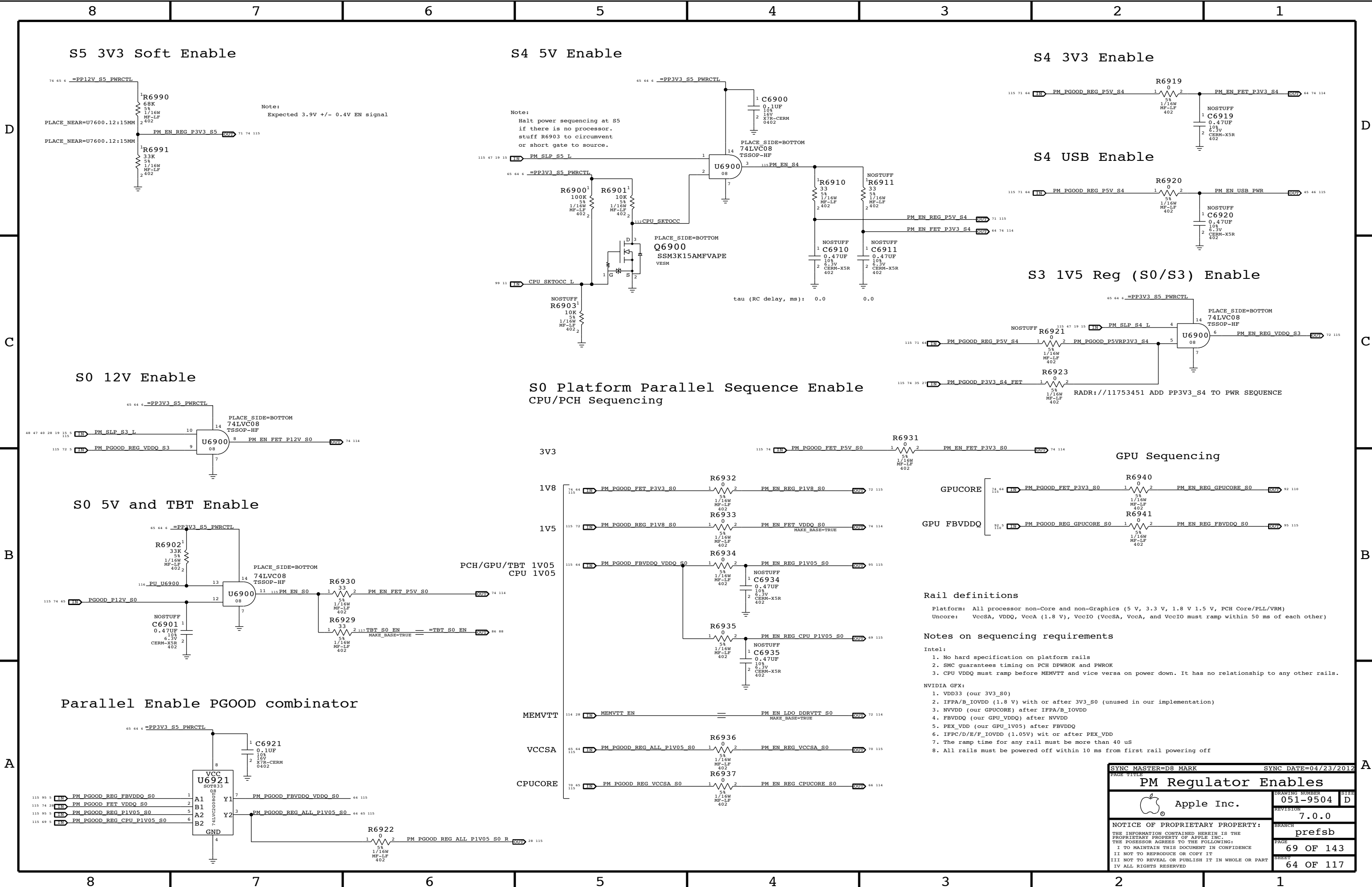
LI Insert Detect (DETECT A)



TBT/DP Audio Enable



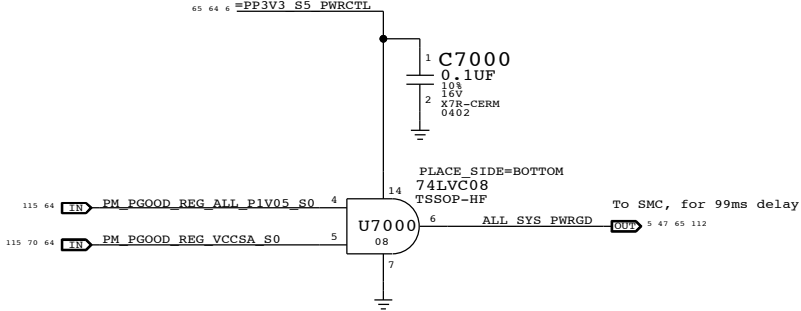
SYNC MASTER=D8 DAVID		SYNC DATE=06/13/2012	
PAGE TITLE			
AUDIO: Detects/Grounding			
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		REVISION	7.0.0
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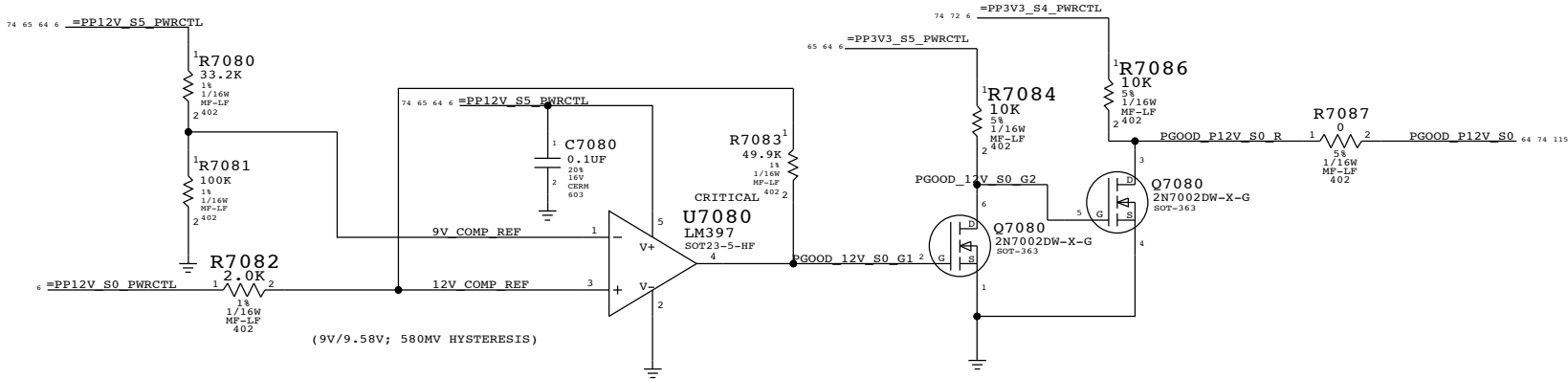
SYNC MASTER=D8 MARK		SYNC DATE=04/23/2012	
PAGE TITLE		PM Regulator Enables	
DRAWING NUMBER		051-9504	SIZE D
REVISION		7.0.0	BRANCH
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Platform Power Good Derive SMC ALL_SYS_PWRGD

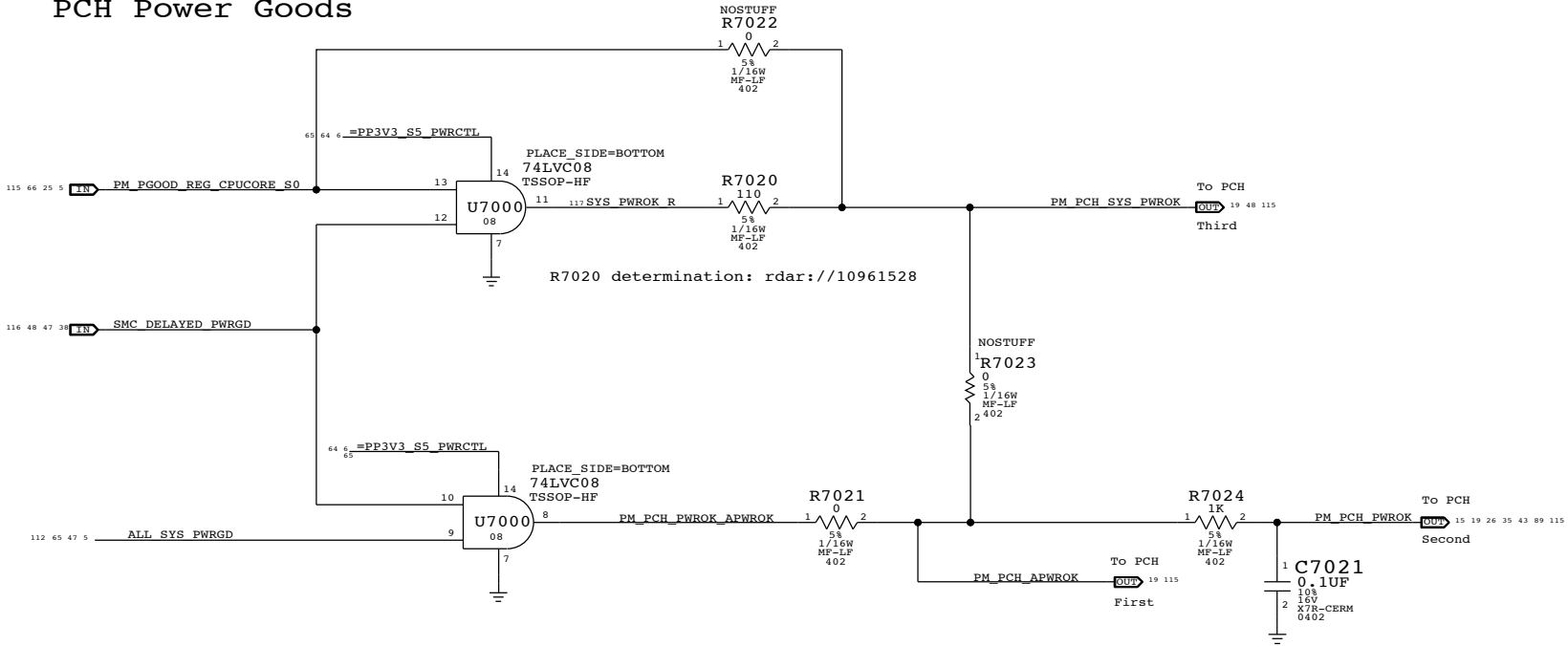
The end of the power sequence for S0 rails except CPU CORE.



PGOOD COMPARATORS FOR PP12V_S0



PCH Power Goods



radar://11043352 Need AND Gate to deassert PM_PCH_PWROK to PCH when unexpected power loss happens

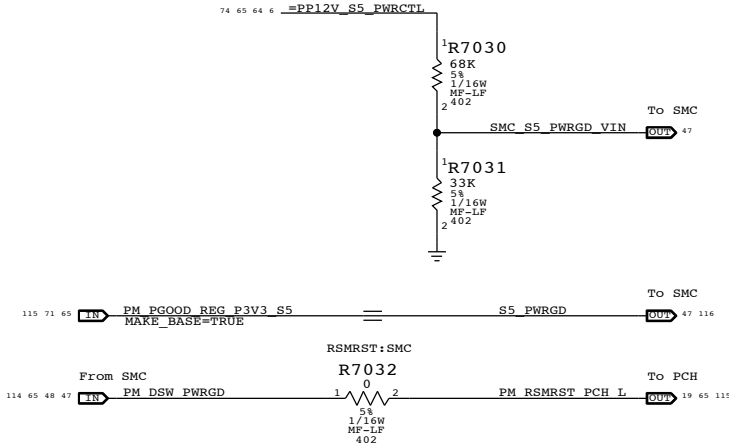
Resume Reset

Intel Doc# 29517 Maho Bay PDG, Section 22.13
Intel Doc# 29562 Panther Point EDS, Section 8.7 and 8.8

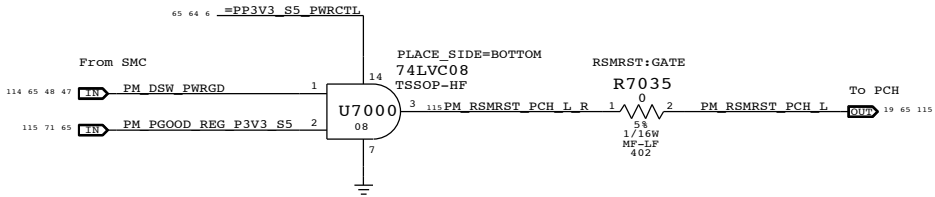
Note:
The iMac K70K72 designs does not support Deep Sx modes so both DPWROK and RSMRST# signals are shorted together

Requirements:
Power on:
Asserted at least 10 ms after all suspend well power is valid
Power off or loss of AC:
Transition to 0.8V or less before VccSUS3_3 drops to 2.90 V
to allow PCH to switch suspend well to battery without excessive loading

Primary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation.
SMC de-asserts RSMRST# (PM_DSW_PWRGD) when S5_PWRGD input is asserted and SMC_S5_PWRGD_VIN input is above comparator input level of 1.5 V.
SMC asserts RSMRST# (PM_DSW_PWRGD) when SMC_S5_PWRGD_VIN input drops from 1.8 V to 1.5 V (as implemented) when 12 V S5 rail drops to 10 V.



Secondary method:
The SMC guarantees proper assertion and de-assertion of RSMRST# for normal operation via PM_DSW_PWRGD.
RSMRST# is asserted when power good from regulator is de-asserted in the event AC is lost. Power good de-assertion should happen quickly enough to meet Intel spec.



SYNC MASTER=D8 MARK		SYNC DATE=04/23/2012	
PAGE TITLE		PM Power Good	
Apple Inc.		DRAWING NUMBER	051-9504
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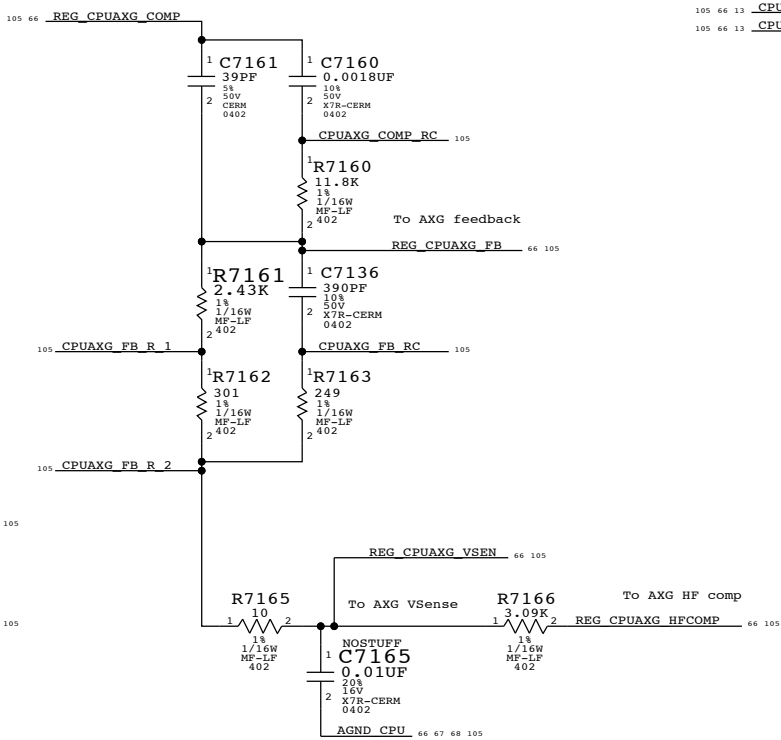
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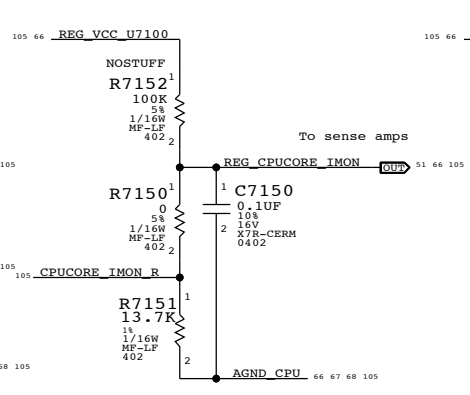
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105 66

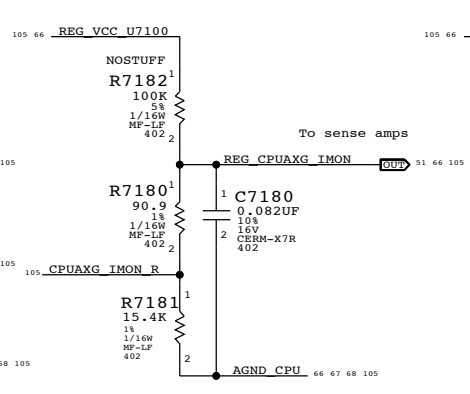
AXG compensation and feedback



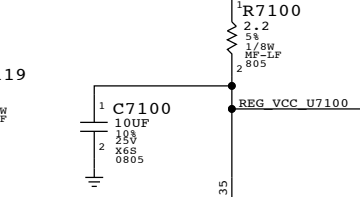
Core temp measurement

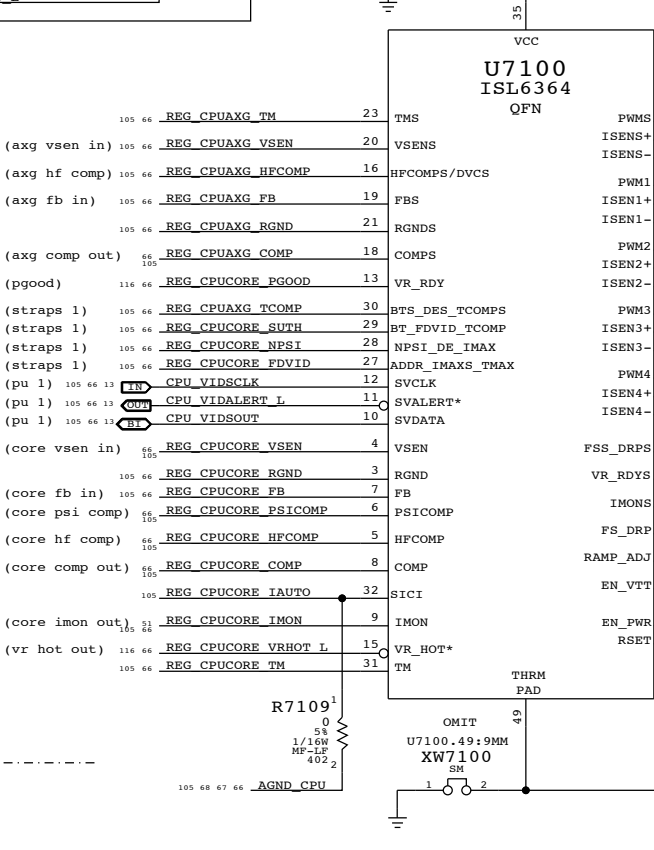


AXG temp measurement

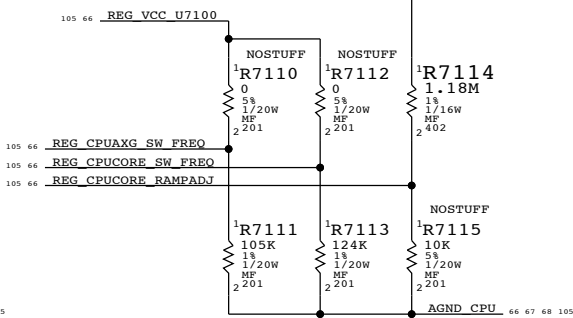


```
6  =PP5V_S0_REG_CPUCORE
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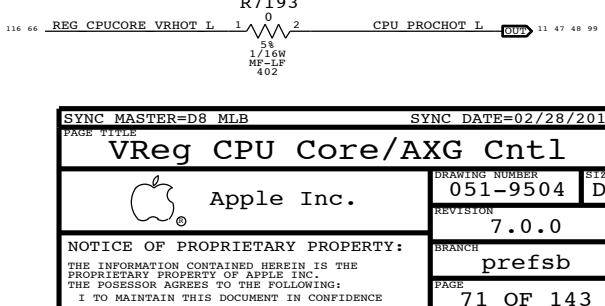


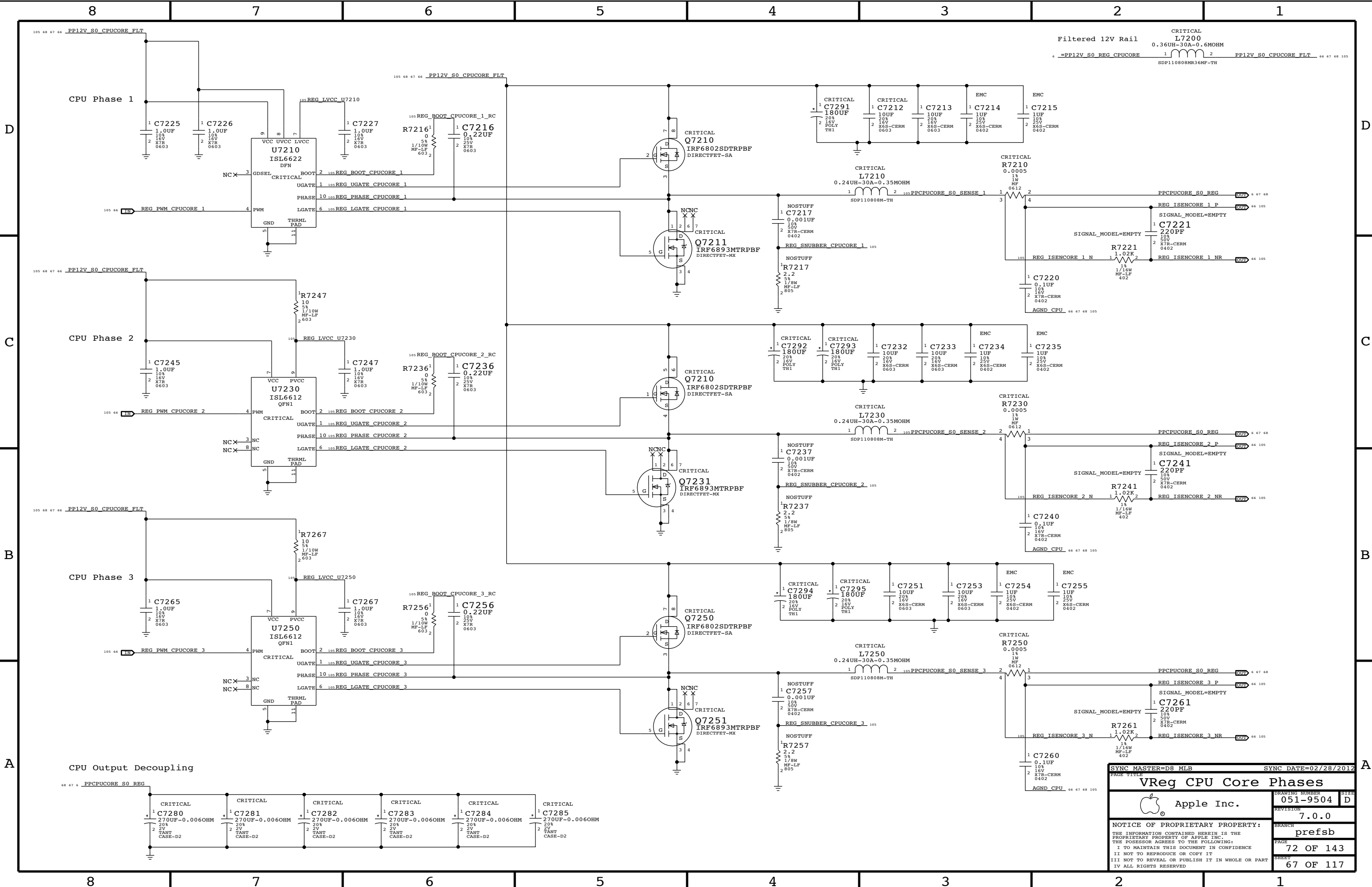


Straps 2 105

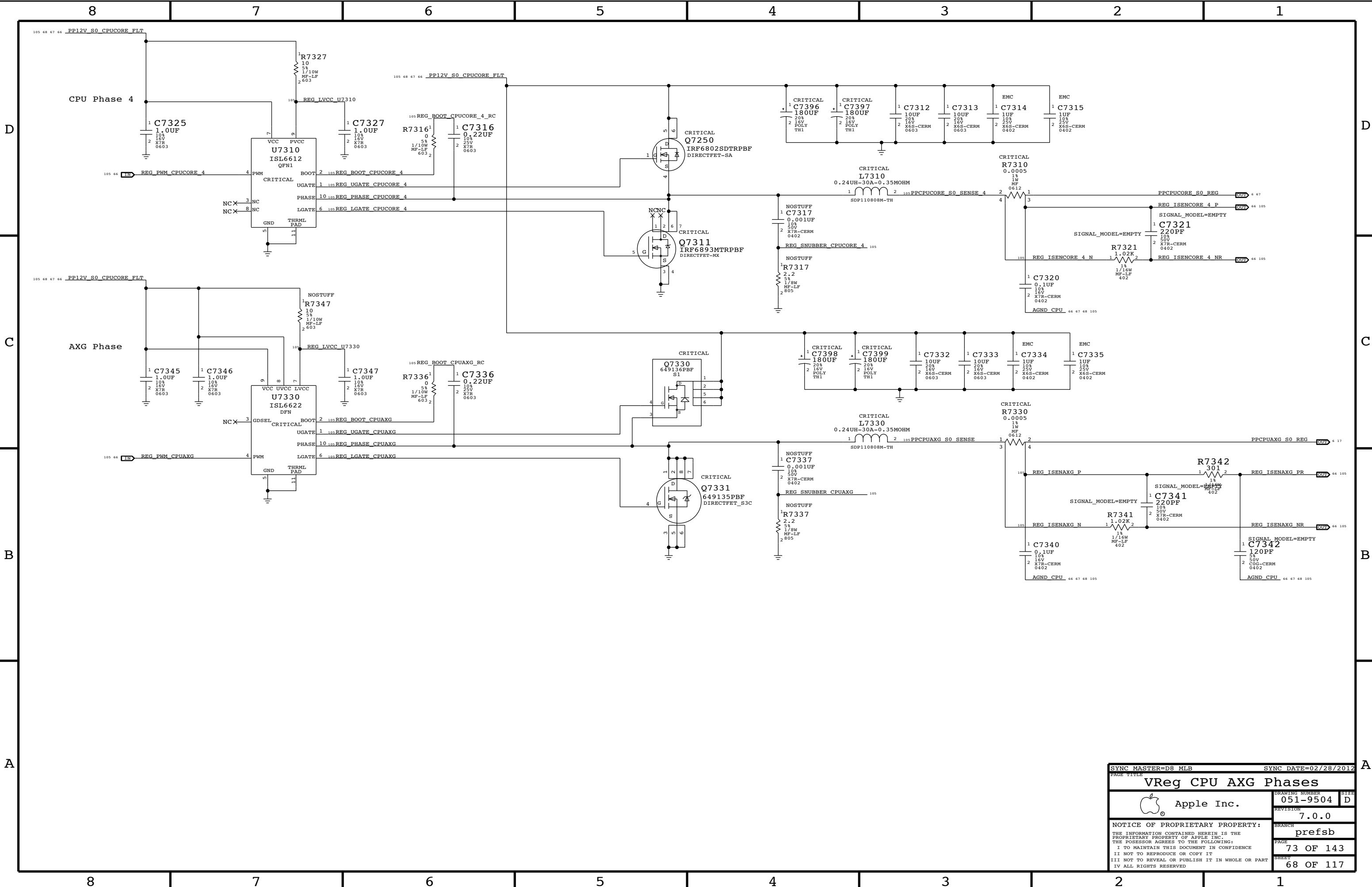


VRHot to ProcHot

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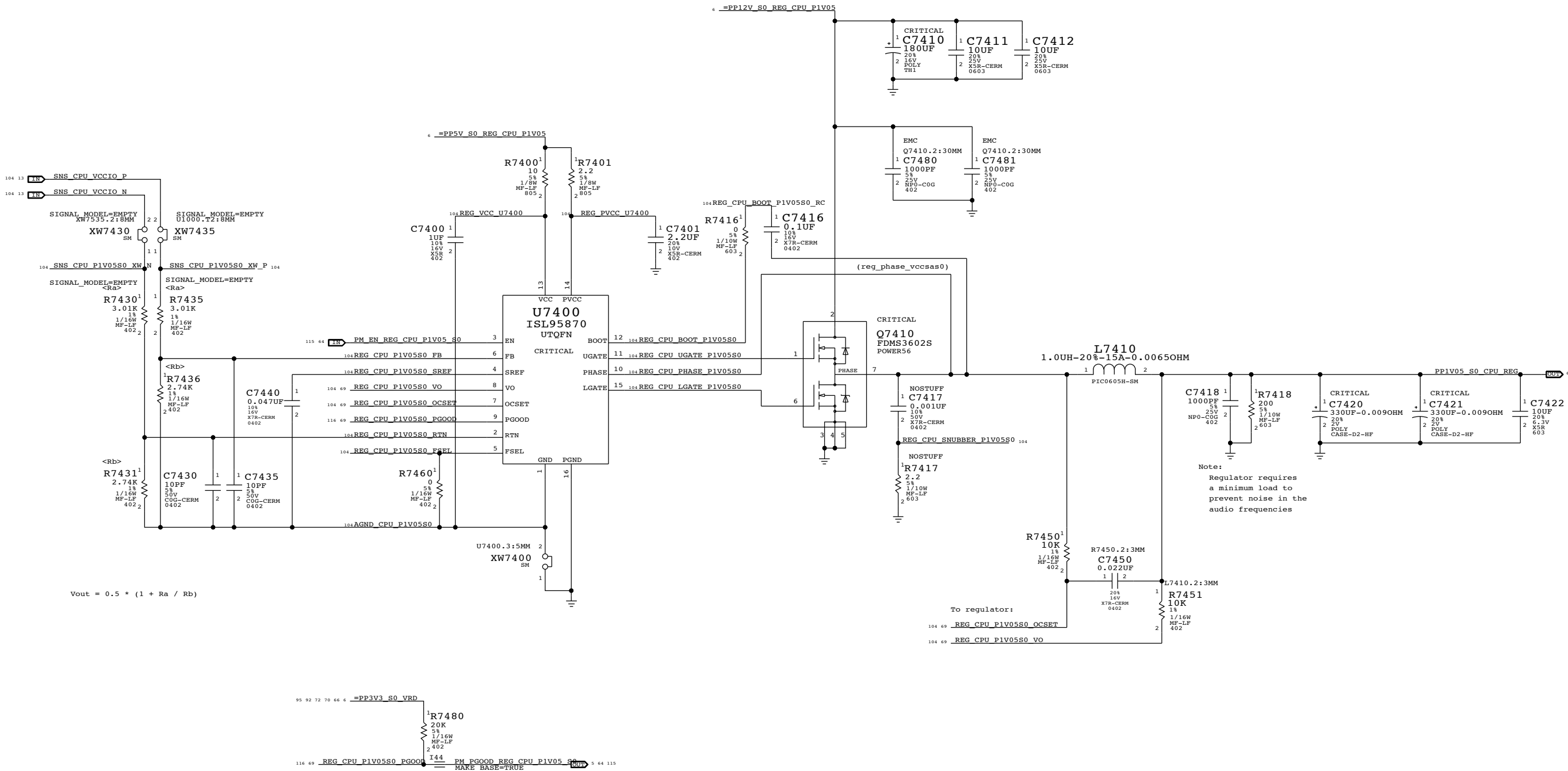


SYNC MASTER=D8 MLB		SYNC DATE=02/28/2012	
PAGE TITLE		VReg CPU Core Phases	
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		REVISION	7.0.0
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CPU VccIO (1.05V) S0 Regulator

Max avg current: 8.10 A (BUDGET)
Max peak current: 8.50 A (BUDGET)
OC trip point: ? A (min)/? A (max)
Switching freq: 500 kHz



$$V_{out} = 0.5 * (1 + R_a / R_b)$$

SYNC MASTER=D8 KOSECOFF		SYNC DATE=02/25/2012	
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VReg CPU 1.05V S0			
 Apple Inc.	DRAWING NUMBER		051-9504
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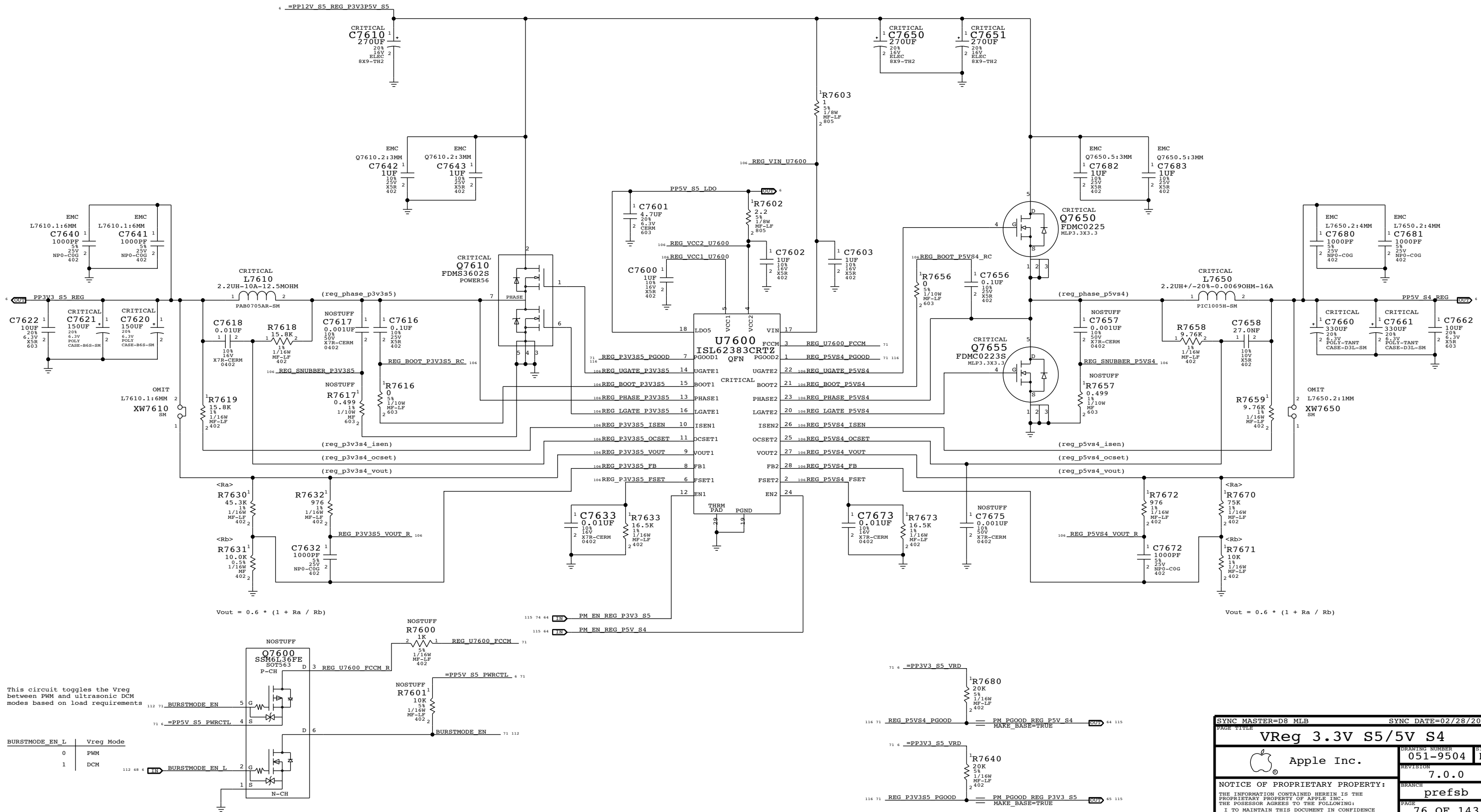
A

3.3V S5 Regulator

Max avg current: 6 A (design)/ 4.85 A (budget)
Max peak current: ? A (design)/ 6.6 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz

5V S4 Regulator

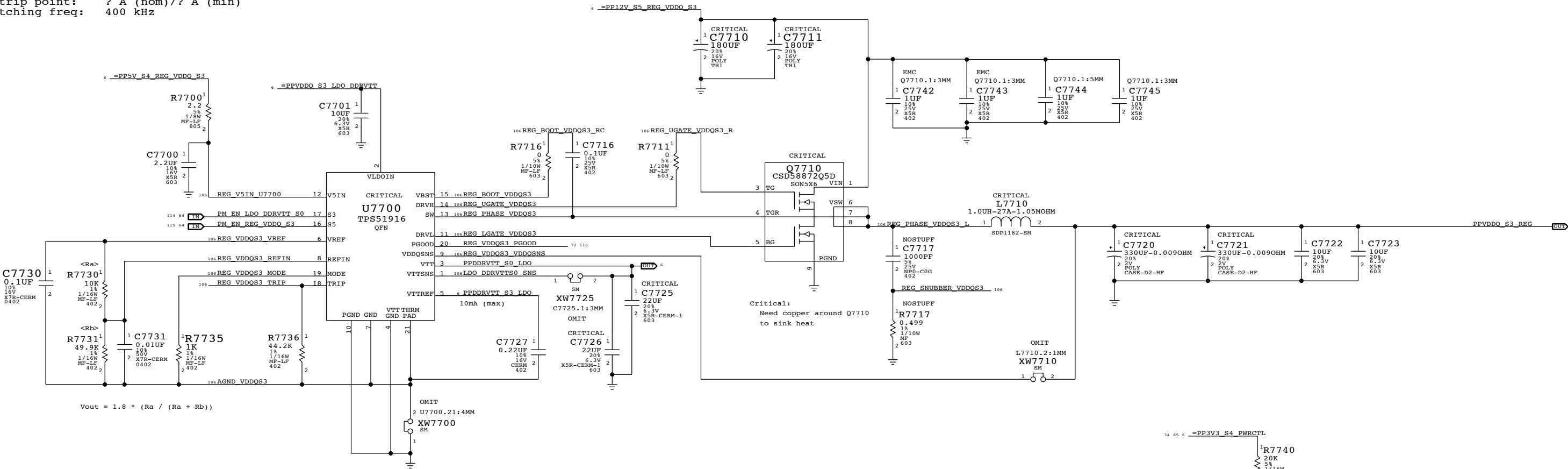
Max avg current: 10 A (design)/ 6.08 A (budget)
Max peak current: ? A (design)/ 6.9 A (budget)
OC trip point: ? A (nom)/? A (min)
Switching freq: 350 kHz



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VDDQ (1.5V) S3 Regulator

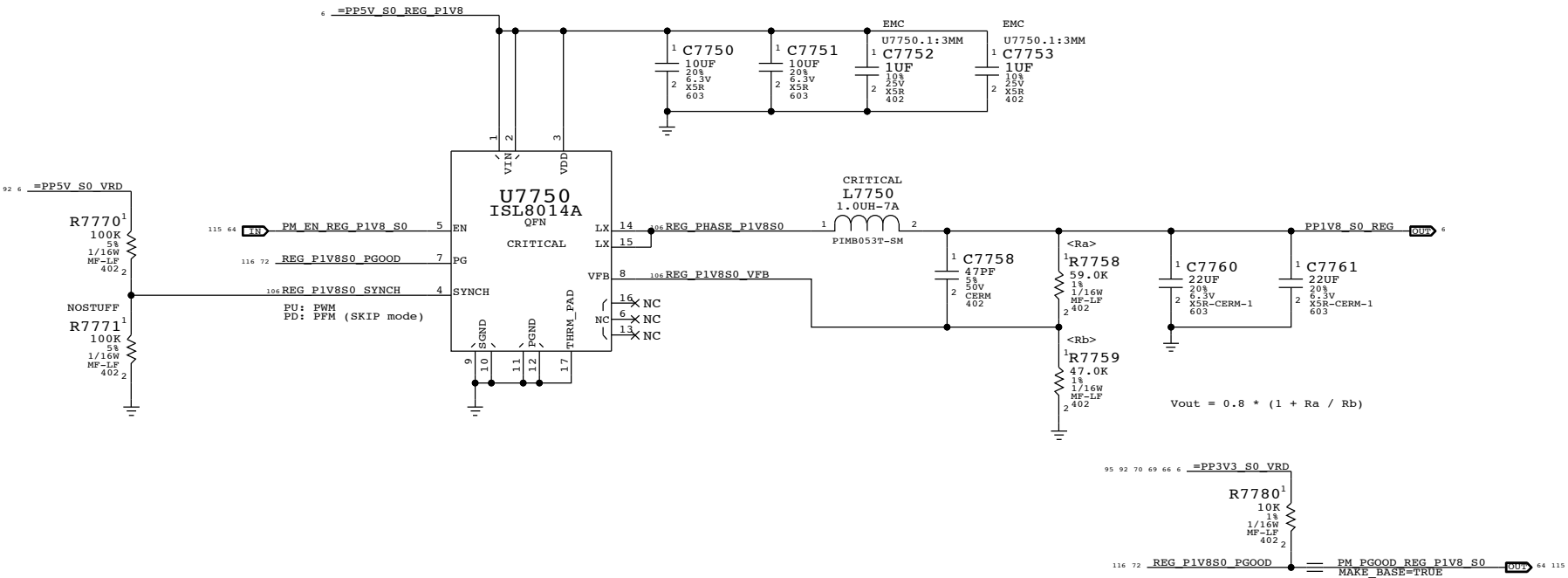
Max avg current: 9.0 A (BUDGET)
Max peak current: 11.3 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: 400 kHz



$$V_{out} = 1.8 * (R_a / (R_a + R_b))$$

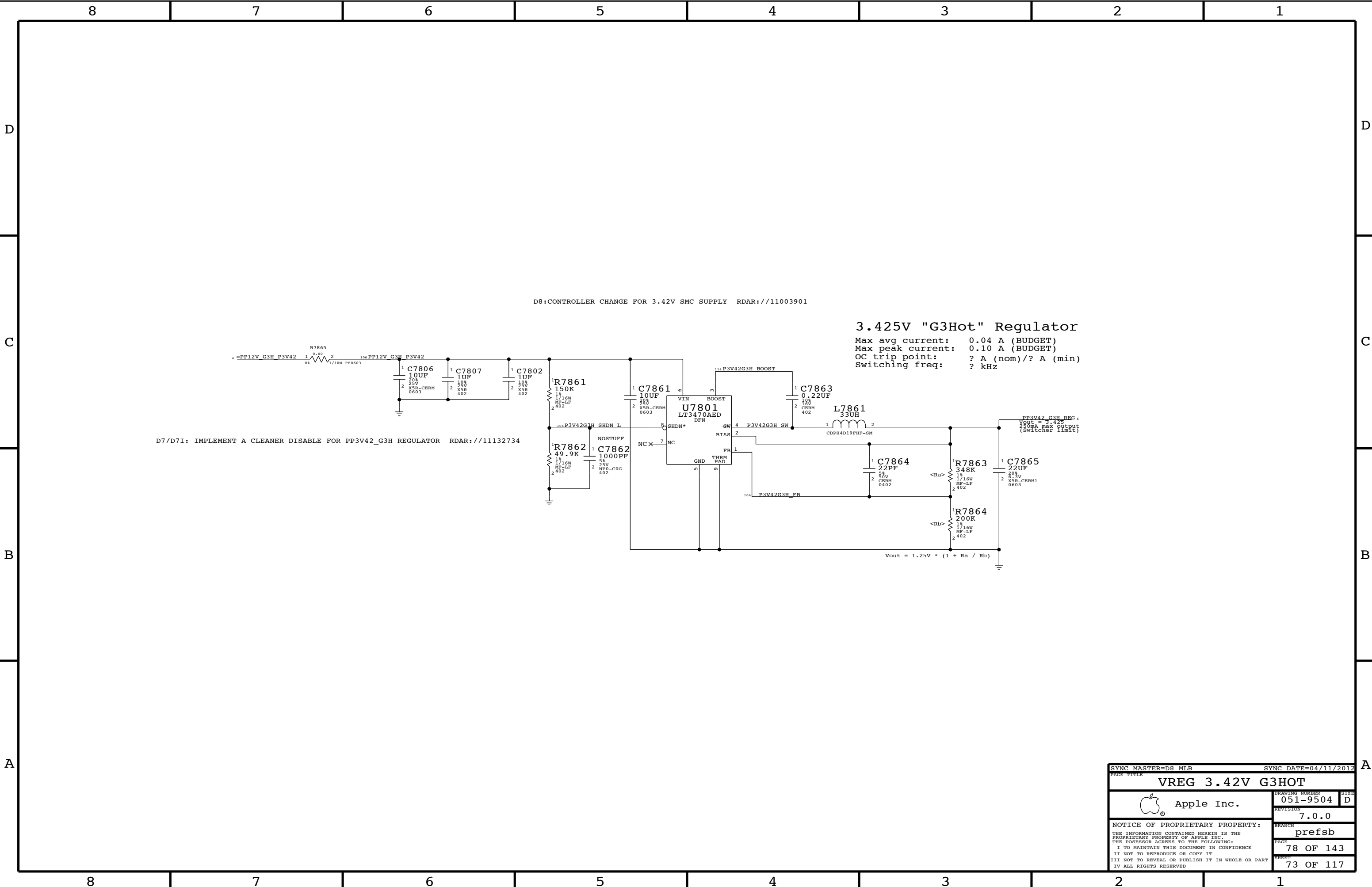
1.8V S0 Regulator

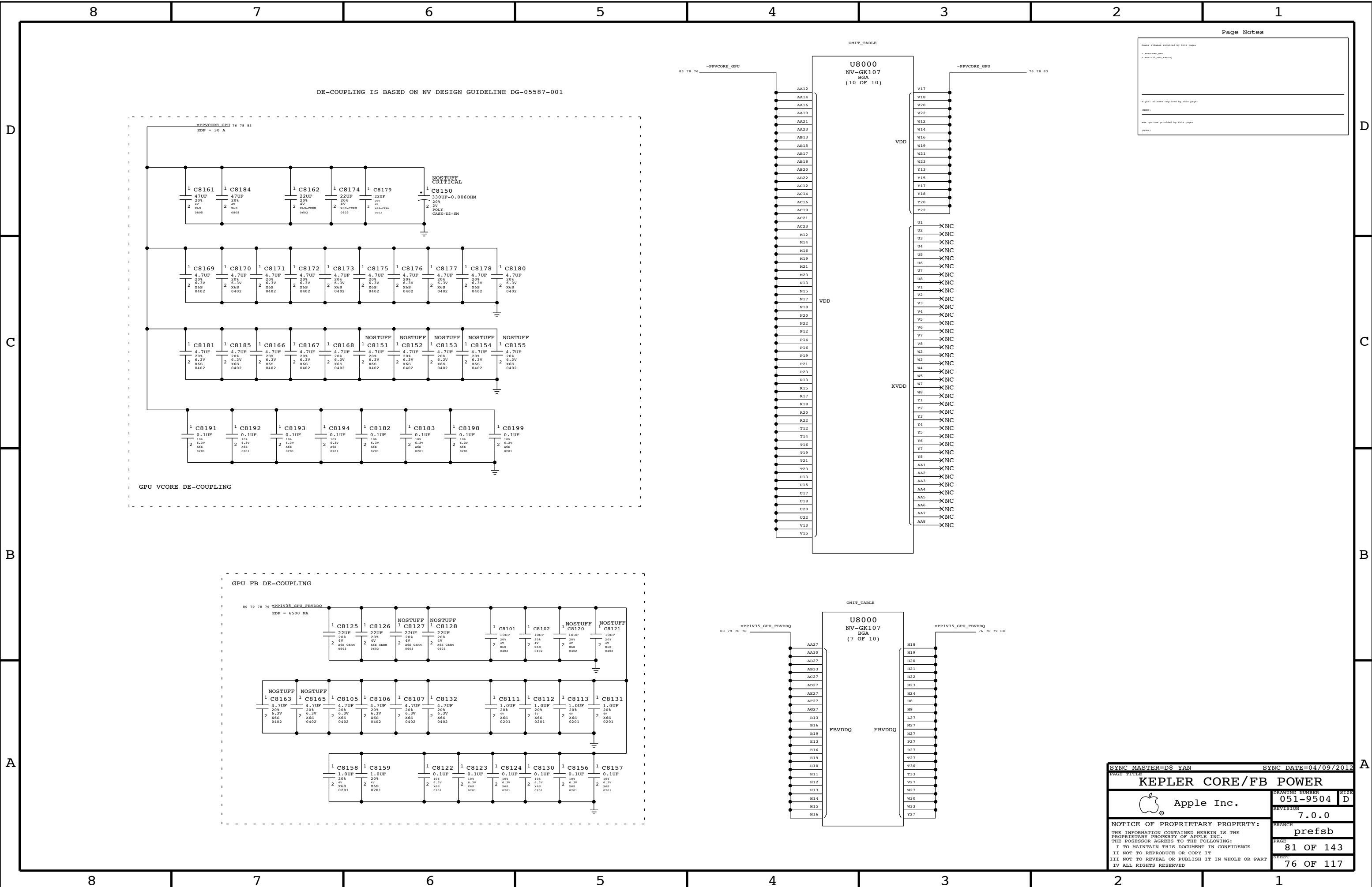
Max avg current: 0.6 A (BUDGET)
Max peak current: 1.7 A (BUDGET)
OC trip point: ? A (nom)/? A (min)
Switching freq: ? kHz



$$V_{out} = 0.8 * (1 + R_a / R_b)$$

PAGE TITLE		PAGE TITLE	
VReg VDDQ and 1.8V S0		VReg VDDQ and 1.8V S0	
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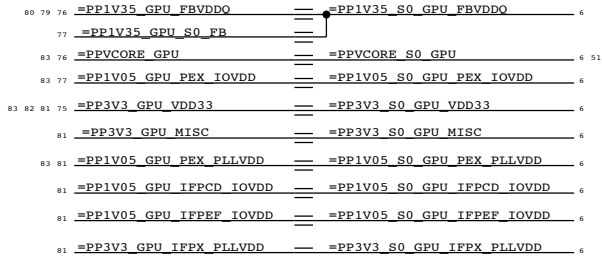




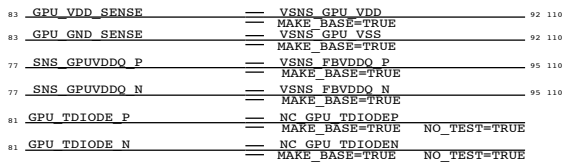
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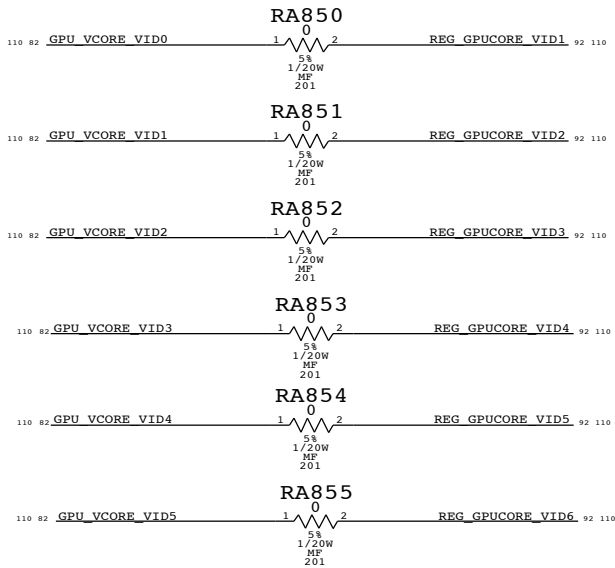
GPU POWER ALIAS



GPU SIGNAL & SENSE ALIAS



GPU VIDS ALIAS (VR VID0 IS TIED LOW)

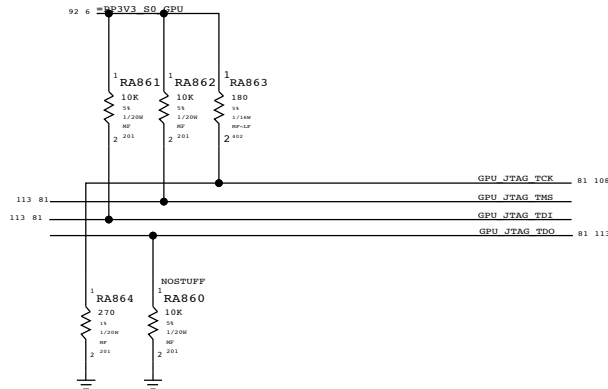


THESE POWER ALIASES ARE CRAETED TO MATCH D8 GK104

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PU/PD IS BASED ON RECOMMENDATION FROM NV FOR NVDIA GPU JTAG DEBUGGER



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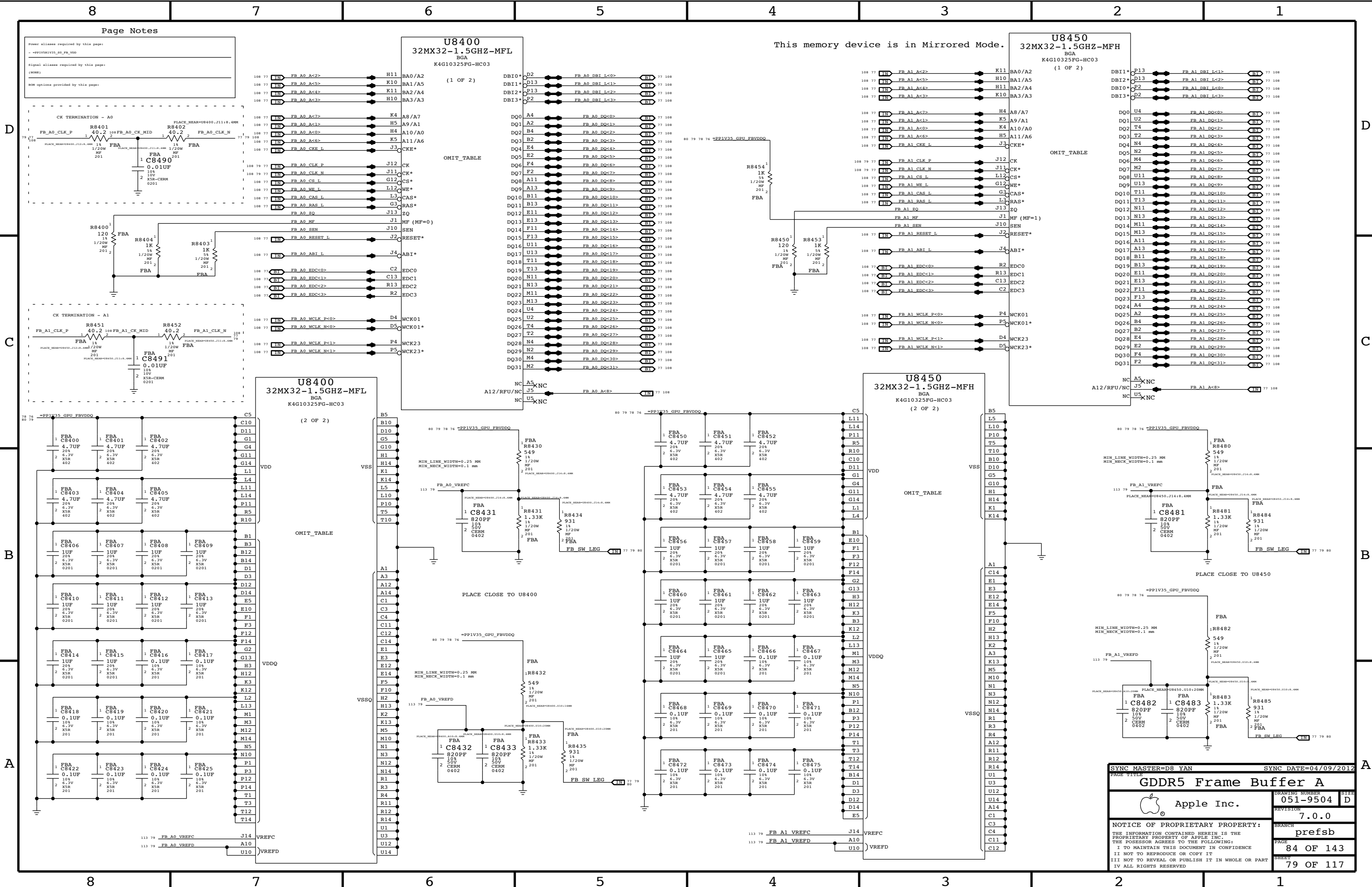
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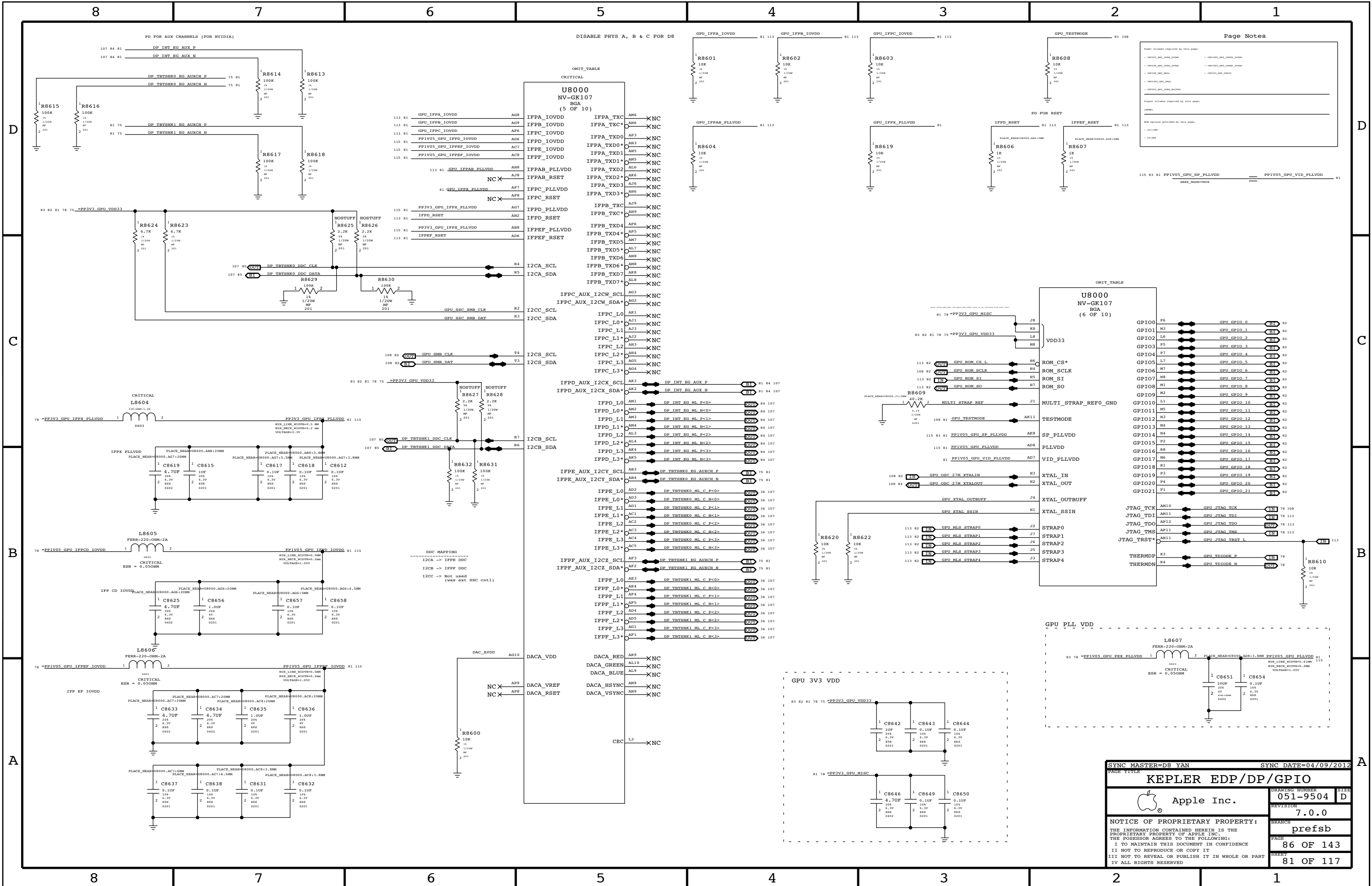
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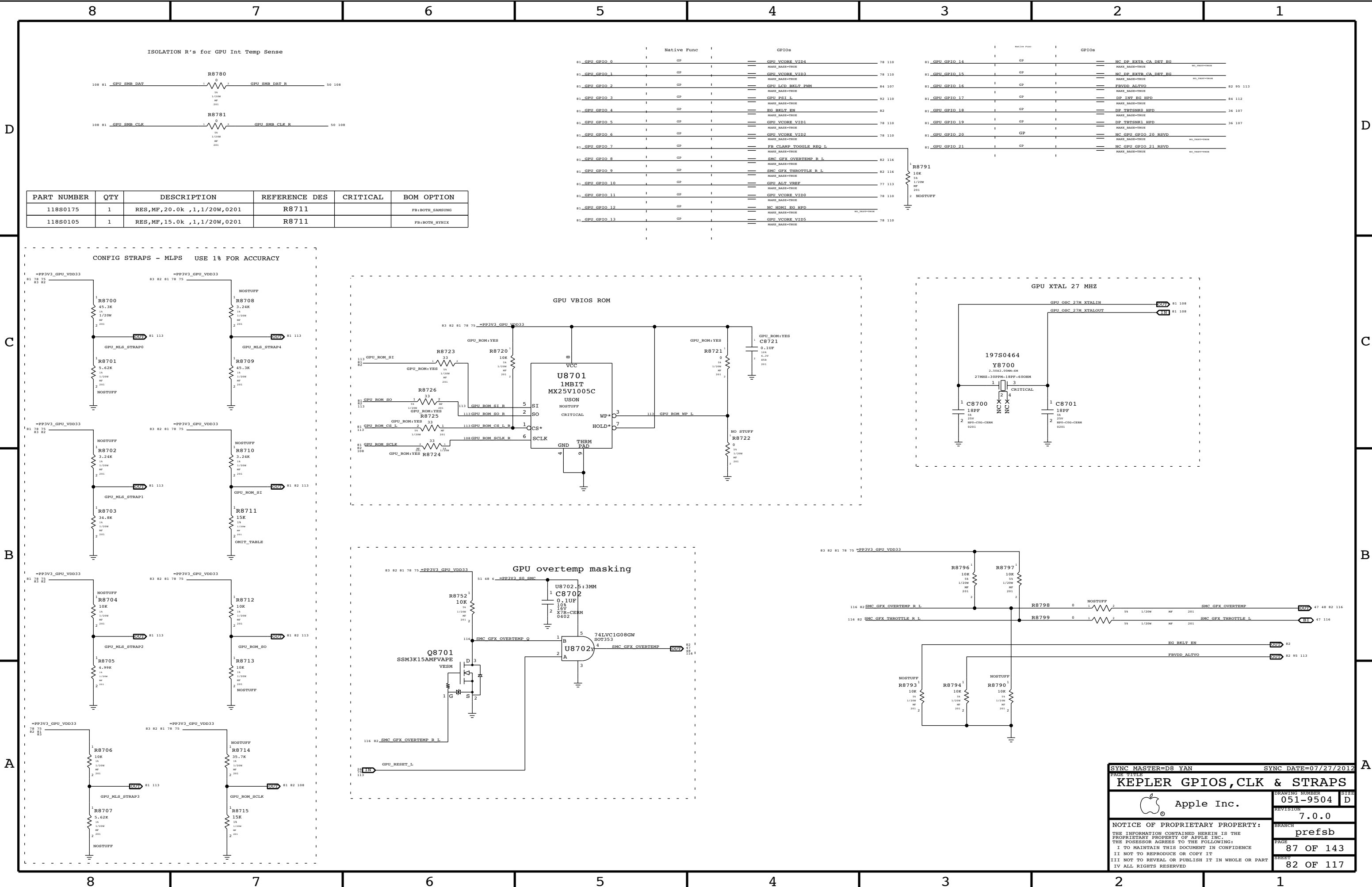
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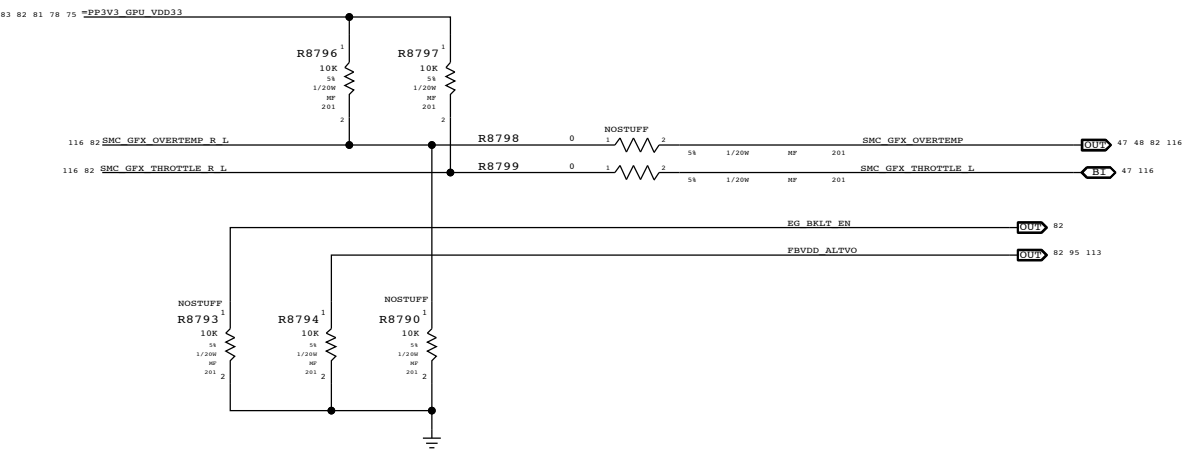
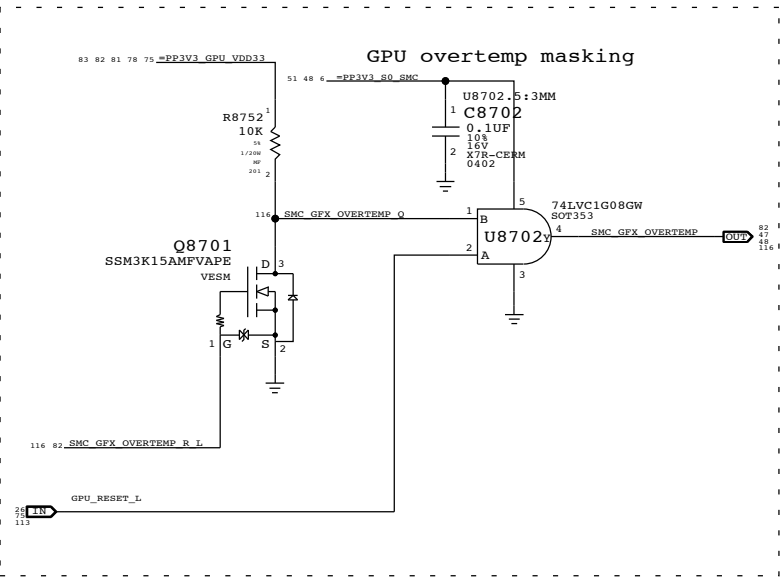
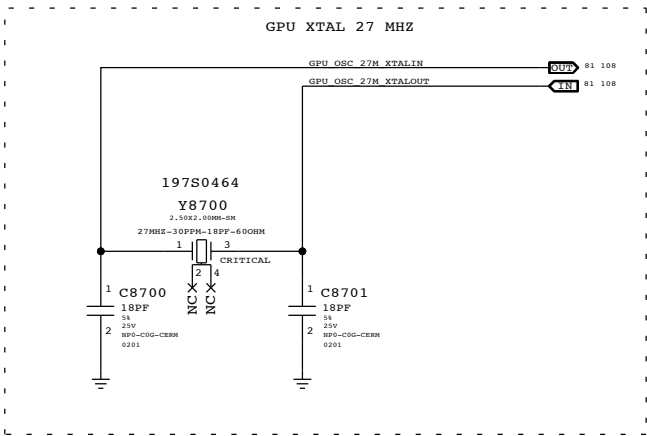
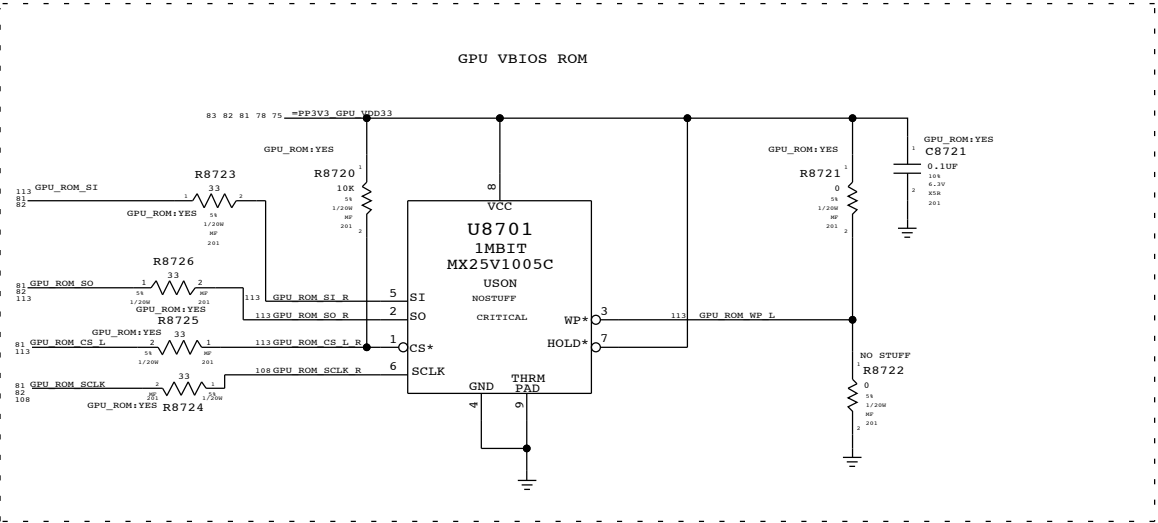




PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S0175	1	RES,MF,20.0k ,1,1/20W,0201	R8711		FB:BOTH_SAMSUNG
118S0105	1	RES,MF,15.0k ,1,1/20W,0201	R8711		FB:BOTH_HYNIX

	Native Func	GPIOs	
81_GPU_GPIO_0	GP	GPU_VCORE_VID4	78 110
81_GPU_GPIO_1	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_2	GP	GPU_VCORE_VID3	78 110
81_GPU_GPIO_3	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_4	GP	GPU_LCD_RKLT_EN	84 107
81_GPU_GPIO_5	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_6	GP	GPU_FBI_L	92 110
81_GPU_GPIO_7	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_8	GP	EG_RKLT_EN	82
81_GPU_GPIO_9	GP	GPU_VCORE_VID1	78 110
81_GPU_GPIO_10	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_11	GP	GPU_VCORE_VID2	78 110
81_GPU_GPIO_12	GP	FB_CLAMP_TOGGLE_REQ_L	82 116
81_GPU_GPIO_13	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_14	GP	SMC GFX OVERTEMP_R_L	82 116
81_GPU_GPIO_15	GP	SMC GFX THROTTLE_R_L	82 116
81_GPU_GPIO_16	GP	GPU_ALT_VREF	77 113
81_GPU_GPIO_17	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_18	GP	GPU_VCORE_VID0	78 110
81_GPU_GPIO_19	GP	NC_HDMI_EG_HPD	84 112
81_GPU_GPIO_20	GP	MAKE_BASE-TRUE	
81_GPU_GPIO_21	GP	GPU_VCORE_VID5	78 110
81_GPU_GPIO_22	GP	MAKE_BASE-TRUE	

	GPIOs	
81_GPU_GPIO_14	GP	NC_DP_EXTA_CA_DET_EG
81_GPU_GPIO_15	GP	MAKE_BASE-TRUE
81_GPU_GPIO_16	GP	NC_DP_EXTRA_CA_DET_EG
81_GPU_GPIO_17	GP	MAKE_BASE-TRUE
81_GPU_GPIO_18	GP	FBVDD_ALTV0
81_GPU_GPIO_19	GP	MAKE_BASE-TRUE
81_GPU_GPIO_20	GP	DP_INF_EG_HPD
81_GPU_GPIO_21	GP	MAKE_BASE-TRUE
81_GPU_GPIO_22	GP	DP_TBTSNK0_HPD
81_GPU_GPIO_23	GP	MAKE_BASE-TRUE
81_GPU_GPIO_24	GP	DP_TBTSNK1_HPD
81_GPU_GPIO_25	GP	MAKE_BASE-TRUE
81_GPU_GPIO_26	GP	NC_GPU_GPIO_20_RSVD
81_GPU_GPIO_27	GP	MAKE_BASE-TRUE
81_GPU_GPIO_28	GP	NC_GPU_GPIO_21_RSVD
81_GPU_GPIO_29	GP	MAKE_BASE-TRUE



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SYNC DATE=07/27/2012

KEPLER GPIOs,CLK & STRAPS

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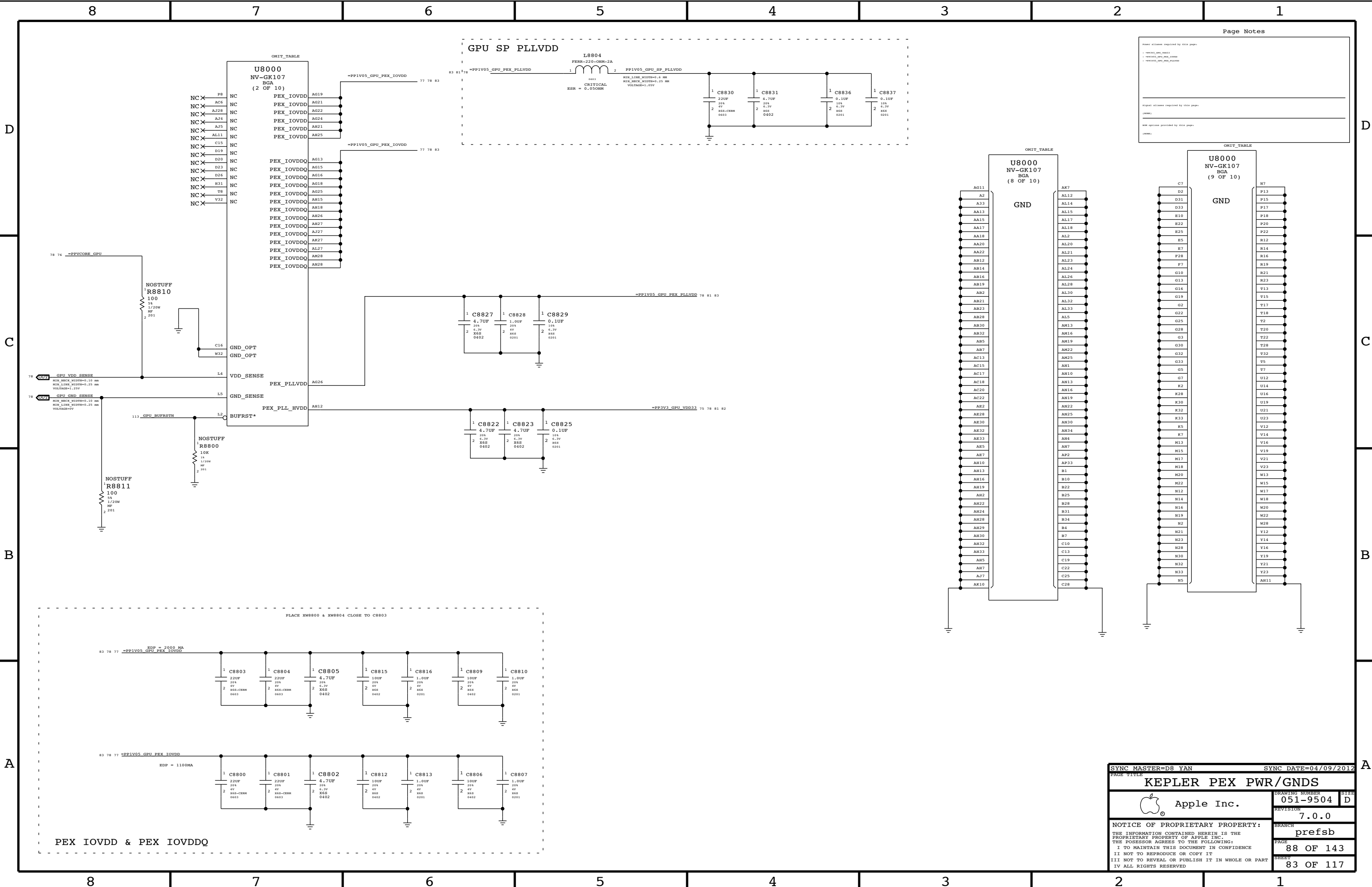
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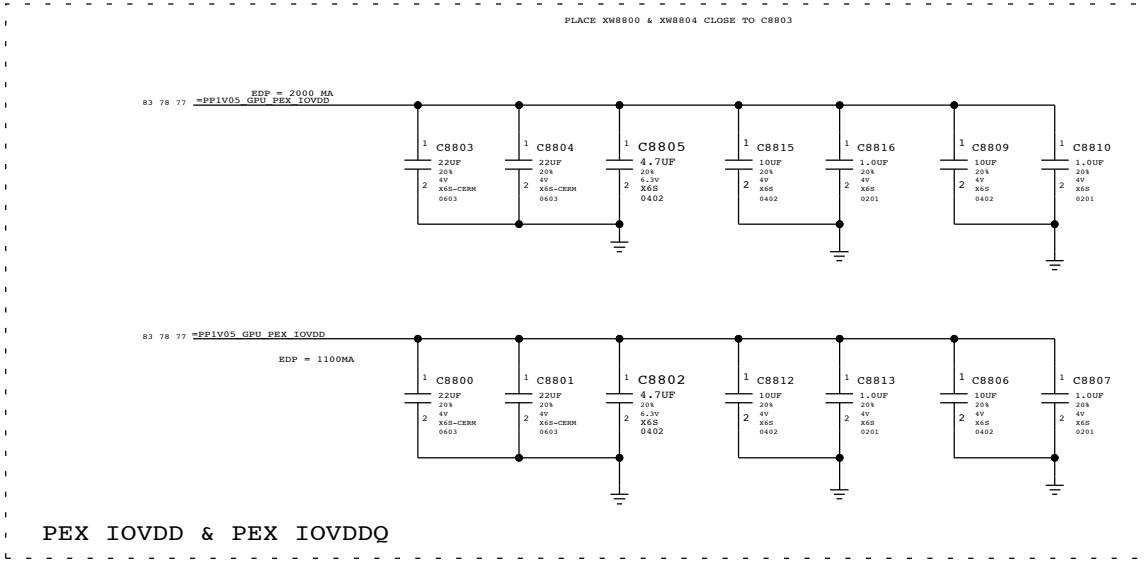
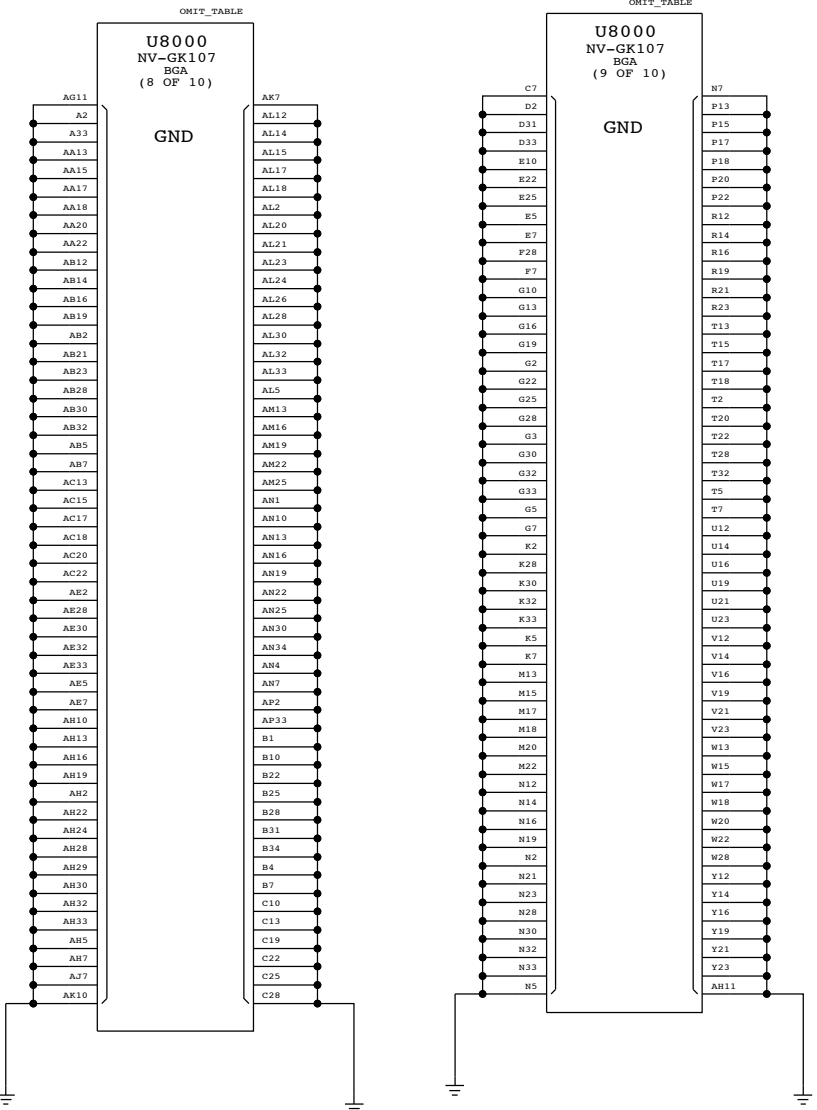
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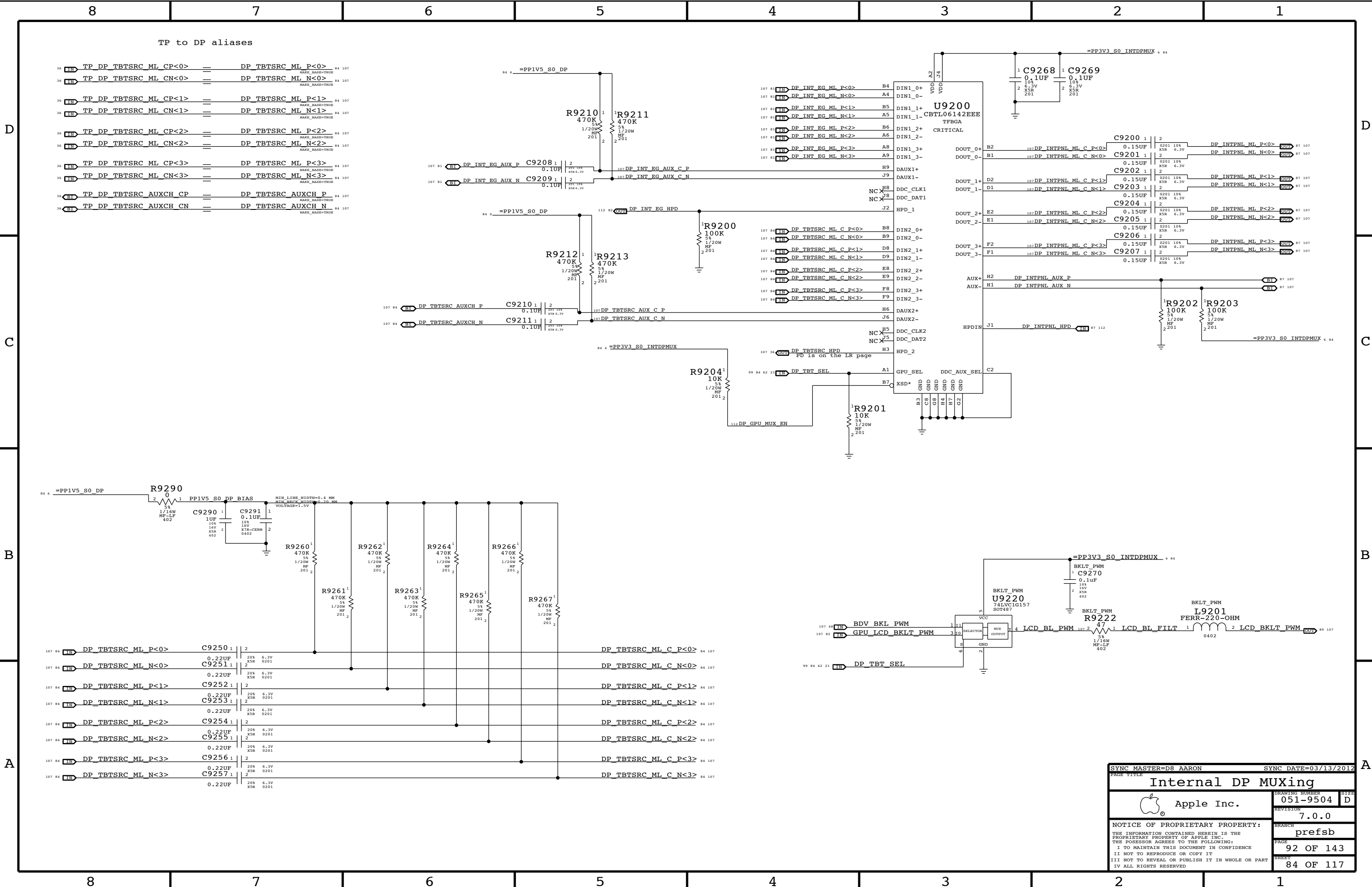
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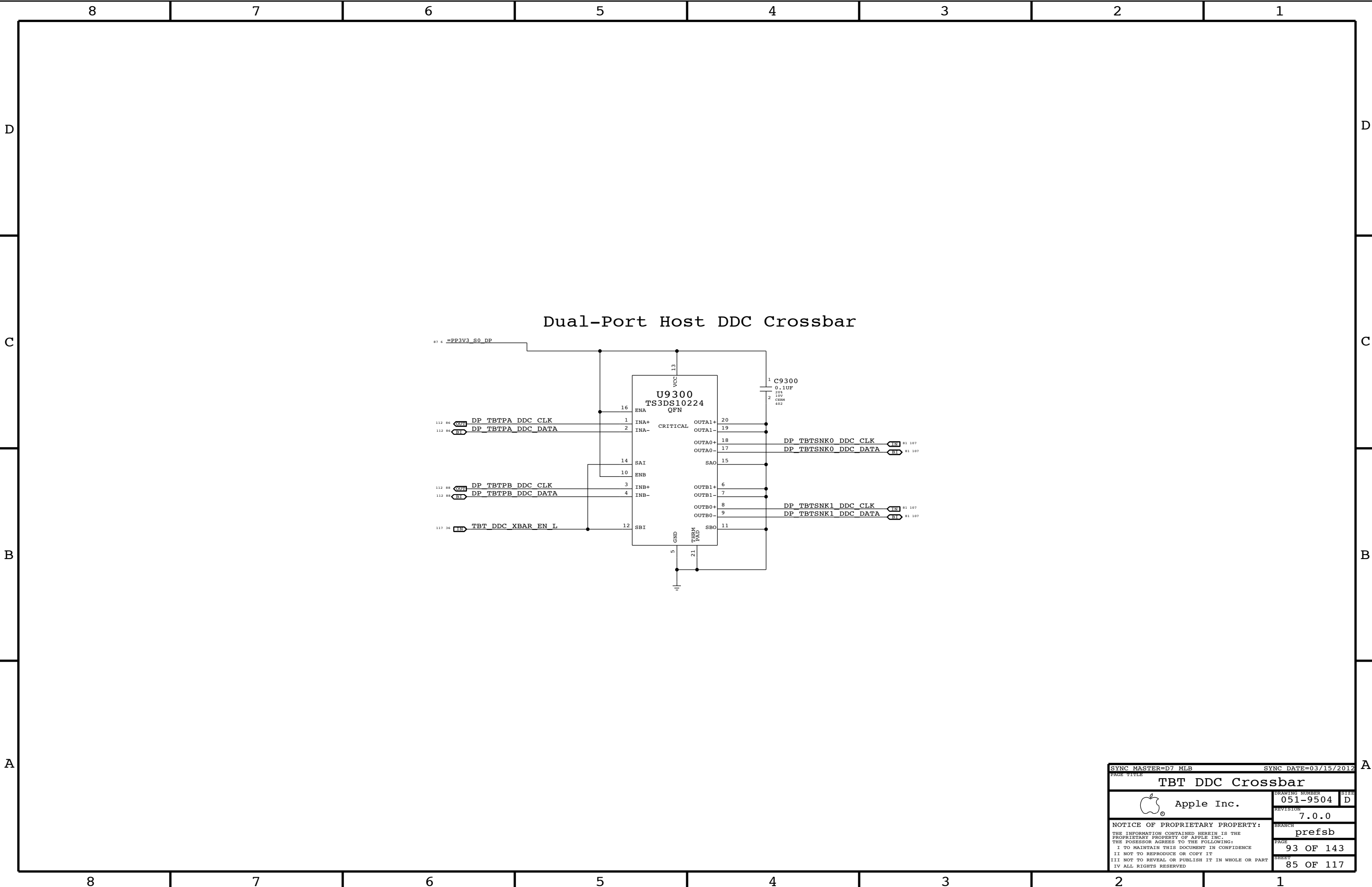


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- PP1V05_GPU_PEX_PLLVDD	
- PP1V05_GPU_PEX_IOVDD	
- PP1V05_GPU_PEX_PLLVDD	
Signal aliases required by this page:	
(none)	
Non options provided by this page:	
(none)	



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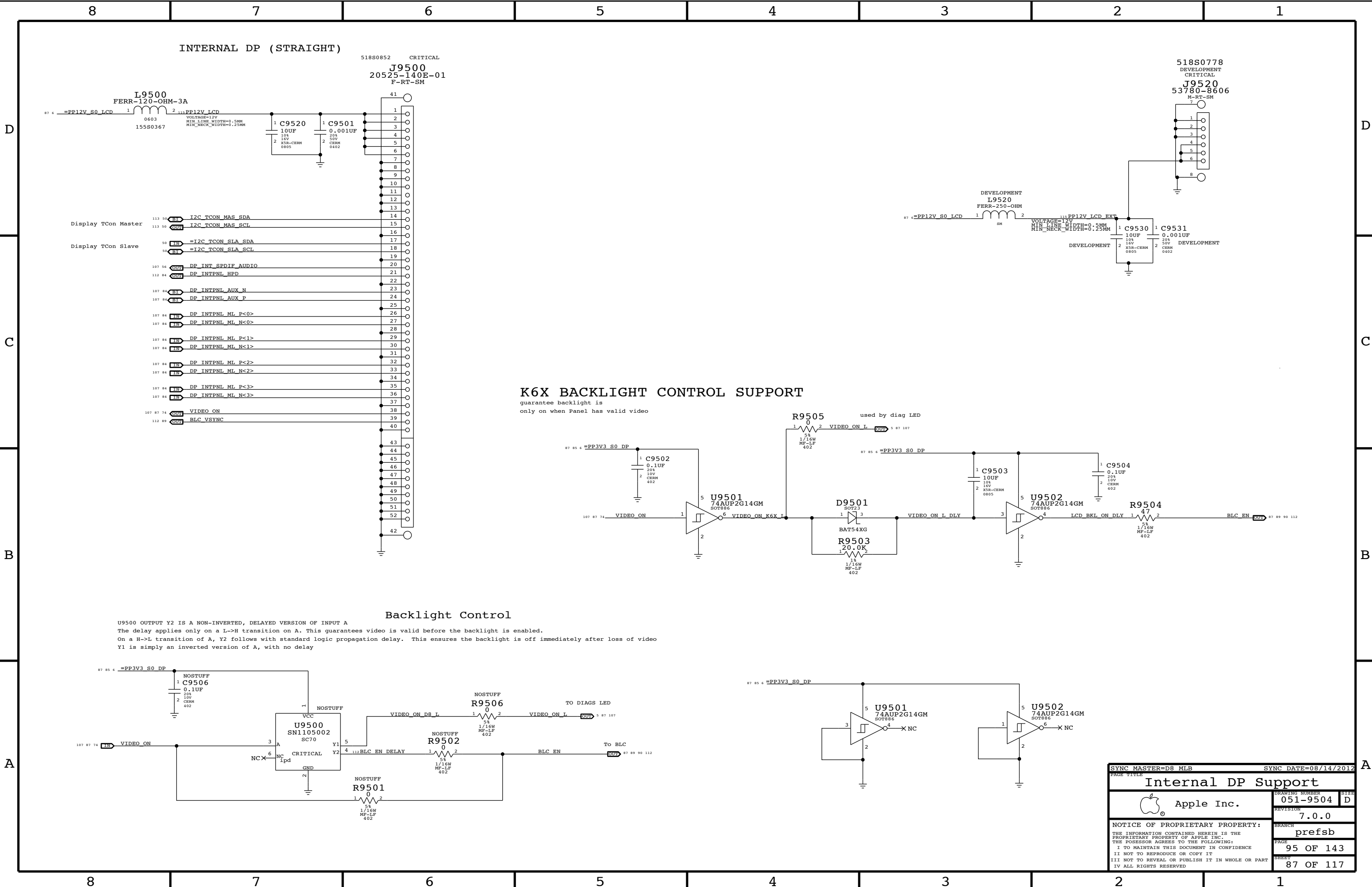
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K6X BACKLIGHT CONTROL SUPPORT

guarantee backlight is
only on when Panel has valid video

Backlight Control

U9500 OUTPUT Y2 IS A NON-INVERTED, DELAYED VERSION OF INPUT A
The delay applies only on a L->H transition on A. This guarantees video is valid before the backlight is enabled.
On a H->L transition of A, Y2 follows with standard logic propagation delay. This ensures the backlight is off immediately after loss of video
Y1 is simply an inverted version of A, with no delay

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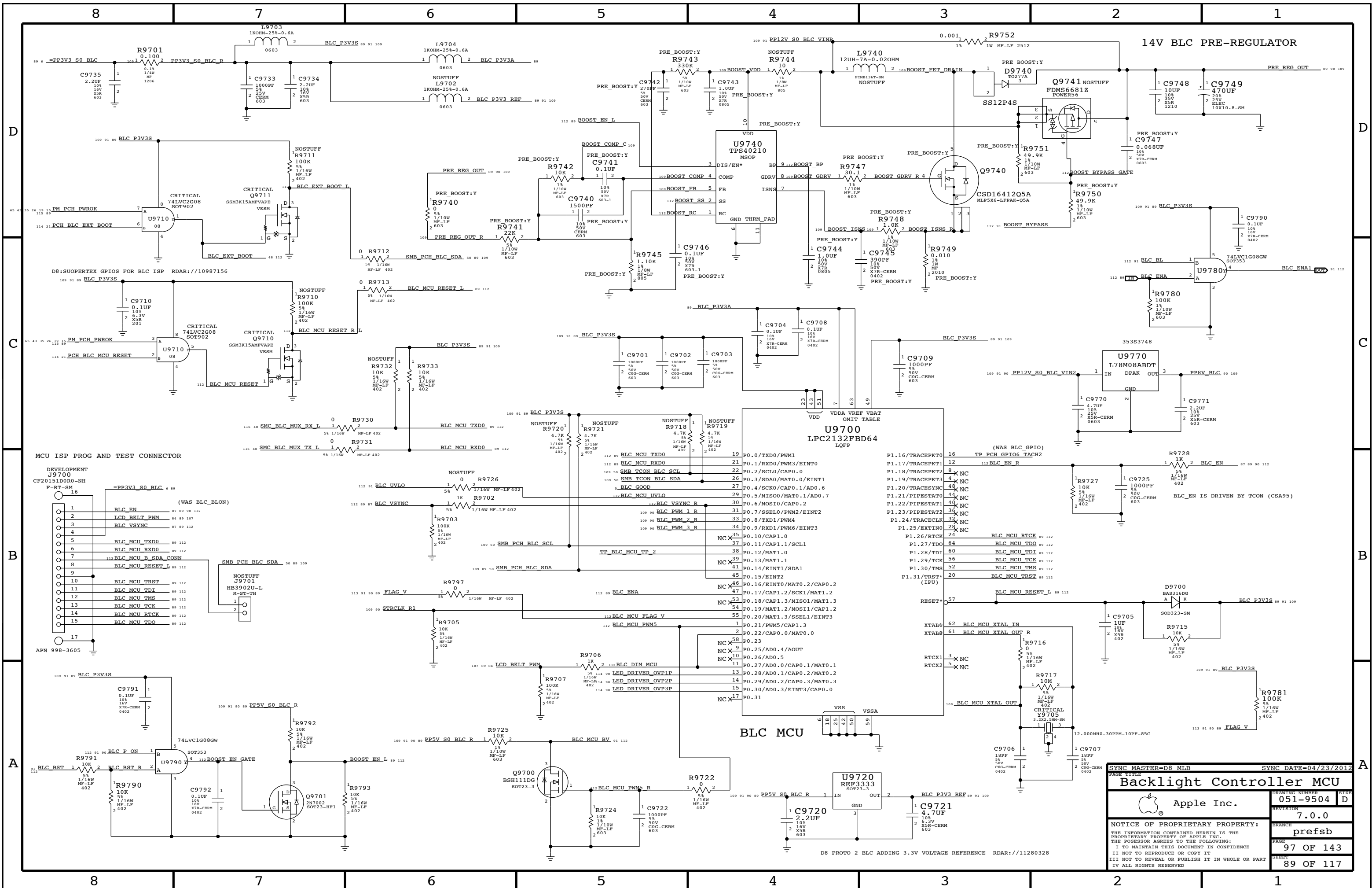
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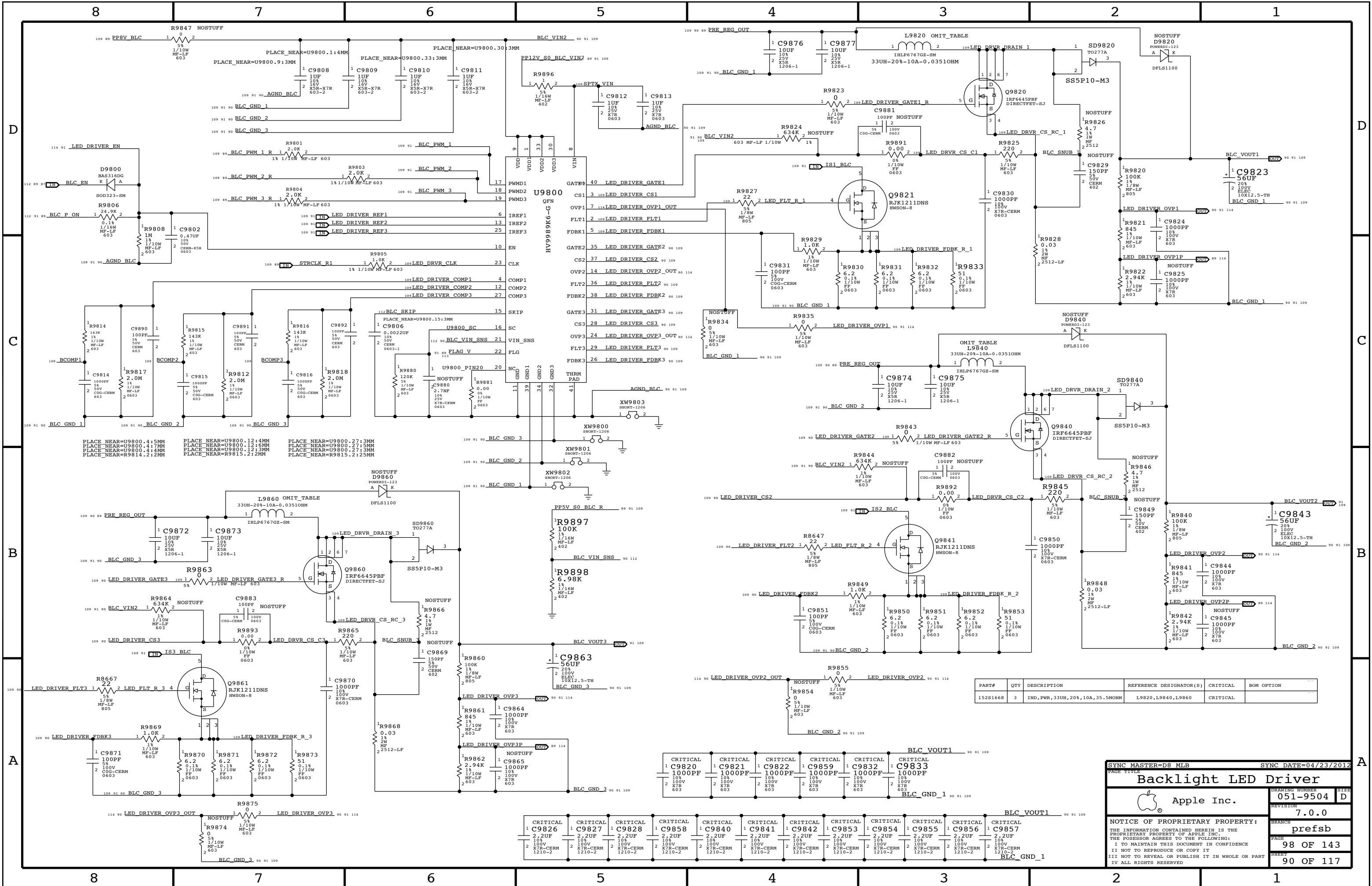
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Backlight Controller MCU		051-9504	
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Backlight LED Driver

Apple Inc.

051-9504

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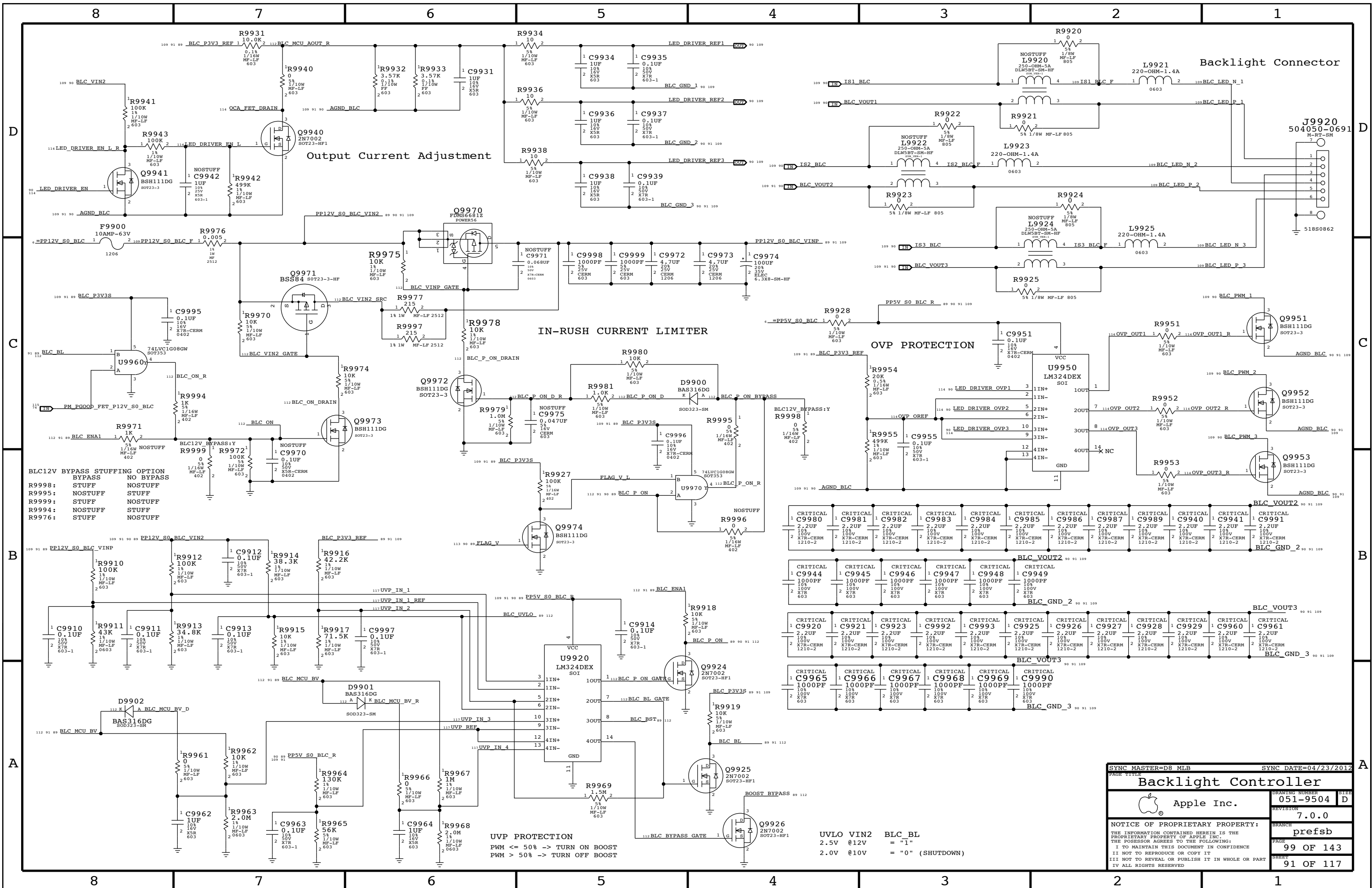
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Output Current Adjustment

IN-RUSH CURRENT LIMITER

OVP PROTECTION

Backlight Connector

BLC12V BYPASS STUFFING OPTION
BYPASS NO BYPASS
R9998: STUFF NOSTUFF
R9995: NOSTUFF STUFF
R9999: STUFF NOSTUFF
R9994: NOSTUFF STUFF
R9976: STUFF NOSTUFF

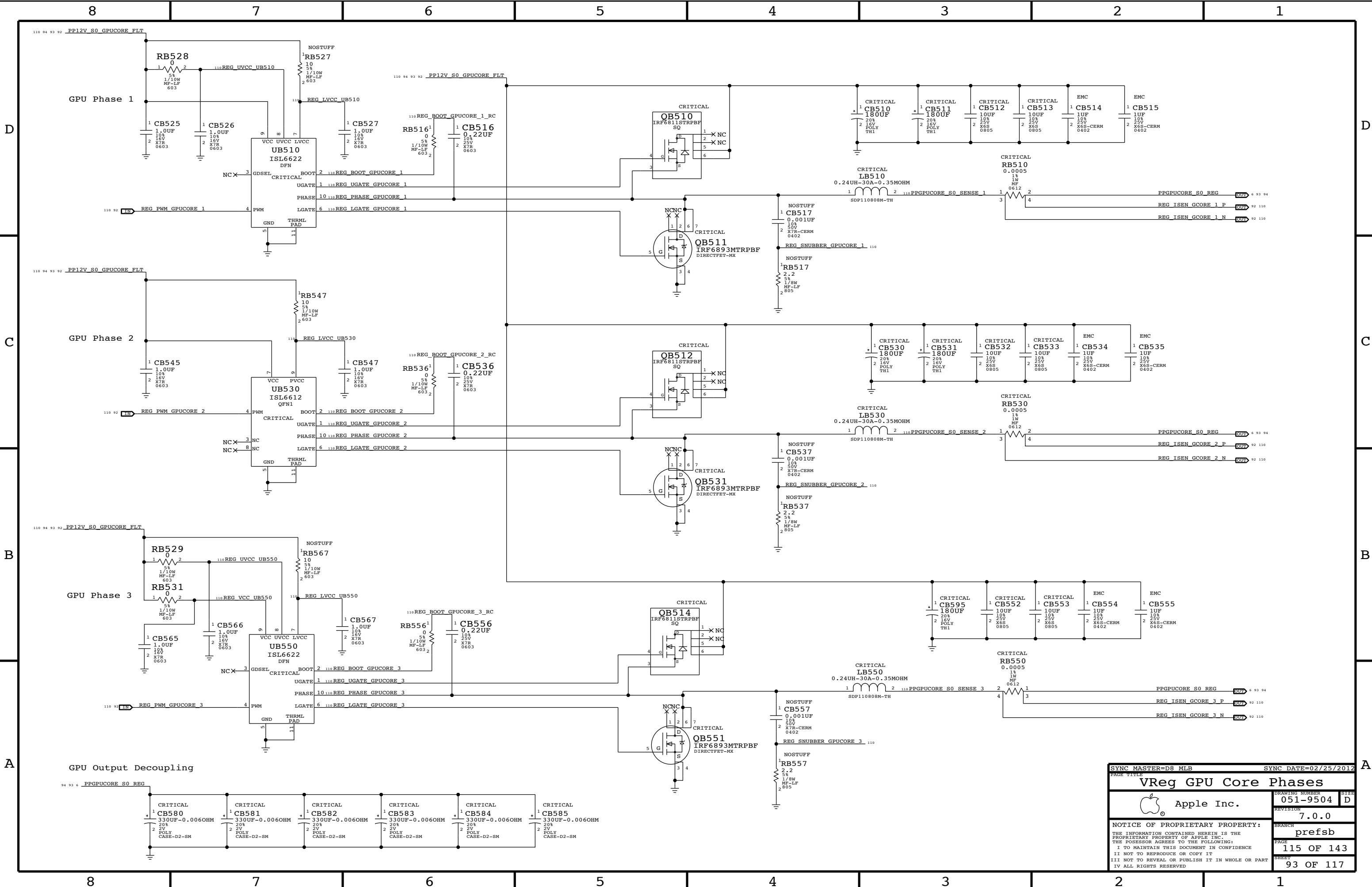
UVP PROTECTION
PWM <= 50% -> TURN ON BOOST
PWM > 50% -> TURN OFF BOOST

UVLO VIN2 BLC_BL
2.5V @12V = "1"
2.0V @10V = "0" (SHUTDOWN)

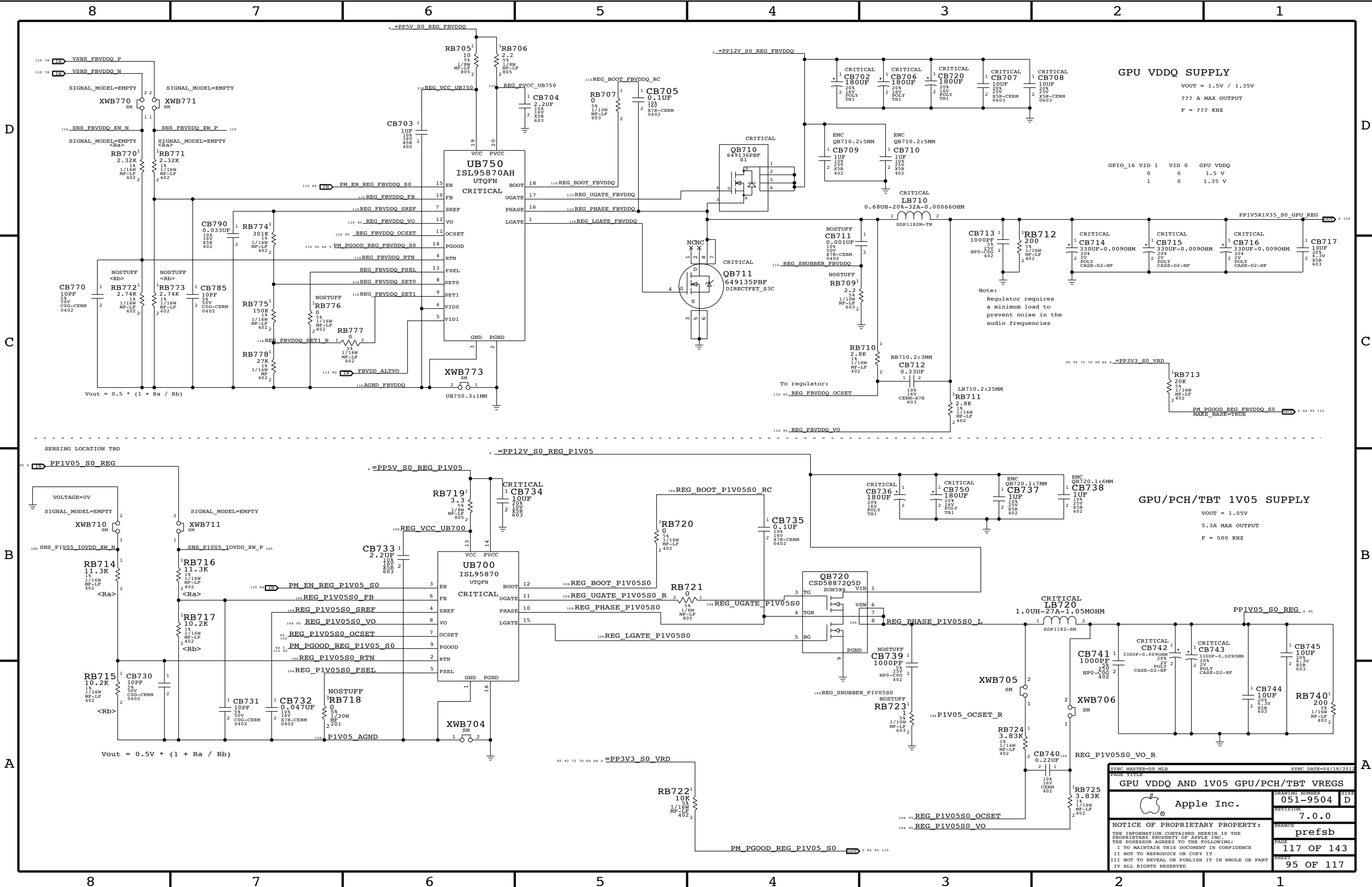
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GPU VDDQ SUPPLY

VOUT = 1.5V / 1.35V
??? A MAX OUTPUT
F = ??? KHZ

GPIO_16 VID 1 VID 0 GPU VDDQ
0 0 1.5 V
1 0 1.35 V

GPU/PCH/TBT 1V05 SUPPLY

VOUT = 1.05V
5.3A MAX OUTPUT
F = 500 KHZ

GPU VDDQ AND 1V05 GPU/PCH/TBT VREGS		DRAWING NUMBER	051-9504	SIZE	D
Apple Inc.		REVISION	7.0.0	BRANCH	prefsb
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DDR3 Constraints

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PCI EXPRESS

PCIe-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
PCIE_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
PCIE_PHY	*	PCIE_80D
CLK_PCIE_PHY	*	PCIE_90D
COMP_PCIE_PHY	*	COMP_SE

PCIe-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_ISO	TOP,BOTTOM	=5X_DIELECTRIC	?
PCIE_ISO	*	=4X_DIELECTRIC	?
CLK_PCIE_ISO	*	=5:1_SPACING	?
COMP_PCIE_ISO	*	=4:1_SPACING	?

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE	*	*	PCIE_ISO
CLK_PCIE	*	*	CLK_PCIE_ISO
COMP_PCIE	*	*	COMP_PCIE_ISO

PCIe (PCH)

Electrical Constraint Set	Physical	Spacing
xl AirPort		
E599 PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
E600 PCIE_GEN2_R2D_CONN_AP	PCIE_PHY	PCIE
E601	PCIE_PHY	PCIE
E602	PCIE_PHY	PCIE
E603 PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
E604 PCIE_GEN2_D2R_CONN_AP	PCIE_PHY	PCIE
E605 PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
E606 PCIE_REF_CLK_CONN	CLK_PCIE_PHY	CLK_PCIE
xl Caesar IV		
E610 PCIE_GEN2_R2D	PCIE_PHY	PCIE
E609 PCIE_GEN2_R2D	PCIE_PHY	PCIE
E618	PCIE_PHY	PCIE
E619	PCIE_PHY	PCIE
E616 PCIE_GEN2_D2R	PCIE_PHY	PCIE
E617 PCIE_GEN2_D2R	PCIE_PHY	PCIE
E615	PCIE_PHY	PCIE
E614	PCIE_PHY	PCIE
E613	PCIE_PHY	PCIE
E612	PCIE_PHY	PCIE
E611 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE
E610 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE

PCIE (PCH - TBT)

Electrical Constraint Set		Physical		Spacing	
x4 Thunderbolt					
ES17	PCIE_GEN2_R2D_P1NV	PCIE_PHY	PCIE	PCIE TBT R2D P<2..0>	no_constraint 36
ES17	PCIE_GEN2_R2D_P1NV	PCIE_PHY	PCIE	PCIE TBT R2D N<2..0>	no_constraint 36
ES20	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE TBT R2D P<3>	no_constraint 36
ES19	PCIE_GEN2_R2D	PCIE_PHY	PCIE	PCIE TBT R2D N<3>	no_constraint 36
ES20		PCIE_PHY	PCIE	PCIE TBT R2D C P<3..0>	no_constraint 36
ES20		PCIE_PHY	PCIE	PCIE TBT R2D C N<3..0>	no_constraint 36
ES30		PCIE_PHY	PCIE	PCIE TBT D2R P<3..0>	no_constraint 36
ES30		PCIE_PHY	PCIE	PCIE TBT D2R N<3..0>	no_constraint 36
ES30	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C P<1..0>	no_constraint 36
ES30	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C N<1..0>	no_constraint 36
ES30	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE TBT D2R C P<2>	no_constraint 36
ES21	PCIE_GEN2_D2R	PCIE_PHY	PCIE	PCIE TBT D2R C N<2>	no_constraint 36
ES21	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C P<3>	no_constraint 36
ES21	PCIE_GEN2_D2R_P1NV	PCIE_PHY	PCIE	PCIE TBT D2R C N<3>	no_constraint 36
ES30	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE CLK100M TBT P	no_constraint 36
ES30	PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	PCIE CLK100M TBT N	no_constraint 36

PCIe (CPU)

[illegible]

PCIe (CPU)

[illegible]

SYNC MASTER=D8 AARON		SYNC DATE=03/13/2012	
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CPU PCIe Constraints			
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DMI

DMI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
DMI_PHY	*	DMI_85D

CPU Misc / FDI

CPU-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
COMP_FDI_SE	*	Y	0.3 MM	0.25 MM	3 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
CPU_PHY	*	CPU_50S
COMP_FDI_PHY	*	COMP_FDI_SE

CPU-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ISO	*	=3:1_SPACING	?
COMP_FDI_ISO	*	=4:1_SPACING	?

FDI Compensation Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Table	Trace	Design	Iso	Design	Comments
6-4	10	11.81	-	15.75	Using PCIe guidelines

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU	*	*	CPU_ISO
COMP_FDI	*	*	COMP_FDI_ISO

XDP

XDP-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
XDP_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
XDP_PHY	*	XDP_50S

XDP-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
XDP_ISO	*	=2:1_SPACING	?
XDP_CLK_ISO	*	=4:1_SPACING	?

Desktop Debug Design Guide (Intel Doc# 430883)

Section	Imp	Design	Iso	Design	Comments
1.5	45-65	50	-	15.75	Isolation is for JTAG clocks. All signals default are 50 Ohm SE.

Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XDP	*	*	XDP_ISO
CLK_JTAG	*	*	XDP_CLK_ISO
OBS_STROBE	*	*	XDP_CLK_ISO

DMI

Electrical Constraint Set	Physical	Spacing	
DMI			
FE00 DMI_N2S	DMI_PHY	PCIE	DMI N2S P<3..0> NO TEST=TRUE 10 19
FE01 DMI_N2S	DMI_PHY	PCIE	DMI N2S N<3..0> NO TEST=TRUE 10 19
FE02 DMI_S2N	DMI_PHY	PCIE	DMI S2N P<3..0> NO TEST=TRUE 10 19
FE03 DMI_S2N	DMI_PHY	PCIE	DMI S2N N<3..0> NO TEST=TRUE 10 19
FE04 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI CLK100M CPU P 11 18
FE05 PCIE_REF_CLK	CLK_PCIE_PHY	CLK_PCIE	DMI CLK100M CPU N 11 18
DMI Compensation			
FE06	COMP_PCIE_PHY	COMP_PCIE	PCH DMI COMP 19
FE07	COMP_PCIE_PHY	COMP_PCIE	PCH DMI2RB1AS 19

CPU Misc.

Electrical Constraint Set	Physical	Spacing	
CPU Misc.			
FE60	CPU_EHY	CPU	CPU SKTOCC L 11 64
FE60	CPU_EHY	CPU	CPU PROC_SEL 11 19
FE60	CPU_EHY	CPU	CPU CATERR L 11 48
FE60	CPU_EHY	CPU	CPU PECT 11 21 47 48
FE60	CPU_EHY	CPU	CPU PROCHOT L 11 47 48 66
FE60	CPU_EHY	CPU	CPU PROCHOT_R L 11
FE60	CPU_EHY	CPU	CPU THRMTRIP L 11 48
FE60	CPU_EHY	CPU	CPU RESET L 11 26
FE60	CPU_EHY	CPU	PLT RESET LS1V05 L 11
FE60	PM_SYNC	CPU	PM SYNC 11 19
FE60	CPU_EHY	CPU	CPU PWRGD 11 21 25 28
FE60	CPU_EHY	CPU	PM MEM_PWRGD 11 19 28
FE60	CPU_EHY	CPU	PM MEM_PWRGD_R 11
FE60	CPU_EHY	CPU	CPU MEM_RESET L 11 28

FDI

Electrical Constraint Set	Physical	Spacing	
FDI Compensation			
FEED	COMP_FDI_PHY	COMP_FDI	CPU_FDI_COMPIO

XDP

Electrical Constraint Set	Physical	Spacing	
XDP_MISC.			
FW00 XDP_BFM_L	XDP_PHY	XDP	XDP_BFM_L<7..0> 11 25
FW01	XDP_PHY	XDP	CPU_CFG<17..16> 10 25
FW02	XDP_PHY	XDP	CPU_CFG<11..0> 10 15 25
FW03	XDP_PHY	XDP	XDP_DBRESET_L 11 25
FW04	XDP_PHY	XDP	XDP_CPU_PWRGD 25
FW05	XDP_PHY	XDP	XDP_CPU_PWRBTN_L 25
FW06	XDP_PHY	XDP	XDP_CPU_CFG<0> 25
FW07	XDP_PHY	XDP	XDP_VR_READY 25
FW08	XDP_PHY	XDP	XDP_CPU_PLTRST_L 25 26
FW09	XDP_PHY	XDP	XDP_PCH_PWRGD 25
FW10	XDP_PHY	XDP	XDP_PCH_PWRBTN_L 25
FW11	XDP_PHY	XDP	XDP_PCH_PLTRST_L 25 26
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Chipset Test Interface

Electrical Constraint Set		Physical	Spacing	
OBS PCH Side				
FE09	XDP_PCH_OBS	XDP_PHV	XDP	USB_EXTN_OC_R_L
FE10	XDP_PCH_OBS	XDP_PHV	XDP	USB_EXTB_OC_R_L
FE11	XDP_PCH_OBS	XDP_PHV	XDP	USB_EXTC_OC_R_L
FE12	XDP_PCH_OBS	XDP_PHV	XDP	USB_EXTD_OC_R_L
FE13	XDP_PCH_OBS	XDP_PHV	XDP	USB_EXTN_OC_EHCI_R_L
FE14	XDP_PCH_OBS	XDP_PHV	XDP	USB_EXTD_OC_EHCI_R_L
FE15	XDP_PCH_OBS	XDP_PHV	XDP	AP_PWR_EN_R
FE16	XDP_PCH_OBS	XDP_PHV	XDP	SDCONN_STATE_CHANGE_R
FE17	XDP_PCH_OBS	XDP_PHV	OBS_STROBE	TBT_CIO_PLUG_EVENT_R
FE18	XDP_PCH_OBS	XDP_PHV	XDP	ISOLATE_CPU_MEM_R_L
FE19	XDP_PCH_OBS	XDP_PHV	XDP	GPU_GOOD_R
FE20	XDP_PCH_OBS	XDP_PHV	XDP	DP_AUXCH_ISOL_R
FE21	XDP_PCH_OBS	XDP_PHV	XDP	SATARDRVN_EN_R
FE22	XDP_PCH_OBS	XDP_PHV	XDP	DP_TBT_SEL_R
FE23	XDP_PCH_OBS	XDP_PHV	XDP	JTAG_TBT_TCK_R
FE24	XDP_PCH_OBS	XDP_PHV	XDP	AUD_IPHS_SWITCH_EN_PCH_R
FE25	XDP_PCH_OBS	XDP_PHV	XDP	ENET_LOW_PWR_PCH_R
FE26	XDP_PCH_OBS_UNUSEN	XDP_PHV	OBS_STROBE	XDP_PIN03
OBS XDP Side				
FE27		XDP_PHV	XDP	XDP_DA0_USB_EXTN_OC_L
FE28		XDP_PHV	XDP	XDP_DA1_USB_EXTB_OC_L
FE29		XDP_PHV	XDP	XDP_DA2_USB_EXTC_OC_L
FE30		XDP_PHV	XDP	XDP_DA3_USB_EXTD_OC_L
FE31		XDP_PHV	XDP	XDP_DB0_USB_EXTB_OC_EHCI_L
FE32		XDP_PHV	XDP	XDP_DB1_USB_EXTD_OC_EHCI_L
FE33		XDP_PHV	XDP	XDP_DB2_AP_PWR_EN
FE34		XDP_PHV	XDP	XDP_DB3_SDCONN_STATE_CHANGE
FE35		XDP_PHV	OBS_STROBE	XDP_FC1_TBT_CIO_PLUG_EVENT
FE36		XDP_PHV	XDP	XDP_DC0_ISOLATE_CPU_MEM_L
FE37		XDP_PHV	XDP	XDP_DC1_GPU_GOOD
FE38		XDP_PHV	XDP	XDP_DC2_DP_AUXCH_ISOL
FE39		XDP_PHV	XDP	XDP_DC3_SATARDRVN_EN
FE40		XDP_PHV	XDP	XDP_DD0_DP_TBT_SEL
FE41		XDP_PHV	XDP	XDP_DD1_JTAG_TBT_TCK
FE42		XDP_PHV	XDP	XDP_DD2_AUD_IPHS_SWITCH_EN_PCH
FE43		XDP_PHV	XDP	XDP_DD3_ENET_LOW_PWR_PCH
FE44		XDP_PHV	OBS_STROBE	XDP_FC0_PCH_GPIO15
Standard usage branch off from OBS				
FE45		XDP_PHV	XDP	USB_EXTN_OC_L
FE46		XDP_PHV	XDP	USB_EXTB_OC_L
FE47		XDP_PHV	XDP	USB_EXTC_OC_L
FE48		XDP_PHV	XDP	USB_EXTD_OC_L
FE49		XDP_PHV	XDP	USB_EXTB_OC_EHCI_L
FE50		XDP_PHV	XDP	USB_EXTD_OC_EHCI_L
FE51		XDP_PHV	XDP	AP_PWR_EN
FE52		XDP_PHV	XDP	SDCONN_STATE_CHANGE
FE53		XDP_PHV	OBS_STROBE	TBT_CIO_PLUG_EVENT
FE54		XDP_PHV	XDP	ISOLATE_CPU_MEM_L
FE55		XDP_PHV	XDP	GPU_GOOD
FE56		XDP_PHV	XDP	DP_AUXCH_ISOL
FE57		XDP_PHV	XDP	SATARDRVN_EN
FE58		XDP_PHV	XDP	DP_TBT_SEL
FE59		XDP_PHV	XDP	JTAG_TBT_TCK
FE60		XDP_PHV	XDP	JTAG_TBT_TCK_ISOL
FE61		XDP_PHV	XDP	AUD_IPHS_SWITCH_EN_PCH
FE62		XDP_PHV	XDP	ENET_LOW_PWR_PCH
FE63		XDP_PHV	XDP	DP_AUXCH_ISOL_EN

SYNC MASTER=D8 ROSITA		SYNC DATE=03/23/2012	
PAGE TITLE			
CPU MISC/DMI/FDI/XDP		Constraints	
 Apple Inc.		DRAWING NUMBER 051-9504	
		REVISION 7.0.0	
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PCH

PCH-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCH_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCH-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCH_ISO	*	=6.5X_DIELECTRIC	?	CLK_PCH	*	*	CLK_PCH_ISO
COMP_PCH_ISO	*	=2:1_SPACING	?	COMP_PCH	*	*	COMP_PCH_ISO
PCH_ISO	*	=3:1_SPACING	?	PCH	*	*	PCH_ISO

PCI

PCI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_PCI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

PCI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCI_ISO	*	=3.6X_DIELECTRIC	?	CLK_PCI	*	*	CLK_PCI_ISO

LPC

LPC-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

LPC-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
LPC_ISO	*	=1.5:1_SPACING	?	LPC	*	*	LPC_ISO
CLK_LPC_ISO	*	=3.6X_DIELECTRIC	?	CLK_LPC	*	*	CLK_LPC_ISO

HDA

HDA-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

HDA-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDA_ISO	*	=2X_DIELECTRIC	?	HDA	*	*	HDA_ISO

Crystal

Crystal-specific Physical Rules

[illegible]

Crystal-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
XTAL_ISO	*	=4X_DIELECTRIC	?	XTAL	*	*	XTAL_ISO

SPI

SPI-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

SPI-specific Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SPI_ISO	*	=2X_DIELECTRIC	?	SPI	*	*	SPI_ISO

PCI

Electrical Constraint Set	Physical	Spacing	
PCI Clock			
	CLK_PCT_50S	CLK_PCT	PCH_CLK33M_PCTIIN 18 26
	CLK_PCT_50S	CLK_PCT	PCH_CLK33M_PCTIOUT 20 26

LPC

Electrical Constraint Set	Physical	Spacing	
LPC			
RE30	LPC_50S	LPC	LPC_AD<3..0> 18 47 49
RE30	LPC_50S	LPC	LPC_R_AD<3..0> 18
RE30	LPC_50S	LPC	LPC_FRAME_L 18 47 49
RE30	LPC_50S	LPC	LFRAME_L 18
LPC Clocks			
RE30	CLK_LPC_50S	CLK_LPC	LPC_CLK33M LPCPLUS 26 49
RE40	CLK_LPC_50S	CLK_LPC	LPC_CLK33M LPCPLUS_R 20 26
RE40	CLK_LPC_50S	CLK_LPC	LPC_CLK33M SMC 26 47
RE40	CLK_LPC_50S	CLK_LPC	LPC_CLK33M SMC_R 20 26

PCH Clocks

Electrical Constraint Set	Physical	Spacing	
PCH Reference Clock			
FE05	CLK_PCH_50S	CLF_PCH	SYSCLK_CLK25M_SB 26
FE06	CLK_PCH_50S	CLF_PCH	PCH_CLK25M_XTALIN 18 26
PCH Ref Clock Comp			
FE04	PCH_50S	COMP_PCH	PCH_XCLK_RCOMP 18
PCH RTC 32K			
FE02	CLK_XTAL	XTAL	PCH_CLK32K_RTCX1 18 26
FE03	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2 18 26
FE09	CLK_XTAL	XTAL	PCH_CLK32K_RTCX2_R 26
SMC 32K			
FE10	CLK_PCH_50S	CLF_PCH	PM_CLK32K_SUSCLK_R 19 48
FE08	CLK_PCH_50S	CLF_PCH	SMC_CLK32K 47 48

25 MHz Reference Clocks

Electrical Constraint Set	Physical	Spacing	
25M Reference Crystal			
 P100	CLK_XTAL	XTAL	SYSCLK_CLK25M_X1 26
 P100	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2 26
 P100	CLK_XTAL	XTAL	SYSCLK_CLK25M_X2_R 26
25M Reference Clocks			
 P100	CLK_ECH_50S	CLK_ECH	SYSCLK_CLK25M_ENET 26 39
 P100	CLK_ECH_50S	CLK_ECH	SYSCLK_CLK25M_ENET_R 26 39
 P100	CLK_ECH_50S	CLK_ECH	SYSCLK_CLK25M_TBT 26 36
 P100	CLK_ECH_50S	CLK_ECH	SYSCLK_CLK25M_TBT_R 26 36

HDA

Electrical Constraint Set	Physical	Spacing	
HDA			
H350	HDA_50S	HDA	HDA_BIT_CLK 18 56
H351	HDA_50S	HDA	HDA_BIT_CLK_R 18
H352	HDA_50S	HDA	HDA_RST_L 18 56
H353	HDA_50S	HDA	HDA_RST_R_L 18
H354	HDA_50S	HDA	HDA_SDOUT 18 56
H355	HDA_50S	HDA	HDA_SDOUT_R 15 18
H356	HDA_50S	HDA	HDA_SYNC 15 18 56
H357	HDA_50S	HDA	HDA_SYNC_R 18
H358	HDA_50S	HDA	HDA_SDI0 18 56
H359	HDA_50S	HDA	AUD_SDI_R 56
H316	HDA_50S	HDA	SPI_DESCRIPTOR_OVERRIDE_R 15
SPDIF			
H330		HDA	AUD_SPDIF_CHIP 56
H331		HDA	AUD_SPDIF_OUT 56 60

SPI Bootrom

Electrical Constraint Set	Physical	Spacing	
SPI ROM			
H032	SPI_50S	SPI	SPI_CLK_R 18 49
H039	SPI_50S	SPI	SPI_CLK 49
H039	SPI_50S	SPI	SPI_ALT_CLK 49
H400	SPI_50S	SPI	SPI_SMC_CLK 47 48
H409	SPI_50S	SPI	SPI_MLB_CLK 48 49
H032	SPI_50S	SPI	SPI_CS0_R_L 18 49
H039	SPI_50S	SPI	SPI_CS0_L 49
H039	SPI_50S	SPI	SPI_ALT_CS_L 49
H400	SPI_50S	SPI	SPI_SMC_CS_L 47 48
H039	SPI_50S	SPI	SPI_MLB_CS_L 48 49
H039	SPI_50S	SPI	SPI_MOSI_R 18 49
H400	SPI_50S	SPI	SPI_MOSI 49
H409	SPI_50S	SPI	SPI_ALT_MOSI 49
H039	SPI_50S	SPI	SPI_SMC_MOSI 47 48
H038	SPI_50S	SPI	SPI_MLB_MOSI 48 49
H400	SPI_50S	SPI	SPI_MISO 18 49
H039	SPI_50S	SPI	SPI_ALT_MISO 49
H039	SPI_50S	SPI	SPI_SMC_MISO 47 48
H038	SPI_50S	SPI	SPI_MLB_MISO 48 49
H400	SPI_50S	SPI	SPIROM_USE_MLB 21 49

SYNC MASTER=D8 MARK		SYNC DATE=02/10/2012	
PAGE TITLE			
PCH and BR Constraints			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-9504	D
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USB

USB-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
USB_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
USB_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
USB2_PHY	*	USB_90D
USB3_PHY	*	USB_85D
USB_HUB_PHY	*	USB_50S

USB Min Spacing Rules (mils) (Maho Bay PDG, Intel Doc# 473718)

Section	Imp	Design	Iso	Comments
12.2.1	90	90	12/2.8 mils = 4.29:1	USB 2.0
13.3.1	85	85	20/2.8 mils = 7.14:1	USB 3.0
			50/2.8 mils = 17.9:1	USB 2.0/3.0

SMSC Hub Application Note 15.17

Single-ended impedance range from 45-80 ohm is acceptable

USB 2.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB2_ISO	*	=4.4X_DIELECTRIC	?
USB2_ISO	TOP,BOTTOM	=4.4X_DIELECTRIC	?
USB2_CLK_ISO	*	=18X_DIELECTRIC	?
USB2_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB_HUB_ISO	*	=21X_SPACING	?

USB 2.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB2	*	*	USB2_ISO
USB2	*CLK*	*	USB2_CLK_ISO
USB2	DISPLAYPORT	*	USB2_CLK_ISO
USB2	*TBT*	*	USB2_CLK_ISO
USB2	*ENET*	*	USB2_CLK_ISO
USB2	*SD*	*	USB2_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB_HUB	*	*	USB_HUB_ISO

USB 3.0 Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_ISO	TOP,BOTTOM	=7.3X_DIELECTRIC	?
USB3_CLK_ISO	*	=18X_DIELECTRIC	?
USB3_CLK_ISO	TOP,BOTTOM	=18X_DIELECTRIC	?
USB3_USB3_ISO	*	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	TOP,BOTTOM	=7.3X_DIELECTRIC	?
USB3_USB3_ISO	ISL10	=5X_DIELECTRIC	?

USB 3.0 Spacing Rules

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3	*	*	USB3_ISO
USB3	*CLK*	*	USB3_CLK_ISO
USB3	DISPLAYPORT	*	USB3_CLK_ISO
USB3	*TBT*	*	USB3_CLK_ISO
USB3	*ENET*	*	USB3_CLK_ISO
USB3	*SD*	*	USB3_CLK_ISO
USB3	PCIE	*	USB3_CLK_ISO
USB3	SATA	*	USB3_CLK_ISO
USB3	USB3	*	USB3_USB3_ISO

CAMERA CONTROLLER

Camera Controller's SMIA Interface & MISC. Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMIA_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
CAM_SE	*	Y	0.2 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SMIA_DIFF_PHY	*	SMIA_100D
CAM_PHY	*	CAM_SE

Camera Controller's SMIA Interface & MISC. Spacing Definitions

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMIA_DIFF_ISO	*	=6:1_SPACING	?
SMIA_DIFF2DIFF	*	=3:1_SPACING	?
CAM_ISO	*	=2:1_SPACING	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SMIA_DIFF	*	*	SMIA_DIFF_ISO
SMIA_DIFF	SMIA_DIFF	*	SMIA_DIFF2DIFF
CAM	*	*	CAM_ISO

USB 3.0 and USB 2.0 Trixies Muxing

Electrical Constraint Set		Physical	Spacing	
External Port A (J4600)				
R490	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P
R491	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N
R492	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_P
R493	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_N
R494	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P
R495	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N
R496	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_F_P
R497	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_F_N
R498	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_C_P
R499	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_C_N
R490	USB2_MUXED_MUXO_CONN	USB2_PHY	USB2	USB_PCH_0_P
R491	USB2_MUXED_MUXO_CONN	USB2_PHY	USB2	USB_PCH_0_N
R492	USB2_PHY	USB2	USB2	USB2_EXTB_MUXED_P
R493	USB2_PHY	USB2	USB2	USB2_EXTB_MUXED_N
R494	USB2_PHY	USB2	USB2	USB2_EXTB_MUXED_F_P
R495	USB2_PHY	USB2	USB2	USB2_EXTB_MUXED_F_N
External Port B (J4610)				
R496	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P
R497	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N
R498	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_P
R499	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_N
R490	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P
R491	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N
R492	USB3_PHY	USB3	USB3	USB3_EXTB_TX_F_P
R493	USB3_PHY	USB3	USB3	USB3_EXTB_TX_F_N
R494	USB3_PHY	USB3	USB3	USB3_EXTB_TX_C_P
R495	USB3_PHY	USB3	USB3	USB3_EXTB_TX_C_N
R496	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_P
R497	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_1_N
R498	USB2_PHY	USB2	USB2	USB_PCH_9_P
R499	USB2_PHY	USB2	USB2	USB_PCH_9_N
R490	USB2_PHY	USB2	USB2	USB2_EXTB_MUXED_P
R491	USB2_PHY	USB2	USB2	USB2_EXTB_MUXED_N
R492	USB2_PHY	USB2	USB2	USB2_EXTB_MUXED_F_P
R493	USB2_PHY	USB2	USB2	USB2_EXTB_MUXED_F_N
External Port C (J4700)				
R494	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_P
R495	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_N
R496	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_P
R497	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTB_RX_F_N
R498	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_P
R499	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTB_TX_N
R490	USB3_PHY	USB3	USB3	USB3_EXTB_TX_F_P
R491	USB3_PHY	USB3	USB3	USB3_EXTB_TX_F_N
R492	USB3_PHY	USB3	USB3	USB3_EXTB_TX_C_P
R493	USB3_PHY	USB3	USB3	USB3_EXTB_TX_C_N
R494	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_P
R495	USB2_CONN	USB2_PHY	USB2	USB_PCH_2_N
R496	USB2_PHY	USB2	USB2	USB2_EXTB_F_P
R497	USB2_PHY	USB2	USB2	USB2_EXTB_F_N
External Port D (J4710)				
R498	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_P
R499	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_N
R490	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_F_P
R491	USB3_RX_CONN	USB3_PHY	USB3	USB3_EXTD_RX_F_N
R492	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_P
R493	USB3_TX_CONN	USB3_PHY	USB3	USB3_EXTD_TX_N
R494	USB3_PHY	USB3	USB3	USB3_EXTD_TX_F_P
R495	USB3_PHY	USB3	USB3	USB3_EXTD_TX_F_N
R496	USB3_PHY	USB3	USB3	USB3_EXTD_TX_C_P
R497	USB3_PHY	USB3	USB3	USB3_EXTD_TX_C_N
R498	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_P
R499	USB2_MUXED_CONN	USB2_PHY	USB2	USB_PCH_3_N
R490	USB2_PHY	USB2	USB2	USB_PCH_10_P
R491	USB2_PHY	USB2	USB2	USB_PCH_10_N
R492	USB2_PHY	USB2	USB2	USB2_EXTD_MUXED_P
R493	USB2_PHY	USB2	USB2	USB2_EXTD_MUXED_N
R494	USB2_PHY	USB2	USB2	USB2_EXTD_MUXED_F_P
R495	USB2_PHY	USB2	USB2	USB2_EXTD_MUXED_F_N
Camera (U4200)				
R496	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_P
R497	USB2_CONN_INT	USB2_PHY	USB2	USB_CAMERA_N
PCH USB Compensation				
R498		PCH_50S	COMP_PCH	PCH_USB_RBIA5

Electrical Constraint Set	Physical	Spacing	Voltage	
Camera Controller Local Ground				
592	GND_PHY	GND	0V	CAM_AGND 42
592	GND_PHY	GND	0V	CAM_PLLGND 42

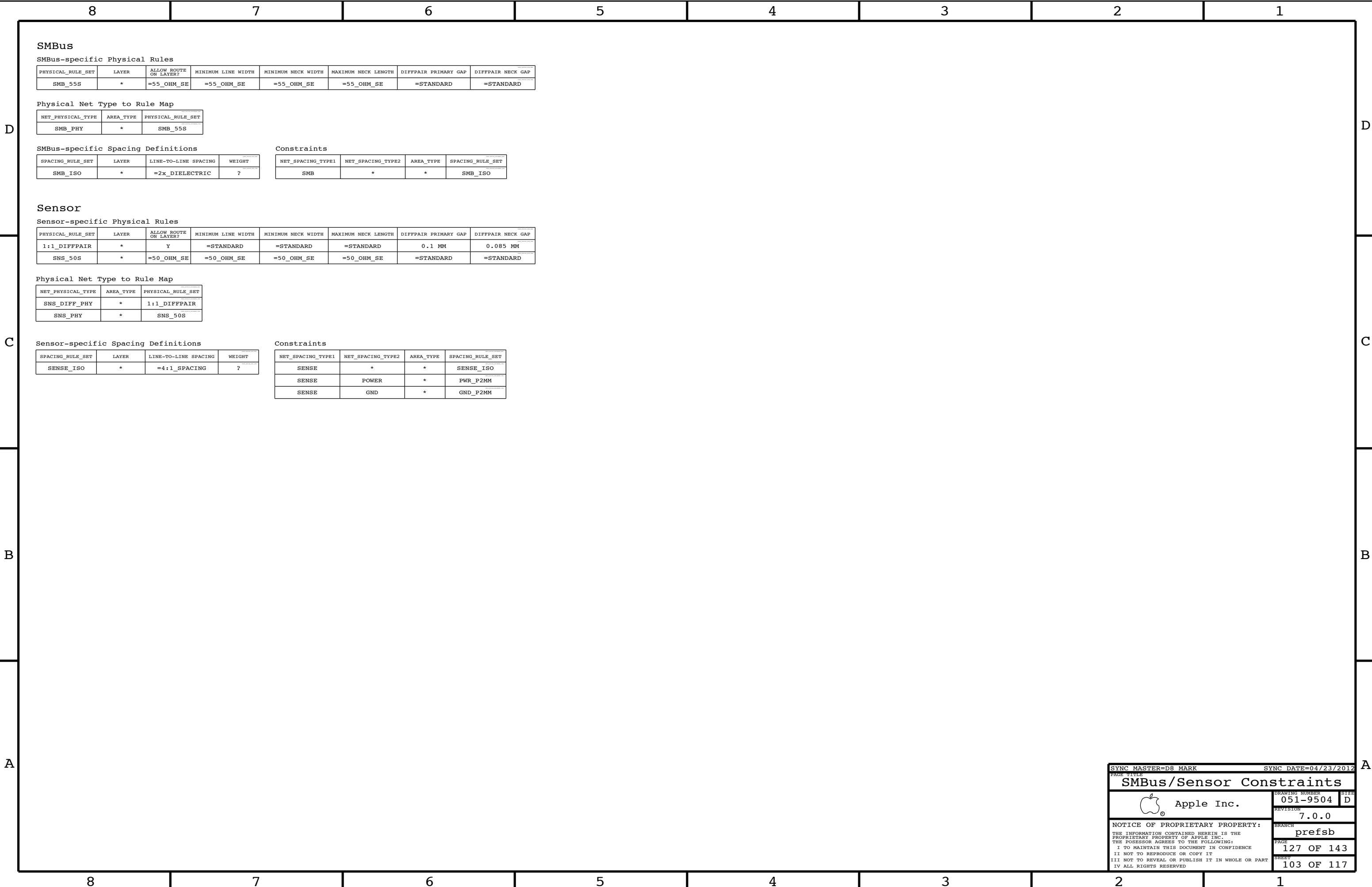
USB Hub

Electrical Constraint Set	Physical	Spacing	
USB 2.0 Hub			
USB2_HUB_PCH	USB2_PHY	USB2	USB_PCH_7_P 20 27
USB2_HUB_PCH	USB2_PHY	USB2	USB_PCH_7_N 20 27
USB2_HUB_BT	USB2_PHY	USB2	USB_BT_P 27 35
USB2_HUB_BT	USB2_PHY	USB2	USB_BT_N 27 35
USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_P 35
USB2_HUB_BT	USB2_PHY	USB2	USB_BT_MUX_N 35
USB_HUB_PHY	USB_HUB_PHY	USB_HUB	USB_HUB_2P 27
USB_HUB_PHY	USB_HUB_PHY	USB_HUB	USB_HUB_2N 27
USB 2.0 Hub Misc.			
USB_HUB_PHY	USB_HUB_PHY	USB_HUB	USB_HUB_RESET_L 27
USB_HUB_PHY	USB_HUB_PHY	USB_HUB	USB_HUB_VBUS_DET 27
USB_HUB_PHY	USB_HUB_PHY	USB_HUB	USB_HUB_NON_REMO 27
USB_HUB_PHY	USB_HUB_PHY	USB_HUB	USB_HUB_NON_REM1 27
USB_HUB_PHY	USB_HUB_PHY	USB_HUB	USB_HUB_HS_IND 27
USB 2.0 Hub Compensation			
COMP_FOS	COMP_FCH		USB_HUB_RBIA5 27
USB 2.0 Hub Crystal			
CLK_XTAL	XTAL		USB_HUB_XTAL1 27
CLK_XTAL	XTAL		USB_HUB_XTAL2 27
CLK_XTAL	XTAL		USB_HUB_XTAL2_R 27

Camera Controller

Electrical Constraint Set	Physical	Spacing	
SMIA			
HSD0 SMIA_DATA	SMIA_DIEFF_PHY	SMIA_DIEFF	SMIA_DATA_P 42
HSD0 SMIA_DATA	SMIA_DIEFF_PHY	SMIA_DIEFF	SMIA_DATA_N 42
HSD0 SMIA_CLK	SMIA_DIEFF_PHY	SMIA_DIEFF	SMIA_CLK_P 42
HSD0 SMIA_CLK	SMIA_DIEFF_PHY	SMIA_DIEFF	SMIA_CLK_N 42
MISC			
HSD0 SPI_50S	SPI	CAM_SF_CLK	42
HSD0 SPI_50S	SPI	CAM_SF_CLK_R	42
HSD0 SPI_50S	SPI	CAM_SF_DIN	42
HSD0 SPI_50S	SPI	CAM_SF_DIN_R	42
HSD0 SPI_50S	SPI	CAM_SF_CS_L	42
HSD0 SPI_50S	SPI	CAM_SF_WP_L	42
HSD0 SPI_50S	SPI	CAM_SF_DOUT	42
HSD0 SPI_50S	SPI	CAM_SF_DOUT_R	42
HSD0 SPI_50S	SPI	CAM_SF_HOLD_L	42
HSD0 CAM_PHY	CAM	CAM_USB_VRES	42
HSD0 CAM_PHY	CAM	MIPI_RESISTOR	42
HSD0 PM	PM	CAM_EXT_BOOT_L	43
HSD0 PM	PM	PCH_CAM_EXT_BOOT_R_L	21 43
HSD0 PM	PM	CAM_PIV2_RST_HOLDOFF	43
HSD0 PM	PM	CAM_PIV2_RST_HOLDOFF_L	43
I2C			
HSD0 SMB_PHY	SMB	I2C_CAMSENSOR_SDA	42
HSD0 SMB_PHY	SMB	I2C_CAMSENSOR_SCL	42
HSD0 SMB_PHY	SMB	SMB_ALS_F_SDA	42
HSD0 SMB_PHY	SMB	SMB_ALS_F_SCL	42
Camera Controller Crystal			
HSD0 CLK_XTAL	XTAL	CAM_XTAL_IN	42
HSD0 CLK_XTAL	XTAL	CAM_XTAL_OUT	42
HSD0 CLK_XTAL	XTAL	CAM_XTAL_OUT_R	42

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DC-DC

Power-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
GND_F3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
GND_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
POWER_P3MM	*	Y	0.300 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P6MM	*	Y	0.600 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD
POWER_P5MM	*	Y	0.500 MM	0.200 MM	3.0 MM	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
VR_CTL_PHY	*	POWER_P5MM
VR_CTL_PHY	BGA	STANDARD
VR_VID_PHY	*	POWER_50S
POWER_PHY	*	POWER_P6MM
GND_PHY	*	GND_P5MM

Constraints
Power and Common

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
POWER	*	*	POWER_ISO
GND	*	*	GND_ISO

Power-specific Spacing Definitions
Power and Common

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
POWER_ISO	*	=STANDARD	?

DC-DC Baddies

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
SWNODE_ISO	*	=8:1_SPACING	1000
SWNODE_SW2SW	*	=1:1_SPACING	?
SWNODE_SW2PWR	*	=2:1_SPACING	?
SWNODE_SW2GND	*	=2:1_SPACING	?
SWNODE_LG2SW	*	=3:1_SPACING	?

DC-DC Control

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
VR_CTL_ISO	*	=3:1_SPACING	?
VR_VID_ISO	*	=4X_DIELECTRIC	?

CPU VccSA

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST
Input Bus					
E24	POWER_PHY	POWER	5V		REG_VCC_U7500
E25	POWER_PHY	POWER	5V		REG_PVCC_U7500
Local Ground					
E16	GND_PHY	GND	0V		AGND_VCCSAS0
VCCIO					
E27	POWER_PHY	VR_SWITCH	12V	TRUE	REG_PHASE_VCCSAS0
E28	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_BOOT_VCCSAS0
E29	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_BOOT_VCCSAS0_RC
E30	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_UGATE_VCCSAS0
E31	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_LGATE_VCCSAS0
E32	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_SNUBBER_VCCSAS0
E33	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_VCCSAS0_OCSET
E34	VR_CTL_PHY	VR_CTL			REG_VCCSAS0_VO
E35					
E36					
E37	VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE		SNS_CPU_VCCSA
E38	VSNS_CPU_VCCSA	SNS_DIFF_PHY	SENSE		SNS_VCCSAS0_XW_P
E39					SNS_VCCSAS0_XW_N
E40					REG_VCCSAS0_FB
E41					REG_VCCSAS0_RTN
E42					REG_VCCSAS0_SREF
E43					REG_VCCSAS0_FSEL
Output Bus					
E44	POWER_PHY	POWER	0.925V		PPVCCSA_S0

CPU VccIO PCH 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST
Input Bus					
E11	POWER_PHY	POWER	5V		REG_VCC_U7400
E12	POWER_PHY	POWER	5V		REG_PVCC_U7400
Local Ground					
E13	GND_PHY	GND	0V		AGND_CPU_P1V05S0
1.05V S0					
E15	POWER_PHY	VR_SWITCH	12V	TRUE	REG_CPU_PHASE_P1V05S0
E17	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_CPU_BOOT_P1V05S0
E18	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_CPU_BOOT_P1V05S0_RC
E19	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_CPU_UGATE_P1V05S0
E20	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_CPU_UGATE_P1V05S0_R
E21	VR_CTL_PHY	VR_LGATE	12V	TRUE	REG_CPU_LGATE_P1V05S0
E22	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_CPU_SNUBBER_P1V05S0
E23	VR_CTL_PHY	VR_CTL			REG_CPU_P1V05S0_OCSET
E24	VR_CTL_PHY	VR_CTL			REG_CPU_P1V05S0_VO
E25					
E26	VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE		SNS_CPU_VCCIO_P
E27	VSNS_CPU_VCCIO	SNS_DIFF_PHY	SENSE		SNS_CPU_VCCIO_N
E28					SNS_CPU_P1V05S0_XW_P
E29					SNS_CPU_P1V05S0_XW_N
E30					REG_CPU_P1V05S0_FB
E31					REG_CPU_P1V05S0_RTN
E32					
E33	VR_CTL_PHY	VR_CTL			REG_CPU_P1V05S0_SREF
E34	VR_CTL_PHY	VR_CTL			REG_CPU_P1V05S0_FSEL
Output Bus					
E22	POWER_PHY	POWER	1.05V		PP1V05_S0_CPU

PCH/GPU/TBT 1.05V S0

Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST
Input Bus					
E45	POWER_PHY	POWER	5V		REG_VCC_U8700
Local Ground					
E46	GND_PHY	GND	0V		P1V05_AGND
1.05V S0					
E47	POWER_PHY	VR_SWITCH	12V	TRUE	REG_PHASE_P1V05S0
E48	POWER_PHY	VR_SWITCH	12V	TRUE	REG_PHASE_P1V05S0_L
E49	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_BOOT_P1V05S0
E50	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_BOOT_P1V05S0_RC
E51	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_UGATE_P1V05S0
E52	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_UGATE_P1V05S0_R
E53	VR_CTL_PHY	VR_LGATE	12V	TRUE	REG_LGATE_P1V05S0
E54	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG_SNUBBER_P1V05S0
E55					
E56	VR_CTL_PHY	VR_CTL			REG_P1V05S0_OCSET
E57	VR_CTL_PHY	VR_CTL			REG_P1V05S0_VO
E58	POWER_PHY	VR_SWITCH	12V		REG_P1V05S0_VO_R
E59	POWER_PHY	VR_SWITCH	12V		P1V05_OCSET_R
E60					
E61					
E62					
E63	VR_CTL_PHY	VR_CTL			REG_P1V05S0_SREF
E64	VR_CTL_PHY	VR_CTL			REG_P1V05S0_FSEL
Output Bus					
E65	POWER_PHY	POWER	1.05V		PP1V05_S0_PCH
E66	POWER_PHY	POWER	1.05V		PP1V05_S0
FET Switched					
E67	POWER_PHY	POWER	1.05V		PP1V05_TBTLIC
E68	POWER_PHY	POWER	1.05V		PP1V05_TBTCIO

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	<table><tr><td>Electrical Constraint Set</td><td>Physical</td><td>Spacing</td><td>Voltage</td><td>DIDT</td><td>NO_TEST</td></tr><tr><td>Input Bus</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>H030</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td>PP12V_S0_CPUCORE FLT</td></tr><tr><td>H030</td><td>POWER_PHY</td><td>POWER</td><td>5V</td><td></td><td>REG VCC U7100</td></tr><tr><td>Local Ground</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>H020</td><td>GND_PHY</td><td>GND</td><td>0V</td><td></td><td>AGND CPU</td></tr><tr><td>Phase 1</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>H000</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td>REG LVCC U7210</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG PWM CPUCORE 1</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG PWM CPUCORE 1 R</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG PHASE CPUCORE 1</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG BOOT CPUCORE 1</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG BOOT CPUCORE 1 RC</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG UGATE CPUCORE 1</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG LGATE CPUCORE 1</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG SNUBBER CPUCORE 1</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td>PPCPUCORE S0 SENSE 1</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>REG ISENCORE 1 P</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>REG ISENCORE 1 N</td></tr><tr><td>H000</td><td></td><td>SENSE</td><td></td><td></td><td>REG ISENCORE 1 NR</td></tr><tr><td>Phase 2</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>H010</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td>REG LVCC U7230</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG PWM CPUCORE 2</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG PWM CPUCORE 2 R</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG PHASE CPUCORE 2</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG BOOT CPUCORE 2</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG BOOT CPUCORE 2 RC</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG UGATE CPUCORE 2</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG LGATE CPUCORE 2</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG SNUBBER CPUCORE 2</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td>PPCPUCORE S0 SENSE 2</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>REG ISENCORE 2 P</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>REG ISENCORE 2 N</td></tr><tr><td>H000</td><td></td><td>SENSE</td><td></td><td></td><td>REG ISENCORE 2 NR</td></tr><tr><td>Phase 3</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>H020</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td>REG LVCC U7250</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG PWM CPUCORE 3</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG PWM CPUCORE 3 R</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG PHASE CPUCORE 3</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG BOOT CPUCORE 3</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG BOOT CPUCORE 3 RC</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG UGATE CPUCORE 3</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG LGATE CPUCORE 3</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG SNUBBER CPUCORE 3</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td>PPCPUCORE S0 SENSE 3</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>REG ISENCORE 3 P</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>REG ISENCORE 3 N</td></tr><tr><td>H000</td><td></td><td>SENSE</td><td></td><td></td><td>REG ISENCORE 3 NR</td></tr><tr><td>Phase 4</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>H030</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td>REG LVCC U7310</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG PWM CPUCORE 4</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG PWM CPUCORE 4 R</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG PHASE CPUCORE 4</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG BOOT CPUCORE 4</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG BOOT CPUCORE 4 RC</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG UGATE CPUCORE 4</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG LGATE CPUCORE 4</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG SNUBBER CPUCORE 4</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td>PPCPUCORE S0 SENSE 4</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>REG ISENCORE 4 P</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>REG ISENCORE 4 N</td></tr><tr><td>H000</td><td></td><td>SENSE</td><td></td><td></td><td>REG ISENCORE 4 NR</td></tr><tr><td>H020</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>SNS CORE XW P</td></tr><tr><td>H020</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>SNS CORE XW N</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>SNS CORE R P</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>SNS CORE R N</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>SNS CPU VAXG P</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>SNS CPU VAXG N</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>SNS CPU VCORE P</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>SNS CPU VCORE N</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>SNS P1V05 IOVDD XW P</td></tr><tr><td>H000</td><td>ISNS_CPU_CORE</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>SNS P1V05 IOVDD XW N</td></tr></table>				Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	Input Bus						H030	POWER_PHY	POWER	12V		PP12V_S0_CPUCORE FLT	H030	POWER_PHY	POWER	5V		REG VCC U7100	Local Ground						H020	GND_PHY	GND	0V		AGND CPU	Phase 1						H000	POWER_PHY	POWER	12V		REG LVCC U7210	H000	VR_CTL_PHY	VR_CTL			REG PWM CPUCORE 1	H000	VR_CTL_PHY	VR_CTL			REG PWM CPUCORE 1 R	H000	POWER_PHY	VR_SWITCH	12V	TRUE	REG PHASE CPUCORE 1	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG BOOT CPUCORE 1	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG BOOT CPUCORE 1 RC	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG UGATE CPUCORE 1	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG LGATE CPUCORE 1	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG SNUBBER CPUCORE 1	H000	POWER_PHY	POWER	1.1V		PPCPUCORE S0 SENSE 1	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		REG ISENCORE 1 P	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		REG ISENCORE 1 N	H000		SENSE			REG ISENCORE 1 NR	Phase 2						H010	POWER_PHY	POWER	12V		REG LVCC U7230	H000	VR_CTL_PHY	VR_CTL			REG PWM CPUCORE 2	H000	VR_CTL_PHY	VR_CTL			REG PWM CPUCORE 2 R	H000	POWER_PHY	VR_SWITCH	12V	TRUE	REG PHASE CPUCORE 2	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG BOOT CPUCORE 2	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG BOOT CPUCORE 2 RC	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG UGATE CPUCORE 2	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG LGATE CPUCORE 2	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG SNUBBER CPUCORE 2	H000	POWER_PHY	POWER	1.1V		PPCPUCORE S0 SENSE 2	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		REG ISENCORE 2 P	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		REG ISENCORE 2 N	H000		SENSE			REG ISENCORE 2 NR	Phase 3						H020	POWER_PHY	POWER	12V		REG LVCC U7250	H000	VR_CTL_PHY	VR_CTL			REG PWM CPUCORE 3	H000	VR_CTL_PHY	VR_CTL			REG PWM CPUCORE 3 R	H000	POWER_PHY	VR_SWITCH	12V	TRUE	REG PHASE CPUCORE 3	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG BOOT CPUCORE 3	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG BOOT CPUCORE 3 RC	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG UGATE CPUCORE 3	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG LGATE CPUCORE 3	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG SNUBBER CPUCORE 3	H000	POWER_PHY	POWER	1.1V		PPCPUCORE S0 SENSE 3	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		REG ISENCORE 3 P	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		REG ISENCORE 3 N	H000		SENSE			REG ISENCORE 3 NR	Phase 4						H030	POWER_PHY	POWER	12V		REG LVCC U7310	H000	VR_CTL_PHY	VR_CTL			REG PWM CPUCORE 4	H000	VR_CTL_PHY	VR_CTL			REG PWM CPUCORE 4 R	H000	POWER_PHY	VR_SWITCH	12V	TRUE	REG PHASE CPUCORE 4	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG BOOT CPUCORE 4	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG BOOT CPUCORE 4 RC	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG UGATE CPUCORE 4	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG LGATE CPUCORE 4	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG SNUBBER CPUCORE 4	H000	POWER_PHY	POWER	1.1V		PPCPUCORE S0 SENSE 4	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		REG ISENCORE 4 P	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		REG ISENCORE 4 N	H000		SENSE			REG ISENCORE 4 NR	H020	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		SNS CORE XW P	H020	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		SNS CORE XW N	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		SNS CORE R P	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		SNS CORE R N	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		SNS CPU VAXG P	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		SNS CPU VAXG N	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		SNS CPU VCORE P	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		SNS CPU VCORE N	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		SNS P1V05 IOVDD XW P	H000	ISNS_CPU_CORE	SNS_DIFF_PHY	SENSE		SNS P1V05 IOVDD XW N	<table><tr><td>Electrical Constraint Set</td><td>Physical</td><td>Spacing</td><td>Voltage</td><td>DIDT</td><td>NO_TEST</td></tr><tr><td>AXG</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>H010</td><td>POWER_PHY</td><td>POWER</td><td>12V</td><td></td><td>REG LVCC U7330</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG PWM CPUAXG</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG PWM CPUAXG R</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG PHASE CPUAXG</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG BOOT CPUAXG</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG BOOT CPUAXG RC</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG UGATE CPUAXG</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG LGATE CPUAXG</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_SWITCH</td><td>12V</td><td>TRUE</td><td>REG SNUBBER CPUAXG</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td>PPCPUAXG S0 SENSE</td></tr><tr><td>H000</td><td>ISNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>REG ISENAXG P</td></tr><tr><td>H000</td><td>ISNS_CPU_AXG</td><td>SNS_DIFF_PHY</td><td>SENSE</td><td></td><td>REG ISENAXG N</td></tr><tr><td>H000</td><td></td><td>SENSE</td><td></td><td></td><td>REG ISENAXG 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R</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG CPUAXG_TM</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG CPUAXG_TCOMP</td></tr><tr><td>H000</td><td>VR_CTL_PHY</td><td>VR_CTL</td><td></td><td></td><td>REG CPUAXG_SW_FREQ</td></tr><tr><td>H000</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td>CPU VIDSCLK</td></tr><tr><td>H000</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td>CPU VIDSCLK R</td></tr><tr><td>H000</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td>CPU VIDALERT L</td></tr><tr><td>H000</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td>CPU VIDALERT R L</td></tr><tr><td>H000</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td>CPU VIDSOUT</td></tr><tr><td>H000</td><td>VR_VID_PHY</td><td>VR_VID</td><td></td><td></td><td>CPU VIDSOUT R</td></tr><tr><td>Output Bus</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>H000</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td>PPVCORE_S0 CPU</td></tr><tr><td>H000</td><td>POWER_PHY</td><td>POWER</td><td>1.1V</td><td></td><td>PPVAXG_S0</td></tr></table>				Electrical Constraint Set	Physical	Spacing	Voltage	DIDT	NO_TEST	AXG						H010	POWER_PHY	POWER	12V		REG LVCC U7330	H000	VR_CTL_PHY	VR_CTL			REG PWM CPUAXG	H000	VR_CTL_PHY	VR_CTL			REG PWM CPUAXG R	H000	POWER_PHY	VR_SWITCH	12V	TRUE	REG PHASE CPUAXG	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG BOOT CPUAXG	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG BOOT CPUAXG RC	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG UGATE CPUAXG	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG LGATE CPUAXG	H000	VR_CTL_PHY	VR_SWITCH	12V	TRUE	REG SNUBBER CPUAXG	H000	POWER_PHY	POWER	1.1V		PPCPUAXG S0 SENSE	H000	ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE		REG ISENAXG P	H000	ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE		REG ISENAXG N	H000		SENSE			REG ISENAXG PR	H000		SENSE			REG ISENAXG NR	H000	ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE		SNS AXG R P	H000	ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE		SNS AXG R N	H000	ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE		SNS AXG XW P	H000	ISNS_CPU_AXG	SNS_DIFF_PHY	SENSE		SNS AXG XW N	ISL6364						H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_COMP	H000	VR_CTL_PHY	VR_CTL			CPUCORE_COMP_RC	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_FB	H000	VR_CTL_PHY	VR_CTL			CPUCORE_FB RC	H000	VR_CTL_PHY	VR_CTL			CPUCORE_FB R 1	H000	VR_CTL_PHY	VR_CTL			CPUCORE_FB R 2	H000	VR_CTL_PHY	VR_CTL			CPUCORE PSICOMP RC	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_PSICOMP	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_HFCOMP	H000		SENSE			REG CPUCORE_VSEN	H000		SENSE			REG CPUCORE_RGND	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_IMON	H000	VR_CTL_PHY	VR_CTL			CPUCORE_IMON R	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_TM	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_SUTH	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_NFSI	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_FDVID	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_IAUTO	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_SW_FREQ	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_RAMPADJ	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_EN_PWR	H000	VR_CTL_PHY	VR_CTL			CPUCORE_EN_PWR R	H000	VR_CTL_PHY	VR_CTL			REG CPUCORE_RSET	H000	VR_CTL_PHY	VR_CTL			REG CPUAXG_COMP	H000	VR_CTL_PHY	VR_CTL			CPUAXG_COMP RC	H000	VR_CTL_PHY	VR_CTL			REG CPUAXG_FB	H000	VR_CTL_PHY	VR_CTL			CPUAXG_FB RC	H000	VR_CTL_PHY	VR_CTL			CPUAXG_FB R 1	H000	VR_CTL_PHY	VR_CTL			CPUAXG_FB R 2	H000	VR_CTL_PHY	VR_CTL			REG CPUAXG_HFCOMP	H000		SENSE			REG CPUAXG_VSEN	H000		SENSE			REG CPUAXG_RGND	H000	VR_CTL_PHY	VR_CTL			REG CPUAXG_IMON	H000	VR_CTL_PHY	VR_CTL			CPUAXG_IMON R	H000	VR_CTL_PHY	VR_CTL			REG CPUAXG_TM	H000	VR_CTL_PHY	VR_CTL			REG CPUAXG_TCOMP	H000	VR_CTL_PHY	VR_CTL			REG CPUAXG_SW_FREQ	H000	VR_VID_PHY	VR_VID			CPU VIDSCLK	H000	VR_VID_PHY	VR_VID			CPU VIDSCLK R	H000	VR_VID_PHY	VR_VID			CPU VIDALERT L	H000	VR_VID_PHY	VR_VID			CPU VIDALERT R L	H000	VR_VID_PHY	VR_VID			CPU VIDSOUT	H000	VR_VID_PHY	VR_VID			CPU VIDSOUT R	Output Bus						H000	POWER_PHY	POWER	1.1V		PPVCORE_S0 CPU	H000	POWER_PHY	POWER	1.1V		PPVAXG_S0
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DRAWING NUMBER
051-9504

REVISION
7.0.0

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SYNC MASTER=D8 MARK

SYNC DATE=02/10/2012

CPU VReg Constraints

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GPU

GPU

GPU

GPU

D

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Constraints

BLC High Voltage Output

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_HV	BLC_HV	*	BLC_CTL_ISO
BLC_HV	*	*	BLC_HV_ISO

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_PHASE	*	*	PHASE_ISO
BLC_PHASE	BLC_PHASE	*	PHASE_SW2SW
BLC_PHASE	POWER	*	PHASE_SW2PWR
BLC_PHASE	GND	*	PHASE_SW2GND

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
BLC_CTL	*	*	BLC_CTL_ISO

	Physical	Spacing	Voltage	DIDT	NO_TEST		
Input	Bus						
E925	POWER_PHY	POWER	12V			PP12V_S0_BLC_VIN2	89 90 91
E926	POWER_PHY	POWER	12V			PP12V_S0_BLC_VINP	89 91
E927	POWER_PHY	POWER	14V			PRE_REG_OUT	89 90 91
E928	POWER_PHY	POWER	3.3V			BLC_P3V3S	89 91
E929	POWER_PHY	POWER	3.3V			BLC_P3V3_REF	89 91
E930	POWER_PHY	POWER	3.3V			BLC_P3V3	
E931	POWER_PHY	POWER	3.3V			PP3V3_S0_BLC_R	89
E932	POWER_PHY	POWER	8V			SPTX_VIN	90
E933	POWER_PHY	POWER	8V			PP8V_BLC	89 90
E934	POWER_PHY	POWER	8V			BLC_VIN2	90 91
E935	POWER_PHY	POWER	14V			BOOST_FET_DRAIN	89
E936	POWER_PHY	POWER	12V			BOOST_VDD	
E937	POWER_PHY	POWER	8V			PP5V_S0_BLC_R	89 90 91
E938	POWER_PHY	POWER	12V			PP12V_S0_BLC_F	91
Local Ground							
E939	BLC_CTL_PHY	BLC_PHASE	8V			BLC_GND_1	90 91
E940	BLC_CTL_PHY	BLC_PHASE	8V			BLC_GND_2	90 91
E941	BLC_CTL_PHY	BLC_PHASE	8V			BLC_GND_3	90 91
E942	GND_PHY	GND	8V			AGND_BLC	90 91
Backlight							
E943	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1	90
E944	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE1_R	90
E945	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2	90
E946	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE2_R	90
E947	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3	90
E948	BLC_CTL_PHY	BLC_PHASE				LED_DRIVER_GATE3_R	90
E949	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_RC_1	90
E950	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_RC_2	90
E951	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_RC_3	90
E952	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS1	90
E953	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS2	90
E954	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_CS3	90
E955	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_C1	90
E956	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_C2	90
E957	BLC_CTL_PHY	BLC_CTL				LED_DRVRS_CS_C3	90
E958	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_1	90
E959	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_2	90
E960	BLC_CTL_PHY	BLC_HV	80V			LED_DRIVER_FDBK_R_3	90
E961	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK1	90
E962	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK2	90
E963	BLC_CTL_PHY	BLC_CTL	80V			LED_DRIVER_FDBK3	90
E964	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1_R	89 90
E965	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2_R	89 90
E966	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3_R	89 90
E967	BLC_CTL_PHY	BLC_CTL				BLC_PWM_1	90 91
E968	BLC_CTL_PHY	BLC_CTL				BLC_PWM_2	90 91
E969	BLC_CTL_PHY	BLC_CTL				BLC_PWM_3	90 91
E970	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF1	90 91
E971	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF2	90 91
E972	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_REF3	90 91
E973	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP1	90
E974	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP2	90
E975	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_COMP3	90
E976	BLC_CTL_PHY	BLC_CTL				BCOMP1	90
E977	BLC_CTL_PHY	BLC_CTL				BCOMP2	90
E978	BLC_CTL_PHY	BLC_CTL				BCOMP3	90
E979	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT1	90
E980	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT2	90
E981	BLC_CTL_PHY	BLC_CTL				LED_DRIVER_FLT3	90
E982	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_1	90
E983	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_2	90
E984	BLC_CTL_PHY	BLC_CTL				LED_FLT_R_3	90
E985	BLC_CTL_PHY	BLC_CTL				PRE_REG_OUT_R	89
E986	BLC_CTL_PHY	BLC_CTL				BOOST_FB	89
E987	BLC_CTL_PHY	BLC_CTL				BOOST_COMP	89
E988	BLC_CTL_PHY	BLC_CTL				BOOST_COMP_C	89
E989	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV	89
E990	BLC_CTL_PHY	BLC_CTL			TRUE	BOOST_GDRV_R	89
E991	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS	89
E992	BLC_CTL_PHY	BLC_CTL				BOOST_ISNS_R	89
E993	BLC_CTL_PHY	BLC_HV	80V			LED_DRVRS_DRAIN_1	90
E994	BLC_CTL_PHY	BLC_HV	80V			LED_DRVRS_DRAIN_2	90
E995	BLC_CTL_PHY	BLC_HV	80V			LED_DRVRS_DRAIN_3	90
OUTPUT BUS							
E996	POWER_BLC_RET	BLC_HV	80V			IS1_BLC_F	91
E997	POWER_BLC_RET	BLC_HV	80V			IS2_BLC_F	91
E998	POWER_BLC_RET	BLC_HV	80V			IS3_BLC_F	91
E999	POWER_BLC_RET	BLC_HV	80V			IS1_BLC	90 91
E1000	POWER_BLC_RET	BLC_HV	80V			IS2_BLC	90 91
E1001	POWER_BLC_RET	BLC_HV	80V			IS3_BLC	90 91
E1002	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_1	91
E1003	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_1	91
E1004	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_2	91
E1005	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_2	91
E1006	POWER_BLC_RET	BLC_HV	80V			BLC_LED_P_3	91
E1007	POWER_BLC_RET	BLC_HV	80V			BLC_LED_N_3	91
E1008	POWER_BLC	BLC_HV	80V			BLC_VOUT1	90 91
E1009	POWER_BLC	BLC_HV	80V			BLC_VOUT2	90 91
E1010	POWER_BLC	BLC_HV	80V			BLC_VOUT3	90 91

SYNC MASTER=D8 MLB		SYNC DATE=12/19/2011	
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Caesar IV (Ethernet/SD)

CIV-specific Physical Rules

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
ENET_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD
ENET_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
SD_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=STANDARD	=STANDARD

Physical Net Type to Rule Map

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
ENET_COMP_PHY	*	ENET_50S
ENET_DIFF_PHY	*	ENET_100D
SD_PHY	*	SD_50S

CIV-specific Spacing Definitions
Ethernet

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENET_DIFF_ISO	*	=6:1_SPACING	?
ENET_DIFF2DIFF	*	=3:1_SPACING	?
ENET_TRANS_ISO	*	1.27 MM	?
COMP_ENET_ISO	*	=4:1_SPACING	?

Constraints
Ethernet

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_DIFF	*	*	ENET_DIFF_ISO
ENET_DIFF	ENET_DIFF	*	ENET_DIFF2DIFF
ENET_TRANS	*	*	ENET_TRANS_ISO
COMP_ENET	*	*	COMP_ENET_ISO
ENET_TRANS	ENET_TRANS	*	ENET_DIFF2DIFF

2 kv isolation

SD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SD_ISO	*	=3:1_SPACING	?

SD

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SD	*	*	SD_ISO

VENI, VIDI, VICI!

Electrical Constraint Set Ethernet		Physical	Spacing	
ENET_MDI_NORM		ENET_DIFF_PHY	ENET_DIFF	ENETCONN MDI P<3..0>
ENET_MDI_NORM		ENET_DIFF_PHY	ENET_DIFF	ENETCONN MDI N<3..0>
		ENET_DIFF_PHY	ENET_TRANS	ENETCONN MDI T P<3..0>
		ENET_DIFF_PHY	ENET_TRANS	ENETCONN MDI T N<3..0>
			ENET_TRANS	ENETCONN MCT3
			ENET_TRANS	ENETCONN MCT2
			ENET_TRANS	ENETCONN MCT1
			ENET_TRANS	ENETCONN MCT0
		ENET_COMP_PHY	COMP ENET	ENET_RDAC
SD				
SD_DATA		SD_PHY	SD	ENET CR DATA<7..0> NO TEST=TRUE
		SD_PHY	SD	SDCONN DATA<7..0>
		SD_PHY	SD	SDCONN DATA R<7..0> NO TEST=TRUE
SD_CMD		SD_PHY	SD	ENET SD CMD NO TEST=TRUE
		SD_PHY	SD	SDCONN CMD NO TEST=TRUE
		SD_PHY	SD	SDCONN CMD R NO TEST=TRUE
SD_CLK		SD_PHY	SD	ENET SD CLK NO TEST=TRUE
		SD_PHY	SD	SDCONN CLK NO TEST=TRUE
		SD_PHY	SD	SDCONN CLK R NO TEST=TRUE
		SD_PHY	SD	ENET MEDIA SENSE
		SD_PHY	SD	ENET SD DETECT_L
		SD_PHY	SD	SDCONN_WP
CIV SPI				
		SPI_50S	SPI	ENET_SCLK
		SPI_50S	SPI	ENET_MISO
		SPI_50S	SPI	ENET_MOSI
		SPI_50S	SPI	ENET_CS_L

AUTO-CONSTRAINTS PG 1

4V5 *

	Physical	Spacing	Netname
U1		PM	4V5_REG_EN
U2	POWER_PHY	POWER	4V5_REG_IN

ACDC_*

	Spacing	Netname
13	PM	ACDC_BURST_EN
14	PM	ACDC_BURST_EN_L

ALL *

	Spacing	Netname
15	PM	ALL SYS PWRGD

AP *

	Spacing	Netname	
16	GENERIC ISO	AP_CLKREQ_L	15 21
17	GENERIC ISO	AP_CLKREQ_L_ISO	15 35
18	GENERIC ISO	AP_CLKREQ_Q_L	35
19	GENERIC ISO	AP_EVENT_L	35 47 48
20	GENERIC ISO	AP_PWR_EN_ISO	15 35
21	PM	AP_RESET_CONN_L	
22	PM	AP_RESET_L	26 35
23	PM	AP_WAKE_L	35

BLC12V_*

	Spacing	Netname
H14	PM	BLC12V_FAULT
H15	PM	BLC12V_FAULT_L
H16	PM	BLC12V_UVLO
H17	PM	BLC12V_UVLO_OUT
H18	PM	BLC12V_UVLO_REF

BLC_*

	Physical	Spacing	Voltage	Netname
H19		GENERIC ISO		BLC_BL
H20		GENERIC ISO		BLC_BL_GATE
H21		GENERIC ISO		BLC_BST
H22		GENERIC ISO		BLC_BST_R
H23		GENERIC ISO		BLC_BYPASS_GATE
H24		GENERIC ISO		BLC_DIM_MCU
H25		PM		BLC_EN
H26		GENERIC ISO		BLC_ENA
H27		GENERIC ISO		BLC_ENA1
H28		PM		BLC_EN_DELAY
H29		PM		BLC_EN_R

BLC *

	Physical	Spacing	Voltage	Netname	
R30		FM		BLC_EXT_BOOT	48 89
R31		FM		BLC_EXT_BOOT_L	89
R32	BLC_CTL_PHY	BLC_CTL	5V	BLC_MCU_AOUT_R	91
R33		GENERIC_ISO		BLC_MCU_BV	89 91
R34		GENERIC_ISO		BLC_MCU_BV_D	91
R35		GENERIC_ISO		BLC_MCU_BV_R	91
R36	SMB_PHY	SMB		BLC_MCU_B_SDA_CONN	89
R37		GENERIC_ISO		BLC_MCU_FLAG_V	89
R38		GENERIC_ISO		BLC_MCU_PWM5	89
R39		GENERIC_ISO		BLC_MCU_PWM5_R	89
R40		FM		BLC_MCU_RESET	89
R41		FM		BLC_MCU_RESET_L	89
R42		FM		BLC_MCU_RESET_R_L	89
R43	XDP_PHY	CLK_JTAG		BLC_MCU_RTCK	89
R44		GENERIC_ISO		BLC_MCU_RXD0	89
R45	XDP_PHY	XDP		BLC_MCU_TCK	89
R46	XDP_PHY	XDP		BLC_MCU_TDI	89
R47	XDP_PHY	XDP		BLC_MCU_TDO	89
R48	XDP_PHY	XDP		BLC_MCU_TMS	89
R49	XDP_PHY	XDP		BLC_MCU_TRST	89
R50	XDP_PHY	XDP		BLC_MCU_TXD0	89
R51		GENERIC_ISO		BLC_MCU_UVLO	89
R52		GENERIC_ISO		BLC_ON	91
R53		GENERIC_ISO		BLC_ON_DRAIN	91
R54		GENERIC_ISO		BLC_ON_R	90 91
R55		GENERIC_ISO		BLC_P_ON	89 90 91
R56		GENERIC_ISO		BLC_P_ON_BYPASS	91
R57		GENERIC_ISO		BLC_P_ON_D	91
R58		GENERIC_ISO		BLC_P_ON_DRAIN	91
R59		GENERIC_ISO		BLC_P_ON_D_R	91
R60		GENERIC_ISO		BLC_P_ON_GATE	91
R61		GENERIC_ISO		BLC_P_ON_R	91
R62		GENERIC_ISO		BLC_SKIP	90
R63		GENERIC_ISO		BLC_SNUB_1	90
R64		GENERIC_ISO		BLC_SNUB_2	90
R65		GENERIC_ISO		BLC_SNUB_3	90
R66		GENERIC_ISO		BLC_UVLO	89 91
R67		GENERIC_ISO		BLC_VIN2_GATE	91
R68		GENERIC_ISO		BLC_VIN2_SRC	91
R69		GENERIC_ISO		BLC_VINP_GATE	91
R70		GENERIC_ISO		BLC_VIN_SNS	90
R71		GENERIC_ISO		BLC_VSYNC	87 89
R72		GENERIC_ISO		BLC_VSYNC_R	89

BT *

	Spacing	Netname	
F80	PM	BT_PWR_EN	35
F81	PM	BT_PWR_RST_L	15 20 35
F82	PM	BT_PWR_RST_L_O	35

BURSTMODE *

	Spacing	Netname
REG	PM	BURSTMODE EN 71
REG	PM	BURSTMODE EN L 6 48 71

CAM_*

	Spacing	Netname	
185	PM	CAM_EXT_BOOT	42 43
186	PM	CAM_PROC_RESET	43
187	PM	CAM_PROC_RESET_L	42 43

CPU_*

	Physical	Spacing	Netname	
P88		PM	CPU_PWRGD_1V05_R	28
P89		PM	CPU_PWRGD_3V3	28
P90		PM	CPU_PWRGD_3V3_R	28
P91		GENERIC_ISO	CPU_SKTOCC	64
P92	CPU_PHY	CPU	CPU_THRMTRIP_3V3	47
P93	CPU_PHY	CPU	CPU_THRMTRIP_R_L	48

DEBUG_*

Spacing	Netname
PM	DEBUG_RESET_L

26 49

DP_*

	Physical	Spacing	Netname	
1855		FM	DP_AUXIO_EN	15 86 88
1856		FM	DP_GPU_MUX_EN	84
1857		GENERIC_ISO	DP_INTPNL_HPD	84 87
1858		GENERIC_ISO	DP_INT_EG_HPD	82 84
1859	TBT_GEN_55S	TBT_GEN	DP_TBTPA_DDC_CLK	85 86
1860	TBT_GEN_55S	TBT_GEN	DP_TBTPA_DDC_DATA	85 86
1861		GENERIC_ISO	DP_TBTPA_HPD	36 86
1862	TBT_GEN_55S	TBT_GEN	DP_TBTPB_DDC_CLK	85 88
1863	TBT_GEN_55S	TBT_GEN	DP_TBTPB_DDC_DATA	85 88
1864		GENERIC_ISO	DP_TBTPB_HPD	36 88

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ENETCONN_*

Spacing	Netname
GENERIC ISO	ENETCONN_TCT

ENET_*

Physical	Spacing	Netname
	GENERIC ISO	ENET_ACT
	GENERIC ISO	ENET_ASF_GPIO
	GENERIC ISO	ENET_CLKREQ_L
	PM	ENET_CLKREQ_L_O
	PM	ENET_CR_1V8_EN
	PM	ENET_CR_1V8_EN_R
	PM	ENET_CR_3V3_EN_L
	PM	ENET_CR_3V3_EN_L_R
	PM	ENET_CR_PWREN
XDP PHY	XDP	ENET_LOW_PWR
	PM	ENET_PWR_EN_L
	PM	ENET_PWR_EN_L_R
	PM	ENET_RESET_L
	PM	ENET_SD_RESET_L
	GENERIC ISO	ENET_SR_DISABLE
	GENERIC ISO	ENET_SR_LX
	PM	ENET_WAKE_L

FAN_*

Spacing	Netname
GENERIC ISO	FAN_0_PWM_FET
GENERIC ISO	FAN_0_PWM_FILT
GENERIC ISO	FAN_0_TACH_FET
GENERIC ISO	FAN_0_TACH_FILT

FBVDD_*

Spacing	Netname
GENERIC ISO	FBVDD_ALTV0

FB_*

Spacing	Netname
GENERIC ISO	FB_A0_VREFC
GENERIC ISO	FB_A0_VREFD
GENERIC ISO	FB_A1_VREFC
GENERIC ISO	FB_A1_VREFD
GENERIC ISO	FB_B0_VREFC
GENERIC ISO	FB_B0_VREFD
GENERIC ISO	FB_B1_VREFC
GENERIC ISO	FB_B1_VREFD
GENERIC ISO	FB_VREF

FET_*

Physical	Spacing	Voltage	Netname
	GENERIC ISO		FET_EN_P12V_S0
	GENERIC ISO		FET_EN_P12V_S0_BLC
	GENERIC ISO		FET_EN_P12V_S0_BLC_R
	GENERIC ISO		FET_EN_P12V_S0_R
	GENERIC ISO		FET_EN_P12V_S5
	GENERIC ISO		FET_EN_P12V_S5_R
	GENERIC ISO		FET_EN_VDDQ_S0
	GENERIC ISO		FET_HDD_SLGSW
POWER PHY	POWER	12V	FET_VCC_U7950
POWER PHY	POWER	12V	FET_VCC_U7970
POWER PHY	POWER	12V	FET_VCC_U7980

FLAG_*

Spacing	Netname
GENERIC ISO	FLAG_V

G3_*

Spacing	Netname
GENERIC ISO	G3_POWERON_L

GND_*

Spacing	Netname
SENSE	GND_SMC_AVSS

GPU_*

Physical	Spacing	Netname
	GENERIC ISO	GPU_ALT_VREF
	GENERIC ISO	GPU_BUFRSTN
	GENERIC ISO	GPU_IFPAB_PLLVDD
	GENERIC ISO	GPU_IFPA_IOVDD
	GENERIC ISO	GPU_IFPB_IOVDD
	GENERIC ISO	GPU_IFPC_IOVDD
XDP PHY	XDP	GPU_JTAG_TDI
XDP PHY	XDP	GPU_JTAG_TDO
XDP PHY	XDP	GPU_JTAG_TMS
XDP PHY	XDP	GPU_JTAG_TRST_L
	GENERIC ISO	GPU_MLS_STRAP0
	GENERIC ISO	GPU_MLS_STRAP1
	GENERIC ISO	GPU_MLS_STRAP2
	GENERIC ISO	GPU_MLS_STRAP3
	GENERIC ISO	GPU_MLS_STRAP4
	GENERIC_ISO	GPU_RESET_L
SPI 50S	SPI	GPU_ROM_CS_L
SPI 50S	SPI	GPU_ROM_CS_L_R
SPI 50S	SPI	GPU_ROM_SI
SPI 50S	SPI	GPU_ROM_SI_R
SPI 50S	SPI	GPU_ROM_SO
SPI 50S	SPI	GPU_ROM_SO_R
SPI 50S	SPI	GPU_ROM_WP_L

HDD_*

Spacing	Netname
GENERIC ISO	HDD_12V_S0_GATE
GENERIC ISO	HDD_OOB_1V00_REF
PM	HDD_PWR_EN
GENERIC ISO	HDD_PWR_EN_L
GENERIC ISO	HDD_PWR_EN_R

I2C_*

Physical	Spacing	Netname
SMB_PHY	SMB	I2C_TCON_MAS_SCL
SMB_PHY	SMB	I2C_TCON_MAS_SDA

IFPD_*

Spacing	Netname
GENERIC ISO	IFPD_RSET

IFPEF_*

Spacing	Netname
GENERIC ISO	IFPEF_RSET

ISNSA_*

Electrical	Physical	Spacing	Netname
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VG3H_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VG3H_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_P1V05_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_P1V05_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_VCCSA_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_CPU_VCCSA_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_FBVDDQ_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_FBVDDQ_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_HDD_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_HDD_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_P1V05_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P12VS0_P1V05_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V05S0_PCH_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V05S0_PCH_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V5S0_CPU_MEM_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P1V5S0_CPU_MEM_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S0_SSD_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S0_SSD_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S4_AP_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P3V3S4_AP_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P5VS0_HDD_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_P5VS0_HDD_P
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_PVDDQS3_DDR_N
SNS_CURRENT	SNS_DIFF_PHY	SENSE	ISNSA_PVDDQS3_DDR_P

8	7	6	5	4	3	2	1
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D

21 49

E44	GENERIC_ISO	LPC_PWRDWN_L	19 26 47 49
E45	GENERIC_ISO	LPC_SERIRO	18 47 49

46	PM	MEMVTT EN	28 64
47	PM	MEMVTT EN L	28

C

E48	GENERIC_ISO	MEM_EVENT_L	29	30	31	32	47	48
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E2E1	XDP_PHY	XDP	JTAG GMUX TMS	20
E2E2	XDP_PHY	XDP	JTAG TBT TDI	15
E2E3	XDP_PHY	XDP	JTAG TBT TDI ISOL	15
E2E4	XDP_PHY	XDP	JTAG TBT TDO	15
E2E5	XDP_PHY	XDP	JTAG TBT TDO ISOL	15
E2E6	XDP_PHY	XDP	JTAG TBT TMS	15
E2E7	XDP_PHY	XDP	JTAG TBT TMS ISOL	15

149	XDP_PHY	XDP	MOJO_RX_L	45 47 48
150	XDP_PHY	XDP	MOJO_TX_L	45 47 48

IS1	GENERIC ISO	OCA FET DRAIN	93
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931	PM	LED DRIVER EN L	90
932	PM	LED DRIVER EN L R	91
933	GENERIC ISO	LED DRIVER OVP1	90
934	GENERIC ISO	LED DRIVER OVP1P	89
935	GENERIC ISO	LED DRIVER OVP1 OUT	90
936	GENERIC ISO	LED DRIVER OVP2	90
937	GENERIC ISO	LED DRIVER OVP2P	91
938	GENERIC ISO	LED DRIVER OVP2 OUT	90
939	GENERIC ISO	LED DRIVER OVP3	91
940	GENERIC ISO	LED DRIVER OVP3P	90
941	GENERIC ISO	LED DRIVER OVP3 OUT	89

929	GENERIC_ISO	OVP_OREF	930
930	GENERIC_ISO	OVP_OUT1	931
931	GENERIC_ISO	OVP_OUT1_R	932
932	GENERIC_ISO	OVP_OUT2	933
933	GENERIC_ISO	OVP_OUT2_R	934
934	GENERIC_ISO	OVP_OUT3	935
935	GENERIC_ISO	OVP_OUT3_R	936

I59	PM	P1V2 S4 EN	43
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I60	PM	P1V8_S4_EN	43
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161	GENERIC ISO	P3V3AP VMON	35
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162	GENERIC ISO	P3V3 S0 EN G	74
163	GENERIC ISO	P3V3 S3 EN G	74

164	GENERIC ISO	P3V42G3H BOOST	73
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GENERIC ISO	P5V S0 EN G	74
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166	GENERIC ISO	PCA9557D RESET L	26 28 34
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PCB		PM	PCB_BLC_EXT_BOOT	21	89
PCB		PM	PCB_BLC_EXT_BOOT_R	15	71
PCB		PM	PCB_BLC_MCU_RESET	21	89
PCB		PM	PCB_BLC_MCU_RESET_R	15	71
PCB		PM	PCB_CAM_RESET	15	21
PCB		PM	PCB_CAM_RESET_R	21	43
PCB		GENERIC_ISO	PCB_DSWMRWM		
PCB	CPU_PHY	CPU	PCB_PECI		21
PCB	CPU_PHY	CPU	PCB_PROCPWRGD		21
PCB		GENERIC_ISO	PCB_RCIN_L		21
PCB		GENERIC_ISO	PCB_RI_L		19
PCB		GENERIC_ISO	PCB_SMBALERT_L		15
PCB		GENERIC_ISO	PCB_SRTCRST_L		18
PCB		GENERIC_ISO	PCB_STRP_TOPBLK_SWP_L		20
PCB		GENERIC_ISO	PCB_SUSWARN_L		15

182	GENERIC_ISO	PCIE_CLKREQ5_GPIO44_L	18
183	PM	PCIE_WAKE_L	19 35 40

184	GENERIC ISO	PEG CLKREQ L	15 18 75
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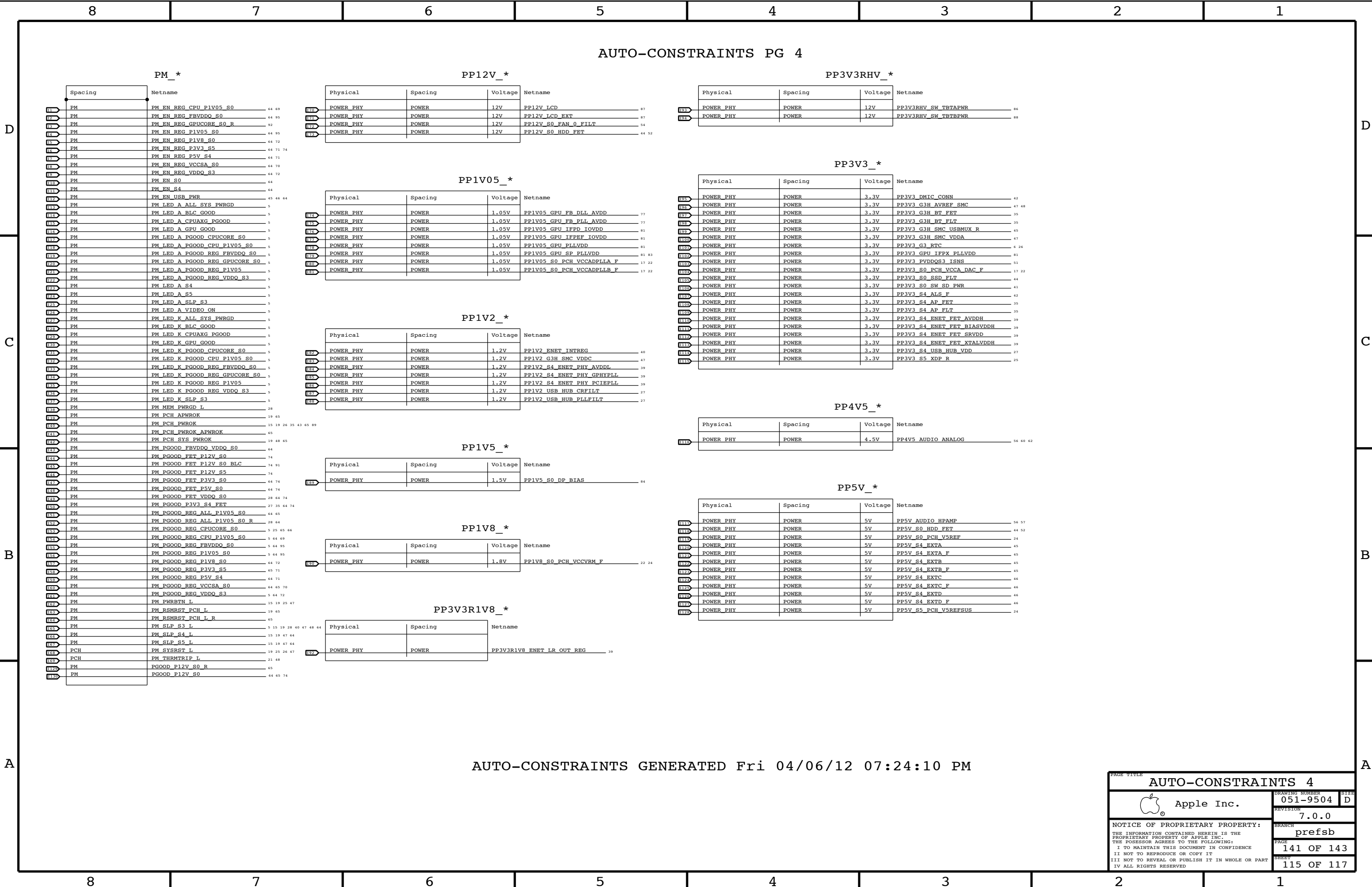
PM	PGOOD P1V5 S0 DLY	28
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186	PCH	PLT RESET L	20
187	PCH	PLT_RST_BUF L	26

PM	PM	PM CLKRUN_L	47	48	49
PM	PM	PM DSW PWRGD	47	48	49
PM	PM	PM EN FET P12V_S0	64	74	
PM	PM	PM EN FET P12V_S0_R	64	74	
PM	PM	PM EN FET P3V3_S0	64	74	
PM	PM	PM EN FET P3V3_S4	64	74	
PM	PM	PM EN FET P5V_S0	64	74	
PM	PM	PM EN FET VDDQ_S0	64	74	
PM	PM	PM EN LDO DDRVTT_S0	64	72	
PM	PM	PM EN REG CPUCORE_S0	64	66	

A

DCBA



PP12V_*

Physical

Spacing

Voltage

Netname

POWER_PHY

POWER

1.05V

PP1V05_GPU_FB_DLL_AVDD

77 77

POWER_PHY

POWER

1.05V

PP1V05_GPU_FB_PLL_AVDD

77 77

POWER_PHY

POWER

1.05V

PP1V05_GPU_IFPD_IOVDD

61 61

POWER_PHY

POWER

1.05V

PP1V05_GPU_IPPEF_IOVDD

61 61

POWER_PHY

POWER

1.05V

PP1V05_GPU_PLLVDD

61 61

POWER_PHY

POWER

1.05V

PP1V05_GPU_SP_PLLVDD

61 83

POWER_PHY

POWER

1.05V

PP1V05_S0_PCH_VCCADPLLA_F

17 22

POWER_PHY

POWER

1.05V

PP1V05_S0_PCH_VCCADPLLB_F

17 22

PP1V2_*

Physical

Spacing

Voltage

Netname

POWER_PHY

POWER

1.2V

PP1V2_ENET_INTREG

40 40

POWER_PHY

POWER

1.2V

PP1V2_G3H_SMC_VDDC

47 47

POWER_PHY

POWER

1.2V

PP1V2_S4_ENET_PHY_AVDDL

39 39

POWER_PHY

POWER

1.2V

PP1V2_S4_ENET_PHY_GPHYPLL

39 39

POWER_PHY

POWER

1.2V

PP1V2_S4_ENET_PHY_PCIEPLL

39 39

POWER_PHY

POWER

1.2V

PP1V2_USB_HUB_CRFLT

27 27

POWER_PHY

POWER

1.2V

PP1V2_USB_HUB_PLLFILT

27 27

PP1V5_*

Physical

Spacing

Voltage

Netname

POWER_PHY

POWER

1.5V

PP1V5_S0_DP_BIAS

64 64

PP1V8_*

Physical

Spacing

Voltage

Netname

POWER_PHY

POWER

1.8V

PP1V8_S0_PCH_VCCVRM_F

22 24

PP3V3R1V8_*

Physical

Spacing

Netname

POWER_PHY

POWER

PP3V3R1V8_ENET_LR_OUT_REG

39 39

PP3V3RHHV_*

Physical

Spacing

Voltage

Netname

POWER_PHY

POWER

12V

PP3V3RHHV_SW_TBTAPWR

86 86

POWER_PHY

POWER

12V

PP3V3RHHV_SW_TBTBPWR

88 88

PP3V3_*

Physical

Spacing

Voltage

Netname

POWER_PHY

POWER

3.3V

PP3V3_DMIC_CONN

42 42

POWER_PHY

POWER

3.3V

PP3V3_G3H_AVREF_SMC

47 48

POWER_PHY

POWER

3.3V

PP3V3_G3H_BT_FET

35 35

POWER_PHY

POWER

3.3V

PP3V3_G3H_BT_FLT

35 35

POWER_PHY

POWER

3.3V

PP3V3_G3H_SMC_USBMUX_R

45 45

POWER_PHY

POWER

3.3V

PP3V3_G3H_SMC_VDDA

47 47

POWER_PHY

POWER

3.3V

PP3V3_G3_RTC

6 26

POWER_PHY

POWER

3.3V

PP3V3_GPU_IPFX_PLLVDD

81 81

POWER_PHY

POWER

3.3V

PP3V3_PVDDQS3_ISNS

51 51

POWER_PHY

POWER

3.3V

PP3V3_S0_PCH_VCCA_DAC_F

17 22

POWER_PHY

POWER

3.3V

PP3V3_S0_SSD_FLT

44 44

POWER_PHY

POWER

3.3V

PP3V3_S0_SW_SD_PWR

41 41

POWER_PHY

POWER

3.3V

PP3V3_S4_ALS_F

42 42

POWER_PHY

POWER

3.3V

PP3V3_S4_AP_FET

35 35

POWER_PHY

POWER

3.3V

PP3V3_S4_AP_FLT

35 35

POWER_PHY

POWER

3.3V

PP3V3_S4_ENET_FET_AVDDH

39 39

POWER_PHY

POWER

3.3V

PP3V3_S4_ENET_FET_BIASVDDH

39 39

POWER_PHY

POWER

3.3V

PP3V3_S4_ENET_FET_SRVDD

39 39

POWER_PHY

POWER

3.3V

PP3V3_S4_ENET_FET_XTALVDDH

39 39

POWER_PHY

POWER

3.3V

PP3V3_S4_USB_HUB_VDD

27 27

POWER_PHY

POWER

3.3V

PP3V3_S5_XDP_R

25 25

PP4V5_*

Physical

Spacing

Voltage

Netname

POWER_PHY

POWER

4.5V

PP4V5_AUDIO_ANALOG

56 60 62

PP5V_*

Physical

Spacing

Voltage

Netname

POWER_PHY

POWER

5V

PP5V_AUDIO_HPAMP

56 57

POWER_PHY

POWER

5V

PP5V_S0_HDD_FET

44 52

POWER_PHY

POWER

5V

PP5V_S0_PCH_V5REF

24 24

POWER_PHY

POWER

5V

PP5V_S4_EXT_A

45 45

POWER_PHY

POWER

5V

PP5V_S4_EXT_A_F

45 45

POWER_PHY

POWER

5V

PP5V_S4_EXTB

45 45

POWER_PHY

POWER

5V

PP5V_S4_EXTB_F

45 45

POWER_PHY

POWER

5V

PP5V_S4_EXTC

46 46

POWER_PHY

POWER

5V

PP5V_S4_EXTC_F

46 46

POWER_PHY

POWER

5V

PP5V_S4_EXTD

46 46

POWER_PHY

POWER

5V

PP5V_S4_EXTD_F

46 46

POWER_PHY

POWER

5V

PP5V_S5_PCH_V5REFSUS

24 24

AUTO-CONSTRAINTS PG 5

PPHV_*

	Physical	Spacing	Voltage	Netname	
18	POWER_PHY	POWER	12V	PPHV_SW_TBTAPWR	86
19	POWER_PHY	POWER	12V	PPHV_SW_TBTBPWR	88

S5_*

	Spacing	Netname	
120	PM	S5_PWRGD	47 65

SMCISNS_*

	Physical	Spacing	Netname	
148	SNS_PHY	SENSE	SMCISNS_CPUAXG	48 51
149	SNS_PHY	SENSE	SMCISNS_CPUCORE	48 51
150	SNS_PHY	SENSE	SMCISNS_GPUCORE	48 51
151	SNS_PHY	SENSE	SMCISNS_P12VG3H	48 51
152	SNS_PHY	SENSE	SMCISNS_P12VS0_CPU_P1V05	48 55
153	SNS_PHY	SENSE	SMCISNS_P12VS0_CPU_VCCSA	48 55
154	SNS_PHY	SENSE	SMCISNS_P12VS0_FBVDDQ	48 51
155	SNS_PHY	SENSE	SMCISNS_P12VS0_HDD	48 51
156	SNS_PHY	SENSE	SMCISNS_P12VS0_P1V05	48 55
157	SNS_PHY	SENSE	SMCISNS_P1V05S0_PCH	48 51
158	SNS_PHY	SENSE	SMCISNS_P1V5S0_CPU_MEM	48 51
159	SNS_PHY	SENSE	SMCISNS_P3V3S0_SSD	48 51
160	SNS_PHY	SENSE	SMCISNS_P3V3S4_AP	48 55
161	SNS_PHY	SENSE	SMCISNS_P5VS0_HDD	48 51
162	SNS_PHY	SENSE	SMCISNS_PVDDQS3_DDR	48 51

PPVBATT_*

	Physical	Spacing	Voltage	Netname	
18	POWER_PHY	POWER	3.3V	PPVBATT_G3_RTC	26
19	POWER_PHY	POWER	3.3V	PPVBATT_G3_RTC_R	26

SATALED_*

	Spacing	Netname	
172	GENERIC_ISO	SATALED_L	44

SDCONN_*

	Spacing	Netname	
173	GENERIC_ISO	SDCONN_DETECT	41
174	GENERIC_ISO	SDCONN_ILIM	41
175	GENERIC_ISO	SDCONN_OC_L	41

SMCVSNS_*

	Physical	Spacing	Netname	
162	SNS_PHY	SENSE	SMCVSNS_CPUAXG	48 51
163	SNS_PHY	SENSE	SMCVSNS_CPUCORE	48 51
164	SNS_PHY	SENSE	SMCVSNS_GPUCORE	48 51
165	SNS_PHY	SENSE	SMCVSNS_P12VG3H	48 51
166	SNS_PHY	SENSE	SMCVSNS_P1V05S0_PCH	48 51
167	SNS_PHY	SENSE	SMCVSNS_P1V5S0_CPU_MEM	48 51
168	SNS_PHY	SENSE	SMCVSNS_P3V3S0	48 51
169	SNS_PHY	SENSE	SMCVSNS_P5VS0_HDD	48 51
170	SNS_PHY	SENSE	SMCVSNS_PVDDQS3_DDR	48 51

PU_*

	Spacing	Netname	
15	GENERIC_ISO	PU_U6900	64

SD_*

	Spacing	Netname	
176	GENERIC_ISO	SD_DETECT_LVL	41

PWR_*

	Spacing	Netname	
16	PM	PWR_BTN	6 48
17	PM	PWR_BTN_R	6

SLG_*

	Spacing	Netname	
177	PM	SLG_ENET_RESET_L	41
178	PM	SLG_ENET_RESET_R_L	41

SMC_*

	Physical	Spacing	Netname	
179		GENERIC_ISO	SMC_ACDC_ID	6 48
180		GENERIC_ISO	SMC_ACDC_ID_R	6
181		PM	SMC_ASSERT_RTCRST	48
182		GENERIC_ISO	SMC_BLC_MUX_RX_L	48 89
183		GENERIC_ISO	SMC_BLC_MUX_TX_L	48 89
184		GENERIC_ISO	SMC_CPU_CATERR_L	47 48
185	CPU_PHY	CPU	SMC_CPU_PECI	48
186		PM	SMC_DELAYED_PWRGD	38 47 48 65
187		GENERIC_ISO	SMC_DP_HPD_L	47 48
188	CLK_XTAL	XTAL	SMC_EXTAL	47 48
189		GENERIC_ISO	SMC_FAN_0_CTL	47 54
190		GENERIC_ISO	SMC_FAN_0_TACH	47 54
191		GENERIC_ISO	SMC_GFX_OVERTEMP	47 48 82
192		GENERIC_ISO	SMC_GFX_OVERTEMP_Q	82
193		GENERIC_ISO	SMC_GFX_OVERTEMP_R_L	82
194		PM	SMC_GFX_THROTTLE_L	47 82
195		PM	SMC_GFX_THROTTLE_R_L	82
196		PM	SMC_LRESET_L	26 47
197		PM	SMC_MANUAL_RST_L	48
198		PM	SMC_ONOFF_L	47 48
199		SENSE	SMC_OOB1_RX_CN	44
200		SENSE	SMC_OOB1_RX_FILT	44 52
201		SENSE	SMC_OOB1_RX_L	47 52
202		SENSE	SMC_OOB1_RX_R	52
203		SENSE	SMC_OOB1_TX_L	47 48
204		SENSE	SMC_OOB2_RX_L	44 48
205		SENSE	SMC_OOB2_TX_L	44 48
206	CPU_PHY	CPU	SMC_PECI_L	47 48
207	CPU_PHY	CPU	SMC_PECI_L_R	48
208		PM	SMC_PME_S4_WAKE_L	35 47 48
209		GENERIC_ISO	SMC_PM_G2_EN	47 48 74
210		PM	SMC_PM_PCH_SYS_PWROK	47 48
211		CPU	SMC_PROCHOT	47 48
212		PM	SMC_RESET_L	47 48 49

REG_*

	Physical	Spacing	Netname	
18		PM	REG_CPUAXG_PGOOD	5 66
19		PM	REG_CPUCORE_PGOOD	66
190	CPU_PHY	CPU	REG_CPUCORE_VRHOT_L	66
191		PM	REG_CPU_P1V05S0_PGOOD	69
192	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET0	95
193	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET1	95
194	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SET1_R	95
195	VR_CTL_PHY	VR_CTL	REG_FBVDDQ_SREF	95
196		PM	REG_P1V8S0_PGOOD	72
197		PM	REG_P3V3S5_PGOOD	71
198		PM	REG_P5VS4_PGOOD	71
199		PM	REG_VCCSA0_PGOOD	70
200		PM	REG_VDDQS3_PGOOD	72

SMBUS_*

	Physical	Spacing	Netname	
190	SMB_PHY	SMB	SMBUS_PCH_CLK	18 50
191	SMB_PHY	SMB	SMBUS_PCH_CLK_R	50
192	SMB_PHY	SMB	SMBUS_PCH_DATA	18 50
193	SMB_PHY	SMB	SMBUS_PCH_DATA_R	50
194	SMB_PHY	SMB	SMBUS_SMC_0_S0_SCL	47 50
195	SMB_PHY	SMB	SMBUS_SMC_0_S0_SDA	47 50
196	SMB_PHY	SMB	SMBUS_SMC_1_S0_SCL	47 50
197	SMB_PHY	SMB	SMBUS_SMC_1_S0_SDA	47 50
198	SMB_PHY	SMB	SMBUS_SMC_2_S4_SCL	47 50
199	SMB_PHY	SMB	SMBUS_SMC_2_S4_SDA	47 50
200	SMB_PHY	SMB	SMBUS_SMC_3_SCL	47 50
201	SMB_PHY	SMB	SMBUS_SMC_3_SDA	47 50
202	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SCL	47 48 50
203	SMB_PHY	SMB	SMBUS_SMC_4_ASF_SDA	47 48 50
204	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SCL	47 48 50
205	SMB_PHY	SMB	SMBUS_SMC_5_G3H_SDA	47 48 50

RTC_*

	Spacing	Netname	
171	PM	RTC_RESET_L	18 26 48
172	PM	RTC_RESET_L_R	48

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F17	SMB_PHY	SMB	SML_PCH_0_CLK	18 50
F18	SMB_PHY	SMB	SML_PCH_0_DATA	18 50
F19	SMB_PHY	SMB	SML_PCH_1_CLK	18 50
F20	SMB_PHY	SMB	SML_PCH_1_DATA	18 50

CSPI_*SSD_*

SYS_*

BTSNS_*UVP_*AVTCLAMP_*

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